eCOA User Requirements Specification (URS)

Specification Version: 1.0 / May 18, 2026

Prepared for Study Sponsor: Terremoto Biosciences, Inc.

Protocol Number: TER-4480-C01

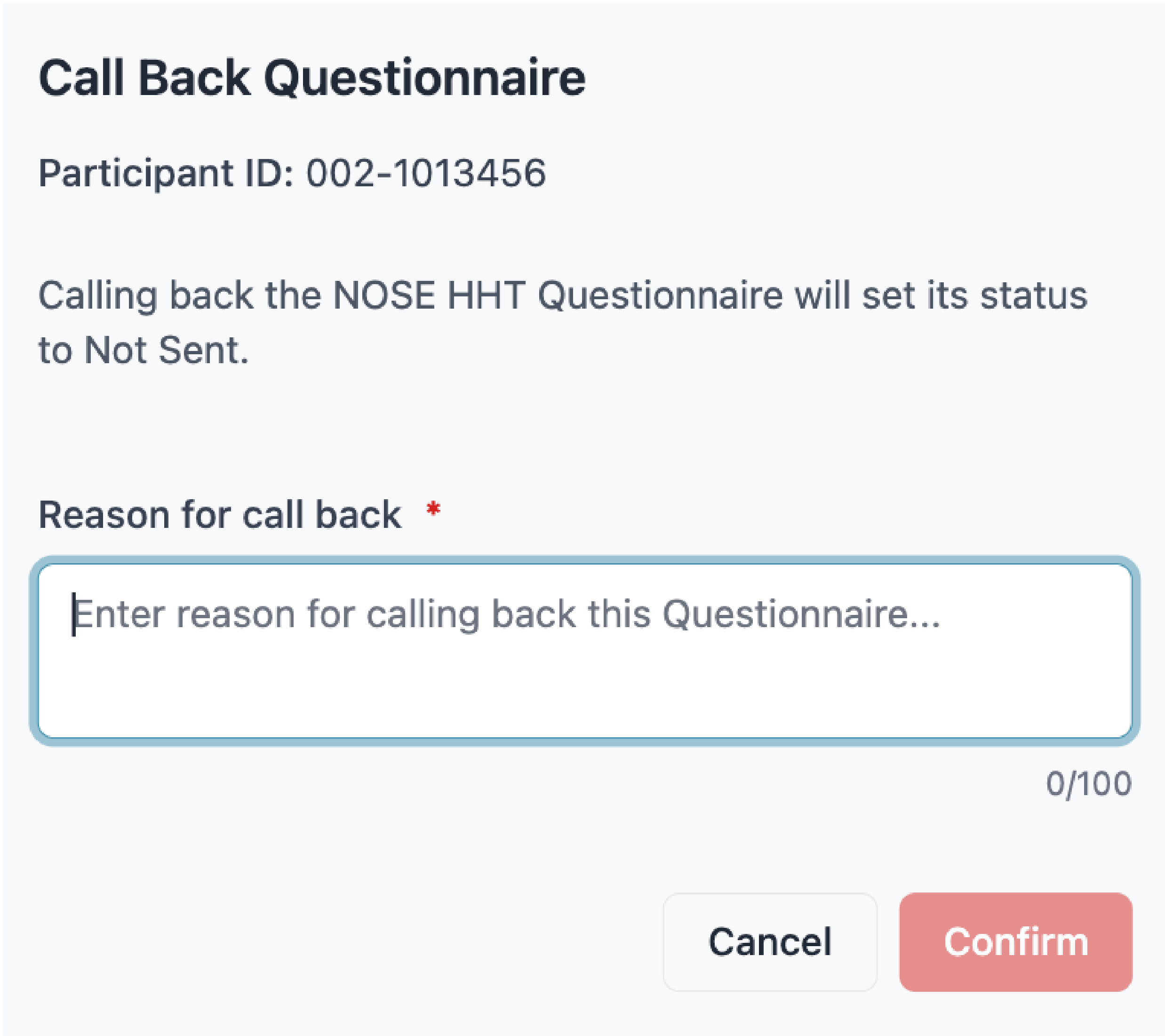
Protocol Version: 1.0

Protocol Date: XX-XX-2026

CONFIDENTIAL

Generated from `spec/` on May 18, 2026

Contents


Call Back Questionnaire
Participant ID: 002-1013456
Calling back the NOSE HHT Questionnaire will set its status to Not Sent.
Reason for call back *
Enter reason for calling back this Questionnaire...
0/100
Cancel
Confirm

Introduction and Purpose

CureHHT eCOA solution is composed of a sponsor portal and a mobile application with available web-based interface, which are integrated with each other and serve as the primary user interfaces for recording of nosebleed events and completion of participant questionnaires for the TER-4480-C01 study.

The Sponsor Portal is the web-based interface used by clinical staff — Study Coordinators, Clinical Research Associates, and Administrators — to manage participant linking, monitor trial progress, distribute questionnaires, and review submitted data. The Mobile Application is the participant-facing interface used to record daily nosebleed events and complete questionnaires sent from the Sponsor Portal. The two components are integrated in such that data captured in the Mobile Application is synchronized to the Sponsor Portal and onward to Rave EDC, and questionnaires initiated in the Sponsor Portal are delivered to the participant's device.

Document Revision History

Version	Date	Author	Description of Changes
1.0	25-Mar-2026	Elvira Koliadina	Initial draft

Signatures

Trial Sponsor Agreement

By my signature below I confirm that the settings have been reviewed and approved within the Trial Sponsor organization.

Approver	Signature
First Name Last Name Study Manager	
First Name Last Name Quality Assurance	
First Name Last Name Data Management	
Print Name & Title	Signature / Date

CureHHT Personnel Agreement

By my signature below I confirm the information contained herein will be implemented as appropriate by CureHHT on behalf of the Trial Sponsor.

Approver	Signature
First Name Last Name Lead Developer	
Elvira Koliadina Project Manager	
Print Name & Title	Signature / Date

SYSTEM-WIDE STANDARDS

4.1 Common User Interface Elements and Actions

The system shall use a consistent set of commonly displayed elements, buttons, and interaction patterns across the Sponsor Portal and Mobile Application. Where these terms are used in subsequent requirements, their behavior shall be interpreted according to the definitions provided in this section, unless specified otherwise.

4.1.1 Buttons

Element	Behavior
Confirm	Confirms a pending action displayed in a dialog. Proceeds with the action upon click.
Cancel	Discards any unsaved changes and returns the user to the previous state. No data is modified.
OK	Acknowledges a system message or notification and dismisses it. No action is confirmed or cancelled.
Submit	Finalizes and sends the current form or questionnaire data. Once submitted, the action cannot be undone unless explicitly permitted by the relevant requirement.
Next	Advances to the next step in a sequence. Data entered in the current step is preserved.
Back	Returns to the previous step in a sequence. Previously entered data is preserved.
Continue	Resumes a flow that was interrupted or paused. Returns the user to the point where they left off.
More details	Opens a detailed view displaying the full record associated with the current row. Represented as a button or link affordance on the row.

Element	Behavior
Close	Dismisses a modal or dialog without performing any action. No data is modified.

4.1.2 UI Elements

Element	Behavior
Tab	A navigation element that filters a list by a category.
Search Input	A single text input that filters a list in real time across specified fields.
Status Badge	An inline label displaying the current status of a record.

4.1.3 Actions

Element	Behavior
Edit	Opens an existing record for modification. Changes are not applied until confirmed or submitted. Represented by a pencil icon.

4.1.4 Interaction Patterns

Element	Behavior
Real-time Filtering	A list updates as the user types without requiring explicit submission.
Preserved Search State	The search input is retained when the user navigates between tabs.

4.1.5 Dialog Patterns

Element	Behavior
Acknowledgement Dialog	Displays a message informing the user of a system event or outcome. Requires the user to click OK to dismiss. No action is cancelled or confirmed.
Confirmation Dialog	Displays the consequence of a pending action. Requires the user to either confirm or cancel before the action is applied.
Reason Dialog — Free Text	Displayed before a high-impact action. Requires the user to enter a free text reason of up to 100 characters before proceeding. The action SHALL NOT proceed until a reason is provided.
Reason Dialog — Predefined List	Displayed before a high-impact action. Requires the user to select a reason from a sponsor-defined list before proceeding. The action SHALL NOT proceed until a reason is selected.

4.1.6 Sponsor Portal User Status

The system maintains a defined set of status values for Sponsor Portal user accounts.

Status	Description
Pending Activation	The account has been created but the user has not yet accepted the invitation or set up their credentials.
Active	The account has been created and the user can log in and perform actions per their role and permissions.
Inactive	The account has been deactivated and the user can no longer log in.

4.1.7 Participant Status

The system maintains a defined set of status values representing the participant's linking, trial participation, device, and data synchronization state.

Status	Description
Not Connected	The participant record has been synced from Rave EDC to the Sponsor Portal but the mobile application has not been linked to the Sponsor Portal.
Pending	An invitation has been sent to the participant but the participant has not yet linked their device.
Expired	An invitation has been sent to the participant but expired before the participant linked their device.
Linked - Awaiting Start	The participant's device is linked to the Sponsor Portal but Diary Data Synchronization has not yet been activated.
Trial Active	Diary Data Synchronization has been activated and diary entries are being transmitted to the Sponsor Portal and Rave EDC.
Disconnected	A Study Coordinator has manually disconnected the participant in the Sponsor Portal. Sponsor-specific rules remain applied to the mobile application but data is not syncing. Data will resume syncing when the participant is reconnected.
Not Participating	The participant has been marked as not participating following disconnection. Sponsor-specific rules are no longer applied to the mobile application.

4.1.8 Participant Status Transitions

The following transitions are valid. All other transitions are prohibited.

From	To	Trigger
Not Connected	Pending	Study Coordinator generates linking code
Pending	Linked - Awaiting Start	Participant successfully enters linking code for the first time

From	To	Trigger
Pending	Expired	The linking code expires as the participant did not use it
Expired	Pending	Study Coordinator regenerates linking code
Linked - Awaiting Start	Trial Active	Study Coordinator starts trial
Linked - Awaiting Start	Disconnected	Study Coordinator disconnects participant
Trial Active	Disconnected	Study Coordinator disconnects participant
Disconnected	Pending	Study Coordinator initiates reconnection
Disconnected	Not Participating	Study Coordinator marks participant as not participating
Not Participating	Disconnected	Study Coordinator reactivates participant

4.1.9 Questionnaire Status

The system maintains a defined set of status values for Questionnaire instances.

Status	Description
Not Sent	No active Questionnaire exists for the current Cycle and a new one may be sent.
Sent	The Questionnaire has been transmitted to the participant's mobile application and is awaiting participant completion.
Delivery Failed	Transmission of the Questionnaire to the participant's mobile application was unsuccessful and the participant has not received it.
Ready to Review	The participant has submitted answers and the Study Coordinator may finalize.

Status	Description
Closed	The Questionnaire for that cycle has been completed by the participant and is finalized by the Study Coordinator. Subsequent Questionnaires for that cycle cannot be generated.

4.2 Status Registry

(No content found for 4.2 — manifest references [])

4.3 User Roles and Permissions

This section defines the role model used across the solution: the generic role templates provided by the platform, the inventory of actions subject to access control, and the sponsor portal deployment's role assignments and permissions. All functional requirements that reference roles or permissions depend on the definitions established here.

4.3.1 Customizable Role-Based Access Control

REQ ID: `DIARY-PRD-rbac-customizable`

Refines: `DIARY-PRD-action-inventory`, `DIARY-PRD-role-definitions`, `DIARY-GUI-role-swi`
`CAL-PRD-role-definitions`

Overview

The system supports multi-role assignment and seamless role switching without disrupting active sessions. Self-privilege-escalation is prevented structurally by disallowing users to modify role assignments of their own account.

Rationale

Clinical trial sponsor portals are routinely staffed by individuals who hold more than one operational role across studies or sites (e.g. a Study Coordinator who is also an Administrator on a sister study). Forcing those users to maintain separate accounts duplicates audit trails, fragments responsibility, and increases the operational burden of account lifecycle. Allowing multi-role assignment with seamless switching keeps a single accountable identity per individual while supporting the legitimate need to perform actions in different capacities. The structural prohibition on self-modification of role assignments is a defense-in-depth control against privilege escalation that does not depend on any single permission check elsewhere in the system.

Assertions

A. The System SHALL allow a **User Account** to be assigned one or more roles.

B. The System SHALL allow an active role to be switched without terminating the user's session.

C. The System SHALL NOT permit a user to modify the role assignments of their own **User Account**.

4.3.2 Action Inventory

REQ ID: `DIARY-PRD-action-inventory`

Refines: `CAL-PRD-role-definitions`

Overview

The Action Inventory enumerates every operation subject to role-based access control. It is the authoritative source for the sponsor-level permissions table — every permission granted or denied in a sponsor deployment must reference an action from this inventory. Actions not listed here cannot be granted or denied. Sponsor deployments may extend the inventory with deployment-specific actions.

Definitions

Action: A discrete operation that a user can perform within the System, subject to role-based access control. An Action either changes system state or retrieves protected data.

Action Inventory: The complete enumerated set of Actions the System recognizes, comprising the platform-level inventory in this requirement and any sponsor-level extensions.

Sponsor-Level Action Extension: An additional Action defined by a specific sponsor deployment beyond the platform Action Inventory.

Action Inventory Table

Category	Action ID	Action
Participant Management	ACT-PAT-001	Link participant
	ACT-PAT-002	Start Trial
	ACT-PAT-003	Disconnect participant
	ACT-PAT-004	Reconnect participant
	ACT-PAT-005	Mark participant as not participating
	ACT-PAT-006	Reactivate participant
	ACT-PAT-007	View participant info
Questionnaire	ACT-QST-001	Send questionnaire
	ACT-QST-002	Call back questionnaire
	ACT-QST-003	Finalize questionnaire
User Account	ACT-USR-001	Create User Account
	ACT-USR-002	Edit User Account
	ACT-USR-003	Deactivate User Account
	ACT-USR-004	Reactivate User Account
	ACT-USR-005	Unlock User Account
	ACT-USR-006	Resend activation email
	ACT-USR-007	Assign role to User Account
	ACT-USR-008	Assign Site to User Account
Site	ACT-SIT-001	View Sites
Audit Log	ACT-AUD-001	View audit log
Administrator Settings	ACT-ADM-001	View Administrator Settings

Rationale

The Action Inventory is the single source of truth for what operations the platform recognizes as access-controlled. Sponsor deployments author their permissions table by referencing these Action IDs; deployments may extend the inventory with sponsor-specific actions but cannot redefine or remove platform actions. Centralizing the inventory ensures that every audit log entry, permission check, and role binding refers to a stable, named operation rather than an ad hoc string, which is essential for FDA 21 CFR Part 11 auditability and for cross-sponsor consistency in the platform's compliance posture.

Assertions

A. The System SHALL enforce role-based access control for every **Action** in the **Action Inventory**.

B. The System SHALL reject any attempt to perform an **Action** by a user whose active role does not have permission for that **Action**.

C. The System SHALL permit an **Action** to proceed when the user's active role has permission for that **Action** and the user is within the permitted scope.

D. The System SHALL support **Sponsor-Level Action Extensions** to the **Action Inventory**.

4.3.3 Role Definitions

REQ ID: `DIARY-PRD-role-definitions`

Refines: `CAL-PRD-role-definitions`

Overview

This requirement defines the generic role templates the platform provides and maps the sponsor portal's named roles to those templates. Sponsor deployments bind permissions to roles via the deployment's permissions table; the platform-side definitions establish the shared vocabulary.

Definitions

Study Coordinator: A portal user responsible for managing day-to-day participant interactions at one or more assigned Sites.

Clinical Research Associate (CRA): A portal user responsible for monitoring Study Coordinator activities and site compliance at one or more assigned Sites.

Administrator: A portal user responsible for managing User Accounts, role assignments, and Site assignments across the study.

Rationale

The sponsor portal's named roles are deployment-facing labels chosen for clarity to clinical staff; the underlying generic role templates (Investigator, Auditor, Administrator) carry the platform-level semantics and are the units of inheritance for cross-sponsor permission patterns. Establishing the mapping at the platform level lets sponsor configurations alter labels and add scope rules without diverging from the platform's shared model, and lets cross-cutting compliance behavior key off the generic template rather than every sponsor's chosen name.

Assertions

A. The System SHALL recognize the following roles in the sponsor portal: **Study Coordinator**, **Clinical Research Associate**, and **Administrator**.

B. The System SHALL map the **Study Coordinator** role to the **Investigator** generic role template.

C. The System SHALL map the **Clinical Research Associate** role to the **Auditor** generic role template.

D. The System SHALL map the **Administrator** role to the **Administrator** generic role template.

4.3.4 Permissions Table

REQ ID: `CAL-PRD-role-definitions`

Overview

The permissions table defines which roles are granted each Action from the Action Inventory (`DIARY-PRD-action-inventory`). It is the sponsor-level binding from platform-defined roles (`DIARY-PRD-role-definitions`) to platform-defined Actions for the Callisto deployment.

Permissions Table

In the following tables, Y indicates the role is granted the Action and `N` indicates the role is denied the Action.

User Account Management

Action ID	Action	Study Coordinator	CRA	Administrator
ACT-USR-001	Create User Account	N	N	Y
ACT-USR-002	Edit User Account	N	N	Y
ACT-USR-003	Deactivate User Account	N	N	Y
ACT-USR-004	Reactivate User Account	N	N	Y
ACT-USR-005	Unlock User Account	N	N	Y
ACT-USR-006	Resend activation email	N	N	Y
ACT-USR-007	Assign role to User Account	N	N	Y

Action ID	Action	Study Coordinator	CRA	Administrator
ACT-USR-008	Assign Site to User Account	N	N	Y

Participant Management

All participant actions are restricted to participants at Sites assigned to the User.

Action ID	Action	Study Coordinator	CRA	Administrator
ACT-PAT-001	Link participant	Y	N	N
ACT-PAT-002	Disconnect participant	Y	N	N
ACT-PAT-003	Reconnect participant	Y	N	N
ACT-PAT-004	Mark participant as not participating	Y	N	N
ACT-PAT-005	Reactivate participant	Y	N	N
ACT-PAT-006	View participant	Y	Y	N

Questionnaire Management

All participant actions are restricted to participants at Sites assigned to the User.

Action ID	Action	Study Coordinator	CRA	Administrator
ACT-QST-001	Send questionnaire	Y	N	N
ACT-QST-002	Call back questionnaire	Y	N	N
ACT-QST-003	Finalize questionnaire	Y	N	N

Sites and Audit Log

Action ID	Action	Study Coordinator	CRA	Administrator
ACT-SIT-001	View All Sites	N	N	Y
ACT-SIT-002	View Assigned Sites	Y	Y	N
ACT-AUD-001	View audit log	Y	Y	Y

Rationale

The Callisto permissions matrix encodes a strict separation between operational roles (Study Coordinator), monitoring roles (CRA), and administrative roles (Administrator). Study

Coordinators perform all participant- and questionnaire-facing actions because the operational workflow is theirs alone; CRAs are read-only by design, with view access to participants and the audit log at their assigned sites so they can monitor compliance without altering trial state; Administrators are confined to User Account lifecycle because their accountability is account governance, not clinical operations. Every cell in the matrix is defensible against the FDA 21 CFR Part 11 principle of separation of duties and against the operational principle that a single mistake by any role should not cascade across boundaries.

Assertions

A. The System SHALL enforce the per-role action permissions defined in the **Permissions Table** below.

4.3.5 Role Switching — Interface Behavior

REQ ID: `DIARY-GUI-role-switching`

Overview

This requirement specifies how the sponsor portal interface exposes role switching to users assigned more than one role.

Definitions

Role Selector: UI element displayed in the header that shows the user's currently active role and allows users with multiple assigned roles to switch their active role.

Display

Interaction

Rationale

Users assigned multiple roles need an unobtrusive, always-visible affordance to confirm and change which role is currently active, since the active role determines visible data and available actions throughout the portal. Hiding the selector from single-role users keeps the header uncluttered for the common case. Omitting a confirmation step keeps role switching fast for users who switch many times per session; the underlying audit log already records role context for every action so the cost of an accidental switch is bounded.

Assertions

A. The interface SHALL display the **Role Selector** in the header for any user assigned two or more roles.

B. The interface SHALL NOT display the **Role Selector** for a user assigned exactly one role.

C. The **Role Selector** SHALL display the user's currently active role.

D. When opened, the **Role Selector** SHALL display the complete list of the user's assigned roles and SHALL visually indicate which role is currently active.

E. When the user selects a role from the **Role Selector**, the interface SHALL set that role as the active role.

F. When the active role changes, the interface SHALL load the default landing view for the selected role.

G. The **Role Selector** SHALL NOT present a confirmation step before switching roles.

4.4 Questionnaire Score Display

4.4.1 Questionnaire Score Display Prohibition

REQ ID: `CAL-PRD-questionnaire-score-display-prohibition`

Overview

The platform does not permit calculated questionnaire scores to be displayed to any user.

Rationale

For the Callisto deployment, calculated questionnaire scores are clinical research outputs that belong in the analysis pipeline, not the user-facing interface. Displaying a score to a Study Coordinator or to a participant could bias subsequent participant responses (priming effects in patient-reported outcome instruments are well documented), undermine the blinding required by the trial protocol, and create the impression that the score is itself a clinical finding for which the portal is the authoritative surface. The score is computed and recorded for downstream analysis; the display surface stays free of it across every role and every screen.

> **Follow-up — configurability**: This requirement currently encodes the only option implemented in code. Future sponsors may require different rules; introduce a configurable seam (e.g. a parameter on the CAL-PRD-* parent, or a new platform-side template the CAL- REQ Satisfies) when the need arises. Until that seam exists, this REQ is normative for the Callisto deployment.

Assertions

A. The System SHALL NOT display calculated Questionnaire Score to any user.

4.5 Audit Log Visibility

This section defines per-role visibility scope of audit log records and the user interface presentation of the **Audit Log View**. The content captured in each **Audit Log Entry**.

4.5.1 Audit Log View — Common Presentation

REQ ID: `DIARY-GUI-audit-log-common`

Refines: `DIARY-GUI-audit-log-study-coordinator`, `DIARY-GUI-audit-log-cra`, `DIARY-GUI-audit-log-administrator`

Overview

The **Audit Log View** uses a consistent presentation across all roles. This requirement defines the shared display, ordering, content, and empty state behavior. Role-specific GUI requirements refine this base by adding role-specific columns, controls, and entry points.

Display

Content

Empty State

Action Display

Detail View

Rationale

A consistent audit log presentation across roles reduces training overhead, allows reviewers comparing entries across roles to read them with the same mental model, and concentrates accessibility, formatting, and column-ordering decisions in one place. Reverse chronological ordering surfaces the most recent activity first, matching the typical investigative workflow of “what just happened?”. The Details column carries the free-text reason that originated with the Action so reviewers do not have to drill into the raw record to understand intent. The raw-text **More details** affordance preserves the full record for inspectors who need every field, while keeping the table dense enough to scan quickly.

Assertions

A. The interface SHALL display the **Audit Log View** as a table containing at minimum the following columns: Timestamp, Action, User, Details.

B. The interface SHALL display Audit Log Entries in reverse chronological order, with the most recent entry first.

C. When an **Audit Log Entry** corresponds to an **Action** that included a free text input from the user (such as a reason, justification, or note), the interface SHALL include that text in the Details column.

D. The interface SHALL display in the Details column a human-readable summary of the **Action** that includes, where applicable, the previous state of the affected record and the parameter or option selected as part of the **Action**.

E. When the **Audit Log View** contains no Audit Log Entries within the active scope or filter selection, the interface SHALL display a message indicating no results were found.

F. The interface SHALL display the **Action** name in the Action column.

G. When an **Action** has parameters that distinguish meaningful variants (such as **Questionnaire Type**), the interface SHALL append the parameter to the **Action** name in the Action column for human readability.

H. The interface SHALL provide a **More details** button on each **Audit Log Entry** that, when selected, displays the full **Audit Log Entry** record as raw text.

4.5.2 Study Coordinator Audit Log View

REQ ID: `DIARY-GUI-audit-log-study-coordinator`

Overview

The **Study Coordinator Audit Log View** presents the **Study Coordinator**'s own Actions. **Participant**-related Actions are displayed alongside a **Participant ID** column.

Display

Controls

Rationale

A Study Coordinator's investigative workflow centers on individual participants — confirming what was done for a specific participant during a specific visit, or assembling documentation in response to a monitoring request. Surfacing **Participant ID** as a first-class column and providing a dedicated search input collapses the most common filter operation to a single keystroke. Limiting the view to the Coordinator's own actions reinforces the separation-of-duties model encoded in the per-role scope: Coordinators are accountable for their own audit trail and do not need (and should not have) visibility into peer actions.

Assertions

A. The interface SHALL display the **Participant ID** column in addition to the columns defined in DIARY-GUI-audit-log-common.

B. The interface SHALL provide a **Participant ID** search input that filters the displayed Audit Log Entries by **Participant ID** in real time.

4.5.3 CRA Audit Log View

REQ ID: `DIARY-GUI-audit-log-cra`

Overview

The **CRA Audit Log View** is accessed by selecting a **Site** from the **CRA**'s list of assigned Sites and presents **Study Coordinator** Actions for the selected **Site**. A **Study Coordinator** filter and a **Participant ID** search input enable focused review.

Display

Controls

Rationale

CRAs perform site monitoring — verifying that Study Coordinator activity at a specific Site is compliant and consistent with protocol. The single-Site scope makes the unit of monitoring explicit (a CRA reviews one Site at a time, not their entire assignment portfolio in a single view), the Site name in the header keeps the active scope visible, and the Coordinator selector with an "all Coordinators" default supports both audit-wide review and Coordinator-specific drill-down without forcing the CRA to apply filters manually. The Participant ID search supports the same per-participant investigation pattern the Study Coordinator view supports, but scoped to the Site under review.

Assertions

A. The interface SHALL display the **Participant ID** and **Site** columns in addition to the columns defined in DIARY-GUI-audit-log-common.

B. The interface SHALL display the name of the selected **Site** in the **Audit Log View** header.

C. The interface SHALL provide a **Study Coordinator** selector that filters the displayed Audit Log Entries by the selected **Study Coordinator**.

D. The interface SHALL provide a default selection in the **Study Coordinator** selector that displays Audit Log Entries for all Study Coordinators at the selected **Site**.

E. The interface SHALL provide a **Participant ID** search input that filters the displayed Audit Log Entries by **Participant ID** in real time.

4.5.4 Administrator Audit Log View

REQ ID: `DIARY-GUI-audit-log-administrator`

Overview

The Administrator audit log view is presented as a dedicated tab inside the Administrator dashboard, alongside the User Account management tab where Administrator actions originate.

Rationale

Administrator actions are scoped to User Account management; placing the audit log view on the same dashboard as the User Account management surface keeps the Administrator's investigative loop tight — review an account, switch to the audit tab, inspect the history of changes to that account, switch back. A separate top-level audit log surface would force navigation away from the Administrator's working context for what is fundamentally an inline review activity.

Assertions

A. The interface SHALL present the **Administrator Audit Log View** on a dedicated tab within the **Administrator** dashboard, alongside the **User Account** management tab.

4.6 Notification Behavior

4.6.1 Notification Behavior

REQ ID: `DIARY-PRD-notification-behavior`

Overview

This section defines system-wide standards for **Push Notifications** delivered by the **Mobile Application** to the **Participant**'s device. The standards defined herein apply to all notifications generated by the **Mobile Application** and shall not be restated within individual notification requirements.

Push Notifications delivered by the **Mobile Application** serve as nudges that prompt the **Participant** to open the application.

Definitions

Push Notification: A notification delivered by the **Mobile Application** to the **Participant**'s device notification center via the operating system's notification service.

Tap Behavior

Offline Delivery

Rationale

Push notifications in the Diary Platform are nudges, not interaction surfaces: their job is to bring the participant back into the app, where the full clinical context, validation, and audit trail are available. Two consequences follow. First, tapping a notification always opens the **Main Screen** — never a deep link into a partial workflow — so the participant resumes from a known, audit-traceable starting point. Second, the platform never offers response actions inside the notification body (e.g. "Submit answer", "Acknowledge"), because actions taken without the in-app context bypass authentication, validation, and the data-attribution evidence chain. Offline delivery semantics ensure that a notification generated while the device is disconnected is not dropped; the operating system's deferred-delivery mechanism is sufficient for the nudge purpose and avoids the complexity of an app-level re-notification scheduler.

Assertions

A. When the **Participant** taps a **Push Notification** delivered by the **Mobile Application**, the System SHALL open the **Mobile Application** to the **Main Screen**.

B. The System SHALL NOT include response actions within the body of a **Push Notification**.

C. When the **Mobile Application** is offline at the time a **Push Notification** is generated, the System SHALL deliver the **Push Notification** when the **Mobile Application** next establishes connectivity.

4.7 Help and Resources

4.7.1 Help and Resources

REQ ID: `DIARY-PRD-help-resources`

Overview

The **Sponsor Portal** provides users with access to in-portal help materials including video tutorials, frequently asked questions, an interactive guided tour, and a downloadable user guide. The materials surfaced are role-aware so that each user is presented with content relevant to their responsibilities. The detailed inventory of materials, tour content, and FAQ entries is maintained in internal documentation; this requirement defines only the platform-level expectation that the help affordance exists and is available to all **Sponsor Portal** roles.

Definitions

Help and Resources: The collection of in-portal help materials available to **Sponsor Portal** users, comprising video tutorials, frequently asked questions, an Interactive Tour, and a downloadable user guide.

Interactive Tour: A guided walkthrough of the **Sponsor Portal** interface presented to a user on first login and replayable on demand from **Help and Resources**.

Availability

Content

Interactive Tour

Rationale

Clinical staff using the **Sponsor Portal** are not full-time users of the system — they engage with it intermittently during their participant-facing workday. In-portal, always-available help is essential to keep training overhead low and to support correct use without external documentation lookups. Role-aware content keeps each user's help surface focused on what they can actually do (a Study Coordinator should not have to filter through Administrator topics, and vice versa). The **Interactive Tour** on first login bootstraps new users into the portal's primary surfaces; replayability supports occasional refreshers without re-running onboarding. This requirement deliberately defines only the affordance and the major content categories — the precise tour script, FAQ entries, and tutorial inventory are operational artifacts maintained in internal documentation, since they evolve continuously with the product.

Assertions

A. The **System** SHALL make **Help and Resources** available to every **Administrator**, **Clinical Research Associate**, and **Study Coordinator** from any screen of the **Sponsor Portal**.

B. The **System** SHALL present **Help and Resources** content scoped to the user's active **Role**.

C. Help and Resources SHALL include video tutorials, frequently asked questions, the **Interactive Tour**, and a downloadable user guide.

D. The **System** SHALL present the **Interactive Tour** to a user on their first successful login to the **Sponsor Portal**.

E. The **System** SHALL allow a user to replay the **Interactive Tour** on demand from **Help and Resources**.

4.8 Configuration

4.8.1 Configuration Precedence

REQ ID: `DIARY-PRD-configuration-precedence`

Refines: `CAL-PRD-administrator-settings-inventory`

Overview

The platform defines a baseline set of behaviors and default values that apply to every deployment. Each sponsor deployment may, where the platform requirement explicitly permits configuration, override those defaults with values appropriate to that study's protocol. This requirement establishes the precedence rule that resolves any apparent conflict between a platform requirement and a sponsor configuration: when both apply to the same behavior, the sponsor configuration governs. The rule is structural — it ensures that sponsor-specific values such as thresholds, reason lists, reminder times, and enabled features take effect deterministically without requiring each individual platform requirement to restate the precedence relationship. The rule applies only within the bounds set by the platform requirement itself; sponsor configuration cannot grant capabilities the platform does not expose, and cannot override platform behavior that is not declared as configurable.

Definitions

Platform Requirement: A requirement that defines behavior, structure, or default values applicable to every deployment of the **System**. Identified by a requirement ID of the form REQ-pXXXXX or GUI-pXXXXX.

Sponsor Configuration Requirement: A requirement that specifies the values applied to a configurable parameter exposed by a **Platform Requirement** for a specific sponsor deployment. Identified by a requirement ID of the form REQ-CAL-pXXXXX for the sponsor-specific deployment, with analogous prefixes for other deployments.

Configurable Parameter: A value, threshold, list, label, enablement flag, or option that a **Platform Requirement** explicitly declares as configurable per deployment.

Configuration Conflict: A state in which a **Sponsor Configuration Requirement** specifies a value for a **Configurable Parameter** that differs from the default value or behavior described in the corresponding **Platform Requirement**.

Rationale

A multi-sponsor platform must absorb deployment-specific differences (reason lists, reminder cadences, lock thresholds, enabled feature flags) without forking the codebase or rewriting platform requirements per sponsor. The conventional pattern — a single config file silently overrides defaults — is unsuitable for an FDA-regulated platform because it loses traceability: when a behavior is wrong in production, the auditor cannot reconstruct from the requirements which value the sponsor actually configured and why. This requirement encodes a structural precedence rule so every sponsor overlay (`REQ-CAL-pXXXXX` and equivalents) becomes a first-class, citable artifact. The two-way containment — sponsor configuration cannot exceed the platform's exposed configurability, and cannot grant capabilities the platform does not implement — keeps the platform requirements authoritative for the set of behaviors that exist at all, and keeps sponsor requirements authoritative for the specific values within that set. The absence-of-configuration rules (E, F) make the default-vs-opt-in semantics explicit so that a missing sponsor requirement is never ambiguous.

Assertions

A. The **System** SHALL apply the value specified in a **Sponsor Configuration Requirement** in place of the default value or behavior described in the corresponding **Platform Requirement** when a **Configuration Conflict** exists.

B. The **System** SHALL apply a **Sponsor Configuration Requirement** only to a **Configurable Parameter** that is explicitly declared as configurable by a **Platform Requirement**.

C. The **System** SHALL NOT permit a **Sponsor Configuration Requirement** to override behavior of a **Platform Requirement** that is not declared as configurable.

D. The **System** SHALL NOT permit a **Sponsor Configuration Requirement** to grant a capability that is not defined by a **Platform Requirement**.

E. When no **Sponsor Configuration Requirement** exists for a given **Configurable Parameter**, the **System** SHALL apply the default value or behavior described in the corresponding **Platform Requirement**.

F. When a **Platform Requirement** specifies that a behavior takes effect only if a corresponding **Configurable Parameter** is configured, and no **Sponsor Configuration Requirement** configures it, the **System** SHALL NOT apply that behavior.

PRODUCT REQUIREMENTS

5.1 Administration

The Sponsor Portal supports the full **User Account** lifecycle: creation, activation, site assignment, activation-email resend, edit, deactivation, and reactivation. A consistent user-management interface — tabs, modals, the Administrator Dashboard, and the shared **Reason Field Constraints** that govern every free-text reason input — exposes these capabilities to the Administrator role.

5.1.1 Create User Account

REQ ID: `DIARY-PRD-user-account-create`

Refines: `DIARY-PRD-user-account-activation-workflow`, `DIARY-PRD-user-account-site-a` `DIARY-PRD-user-account-edit`, `DIARY-PRD-user-account-deactivate`, `DIARY-GUI-user-man` `DIARY-GUI-user-information-modal`

Overview

This requirement defines the minimum identity and authorization data the **System** captures when an **Administrator** creates a new **User Account**. Subsequent requirements in this file refine activation, site assignment, edits, and lifecycle transitions on the account established here.

Definitions

User Account: The complete set of identity, credential, and authorization data that the **Sponsor Portal** maintains for a single human user.

Full Name: A common name, which is **PII**, of the person associated with a **User Account** that identifies a specific individual in the context of the **Sponsor Portal**.

Email Address: A unique technical identifier used as a destination for system notifications and as a username for authentication to access the system.

Rationale

A **User Account** is the unit of authorization and audit attribution in the **Sponsor Portal**: every action recorded in the audit trail is bound to the account that performed it, and every permission check resolves against the account's assigned roles and Sites. Requiring **Full Name**, **Email Address**, role, and (for non-Administrators) at least one Site at creation time ensures the account is fully addressable, fully scoped, and fully attributable from the moment it exists; partial accounts that lack one of these fields would either be unable to receive activation email, unable to participate in audit reporting, or unable to be authorized for any operation. Validating the email format at the creation boundary catches obvious typos before the activation flow consumes an invalid address.

Assertions

A. The **System** SHALL require a **Full Name**, **Email Address**, at least one **Role**, and — for non-**Administrator** accounts — at least one **Site** to be selected before a **User Account** is created.

B. The **System** SHALL validate the format of the email address provided (e.g., presence of an "@" symbol).

C. The **System** SHALL allow assignment of the **Administrator**, **CRA** and **Study Coordinator** role during User Account creation.

5.1.2 Account Activation Workflow

REQ ID: `DIARY-PRD-user-account-activation-workflow`

Refines: `DIARY-PRD-user-account-activation-resend`

Overview

The account activation process starts when a **User Account** is created, reactivated, or when an activation email is sent again. Activation links are time-limited and can only be used once, which helps protect account security. A user cannot access the system until their account has been successfully activated.

Definitions

Verification Link: A unique, time-limited, single-use secured URL delivered to an email address to confirm that the recipient controls that address.

Activation Webpage: A **Verification Link** associated with a **User Account** that provides a mechanism for the **Account Owner** to activate their account and configure their password and 2FA.

Account States

Activation Webpage Lifecycle

Email Address Verification Lifecycle

Verification Link Behavior

Rejection Behavior

Rationale

Account activation is the boundary at which a candidate **User Account** (created or restored by an Administrator) becomes a credentialed account that can authenticate. Two security properties are essential at this boundary: control of the registered email address (verified by delivering a single-use link to that address), and a bounded window in which the verification holds (14 days). Single-use enforcement prevents the link from being replayed if it leaks; expiry caps the window during which a stolen or stale link could be exploited; invalidating prior links when a new one is issued ensures that resend operations do not leave multiple valid paths open simultaneously. The mid-lifecycle email-update path uses the same verification mechanism so that an Administrator-initiated address change cannot redirect login to an address the user does not control. The rejection message is deliberately generic — it directs the user to the Administrator without disclosing whether the underlying account exists, which preserves user-enumeration resistance.

Assertions

A. When the status of the **User Account** is **Pending Activation**, the user SHALL NOT be able to access the system.

B. When a **User Account** is set to **Pending Activation**, the **System** SHALL create a unique **Activation Webpage** for that **User Account**.

C. When an **Activation Webpage** is created, the **System** SHALL deliver it to the associated **User Account** owner's registered **Email Address**. The username of the user is the **Email Address** that has been entered during user creation.

D. When a **User Account** owner has activated their account and configured a valid password and 2FA, the **System** SHALL set its status to **Active**.

E. When an **Administrator** updates the email address of a user, a new **Verification Link** is generated and sent to that updated email address. When the **Verification Link** is used, the **System** SHALL replace the previous **Email Address** with the new **Email Address** as the **User Account**'s username.

F. When a **Verification Link** is used or 14 days have passed from generation, whichever occurs first, the **System** SHALL invalidate the **Verification Link**.

G. When a new **Verification Link** is issued for a **User Account**, the **System** SHALL invalidate any previously issued **Verification Link** for that **User Account**.

H. When a user attempts to access an invalidated **Verification Link**, the **System** SHALL reject the attempt and display the following message: "This link is no longer valid. Please contact your Administrator to request a new activation email."

5.1.3 Site Assignment

REQ ID: `DIARY-PRD-user-account-site-assignment`

Overview

Sites are managed in **Rave EDC**. The portal displays **Sites** for assignment purposes only and does not allow modification, to maintain data integrity. Near real-time synchronization ensures that the site list reflects the data within **Rave EDC** at all times.

Definitions

Site: A clinical research location authorized by the Sponsor to enroll **Participants** and conduct trial activities. Each **Site** is uniquely identified and synchronized from **Rave EDC**.

Rationale

Sites are clinical trial entities owned by the EDC system of record, not by the portal; replicating Site authorship in the portal would create a divergent inventory and break the EDC-to-portal data-integrity contract. The portal therefore reads the Site list and never edits it, refreshing on Administrator login so that newly-created Sites are available promptly for assignment without requiring an explicit sync action. Administrators are granted study-wide Site visibility because their accountability for **User Account** lifecycle requires they be able to assign any Site to any account; restricting Administrators to a Site subset would block legitimate account configuration and create operational deadlocks during staff onboarding.

Assertions

A. The system SHALL include only **Sites** that are created in **Rave EDC**.

B. The system SHALL refresh the list of valid **Sites** each time an **Administrator** logs in.

C. An **Administrator** SHALL have access to all **Sites**.

D. The system SHALL allow an **Administrator** to assign any **Site** to a **User Account**.

5.1.4 Resend Activation Email

REQ ID: `DIARY-PRD-user-account-activation-resend`

Overview

A user may not receive or may lose their activation email before completing setup. Resending generates a new **Activation Webpage** and invalidates the previous one, ensuring only one valid link exists at any time.

Rationale

The activation email is the only delivery channel for the **Activation Webpage**, and legitimate failure modes (spam filtering, address typos surfaced after creation, transient email infrastructure issues) are common. Allowing unlimited resends keeps the recovery path open for the Administrator without requiring an escalation channel; the resend operation is itself audited and rate-limited only by the Administrator's manual cadence, so abuse potential is low. Invalidating the previous link on each resend keeps the single-valid-link invariant from the parent activation workflow intact: at most one **Verification Link** exists per **User Account** at any moment, regardless of how many times the activation email has been resent.

Assertions

A. While the status of the **User Account** is **Pending Activation**, the system SHALL allow the **Administrator** to resend a new activation email.

B. The system SHALL NOT impose a limit on the number of resend attempts for a **Pending Activation User Account**.

C. When an activation email is resent, the system SHALL create a new **Activation Webpage** for that **User Account** and invalidate the previous one.

5.1.5 Edit User Account

REQ ID: `DIARY-PRD-user-account-edit`

Refines: `DIARY-GUI-user-management-tabs`, `DIARY-GUI-user-information-modal`

Overview

This functionality allows user access to be updated as roles or responsibilities change. If access is removed or changed, the user is immediately logged out by the system to prevent access to information they should no longer see. When an email address is updated, the new one must be verified by the user again before using their user account.

Rationale

User edits encode three risks the platform contains structurally. First, an Administrator editing their own account could escalate privilege or remove their own deactivation safeguard; prohibiting self-edit closes that channel. Second, role or site changes that take effect on the next login leave a window in which a user retains stale authorization; enforcing the change immediately collapses that window. Third, an email change is effectively a credential change (the email is the username and the activation channel), so the new address must be verified by the same mechanism used at account creation, and the prior address must be notified so that an unauthorized change initiated against an unattended Administrator session is visible to the legitimate account owner.

Assertions

A. The **System** SHALL allow an **Administrator** to edit any **User Account** other than their own.

B. The **System** SHALL require at least one **Role** for non-**Administrator** users.

C. The **System** SHALL require at least one **Site** for non-**Administrator** users.

D. The **System** SHALL validate **Email Address** uniqueness across all existing **User Accounts**.

E. When an **Administrator** changes a **User Account**'s **Role** or **Site**, the **System** SHALL enforce the change immediately.

F. When an **Administrator** changes a **User Account**'s **Email Address**, the **System** SHALL deliver a **Verification Link** to the new **Email Address**.

G. When an **Administrator** initiates an **Email Address** change, the **System** SHALL notify the **User Account** owner at the original **Email Address** that a change has been initiated.

5.1.6 Deactivate User Account

REQ ID: `DIARY-PRD-user-account-deactivate`

Refines: `DIARY-PRD-user-account-reactivate`, `DIARY-GUI-user-account-deactivate`

Overview

Deactivation is the action of revoking a **User Account**'s ability to access the system without permanently removing the account or its associated data. A deactivated account retains all historical data and audit trail but cannot be used to log in or maintain active sessions.

Definitions

Deactivation: The action of revoking a **User Account**'s ability to access the system without permanently removing the account or its associated data. A deactivated account retains all historical data and **Audit Trail** but cannot be used to log in or maintain active sessions.

Rationale

Deactivation is the standard off-boarding mechanism for users who should no longer access the system — completion of a study assignment, role change off the study, departure from the sponsor or site organization. Retaining the account record and its full **Audit Trail** is required by FDA 21 CFR Part 11: every historical action attributed to that account must remain auditable indefinitely, even after the account is no longer usable. Terminating active sessions immediately is necessary because any session in flight at deactivation time was authenticated under the about-to-be-revoked credentials and must be invalidated to honor the access decision. Prohibiting self-deactivation closes a denial-of-service vector against the Administrator workforce (an Administrator deactivating themselves and leaving no other active Administrator could lock out the deployment). The free-text reason is captured to support audit reviews of why each account was deactivated.

Assertions

A. The System SHALL allow the Administrator to deactivate any **User Account** except their own.

B. The System SHALL terminate all active sessions associated with a **User Account** immediately upon **deactivation**.

C. The System SHALL prevent login to a deactivated **User Account**.

D. The System SHALL preserve all data and **Audit Trail** associated with a deactivated **User Account**.

E. The System SHALL set the status of a deactivated **User Account** to **Deactivated**.

F. When an **Administrator** deactivates a **User Account**, the **System** SHALL require a free text reason before applying the change. The reason SHALL be captured in the audit trail.

5.1.7 Reactivate User Account

REQ ID: `DIARY-PRD-user-account-reactivate`

Refines: `DIARY-GUI-user-account-reactivate`

Rationale

Reactivation is the inverse of deactivation and serves the case where a previously off-boarded user resumes a role on the study — a returning Study Coordinator, an Administrator restored after a leave of absence. Restoring previously assigned roles and **Site** assignments is the operational expectation: the reactivation event is logically “this same person, same responsibilities, again”, not a fresh account creation. Routing the reactivated account through **Pending Activation** rather than directly to **Active** is a security floor: the user must complete a fresh activation flow, which proves they still control the registered email address and produces a current credential (password, 2FA) rather than reviving stale credentials that may have been compromised during the deactivated interval. The free-text reason is captured for the same audit purpose as deactivation.

Assertions

A. The System SHALL support reactivation of any deactivated **User Account**.

B. When a **User Account** is reactivated, the system SHALL restore all previously assigned roles and **Site** assignments.

C. Upon reactivation, the status of the **User Account** SHALL be set to **Pending Activation** state.

D. When an **Administrator** reactivates a **User Account**, the **System** SHALL require a free text reason before applying the change. The reason SHALL be captured in the audit trail.

5.1.8 Reason Field Constraints

REQ ID: `DIARY-PRD-reason-field-constraints`

Overview

Where a reason is required before an action proceeds, consistent input constraints ensure reasons are meaningful and auditable. Empty or whitespace-only submissions are rejected to ensure every high-impact action is accompanied by a documented reason for compliance purposes. This requirement is the platform-wide template for free-text reason inputs and is referenced by every Reason Dialog (Free Text) defined elsewhere in the specification.

Rationale

Free-text reason inputs accompany high-impact, irreversible actions (deactivation, disconnection, mark-as-not-participating, record deletion) and are the principal narrative artifact the audit trail carries forward for those actions. Two failure modes degrade their audit value: empty or whitespace-only submissions (which let users bypass the reason requirement while passing the input control) and unbounded length (which lets users paste pages of irrelevant text that obscure the actual rationale). The whitespace rejection rule eliminates the former; the 100-character cap addresses the latter while still leaving room for a usefully descriptive sentence. Both constraints are platform-wide so reviewers see consistent reason-field behavior across every action that requires one, and so the audit log's free-text field has predictable bounds.

Assertions

A. Where the **System** requires a reason, the **System** SHALL reject a submission where the reason is empty or contains only whitespace.

B. Where the **System** requires a reason, the **System** SHALL enforce a maximum input length of 100 characters.

5.1.9 User Management Tabs

REQ ID: `DIARY-GUI-user-management-tabs`

Overview

The user management interface separates active and inactive accounts into distinct tabs to reduce cognitive load and prevent accidental actions on the wrong account. Real-time search and preserved search state allow Administrators to work efficiently across tabs without losing context.

Definitions

User Information Modal: A modal dialog displaying the details and available actions for a selected **User Account**.

Tab Display

Search Behavior

Empty State

Row Display

Rationale

Active and inactive accounts are operationally distinct surfaces: actions available on one are not available on the other, and the most common Administrator error in unified lists is acting on an inactive account believing it was active. Splitting into two tabs makes the active/inactive distinction structural rather than visual, and immediate row-movement on status change keeps each tab self-consistent without manual refresh. Real-time search across **Full Name** and **Email Address** matches the two identifiers Administrators use in practice (a Full Name supplied by someone reporting an issue, or an Email Address supplied by the user themselves). Preserving search state across tabs supports the recurring workflow of “search for a user, check Active, then check Inactive” without forcing the Administrator to retype the query. Row selection opens the **User Information Modal** rather than navigating away, so the table context remains visible behind the modal and consecutive lookups are quick.

Assertions

A. The interface SHALL display two tabs: Active Users and Inactive Users.

B. The Active Users tab SHALL display all User Accounts with a status of Active or Pending Activation.

C. The Inactive Users tab SHALL display all User Accounts with a status of Deactivated.

D. The interface SHALL display the Active Users tab by default.

E. The interface SHALL highlight the currently active tab.

F. Each tab SHALL display the total count of User Accounts it contains inline with the tab label.

G. When a User Account status changes, the interface SHALL move the account to the appropriate tab immediately.

H. The interface SHALL provide a single search input that filters User Accounts by Full Name and Email Address.

I. The interface SHALL update search results in real time as the user types.

J. The interface SHALL preserve the search state when the user switches between tabs.

K. When a tab contains no User Accounts matching the current search, the interface SHALL display a message indicating no results were found.

L. The interface SHALL display each **User Account** as a row showing **Full Name**, **Email Address**, **Role(s)**, **Assigned Sites**, and **Status**.

M. Each **User Account** row SHALL be selectable and open the **User Information Modal**.

N. The interface SHALL display a **Create User** action.

5.1.10 Deactivate User Account

REQ ID: `DIARY-GUI-user-account-deactivate`

Overview

The deactivation action is available from the Active Users tab only, since deactivated accounts are no longer present there after the action completes.

Availability

Interaction Flow

Rationale

The deactivation surface is anchored to the **Active Users** tab because that is where the Administrator finds candidates for deactivation; offering the same action from the Inactive

Users tab would be either a no-op (the account is already deactivated) or a confusing alternative entry point. Excluding the Administrator's own account from the action list closes the self-deactivation channel from the GUI side, complementing the PRD-level prohibition and making the unavailable state visible rather than producing a back-end rejection after the Administrator has invested in the workflow. Requiring the reason at the confirmation step rather than after the action commits gives the Administrator a final opportunity to back out and ensures the reason is captured before any state change is applied.

Assertions

A. The interface SHALL make the deactivation action available for every **User Account** displayed in the **Active Users** tab, with the exception of the current user's own account.

B. When the deactivation action is initiated, the interface SHALL display the confirmation dialog and require a free text reason before proceeding.

C. The interface SHALL not allow the deactivation action to proceed until a reason has been provided.

D. If the user cancels the confirmation, the **User Account** will remain unchanged.

E. When deactivation is confirmed, the interface SHALL move the **User Account** to the **Inactive Users** tab immediately.

5.1.11 Reactivate User Account

REQ ID: `DIARY-GUI-user-account-reactivate`

Overview

Reactivation is initiated from the Inactive Users tab and immediately returns the account to the Active Users tab with **Pending Activation** status, requiring the user to complete the activation workflow before regaining access.

Availability

Interaction Flow

Rationale

Reactivation is anchored to the **Inactive Users** tab to mirror the deactivation pattern: the candidate set lives in the inactive tab, and the action there is the only one available. Moving the account to the **Active Users** tab with **Pending Activation** status on confirmation makes the next-step responsibility (the user must complete the activation flow before logging in) immediately visible — the account is back in the active surface and its status badge communicates that login is gated on activation. The free-text reason is gathered at the same confirmation step as deactivation for symmetry and to keep the audit trail's reason fields populated for every reactivation event.

Assertions

A. The interface SHALL make the reactivation action available for every **User Account** displayed in the **Inactive Users** tab.

B. When the reactivation action is initiated, the interface SHALL display the confirmation dialog and require a free text reason before proceeding.

C. The interface SHALL NOT allow the reactivation action to proceed until a reason has been provided.

D. If the user cancels the confirmation, the **User Account** will remain unchanged.

E. When reactivation is confirmed, the interface SHALL move the **User Account** to the **Active Users** tab immediately with a status of **Pending Activation**.

5.1.12 User Information Modal

REQ ID: `DIARY-GUI-user-information-modal`

Overview

The **User Information Modal** provides a summary view of a **User Account** and the primary actions available for that account. All account management actions are initiated from this modal.

Display

Actions

Rationale

The **User Information Modal** is the per-account hub: every lifecycle action (edit, deactivate, reactivate) launches from here, and every account-related question (who, what role, which Sites) is answered here. Scoping the Site list by Role addresses the multi-role case directly — a user holding both Study Coordinator and CRA roles on overlapping but distinct Site sets is shown the correct subset for whichever role context the Administrator is reasoning about. Surfacing **Sites** by Role rather than as a flat union prevents the inverse error in which an Administrator believes a user has Study Coordinator access at a Site that they actually have only as a CRA. Suppressing **Deactivate User** on the current user's own account mirrors the PRD-level self-deactivation prohibition and the GUI-level Active Users tab rule, so the unavailable state is consistent across every surface a self-deactivation attempt could originate from.

Assertions

A. The **User Information Modal** SHALL display the **Full Name**, **Email Address**, **Status**, assigned **Role**(s), and **Site** assignments of the selected **User Account**.

B. The **User Information Modal** SHALL display every **Role** assigned to the **User Account**.

C. When the **User Account** is an **Administrator**, the interface SHALL indicate that the **User Account** has access to all **Sites**.

D. When the **User Account** holds one or more non-**Administrator Roles**, the interface SHALL allow the viewer to select a single **Role** to scope the **Site** list.

E. When a **Role** is selected, the interface SHALL display only the **Site(s)** assigned to the **User Account** under that **Role**.

F. The interface SHALL display the total count of **Sites** assigned to the selected **Role**.

G. The interface SHALL display each **Site** entry using the **Site** identifier and **Site** name.

H. The interface SHALL indicate which **Role** is currently selected.

I. The **User Information Modal** SHALL present an **Edit User** action, a **Deactivate User** action, and a **Close** action for accounts displayed in the **Active Users** tab.

J. The **User Information Modal** SHALL present a **Reactivate User** action and a **Close** action for accounts displayed in the **Inactive Users** tab.

K. The **User Information Modal** SHALL NOT present a **Deactivate User** action for the currently authenticated user's own account.

L. When the user selects **Close**, the interface SHALL dismiss the modal without making any changes.

M. When the user selects **Edit User**, the interface SHALL open the edit workflow for that **User Account**.

5.1.13 Administrator Dashboard

REQ ID: `DIARY-GUI-administrator-dashboard`

Overview

The **Administrator Dashboard** is the primary surface for an **Administrator** to manage **User Accounts** and review audit log activity. Organising the surface as a container with two top-level tabs separates **User Account** management from audit log review while keeping both areas reachable from a single entry point. The dashboard scaffolding defined here establishes the header, the top-level tab navigation, and the default landing tab; the contents of each tab are governed by the requirements referenced from those tabs.

Definitions

Administrator Dashboard: The default surface presented to an **Administrator** upon successful authentication to the **Sponsor Portal**, containing top-level navigation to **User Account** management and audit log review.

Header

Top-Level Tabs

Logout

Rationale

The **Administrator Dashboard** consolidates the two responsibilities of the **Administrator** role — **User Account** management and audit review — into adjacent tabs of a single surface so the Administrator can switch between investigating an account and reviewing the audit log without navigating away from a shared context. A persistent header keeps identity (who is logged in, in what role), system context (which Sponsor Portal), settings access, and logout reachable from any screen of the dashboard, which is necessary because an Administrator may need to log out from any state without first navigating back to a landing page. Defaulting to the **Users** tab on login matches the most common Administrator entry point (open the dashboard to perform account actions); the Audit Logs tab is one click away when investigation is the goal.

Assertions

A. The interface SHALL display a header at the top of the **Administrator Dashboard** containing the **Sponsor Portal** name, the **Full Name** of the authenticated user, the user's active **Role**, a **Settings** action, and a **Logout** action.

B. The header SHALL remain visible on every screen of the **Administrator Dashboard**.

C. When the **Administrator** selects **Settings**, the interface SHALL navigate to the **Administrator Settings** surface defined in DIARY-GUI-administrator-settings.

D. The **Administrator Dashboard** SHALL display two top-level tabs: **Users** and **Audit Logs**.

E. The **Users** tab SHALL display the User Management interface defined in DIARY-GUI-user-management-tabs.

F. The **Audit Logs** tab SHALL display the Administrator Audit Log View defined in DIARY-GUI-audit-log-administrator.

G. The interface SHALL display the **Users** tab by default upon login.

H. The interface SHALL display a visual indicator identifying the currently active top-level tab.

I. When the **Administrator** selects **Logout**, the **System** SHALL terminate the **Session** and return the **Administrator** to the **Sponsor Portal** login interface.

5.3 Questionnaire Management

The platform's **Questionnaire** system rests on a clinical questionnaire foundation, sponsor-specific configuration surfaces, a controlled submission gate that enforces approved structures, and change-control discipline that keeps questionnaires stable within a trial.

5.3.1 Clinical Questionnaire System

REQ ID: `DIARY-PRD-questionnaire-system`

Refines: `DIARY-PRD-questionnaire-sponsor-configuration`, `DIARY-PRD-questionnaire-su`
`DIARY-PRD-questionnaire-change-control`

Overview

Clinical trials use structured questionnaires to collect reliable data in a consistent and compliant way. The questionnaires are carefully built to ensure accuracy and a good participant experience. New questionnaires can be added as needed, and the content always comes from approved documents, not edited directly in the app. This requirement is the foundation that all other questionnaire-handling requirements refine.

Definitions

Questionnaire: A validated data collection instrument presented to a participant to capture self-reported clinical outcomes at protocol-defined points in a clinical trial.

Questionnaire Type: A named category of **Questionnaire**. Each **Questionnaire Type** is implemented as an individual coded component.

Trial Start: The event that formally begins a participant's active trial participation. The trigger and label for **Trial Start** are sponsor-configurable.

Completion Status: The current state of a **Questionnaire** instance for a given participant. Valid values are defined per **Questionnaire** workflow.

Diary Data Synchronization: The continuous transmission of participant diary entries from the participant's mobile application to the **Sponsor Portal** and **Rave EDC**.

Questionnaire Display Name: The participant-facing name of a **Questionnaire** as presented in the mobile application. Distinct from the instrument name used for internal identification and clinical reference.

Trial: A clinical study conducted under a Sponsor's protocol in which **Participants** are enrolled to capture self-reported clinical outcomes through the **Mobile Application**. The **Trial** begins at **Trial Start**.

Questionnaire Registry

Data Synchronization

Configuration

Rationale

The questionnaire subsystem is the principal source of trial-grade clinical data in the platform, and its design choices reflect that role. Implementing each **Questionnaire Type** as an individual coded component is a defense against ad hoc clinical content: every questionnaire that ships is a deliberate engineering artifact reviewed against an approved source document, not a configuration table that a sponsor could edit without change control. **Completion Status** tracking is the foundation for the workflow rules that govern how a questionnaire moves from sent to answered to finalized; without per-instance status the platform could not distinguish "answered but not finalized" from "answered and locked", which is a clinical and regulatory distinction. **Trial Start** is the gate at which **Diary Data Synchronization** activates because pre-trial diary data is participant-private and must not be promoted to the sponsor portal or Rave EDC until the sponsor has acknowledged that the participant is in trial; making the trigger sponsor-configurable keeps the platform protocol-neutral.

Assertions

A. The System SHALL allow implementation of multiple **Questionnaire Types**, each as an individual coded component.

B. The System SHALL track **Completion Status** for each **Questionnaire** instance.

C. The System SHALL activate **Diary Data Synchronization** upon **Trial Start**.

D. The System SHALL deactivate **Diary Data Synchronization** when a participant is disconnected or is marked as Not Participating.

E. The System SHALL support sponsor-configurable configuration of the trigger and workflow that activates **Trial Start**.

F. Each **Questionnaire** SHALL have a **Questionnaire Display Name**.

5.3.2 Sponsor Questionnaire Configuration

REQ ID: `DIARY-PRD-questionnaire-sponsor-configuration`

Overview

A configuration-driven approach allows the sponsor to tailor questionnaire behavior to their study protocol without requiring platform code changes, ensuring that mid-study modifications are managed deliberately and do not unintentionally affect data collection.

Rationale

The five configuration axes named here — enablement, reminder behavior, post-submission processing, post-submission editability, and language — are the dimensions along which clinical protocols typically vary between sponsors using the same underlying **Questionnaire Type**. Encoding them as sponsor configuration rather than platform code keeps the platform's questionnaire library stable across deployments while letting each sponsor express the protocol-specific decisions that their IRB and operations teams require. The configuration approach is bounded — sponsors choose from the offered axes, they do not edit questionnaire content — which preserves the change-control discipline established in the foundation requirement.

Assertions

A. The System SHALL support sponsor-specific configuration of which **Questionnaires** are enabled per deployment.

B. The System SHALL support sponsor-specific configuration of participant reminder behavior for each enabled **Questionnaire**.

C. The System SHALL support sponsor-specific configuration of the processing steps that occur between participant submission and **Rave EDC** synchronization for each **Questionnaire**.

D. The System SHALL support sponsor-specific configuration of whether a participant may edit their answers after submission and under what conditions.

E. The System SHALL support sponsor-specific configuration of which languages are enabled per **Questionnaire**.

5.3.3 Clinical Data Submission Control

REQ ID: `DIARY-PRD-questionnaire-submission-control`

Overview

Clinical data integrity requires that the system only accepts data conforming to approved, predefined structures. A controlled registry of approved data collection instruments ensures that unrecognized or malformed data is rejected at the point of entry, supporting audit trail integrity and validation requirements under 21 CFR Part 11.

Rationale

A clinical trial's evidence chain begins at the data-ingest boundary: every submission that reaches the sponsor portal or **Rave EDC** must be traceable to an approved **Questionnaire** definition, because data that does not conform to a known instrument has no defensible interpretation and corrupts the trial's analyzability. The controlled registry is the platform's enforcement point — submissions are matched against it at the point of entry and rejected if they do not conform, so malformed or unrecognized data never enters the data store. This is a structural prerequisite for 21 CFR Part 11 audit trail integrity: every data point in the audit trail must reference a definition that existed at the time the data was submitted.

Assertions

A. The System SHALL maintain a controlled list of approved **Questionnaire** definitions.

B. The System SHALL reject any data submission that does not conform to an approved **Questionnaire** definition.

5.3.4 Questionnaire Change Control

REQ ID: `DIARY-PRD-questionnaire-change-control`

Overview

Questionnaires must stay the same throughout an active trial so the data remain consistent and traceable. Any updates must follow a formal change process to protect the integrity of the system.

Rationale

Mid-trial **Questionnaire** changes would silently alter the meaning of the data being collected: a question reworded between Day 30 and Day 60 of a participant's enrollment produces two non-comparable answer streams under the same instrument name. The change-control discipline encoded here prevents that class of error by treating every **Questionnaire** modification as a deliberate engineering act subject to review and approval. The new-Questionnaire-Type rule (rather than in-place edit) keeps the data semantics for any given **Questionnaire Type** stable for the lifetime of any trial that uses it. Backward-compatible viewing is essential because trials regularly span multiple **Questionnaire** versions when long-running studies adopt updated instruments part-way through; reviewers must be able to inspect historical data captured under the previous version without that data being silently re-interpreted under the new version's rules.

Assertions

A. The System SHALL use predefined **Questionnaires** for all clinical data collection.

B. The System SHALL NOT allow **Questionnaires** to be created, modified, or deleted during trial operation. If any new **Questionnaire** is required, a new **Questionnaire Type** will be implemented. Modifications for an existing **Questionnaire** will follow the process outlined in the below two assertions.

C. Changes to **Questionnaires** SHALL be subject to formal change control procedures before deployment.

D. The System SHALL maintain backward compatibility to allow viewing of data collected with previous **Questionnaire** versions.

5.3.5 Start Trial Workflow

REQ ID: `CAL-PRD-trial-start-workflow`

Refines: `CAL-GUI-trial-start-workflow`

Overview

Before the trial is started, the diary entries remain on the **Participant**'s device and are not transmitted to the **Sponsor Portal** or **Rave EDC**. This requirement defines the **Start Trial** action that activates **Diary Data Synchronization** and gates the **NOSE HHT** and **HHT-QoL Questionnaires** for the Callisto deployment.

Definitions

Start Trial: The action available to a Study Coordinator for a participant with **Linked - Awaiting Start** status that activates **Diary Data Synchronization** and begins the participant's active trial participation.

Rationale

Start Trial is the boundary at which a **Participant** transitions from pre-trial (their device is linked but their diary data and clinical questionnaires remain device-local) to active trial (their data flows to the **Sponsor Portal** and **Rave EDC**, and clinical questionnaires can be administered). Anchoring synchronization activation and clinical questionnaire eligibility to a single, Study-Coordinator-confirmed event ensures that the trial-active state is an explicit clinical decision rather than an emergent consequence of device linking. The gate on clinical questionnaire delivery (NOSE HHT and HHT-QoL) is necessary because those questionnaires generate trial-grade clinical data; sending them before Start Trial would either produce data with no trial context or force a retroactive trial-start backdate that the audit trail cannot defensibly represent.

Assertions

A. When a Study Coordinator confirms **Start Trial** for a participant with **Linked - Awaiting Start** status, the System SHALL change the Participant Status to **Trial Active** and activate **Diary Data Synchronization**.

B. The System SHALL NOT activate **Diary Data Synchronization** for a participant before **Start Trial** is confirmed.

C. The System SHALL NOT permit **NOSE HHT** or **HHT-QoL** Questionnaires to be sent to a participant before **Start Trial** is confirmed.

5.3.6 Start Trial Workflow Interface

REQ ID: `CAL-GUI-trial-start-workflow`

Rationale

The **Start Trial** button is placed in the Action column of the participant row so it is reachable in a single click from the **Participant Dashboard**, matching the daily workflow of a Study Coordinator who steps a series of newly-linked participants into trial-active state. The Confirmation Dialog is gated by an explicit **Send EQ** confirmation rather than a yes/no toggle because the action is consequential (it commits the **Participant** to clinical questionnaire administration and trial-grade data synchronization) and the labelling makes the next operational step (send the first clinical questionnaire) visible. Replacing **Start Trial** with **Manage Questionnaires** on confirmation is the visual cue that the **Participant** has moved into the trial-active workflow and the next set of actions is the per-questionnaire workflow.

Assertions

A. The interface SHALL present a **Start Trial** button to the Study Coordinator in the Action column for participants with **Linked - Awaiting Start** status.

B. When the Study Coordinator selects **Start Trial**, the interface SHALL display a Confirmation Dialog.

C. The Confirmation Dialog SHALL present a **Send EQ** button and a cancel action.

D. When the Study Coordinator selects **Send EQ**, the interface SHALL update the participant's status to **Trial Active** and replace the **Start Trial** button with a **Manage Questionnaires** button in the Action column.

E. If the Study Coordinator cancels the Confirmation Dialog, the participant's status SHALL remain unchanged.

5.3.7 NOSE HHT and HHT-QoL Questionnaire Workflow

REQ ID: `CAL-PRD-nose-hht-qol-workflow`

Refines: `CAL-GUI-manage-questionnaires-modal`, `CAL-GUI-questionnaire-finalization-w` `CAL-PRD-questionnaire-cycle-tracking`

Overview

NOSE HHT and **HHT-QoL** questionnaires are administered to participants on Day 1 of each treatment **Cycle**. A Study Coordinator initiates each cycle by sending the questionnaire(s) to the **Participant**'s **Mobile Application**. This requirement governs the per-questionnaire workflow from send through finalization.

Sending

Deleting

Participant Completion

Site Visit Review

Finalizing

Rationale

NOSE HHT and HHT-QoL are administered once per treatment **Cycle**, and the workflow encoded here reflects that cadence: at most one active **Questionnaire** per **Cycle** per **Type**, with the **Cycle** advancing only after finalization completes. The send/edit/submit/review/finalize sequence mirrors the clinical workflow — the participant fills out the instrument at home (with unlimited edits before submission to support honest answer revision), the Study Coordinator reviews the submitted answers during a site visit, and the Coordinator finalizes only after the review. Finalization is the locking event: it transmits to Rave EDC, calculates the score, and freezes the answers, after which call-back is no longer available. The retry-on-failure pattern with the participant-facing message is operationally important because Rave EDC connectivity can be transient, and a failure during finalization should not block the workflow or force the Coordinator to wait at the dialog. Returning the card to **Not Sent** on successful finalization is the signal that the next **Cycle** can be initiated.

> **Follow-up — configurability**: This requirement currently encodes the only option implemented in code. Future sponsors may require different rules; introduce a configurable seam (e.g. a parameter on the CAL-PRD-* parent, or a new platform-side template the CAL- REQ Satisfies) when the need arises. Until that seam exists, this REQ is normative for the Callisto deployment.

Assertions

A. The System SHALL enforce a limit of one active **Questionnaire** per **Cycle** per **Questionnaire Type**.

B. When a **Questionnaire** fails to reach the participant's mobile application, the System SHALL change its status to **Delivery Failed**.

C. After the first **Cycle** has been finalized, the System SHALL make the **Start Next Cycle** available in place of **Send Now** action for that **Questionnaire Type**.

D. The System SHALL allow a Study Coordinator to call back a **Questionnaire** with **Sent**, **Delivery Failed** or **Ready to Review** status.

E. The System SHALL NOT allow a Study Coordinator to call back a **Questionnaire** when it is already completed by the participant and finalized by the Study Coordinator.

F. The System SHALL allow the participant to edit their answers at any time before Finalization (i.e., while the **Questionnaire** has **Sent** or **Ready to Review** status). There is no limit on how many times a participant may edit their answers.

G. When a participant submits a **Questionnaire**, the System SHALL change its status to **Ready to Review**.

H. The System SHALL allow a Study Coordinator to review submitted **Questionnaire** answers before finalization.

I. The System SHALL allow a Study Coordinator to finalize a **Questionnaire** with **Ready to Review** status.

J. Upon finalization, the System SHALL lock the answers, calculate the **Questionnaire** score, and transmit answers to **Rave EDC**.

K. When the finalization operation fails, the System SHALL automatically retry the operation in the background. The **Questionnaire** SHALL remain in **Ready to Review** status throughout.

L. While a finalization retry is in progress, the System SHALL display the following message to the Study Coordinator: "Submission is taking longer than expected. The system is retrying. No action is required."

M. Upon successful finalization, the System SHALL change the **Questionnaire** status to **Not Sent**.

5.3.8 Manage Questionnaires Modal

REQ ID: `CAL-GUI-manage-questionnaires-modal`

Definitions

Questionnaire Card: The UI component displayed within the **Manage Questionnaires Modal** representing a single **Questionnaire Type** for a participant, showing its current status and available actions.

Select Starting Cycle Dialog: The dialog displayed when a Study Coordinator selects **Send Now** for the first time for a given **Questionnaire Type**, allowing selection of the **Starting Cycle** before sending.

Modal

Questionnaire Card — Display and Actions

Delete Interaction

Send Now

Start Next Cycle

Delivery Failed Troubleshooting

Rationale

The **Manage Questionnaires Modal** is the per-participant control surface for the clinical questionnaire workflow, and its design favors a card-per-type layout because the Study Coordinator's mental model is "what is the state of NOSE HHT for this participant, and what is the state of HHT-QoL?" rather than a flat list of every questionnaire ever sent. The status-driven action table collapses the workflow's branch logic into a visible matrix: each status has exactly the actions it can take, no others appear, and the status badge sits next to the cycle it describes so the **Cycle** context is never ambiguous. The Select Starting Cycle Dialog appears only on the first send because that is the only moment at which the **Starting Cycle** is a Coordinator choice — every subsequent cycle is auto-incremented per `CAL-PRD-questionnaire-cycle-tracking`. The information-icon Troubleshooting Popover handles the **Delivery Failed** state inline without requiring a separate screen, because the Coordinator's diagnostic loop typically completes (try resending, check connectivity) without leaving the modal.

> **Follow-up — configurability**: This requirement currently encodes the only option implemented in code. Future sponsors may require different rules; introduce a configurable seam (e.g. a parameter on the CAL-PRD-* parent, or a new platform-side template the CAL- REQ Satisfies) when the need arises. Until that seam exists, this REQ is normative for the Callisto deployment.

Assertions

A. The interface SHALL display the **Manage Questionnaires Modal** with the participant ID in the modal header.

B. The interface SHALL display one **Questionnaire Card** per **Questionnaire Type**.

C. The interface SHALL present a close action that dismisses the modal without making any changes.

D. Each **Questionnaire Card** SHALL display the **Questionnaire Type** name and the current **Questionnaire Status**. Date and time information SHALL be paired inline with the **Cycle** it describes (e.g., “Cycle 1 . Apr 24, 2026, 11:34 AM”), not as a standalone field.

E. The interface SHALL present cycle information, **Status Badge** placement, and available actions on each **Questionnaire Card** according to the following table: | Questionnaire Status | Cycle Fields Displayed | Status Badge Placement | Actions Available | | —– | —– | —– | —– | | **Not Sent** — never sent | None | Below **Questionnaire Type** name | **Send Now** | | **Not Sent** — after finalize or call back | **Finalized Cycle** (with date), **Next Cycle** | Inline with **Finalized Cycle** row | **Start Next Cycle** | | **Sent** | **Current Cycle** | Inline with **Current Cycle** row | **Call Back** (Trash icon) | | **Delivery Failed** | **Current Cycle** | Inline with **Current Cycle** row, with information icon | **Finalize**, **Call Back** (Trash icon) | | **Ready to Review** | **Current Cycle** (with completion date) | Inline with **Current Cycle** row | **Finalize**, **Call Back** (Trash icon) | | **Closed** | **Finalized Cycle** (with date) | Inline with **Finalized Cycle** row, as combined badge (“Closed . End of Treatment” or “Closed . End of Study”) | None |

F. When a Study Coordinator selects the call back (Trash icon) action, the interface SHALL display a Reason Dialog — Free Text before proceeding.

G. When the Study Coordinator confirms the call back action, the interface SHALL update the **Questionnaire Card** status to **Not Sent**.

H. When the Study Coordinator cancels the Reason Dialog, the **Questionnaire** SHALL remain unchanged.

I. The **Send Now** action is displayed only when the first **Questionnaire** of each **Questionnaire Type** will be sent to the participant. When a Study Coordinator selects **Send Now**, the interface SHALL display the **Select Starting Cycle Dialog**.

J. The **Select Starting Cycle Dialog** SHALL present a **Confirm and Send** button and a Cancel action.

K. When the Study Coordinator confirms the **Select Starting Cycle Dialog**, the interface SHALL update the **Questionnaire Card** status to **Sent**.

L. If the Study Coordinator cancels the **Select Starting Cycle Dialog**, the **Questionnaire Card** SHALL remain unchanged.

M. When a Study Coordinator selects **Start Next Cycle**, the interface SHALL update the

Questionnaire Card status to **Sent** without displaying the **Select Starting Cycle Dialog**.

N. When the **Questionnaire Status** is **Delivery Failed**, the interface SHALL display an information icon adjacent to the **Status Badge**. When the **Study Coordinator** selects this icon, the interface SHALL display a **Troubleshooting Popover** with guidance on resolving delivery issues.

O. The **Troubleshooting Popover** SHALL dismiss when the **Study Coordinator** clicks outside its bounds or selects a close action within it.

5.3.9 Questionnaire Finalization Workflow

REQ ID: `CAL-GUI-questionnaire-finalization-workflow`

Definitions

Finalization Dialog: The Confirmation Dialog displayed to the Study Coordinator when the **Finalize** button is selected, containing a **Cycle** dropdown for selecting the **Cycle** value before confirming finalization.

Finalization Dialog

Finalization Outcomes

Rationale

Finalization is the action that locks a questionnaire's answers and commits its score to Rave EDC, and the **Finalization Dialog** captures the two pieces of information that must be set at lock time: the **Cycle** value being finalized and the Coordinator's confirmation. Presenting the **Current Cycle** N Day 1 plus the two terminal options (**End of Treatment**, **End of Study**) covers every legitimate finalization case — most cycles finalize to their own N Day 1, and the two terminal cases mark the **Participant**'s endpoint for this **Questionnaire Type**. The Terminal Cycle Warning Dialog (referenced as a cancel path here) exists because terminal finalization is irreversible — no further **Questionnaires** of this type can be sent — and warrants a confirmation step distinct from the standard finalization confirmation. Cancelling either dialog returns the **Questionnaire** to its prior state so the Coordinator can revisit the choice without unintended side effects.

> **Follow-up — configurability**: This requirement currently encodes the only option implemented in code. Future sponsors may require different rules; introduce a configurable seam (e.g. a parameter on the CAL-PRD-* parent, or

a new platform-side template the CAL- REQ Satisfies) when the need arises. Until that seam exists, this REQ is normative for the Callisto deployment.

Assertions

A. When a Study Coordinator selects **Finalize**, the interface SHALL display the **Finalization Dialog** containing a **Cycle** dropdown.

B. The **Cycle** dropdown SHALL present the following selectable options: the **Current Cycle N Day 1** value, **End of Treatment**, and **End of Study**.

C. The **Finalization Dialog** SHALL present a **Finalize Questionnaire** button and a Cancel action.

D. When the **Study Coordinator** finalizes the **Questionnaire** with a **Cycle** N Day 1 value selected, the interface SHALL update the **Questionnaire Card** status to **Not Sent** and display the **Finalized Cycle** (paired with the finalization date and time) and the **Next Cycle** on the card.

E. When the Study Coordinator finalized the **Questionnaire** with a **Terminal Cycle** value (**End of Treatment** or **End of Study**) selected, the interface SHALL update the **Questionnaire Card** status to **Closed**, display the **Terminal Cycle** value alongside the **Closed** status as a combined badge.

F. When the Study Coordinator cancels the **Finalization Dialog**, the **Questionnaire** SHALL remain unchanged.

G. When the **Study Coordinator** cancels the **Terminal Cycle Warning Dialog**, the **Questionnaire** SHALL remain unchanged and the **Study Coordinator** SHALL be returned to the **Finalization Dialog**.

5.3.10 Questionnaire Cycle Tracking

REQ ID: `CAL-PRD-questionnaire-cycle-tracking`

Overview

Each questionnaire must be traceable to a specific protocol **Cycle** for regulatory compliance and data integrity. Automatic cycle incrementing reduces manual error. Because the protocol allows only one questionnaire of each type per cycle, no two completed questionnaires of the same type for a given participant may share the same **Cycle** value. Some participants start the study using paper records, meaning their first electronic questionnaire may not be **Cycle 1 Day 1**. A Study Coordinator may select the starting cycle when sending the first questionnaire of each type. Cycle tracking is independent per **Questionnaire Type**.

Definitions

Cycle: The protocol event identifier assigned to a **Questionnaire** representing the treatment cycle during which it was administered. Valid values are **Cycle N Day 1** (where N is a positive whole number), **End of Treatment**, and **End of Study**.

Current Cycle: The **Cycle** value assigned to the most recently sent **Questionnaire** of a given **Questionnaire Type** for a participant.

Finalized Cycle: The **Cycle** value of the most recently finalized **Questionnaire** of a given **Questionnaire Type** for a participant.

Next Cycle: The **Cycle** value that will be assigned when the next **Questionnaire** of that **Questionnaire Type** is sent. Always the next sequential **Cycle N Day 1** value following the **Finalized Cycle** (the **Finalized Cycle** N incremented by 1).

Starting Cycle: The **Cycle** value assigned to the first **Questionnaire** of a given **Questionnaire Type** sent to a participant.

Terminal Cycle: A **Cycle** value of **End of Treatment** or **End of Study** indicating no further **Questionnaires** of that **Questionnaire Type** will be sent to the participant.

Cycle N Day 1: A **Cycle** value identifying the first day of the Nth treatment cycle, where N is a positive whole number (Cycle 1 Day 1, Cycle 2 Day 1, etc.).

Cycle Assignment

Terminal Cycles

Configuration

Rationale

The **Cycle** value is the protocol coordinate that lets Rave EDC and downstream analyses know which treatment cycle a given **Questionnaire** answer corresponds to. Two **Questionnaires** of the same **Questionnaire Type** sharing a **Cycle** value would break that coordinate (which cycle did the participant actually answer for?), so the uniqueness rule is structural rather than advisory. Automatic incrementing of **Next Cycle** removes a category of manual data-entry error from the Coordinator's workflow; the **Starting Cycle** Coordinator-selection at first-send accommodates the paper-then-electronic pattern in which the first electronic cycle is not Cycle 1. Reassigning the same **Cycle** value to the next **Questionnaire** after a call-back preserves the protocol coordinate across the call-back/resend round-trip — the call-back is operationally an "undo and redo" rather than a "skip and advance". Terminal cycles end the **Questionnaire Type** stream for the **Participant**; uniqueness of terminal cycles per **Type** per **Participant** prevents the ambiguous case of two End-of-Treatment markers for the same instrument.

Assertions

A. The System SHALL assign a **Cycle** value to each **Questionnaire** at the time it is sent.

B. The System SHALL NOT allow two **Questionnaires** of the same **Questionnaire Type** for the same participant to share the same **Cycle** value.

C. When a Study Coordinator sends the first **Questionnaire** of a given **Questionnaire Type** to a participant, the System SHALL require the Study Coordinator to select a **Starting Cycle** before the **Questionnaire** is sent.

D. For each subsequent **Questionnaire** of a given **Questionnaire Type**, the System SHALL assign the **Next Cycle** value automatically by incrementing the **Cycle N** value by 1.

E. When a **Questionnaire** is called back before finalization, the System SHALL reassign the same **Cycle** value to the next **Questionnaire** of that **Questionnaire Type** for that participant.

F. When a **Questionnaire** is finalized with a **Terminal Cycle** value, the System SHALL change the **Questionnaire** status to **Closed**.

G. The System SHALL NOT allow a **Terminal Cycle** value to be assigned to more than one **Questionnaire** of the same **Questionnaire Type** per participant.

H. When the **Questionnaire** status is **Closed**, the System SHALL NOT permit a new **Questionnaire** of that **Questionnaire Type** to be sent to that participant.

I. The System SHALL support per-sponsor configuration request of whether cycle tracking is enabled for a given deployment.

J. When cycle tracking is enabled, the System SHALL support per-sponsor configuration request of whether the Study Coordinator is required to select a **Starting Cycle** when sending the first **Questionnaire** of a given **Questionnaire Type** to a participant.

K. When the **Starting Cycle** selection is disabled, the System SHALL assign **Cycle 1 Day 1** as the **Starting Cycle** for all participants.

5.3.11 Questionnaire Session Resume and Timeout

REQ ID: `CAL-PRD-questionnaire-session-resume-timeout`

Overview

Session resume, expiry behavior, state preservation, and clock semantics are defined by the platform-side `DIARY-PRD-questionnaire-session-timeout` requirement. This requirement specifies only the sponsor portal-specific configuration values for the Callisto deployment.

Rationale

The 30-minute **Session Timeout** matches the typical clinical-portal completion time for the NOSE HHT and HHT-QoL instruments (10-20 minutes for a Participant working through the questions, with comfortable headroom for interruption). The 5-minute warning threshold gives the Participant enough time to either complete the session or actively dismiss the warning before expiry without being so early that the warning becomes background noise. Both values are sponsor-configurable at the platform level (`DIARY-PRD-questionnaire-session-tim` this Callisto overlay pins the specific values that the deployment is operating under.

Assertions

A. The **NOSE HHT** and **HHT-QoL** questionnaires SHALL be configured with a **Session Timeout** of 30 minutes.

B. The **NOSE HHT** and **HHT-QoL** questionnaires SHALL be configured with a **Timeout Warning Notification** threshold of 5 minutes before expiry.

5.4 Participant Workflows

The **Participant** lifecycle in the **Sponsor Portal** is governed by linking-code generation and use, participant registration, the link / disconnect / reconnect / mark-not-participating / reactivate workflows, and the **Participant Dashboard** GUI through which the **Study Coordinator** operates them.

5.4.1 Linking Code Lifecycle Management

REQ ID: `DIARY-PRD-linking-code-lifecycle`

Refines: `CAL-PRD-linking-code-lifecycle-configuration`

Overview

The **Mobile Linking Code** mechanism provides a secure, time-limited channel through which clinical staff can link a **Participant**'s device to the **Sponsor Portal** without requiring the **Participant** to authenticate directly. Single-use enforcement and unpredictability ensure that codes cannot be reused or guessed, while configurable expiry balances security with the practical time **Participants** need to complete the linking process.

Definitions

Mobile Linking Code: A unique, time-limited, single-use code generated by the **System** and provided to a **Participant** by clinical staff. The **Participant** enters the code in the **Mobile Application** to establish a connection between their device and the **Sponsor Portal**.

Participant Linking Code: The **Mobile Linking Code** that was used to establish the connection between a **Participant**'s device and the **Sponsor Portal**. Displayed for reference and troubleshooting purposes. Cannot be used to establish a new connection.

Generation

Expiry

Single-Use

Rejection

Visibility

Rationale

The **Mobile Linking Code** is the bridge between two systems that otherwise do not share a credential surface: the **Sponsor Portal** (operated by clinical staff with managed accounts) and the **Mobile Application** (operated by a **Participant** with no account). Three properties are essential to that bridge. Unpredictability prevents a code from being guessed or derived in the gap between Study Coordinator generation and **Participant** entry. Single-use enforcement prevents a code that has been written down, photographed, or otherwise leaked from being replayed by a second device. Expiry caps the window during which a leaked or stale code can be used at all. Keeping the code short and alphanumeric is a usability requirement — clinical staff dictate codes to **Participants** over phone or in person, and codes that require strict capitalization or special characters fail in practice. Retaining the **Participant Linking Code** after use (under the **Participant Linking Code** definition rather than the active **Mobile Linking Code**) lets clinical staff confirm which code was used for troubleshooting and audit purposes, without that retention re-enabling use of the spent code.

Assertions

A. When a Study Coordinator initiates the participant linking workflow, the System SHALL generate a unique **Mobile Linking Code** for that participant.

B. The System SHALL generate each **Mobile Linking Code** such that it cannot be predicted or derived from any other known value.

C. The System SHALL generate **Mobile Linking Codes** in a format that is short and alphanumeric to support easy communication to the participant.

D. The System SHALL expire a **Mobile Linking Code** after a configurable duration from the time of generation.

E. When a new **Mobile Linking Code** is generated for a participant, the System SHALL immediately invalidate any previously active **Mobile Linking Code** for that participant.

F. The System SHALL invalidate a **Mobile Linking Code** immediately upon successful use.

G. The System SHALL reject any attempt to link using a **Mobile Linking Code** that is expired, already used, or does not exist.

H. The System SHALL retain the **Participant Linking Code** against the Participant record after successful use for reference and troubleshooting purposes.

5.4.2 Linking Code Lifecycle Configuration

REQ ID: `CAL-PRD-linking-code-lifecycle-configuration`

Rationale

The 72-hour expiry window balances two operational realities: a **Participant** may receive their **Mobile Linking Code** from clinical staff at a site visit and then have several days before they next have the opportunity to install the **Mobile Application** and complete the linking process; and the longer a code remains valid the larger the window in which a leaked or photographed code could be misused. Three days is sufficient time for the common participant onboarding cadence (clinic visit on Monday, home setup by Thursday) without leaving the code valid for a full week, which would exceed any plausible legitimate use window.

Assertions

A. The **Mobile Linking Code** expiry duration SHALL be configured to 72 hours from the time of generation.

5.4.3 Participant Registration

REQ ID: `DIARY-PRD-participant-registration`

Refines: `CAL-PRD-participant-registration-configuration`

Overview

Clinical trial portals require a defined mechanism for making **Participant** records available to clinical staff. The platform exposes this as a sponsor-configurable hook; concrete bindings (e.g. **Rave EDC** synchronization) live in sponsor overlay requirements.

Definitions

Participant: An individual whose **Mobile Application** has been linked to the **Sponsor Portal** for the purpose of clinical trial data collection. A **Participant** has been registered in the **Sponsor Portal** as a clinical trial subject.

Rationale

Participant records originate outside the **Sponsor Portal** in the systems that own clinical trial enrollment (EDC, IRB-approved enrollment workflows). Modelling registration as a sponsor-configurable mechanism rather than as a platform-fixed hook lets each sponsor wire the portal to the system of record they actually use, while keeping the platform's downstream workflows — linking, disconnection, finalization — agnostic to the upstream source. The sponsor-configurable initial status accommodates protocols that begin participants in different states (paper-then-electronic, electronic-from-day-one) without hard-coding a single starting point.

Assertions

A. The System SHALL support a sponsor-configurable mechanism for adding **Participant** records to the **Sponsor Portal**.

B. When a **Participant** record is added to the **Sponsor Portal**, the System SHALL assign an initial **Participant** status as defined by the sponsor configuration.

5.4.4 Participant Registration Configuration

REQ ID: `CAL-PRD-participant-registration-configuration`

Overview

The sponsor portal uses **Rave EDC** as the participant registration source. Near real-time synchronization ensures that participant records are available in the **Sponsor Portal** as soon as they are created in **Rave EDC**, without requiring manual intervention from clinical staff.

Definitions

Rave EDC: The electronic data capture system used by the portal as the authoritative source of participant and site records.

Participant ID: The unique identifier assigned to a **Participant** in **Rave EDC** and synchronized to the **Sponsor Portal**. The **Participant ID** is the primary reference used to identify a **Participant** across the System.

Rationale

Rave EDC is the clinical system of record for the Callisto deployment, and binding the **Sponsor Portal**'s participant registration to it preserves the single-source-of-truth property: every **Participant** in the portal corresponds to a Rave EDC enrollment, and the **Participant ID** carries the same value across both systems. Near real-time synchronization removes the operational delay that would otherwise force clinical staff to wait for a manual sync after enrollment. Initial **Not Connected** status reflects the lifecycle reality that EDC enrollment precedes device linking by hours or days; the portal-side **Not Connected** state is what makes the participant a candidate for the **Link New Participant Workflow**. Prohibiting manual create/edit/delete in the portal closes the divergence risk: if portal-side authorship were allowed, the portal could hold records that **Rave EDC** does not, breaking the cross-system identity match.

Assertions

A. The System SHALL synchronize **Participant** records from **Rave EDC** to the **Sponsor Portal** in near real-time.

B. The System SHALL synchronize the **Participant** ID and assigned **Site** for each **Participant** record.

C. When a **Participant** record is synchronized from **Rave EDC** for the first time, the System SHALL assign an initial **Participant** status of **Not Connected**.

D. The System SHALL NOT permit **Participant** records to be created, edited, or deleted manually in the **Sponsor Portal**.

5.4.5 Link New Participant Workflow

REQ ID: `DIARY-PRD-participant-link-new`

Overview

Linking connects a **Participant** record in the **Sponsor Portal** to the **Participant**'s **Mobile Application**, enabling data synchronization and questionnaire distribution. The workflow is initiated by clinical staff and completed by the **Participant** entering the **Mobile Linking Code** in the **Mobile Application**.

Rationale

The first-link workflow has two halves separated by a human-physical handoff: the Study Coordinator generates a **Mobile Linking Code** in the portal, and the **Participant** enters that code in the **Mobile Application**. The platform's responsibilities are bounded by that handoff: the portal generates an unguessable, single-use code (per `DIARY-PRD-linking-code-lifec` and accepts the corresponding entry on the mobile side. Gating the initiation on **Not Connected** status keeps the workflow's precondition explicit — a **Participant** record must exist (typically via the sponsor registration mechanism) and must not already be linked. The first-time qualifier on the **Mobile Application** entry distinguishes initial linking from subsequent reconnection, which uses the same mechanism but a different workflow (`DIARY-PRD-participant-rec`

Assertions

A. The System SHALL allow a Study Coordinator to initiate the linking workflow for a **Participant** with **Not Connected** status.

B. When a Study Coordinator initiates the linking workflow, the System SHALL generate a **Mobile Linking Code**.

C. When a **Participant** successfully enters a valid **Mobile Linking Code** in the **Mobile Application** for the first time, the System SHALL link the **Participant**'s device to the **Sponsor Portal**.

5.4.6 Participant Disconnection Workflow

REQ ID: `DIARY-PRD-participant-disconnection`

Refines: `CAL-PRD-participant-disconnection-reason-options`

Overview

Disconnection temporarily severs the link between a **Participant**'s device and the **Sponsor Portal** without removing the **Participant** record or their historical data. Clinical staff may disconnect a **Participant** when there are issues with the device, technical problems, or other circumstances prevent normal trial participation.

Rationale

Disconnection sits between linking (the **Participant** is in trial) and mark-as-not-participating (the **Participant** has left the trial). It encodes the case where the link is operationally broken — a device fault, a connectivity issue, a temporary withdrawal — but the **Participant** is still expected to resume. Preserving all data and history is the audit-trail requirement; the disconnection event must not erase the historical record. Continuing to apply sponsor-specific rules to the **Mobile Application** is the operational requirement; the **Participant** should still see locked records, validation rules, and questionnaire restrictions that apply to their trial role, because they may reconnect at any time and the **Mobile Application** must not silently revert to personal-use behavior in the interim. The reason format is sponsor-configurable because some sponsors require a controlled vocabulary for downstream analysis (CAL- overlay encodes one such list) while others want free-text capture.

Assertions

A. The System SHALL allow a Study Coordinator to disconnect a **Participant** whose current status permits disconnection.

B. The System SHALL require a reason before disconnection is applied.

C. The System SHALL support sponsor-configurable reason format for disconnection, either as a free text field or a predefined list.

D. The System SHALL preserve all **Participant** data and history upon disconnection.

E. When a **Participant** is disconnected, the System SHALL stop synchronizing data from the **Participant**'s **Mobile Application** to the **Sponsor Portal**.

F. When a **Participant** is disconnected, sponsor-specific rules SHALL remain applied to the **Participant**'s **Mobile Application**.

5.4.7 Disconnection Reason Options

REQ ID: `CAL-PRD-participant-disconnection-reason-options`

Rationale

The Callisto deployment elects the predefined-list option for disconnection reasons (over the free-text alternative the platform supports) because operational reporting at sponsor and site level benefits from a controlled vocabulary: trend analysis of disconnection causes requires categories that aggregate consistently across many disconnect events. The three categories cover the realistic distribution of disconnect causes — hardware faults (**Device Issues**), connectivity or software faults (**Technical Issues**), and the unavoidable residual (**Other**) — without forcing the Study Coordinator to guess between fine-grained distinctions that would not survive downstream aggregation anyway.

> **Follow-up — configurability**: This requirement currently encodes the only option implemented in code. Future sponsors may require different rules; introduce a configurable seam (e.g. a parameter on the CAL-PRD-* parent, or a new platform-side template the CAL- REQ Satisfies) when the need arises. Until that seam exists, this REQ is normative for the Callisto deployment.

Assertions

A. The disconnection reason SHALL be selected from a predefined list.

B. The predefined disconnection reason options SHALL be: **Device Issues**, **Technical Issues**, and **Other**.

5.4.8 Participant Reconnection Workflow

REQ ID: `DIARY-PRD-participant-reconnection`

Overview

Reconnection restores a disconnected **Participant**'s link to the **Sponsor Portal**, enabling them to resume trial participation. A new **Mobile Linking Code** is generated to ensure the previous invalidated code cannot be reused.

Rationale

Reconnection re-uses the link-code mechanism deliberately: the same unguessable, single-use, time-bounded code that establishes a first link is also the credential that re-establishes one. Generating a fresh code (rather than re-issuing the original) closes any window in which a leaked or remembered code from the prior link could be exploited; the code that was originally used has long since been invalidated by the single-use rule, and the new code carries its own independent expiry. Synchronizing the disconnected-period data on link restoration is the principal value of the disconnect/reconnect distinction over mark-as-not-participating: the **Participant** continued recording diary entries locally while disconnected, and reconnection promotes that buffered data to the portal so the trial's clinical record is complete. The free-text reason default with sponsor override matches the pattern used by disconnection.

Assertions

A. The System SHALL allow a Study Coordinator to reconnect a **Participant** with **Disconnected** status.

B. The System SHALL require a reason before reconnection is applied.

C. The reconnection reason SHALL be captured via a **Reason Dialog — Free Text** by default and is sponsor-configurable.

D. When reconnection is confirmed, the System SHALL generate a new **Mobile Linking Code**.

E. When a **Participant** successfully enters the new **Mobile Linking Code**, the System SHALL restore the **Participant**'s link to the **Sponsor Portal**.

F. When a **Participant**'s link is restored, the System SHALL synchronize all data recorded on the **Participant**'s **Mobile Application** during the disconnected period to the **Sponsor Portal**.

5.4.9 Mark as Not Participating

REQ ID: `DIARY-PRD-participant-mark-not-participating`

Refines: `CAL-PRD-participant-not-participating-reason-options`

Overview

A Study Coordinator may mark a **Participant** as not participating when they complete or withdraw from the trial. This is distinct from disconnection — sponsor-specific rules are no longer applied to the **Participant**'s **Mobile Application** and the **Participant** is not expected to resume participation unless explicitly reactivated.

Rationale

Mark-as-not-participating is the off-trial state: the **Participant** has either completed the protocol or withdrawn, and the platform no longer needs to enforce trial-specific rules against their device. Routing the action through **Disconnected** as a precondition keeps the lifecycle linear (Linked → Disconnected → Not Participating) and prevents the case where a still-actively-linked device suddenly loses its trial constraints without the Coordinator first severing the link. Removing sponsor-specific rules from the **Mobile Application** lets a former **Participant** retain personal use of the app post-trial without remaining subject to record-locking, validation thresholds, or reminder schedules that were specific to their trial role. Reason capture mirrors disconnection's pattern, with the controlled-vocabulary case being more common for end-of-trial reporting (CAL- overlay encodes the predefined list this deployment uses).

Assertions

A. The System SHALL allow a Study Coordinator to mark a **Participant** with **Disconnected** status as **Not Participating**.

B. The System SHALL require a reason before the action is applied.

C. The System SHALL support sponsor-configurable reason format for marking as not participating, either as a free text field or a predefined list.

D. When the action is confirmed, the System SHALL stop applying sponsor-specific rules to the **Participant**'s **Mobile Application**.

5.4.10 Mark as Not Participating Reason Options

REQ ID: `CAL-PRD-participant-not-participating-reason-options`

Rationale

End-of-trial reporting is sensitive to the categorical distinction between participant-driven exit (**Subject Withdrawal**), participant death (**Death**, which carries distinct safety-reporting implications), planned protocol completion (**Protocol Treatment/Study Complete**), and the residual case (**Other**). Encoding these as a predefined list ensures the audit trail and downstream analyses can aggregate exit reasons consistently and meet the reporting obligations Callisto's regulatory submissions depend on. The categories are exhaustive enough that **Other** should be rare; when it does occur, the audit trail's free-text reason narrative carries the specific explanation alongside the categorical tag.

> **Follow-up — configurability**: This requirement currently encodes the only option implemented in code. Future sponsors may require different rules; introduce a configurable seam (e.g. a parameter on the CAL-PRD-* parent, or a new platform-side template the CAL- REQ Satisfies) when the need arises. Until that seam exists, this REQ is normative for the Callisto deployment.

Assertions

A. The reason SHALL be selected from a predefined list.

B. The predefined reason options SHALL be: **Subject Withdrawal**, **Death**, **Protocol Treatment/Study Complete**, and **Other**.

5.4.11 Reactivate Participant

REQ ID: `DIARY-PRD-participant-reactivate`

Overview

A Study Coordinator may reactivate a **Participant** who was incorrectly marked as not participating or who has re-enrolled in the trial. Reactivation restores sponsor-specific rules to the **Participant**'s **Mobile Application** and makes the standard reconnection workflow available.

Rationale

Reactivation is the inverse of mark-as-not-participating: the **Participant** is returning to active trial status, and the platform must restore the trial-specific rules that were lifted at off-trial time. Routing reactivation back through the standard reconnection workflow (rather than skipping straight to **Trial Active**) means the device-side link must be re-established through the same link-code mechanism used initially; this is both an audit affordance and a defense against reactivating a **Participant** whose device is no longer in their possession. The free-text reason captures the rationale for what is operationally an unusual event (Mark-as-Not-Participating is intended to be terminal); requiring the reason here gives the audit trail the context reviewers need to interpret the reactivation.

Assertions

A. The System SHALL allow a Study Coordinator to reactivate a **Participant** with **Not Participating** status.

B. The System SHALL require a free text reason before the action is applied.

C. When the action is confirmed, the System SHALL re-apply sponsor-specific rules to the **Participant**'s **Mobile Application**.

5.4.12 Participant Dashboard

REQ ID: `DIARY-GUI-participant-dashboard`

Refines: `DIARY-GUI-show-linking-code`, `DIARY-GUI-link-participant-flow`, `CAL-GUI-part`

Overview

The **Participant Dashboard** is the primary surface for clinical staff to manage participant linking, trial status, and questionnaires. It consists of a participant list organised by tabs, a search input, and a **Participant Actions Modal** accessible from each participant row.

Definitions

Participant Actions Modal: A modal dialog displayed when a Study Coordinator selects a participant row, presenting the **Participant** ID and the secondary actions available for that participant's current status.

Tab Organization

Search

Participant List

Participant Actions Modal

Action Availability

Rationale

The **Participant Dashboard** is the Study Coordinator's daily working surface, and its design choices favor speed and accuracy over visual variety. Tab-by-status organisation keeps the candidate set for any given action small and self-consistent — a Coordinator looking to disconnect a **Participant** finds them under the Active tab, never mixed with **Not Connected** rows. Cross-tab partial-match search supports the recurring "find this specific **Participant** ID" workflow without forcing the Coordinator to first guess which tab they live under. Tab counts that update with the search reflect the same scoping rule, so the Coordinator can see at a glance whether the search matched within each status grouping. Row selection opens the modal rather than navigating away because the dashboard context (other **Participants**, current tab, search state) must remain visible behind the modal for the common workflow of consecutive lookups.

Assertions

A. The System SHALL organise participants into tabs based on their current **Participant** status.

B. Each tab SHALL display a count of participants matching its status grouping.

C. The System SHALL provide a search input that filters participants by **Participant** ID.

D. The search SHALL apply across all tabs regardless of the currently selected tab.

E. The search SHALL support partial matching.

F. The search SHALL update results in real time as the user types.

G. When the search is active, each tab count SHALL update to reflect matching participants within that tab's status grouping.

H. When no participants match the search, the **System** SHALL display the message "No results found."

I. The interface SHALL display the **Participant** ID, **Site**, current status as a **Status Badge**, and a primary action in the Action column for each participant row.

J. The interface SHALL display the currently active tab with a visual indicator.

K. When a **Participant**'s status changes, the interface SHALL move the participant row to the appropriate tab immediately.

L. The interface SHALL display the **Participant Actions Modal** when a Study Coordinator selects a participant row.

M. The **Participant Actions Modal** SHALL display the **Participant** ID.

N. The interface SHALL present the actions available in the **Participant Actions Modal** according to the action availability table.

O. The interface SHALL display the primary action in the Action column and the secondary actions in the **Participant Actions Modal** according to the action availability table defined in CAL-GUI-participant-dashboard-configuration.

5.4.13 Participant Dashboard Configuration

REQ ID: `CAL-GUI-participant-dashboard-configuration`

Overview

This requirement defines the Callisto-specific configuration of the **Participant Dashboard** — the concrete tab set and the per-status action availability table. The platform-side dashboard scaffolding (tabs, search, modal) is established by `DIARY-GUI-participant-dashboard`; this requirement specifies how the Callisto deployment fills it in.

Definitions

Participant Summary: The participant list section of the **Sponsor Portal** dashboard displaying all participants across the Study Coordinator's assigned sites.

Tabs

Action Availability Table

Rationale

The three-tab structure (**Not Connected**, **Active**, **Inactive**) groups participants by operational phase rather than by individual status, which keeps the dashboard scannable as the participant population grows: a Study Coordinator looking for a participant who is "somewhere in onboarding" finds them under one tab without first guessing whether they are

Pending, Expired, or Not Connected. The action availability table is the operational core of the Coordinator's workflow — it encodes which actions are valid in which state, and the primary-action column makes the next-expected action reachable in a single click for the common path.

> **Follow-up — configurability**: This requirement currently encodes the only option implemented in code. Future sponsors may require different rules; introduce a configurable seam (e.g. a parameter on the CAL-PRD-* parent, or a new platform-side template the CAL- REQ Satisfies) when the need arises. Until that seam exists, this REQ is normative for the Callisto deployment.

Assertions

A. The interface SHALL display three tabs: **Not Connected**, **Active**, and **Inactive**.

B. The **Not Connected** tab SHALL display participants with status **Not Connected**, **Pending**, or **Expired**.

C. The **Active** tab SHALL display participants with status **Linked - Awaiting Start** or **Trial Active**.

D. The **Inactive** tab SHALL display participants with status **Disconnected** or **Not Participating**.

E. The interface SHALL display the **Not Connected** tab by default.

F. The interface SHALL display the primary action in the Primary Action column and the secondary actions in the **Participant Actions Modal** according to the following table: | Status | Primary Action Column | Participant Actions Modal | | :—- | :—- | :—- | | **Not Connected** | Link Participant | Link Participant | | **Pending** | Show Linking Code | Show Linking Code | | **Expired** | Regenerate Code | Regenerate Code | | **Linked - Awaiting Start** | Start Trial | Show Linking Code, Disconnect Participant | | **Trial Active** | Manage Questionnaires | Show Linking Code, Disconnect Participant | | **Disconnected** | Reconnect Participant | Reconnect Participant, Mark as Not Participating, Show Linking Code | | **Not Participating** | Reactivate Participant | Reactivate Participant, Show Linking Code |

5.4.14 Show Linking Code

REQ ID: `DIARY-GUI-show-linking-code`

Overview

Show Linking Code is not a new Action — **Participant** linking-code visibility is already governed by the view-participant permission. This requirement specifies how the **Partici-**

pant Dashboard surfaces that visibility as a pseudo-action and what the interface displays when a Study Coordinator selects it.

Linking Code Display

Rationale

Show Linking Code is a pseudo-action: it does not change participant state, it surfaces the existing view-participant permission in a place the Coordinator already works. The split between **Mobile Linking Code** (Pending) and **Participant Linking Code** (every other status) reflects the linking-code lifecycle — Pending participants need the active code to complete linking; everyone else gets the historical reference for troubleshooting per `DIARY-PRD-linking-code-lifecycle` assertion H. The Save-as-PDF affordance supports the common workflow of giving the participant a physical copy of their linking code at a site visit.

Assertions

A. When a Study Coordinator selects Show Linking Code for a **Participant** with **Pending** status, the interface SHALL display the **Mobile Linking Code** with a Copy action and a Save as PDF action.

B. The Save as PDF action SHALL generate a PDF containing the **Mobile Linking Code** and participant instructions.

C. When a Study Coordinator selects Show Linking Code for a **Participant** with any status other than **Pending**, the interface SHALL display the **Participant Linking Code**.

5.4.15 Link Participant Flow

REQ ID: `DIARY-GUI-link-participant-flow`

Overview

This requirement declares the Link Participant Action (`ACT-PAT-001`) and specifies the dialog flow a Study Coordinator follows to issue a **Mobile Linking Code** for a **Participant** in **Not Connected** status.

Definitions

ACT-PAT-001 — Link Participant: The Action a Study Coordinator performs to issue a **Mobile Linking Code** for a **Participant** in **Not Connected** status, transitioning the **Participant** to **Pending**.

Link Participant Flow

Rationale

The Confirmation/Acknowledgement two-step captures both halves of what the Coordinator must accomplish in a single workflow: the Confirmation Dialog records the Coordinator's deliberate intent to issue a code (with the expiry duration shown so they can set participant expectations), and the Acknowledgement Dialog gives the visible delivery of the generated code together with a Copy affordance and a live expiry timer. Splitting the two means the code is only generated after explicit confirmation, and the Coordinator cannot leave the dashboard without having seen the code at least once.

Assertions

A. When a Study Coordinator selects Link Participant, the interface SHALL display a Confirmation Dialog showing the **Participant** ID and the configured **Mobile Linking Code** expiry duration.

B. When the Study Coordinator confirms, the interface SHALL display an Acknowledgement Dialog showing the generated **Mobile Linking Code**, a Copy action, and the remaining time until expiry.

C. When the Study Coordinator dismisses the Acknowledgement Dialog, the interface SHALL update the **Participant**'s **Status Badge** to **Pending**.

5.5 Sponsor Portal Authentication

Sponsor Portal authentication comprises password composition and reuse rules, two-factor authentication, forgot-password recovery (PRD plus the workflow GUI), and session management.

5.5.1 Password Requirements

REQ ID: `DIARY-PRD-password-requirements`

Definitions

Password: A secret string of characters used to authenticate a user's identity when accessing the **System**.

Password Reuse Limit: The configurable number of the user's most recent **Passwords** that a new **Password** must not match.

Rationale

The composition rules (length, character classes) and the common-password rejection are baseline defenses against credential-guessing attacks: each rule independently raises the cost of a successful brute-force or dictionary attempt, and together they place the minimum acceptable password well above the threshold at which automated attacks succeed against unprotected accounts. The NIST SP 800-63B reference is the authoritative source for the common-password list and is named explicitly so the deployment can update the list as NIST updates its guidance. The 90-day expiry is a sponsor-overridable default; the override exists because some sponsor deployments operate under regulatory regimes that mandate a different interval. The reuse-limit rule prevents the common operator behavior of cycling between two passwords (defeats the purpose of expiry); making the limit configurable lets a deployment choose how aggressive its reuse defense should be.

Assertions

A. The **System** SHALL require a **Password** to be a minimum of 12 characters in length.

B. The **System** SHALL require a **Password** to contain at least one uppercase letter, one lowercase letter, one numeric character, and one special character.

C. The **System** SHALL reject a **Password** that is commonly used or easily guessable as per NIST SP 800-63B commonly used password list.

D. The **System** SHALL require a **Password** to be changed after 90 days, unless a different interval has been configured for the deployment.

E. The **System** SHALL prevent a user from accessing the **System** until their **Password** has been changed upon expiry.

F. The **System** SHALL reject a new **Password** that matches any of the user's previous **Password Reuse Limit Passwords**, where **Password Reuse Limit** is configurable per deployment.

5.5.2 Two-Factor Authentication

REQ ID: `DIARY-PRD-two-factor-authentication`

Refines: `CAL-PRD-two-factor-authentication-configuration`

Overview

Two-Factor Authentication adds a second independent factor beyond the **Password** to prevent unauthorized access in the event of credential compromise. The platform supports configurable second factors so each sponsor deployment can select the method that best fits its operational and regulatory context.

Definitions

Second Factor: The independent authentication factor required in addition to the **Password** during login. The specific method is sponsor-configurable per deployment.

Verification Code: A single-use, time-limited code presented as the **Second Factor** during login.

Code Expiry: The configurable duration after which an unused **Verification Code** becomes invalid.

Enforcement

Verification Code Lifecycle

Configuration

Rationale

The single password is no longer a sufficient credential for clinical trial portal access: credential leakage from unrelated services, phishing, and shared-workstation compromise are all common, and any of them can hand an attacker a working password without the user noticing. The second factor breaks that single-credential failure mode by requiring possession of an independent channel (email, authenticator app, or SMS, depending on sponsor configuration) before access is granted. The **Verification Code** is single-use and time-limited because a code that survived either property would inherit the same replay vulnerability the second factor exists to prevent. Sponsor-configurability of the method and expiry duration recognises that the operational tradeoffs vary by deployment: a sponsor with strict email infrastructure may choose a longer expiry; a sponsor with authenticator-app adoption may choose a much shorter one.

Assertions

A. The **System** SHALL require Two-Factor Authentication for every login to the **Sponsor Portal**.

B. The **System** SHALL NOT grant access to the **Sponsor Portal** until both the **Password** and a valid **Verification Code** have been successfully validated.

C. When a **User Account** owner submits a valid **Password**, the **System** SHALL generate a **Verification Code** and deliver it via the configured **Second Factor** method.

D. The **System** SHALL ensure each **Verification Code** is single-use and is invalidated immediately upon successful use.

E. When a **Verification Code** has not been used within the **Code Expiry** duration, the **System** SHALL invalidate the **Verification Code**.

F. When a **Verification Code** has been invalidated, the **System** SHALL require the **User Account** owner to restart the login process to receive a new **Verification Code**.

G. The **System** SHALL support sponsor-configurable selection of the **Second Factor** method per deployment.

H. The **System** SHALL support sponsor-configurable **Code Expiry** per deployment.

5.5.3 Two-Factor Authentication Configuration

REQ ID: `CAL-PRD-two-factor-authentication-configuration`

Rationale

Email delivery of the **Verification Code** is the Callisto deployment's chosen **Second Factor** because the registered **Email Address** is already established as the authoritative communication channel for every **User Account** (it is the activation channel, the password-reset channel, and the username), and using it as the 2FA channel keeps the authentication surface coherent without introducing additional credential infrastructure (authenticator apps, SMS gateway). The 10-minute **Code Expiry** is short enough that a leaked code becomes useless quickly while long enough to absorb realistic email-delivery latency (typical clinical-environment email lag is under 2 minutes; the 10-minute window accommodates the long-tail cases).

Assertions

A. The **Second Factor** method SHALL be configured to email delivery to the **User Account** owner's registered **Email Address**.

B. The **Code Expiry** SHALL be configured to 10 minutes.

5.5.4 Forgot Password

REQ ID: `DIARY-PRD-password-forgot`

Refines: `DIARY-GUI-password-forgot-workflow`

Overview

A **User Account** owner who has forgotten their **Password** must be able to reset it without Administrator intervention. The reset mechanism uses a time-limited, single-use **Verification Link** delivered to the registered **Email Address** to confirm the requester controls the account.

Initiation

Verification Link Behavior

Reset Completion

Rejection Behavior

Rationale

Self-service password reset is a usability requirement (users who lose their **Password** must not be blocked from the system pending an Administrator escalation) and a security requirement (the reset mechanism must not become an attack channel). The verification-link mechanism mirrors the activation workflow: a single-use, time-bounded URL delivered to the registered email address is the credential that proves control of the account. The display-confirmation-regardless rule prevents user enumeration via the reset form — an attacker submitting addresses learns nothing about which ones exist in the system. Terminating all active sessions on successful reset and requiring 2FA on the next login closes the window in which a stolen session or stolen 2FA token from the pre-reset interval could still operate against the now-changed account. The shorter expiry (24 hours, versus 14 days for activation) reflects the higher attack value of a password-reset link compared to an activation link.

Assertions

A. The **System** SHALL allow a **User Account** owner to initiate a **Password** reset from the **Sponsor Portal** login interface by providing their **Email Address**.

B. When a **Password** reset is initiated, the **System** SHALL generate a **Verification Link** and deliver it to the registered **Email Address**.

C. The **System** SHALL display a confirmation that an email has been sent regardless of whether the **Email Address** matches an existing **User Account**.

D. When a **Verification Link** issued for **Password** reset is used or 24 hours have passed from generation, whichever occurs first, the **System** SHALL invalidate the **Verification Link**.

E. When a new **Password** reset **Verification Link** is issued for a **User Account**, the **System** SHALL invalidate any previously issued **Password** reset **Verification Link** for that **User Account**.

F. When a **User Account** owner accesses a valid **Verification Link**, the **System** SHALL allow them to set a new **Password**.

G. Upon successful **Password** reset, the **System** SHALL terminate all active sessions associated with that **User Account**.

H. The **System** SHALL require Two-Factor Authentication on the next login following a successful **Password** reset.

I. When a **User Account** owner attempts to access an invalidated **Verification Link**, the **System** SHALL reject the attempt and prompt the user to initiate a new **Password** reset.

5.5.5 Forgot Password Workflow Interface

REQ ID: `DIARY-GUI-password-forgot-workflow`

Overview

The Forgot Password workflow spans four screens: the request screen where the **User Account** owner enters their **Email Address**, the confirmation screen shown after submission, the reset screen reached through a valid **Verification Link**, and the invalid link screen reached through an invalidated **Verification Link**. Consistent screen behavior across the flow ensures the **User Account** owner can complete recovery or recognize when to start over.

Entry Point

Forgot Password Request Screen

Confirmation Screen

Reset Screen

Invalid Link Screen

Rationale

The four-screen structure mirrors the four states the workflow can reach: requesting reset, awaiting email, completing reset, recovering from an invalidated link. Identical confirmation content regardless of whether the email matches is the GUI-level enforcement of the user-enumeration resistance established at the PRD level — divergent UI between matching and non-matching cases would leak the same information the PRD assertion is trying to hide. The inline composition-rule error on Submit is necessary because users who submitted an invalid **Password** must learn which rule failed without losing the typed values, otherwise the show/hide toggle gains a punitive UX where users repeatedly retype the same long password. The Invalid Link screen's link back to the request screen restarts the recovery loop in a single click, recognising that the most common reason for an invalidated link is the 24-hour expiry catching a user who opened the email belatedly.

Assertions

A. The **Sponsor Portal** login interface SHALL present a Forgot Password action that navigates to the Forgot Password Request screen.

B. The Forgot Password Request screen SHALL present a single **Email Address** input field and a Submit action.

C. The Forgot Password Request screen SHALL present a Back to Login action that returns to the login interface.

D. The interface SHALL not enable the Submit action until a value has been entered in the **Email Address** field.

E. The interface SHALL validate that the entered value conforms to a valid email address format before submission is accepted.

F. When the Submit action is invoked, the interface SHALL navigate to the Confirmation screen.

G. The Confirmation screen SHALL display a message indicating that an email has been sent if the submitted **Email Address** is associated with an account.

H. The Confirmation screen SHALL display the duration after which the **Verification Link** expires.

I. The Confirmation screen SHALL display guidance to check the spam folder if the email is not received.

J. The Confirmation screen SHALL present a Back to Login action that returns to the login interface.

K. The Confirmation screen SHALL display the same content regardless of whether the submitted **Email Address** matches an existing **User Account**.

L. The Reset screen SHALL be reachable only by accessing a valid **Verification Link** delivered to the **User Account** owner's registered **Email Address**.

M. The Reset screen SHALL present a New **Password** field and a Confirm **Password** field.

N. Each **Password** field SHALL present a show/hide toggle that controls whether the entered value is displayed in plain text or masked.

O. The Reset screen SHALL present a Submit action.

P. The interface SHALL not enable the Submit action until both **Password** fields have been populated and contain matching values.

Q. When the Submit action is invoked and the **Password** is rejected for failing composition or reuse rules, the interface SHALL display an inline message identifying which rule was violated and SHALL not navigate away from the Reset screen.

R. When the Submit action is invoked and the **Password** is accepted, the interface SHALL navigate to the **Sponsor Portal** login interface and display a confirmation that the **Password** has been changed.

S. When a **User Account** owner accesses an invalidated **Verification Link**, the interface SHALL display the Invalid Link screen.

T. The Invalid Link screen SHALL display a message indicating that the **Verification Link** is no longer valid.

U. The Invalid Link screen SHALL present an action that navigates to the Forgot Password Request screen.

5.5.6 Session Management

REQ ID: `DIARY-PRD-session-management`

Refines: `CAL-PRD-login-rate-limiting-configuration`

Overview

The **Sponsor Portal** terminates inactive sessions to limit the window during which an unattended authenticated session could be exploited. Session limits also ensure that role and permission changes take effect within a bounded time.

Definitions

Session: An authenticated period during which a **User Account** owner can access the **Sponsor Portal** without re-entering credentials.

Session Idle Timeout: The configurable maximum duration of inactivity allowed during a **Session** before the **System** terminates it.

Session Establishment

Idle Timeout

Termination

Configuration

Rationale

A **Session** in the **Sponsor Portal** is a high-value authentication artifact — it represents a successful two-factor login and confers access to clinical data and User Account management capabilities for its duration. The idle timeout caps the window in which an unattended workstation could be exploited; tracking inactivity from the user's most recent interaction (rather than from session creation) is the standard pattern that balances security against operational disruption. The cascade rules (deactivation, role change, Site change immediately terminate sessions) ensure that authorization changes take effect synchronously rather than waiting for the next login: a Coordinator who has lost their role for cause cannot continue acting under the old role until their **Session** happens to time out. Sponsor-configurability of the timeout duration acknowledges that the right tradeoff between security and operator disruption varies by deployment; the 10-minute default reflects clinical-portal industry baseline.

Assertions

A. The **System** SHALL establish a **Session** when a **User Account** owner successfully completes Two-Factor Authentication.

B. The **System** SHALL track elapsed inactivity from the **User Account** owner's most recent interaction with the **Sponsor Portal**.

C. When the **Session Idle Timeout** is exceeded, the **System** SHALL terminate the **Session** and require re-authentication.

D. The **System** SHALL allow a **User Account** owner to explicitly terminate their **Session** by logging out.

E. When a **User Account** is deactivated, the **System** SHALL terminate all active **Sessions** associated with that **User Account** immediately.

F. When a **User Account**'s **Role** or **Site** assignment is changed, the **System** SHALL terminate all active **Sessions** associated with that **User Account** immediately.

G. The **System** SHALL support sponsor-configurable **Session Idle Timeout** per deployment, with a default of 10 minutes.

5.5.7 Login Attempt Rate Limiting Configuration

REQ ID: `CAL-PRD-login-rate-limiting-configuration`

Rationale

Five consecutive failures is well above the rate at which a legitimate user mistyping their **Password** would trigger the rate limit (most legitimate users notice the mistake within two or three attempts), but well below the rate at which an automated brute-force attempt would be useful. The 60-second cooldown forces an attacker to amortize attempts across many waits, raising the cost of any brute-force campaign to impractical levels while imposing only a minor inconvenience on a legitimate user who has genuinely forgotten their password (they would typically initiate a Forgot Password reset rather than retry). The exact-text error message is specified so that the user-visible behavior is consistent across the deployment and matches the cooldown duration.

Assertions

A. The **Login Rate Limit Threshold** SHALL be configured to 5 consecutive failed login attempts.

B. The **Login Rate Limit Cooldown** SHALL be configured to 60 seconds.

C. The Login Rate Limit error message SHALL display the text: "Too many failed attempts. Please wait 60 seconds before trying again."

5.6 Administrator Settings

This section defines the **Administrator Settings** surface, a read-only view available to Administrators that displays the current values of sponsor-configured platform parameters. The surface provides Administrators with visibility into how the deployment is configured without granting modification capability.

5.6.1 Administrator Settings Surface

REQ ID: `DIARY-PRD-administrator-settings`

Refines: `DIARY-GUI-administrator-settings`, `CAL-PRD-administrator-settings-inventor`

Overview

The **Sponsor Portal** maintains a body of sponsor-configured platform parameters that govern System behavior. These parameters are defined in individual requirements throughout the specification and are typically static for the duration of a deployment. **Administrators** require visibility into the current values of these parameters to support troubleshooting, audit response, and operational verification, without needing to consult the specification or contact CureHHT personnel. The **Administrator Settings** surface consolidates these values into a single read-only view scoped to the **Administrator** role.

Definitions

Configuration Parameter: A named, sponsor-configured value that governs a specific System behavior, defined in a single source requirement and surfaced to **Administrators** for visibility.

Configuration Category: A grouping of related **Configuration Parameters** presented together within the **Administrator Settings** surface.

Administrator Settings: The read-only surface within the **Sponsor Portal** that displays **Configuration Parameters** to **Administrators**, organised by **Configuration Category**.

Availability

Content

Read-Only Behavior

Configuration

Rationale

Sponsor-configured parameters (session timeouts, code expiries, threshold values, reminder schedules) shape the system's runtime behavior but are not visible to the **Administrator** who is operationally responsible for the deployment. Without a surface that exposes these values, troubleshooting and audit response require the **Administrator** to either read the specification or escalate to CureHHT, neither of which scales. Consolidating the visible parameters under the **Administrator** role, with one **Configuration Category** per logical grouping and one source-requirement reference per parameter, gives the **Administrator** a single place to confirm "what is this deployment configured to do?" without granting modification capability. Modification stays out of band (the parameters are set during sponsor onboarding and changed via CureHHT change-control) because their values are tightly coupled to compliance review; a runtime modification surface would bypass that review.

Assertions

A. The **System** SHALL provide an **Administrator Settings** surface within the **Sponsor Portal**.

B. The **System** SHALL display each **Configuration Parameter** in **Administrator Settings** with its name, current value, and unit of measurement where applicable.

C. The **System** SHALL display each **Configuration Parameter** under exactly one **Configuration Category**.

D. The **System** SHALL display the current value of each **Configuration Parameter** as defined by the source requirement that establishes it.

E. The **System** SHALL NOT permit modification of any **Configuration Parameter** value from the **Administrator Settings** surface.

F. The **System** SHALL support sponsor-configurable selection of which **Configuration Parameters** are surfaced in **Administrator Settings** per deployment.

G. The **System** SHALL support sponsor-configurable definition of **Configuration Categories** per deployment.

5.6.2 Administrator Settings Configuration Inventory

REQ ID: `CAL-PRD-administrator-settings-inventory`

Overview

This requirement defines the **Configuration Parameters** surfaced in **Administrator Settings** for the Callisto **Sponsor Portal** deployment, the **Configuration Categories** under which they are grouped, and the source requirement establishing each value.

Rationale

The configuration inventory is the per-deployment realisation of the platform-level **Administrator Settings** surface: each row binds a user-visible parameter name to the source requirement that establishes its current value, so the Administrator surface can render the current value by reading the named requirement rather than maintaining a parallel value store. Six categories were chosen to reflect the operational domains an Administrator reasons about — authentication, mobile security, entry validation, questionnaire timing, notifications, and rate limits — and the order from authentication outward to rate limiting follows the rough operational frequency with which an Administrator might consult each category. Cross-repo refines targets are used for the platform-side parents (session management, password rules, administrator settings, configuration precedence) and for the Callisto-side overlays in this and sibling files; binding the surface to the source requirements (rather than to a free-text value table) keeps the surface in sync with whatever values future change-control rounds adopt.

Assertions

A. The **Administrator Settings** surface SHALL display the following **Configuration Categories** in the order listed: **Authentication and Sessions**, **Mobile Application Security**, **Diary Entry Rules**, **Questionnaire Sessions**, **Notifications and Reminders**, and **Rate Limiting**.

B. The **Administrator Settings** surface SHALL display the following **Configuration Parameters** under their respective **Configuration Categories** with the current values shown: | Configuration Category | Parameter | Source Requirement | | — | — | — | | Authentication and Sessions | Sponsor Portal Session Idle Timeout | DIARY-PRD-session-management | | Authentication and Sessions | Password Expiry Interval | DIARY-PRD-password-requirements | | Authentication and Sessions | Two-Factor Authentication Code Expiry | CAL-PRD-two-factor-authentication-configuration | | Mobile Application Security | Mobile Linking Code Expiry | CAL-PRD-linking-code-lifecycle-configuration | | Mobile Application Security | Application Lock Idle Timeout | CAL-PRD-application-lock-configuration | | Mobile Application Security | Application Lock PIN Length | CAL-PRD-application-lock-configuration | | Mobile Application Security | Application Lock Failed Attempt Threshold | CAL-PRD-application-lock-configuration | | Diary Entry Rules | Justification Threshold | CAL-PRD-entry-time-restrictions-configuration | | Diary Entry Rules | Lock Threshold | CAL-PRD-entry-time-restrictions-configuration | | Diary Entry Rules | Short Duration Threshold | CAL-PRD-entry-duration-check-configuration | | Diary Entry Rules | Long Duration Threshold | CAL-PRD-entry-duration-check-configuration | | Questionnaire Sessions | NOSE HHT and HHT-QoL Session Timeout | CAL-PRD-questionnaire-session-resume-timeout | | Questionnaire Sessions | NOSE HHT and HHT-QoL Timeout Warning Threshold | CAL-PRD-questionnaire-session-resume-timeout | | Notifications and Reminders | Incomplete Record Lock Warning Offset | CAL-PRD-notification-incomplete-record-lock-configuration | | Notifications and Reminders | Yesterday Entry Reminder Time | CAL-PRD-notification-yesterday-entry-configuration | | Notifications and Reminders | Ongoing Epistaxis Event Reminder Interval | CAL-PRD-notification-ongoing-epistaxis-configuration | | Rate Limiting | Login Rate Limit Threshold | CAL-PRD-login-rate-limiting-configuration | | Rate Limiting | Login Rate Limit Cooldown | CAL-PRD-login-rate-limiting-configuration | | Rate Limiting | Linking Code Rate Limit Threshold | CAL-PRD-entry-time-restrictions-configuration | | Rate Limiting | Linking Code Rate Limit Cooldown | CAL-PRD-entry-time-restrictions-configuration |

5.6.3 Administrator Settings Interface

REQ ID: `DIARY-GUI-administrator-settings`

Overview

The **Administrator Settings** surface is reached from the **Settings** action in the **Administrator Dashboard** header. The surface presents **Configuration Parameters** as a grouped list, with each **Configuration Category** as a section heading and each **Configuration Parameter** as a row beneath it. The interface communicates the read-only nature of the view so the **Administrator** does not expect to edit values directly.

Display

Read-Only Indication

Rationale

Settings is a low-frequency surface — an **Administrator** opens it during troubleshooting or audit response, not during routine workflow — so it is reached via a single header action rather than a tab. Presenting it as a full-page view that replaces the tab content (rather than a modal or a side-panel) reflects the surface's information density (multiple categories, many parameters per category) and lets the **Administrator** scan the entire configuration without scrolling between panels. The explicit read-only notice and the absence of any input control communicate the read-only contract at the GUI level: an **Administrator** who reaches this screen should not waste time looking for an edit affordance, and should know who to contact (CureHHT personnel) when a value needs to change. The back action returns to the previously active tab rather than always defaulting to **Users** because Administrators frequently reach Settings from the audit log during an investigation and should be returned to the same investigation context.

Assertions

A. The interface SHALL display the **Administrator Settings** surface as a full-page view that replaces the **Administrator Dashboard** top-level tab content.

B. The interface SHALL display a header on the **Administrator Settings** surface containing the title **Settings** and a back action that returns the **Administrator** to the previously active top-level tab.

C. The interface SHALL display each **Configuration Category** as a labeled section.

D. The interface SHALL display each **Configuration Parameter** as a row within its **Configuration Category** section, showing the parameter name, the current value, and the unit of measurement where applicable.

E. The interface SHALL display **Configuration Parameters** in the order defined by the deployment configuration.

F. The interface SHALL display a notice on the **Administrator Settings** surface indicating that values are read-only and that changes require contacting CureHHT personnel.

G. The interface SHALL NOT present any input control, edit action, or save action on the **Administrator Settings** surface.

MOBILE APPLICATION

6.1 Mobile Application Foundation

The **Mobile Application** foundation comprises the dual-mode (personal vs. linked) operating model, offline-first data entry, the **Diary Start Day** invariant, the sponsor-configurable **Clinical Trial Privacy Policy**, and the **Application Lock**.

6.1.1 Diary Mobile Application

REQ ID: `DIARY-PRD-mobile-application`

Refines: `DIARY-PRD-mobile-offline-first`

Overview

The mobile application serves dual purposes: personal health tracking for individual **Users** and compliant data capture for clinical trials. In personal use mode the application requires no account and stores data locally. In linked use mode the application synchronizes data with the sponsor portal for clinical trial participation.

Definitions

Mobile Application: The iOS and Android application provided by the System for participant-reported diary data capture and questionnaire completion. Operates in personal use mode without an account or in linked use mode connected to a **Sponsor Portal** deployment.

User: An individual using the **Mobile Application** in personal use mode, without an account and without a link to a clinical trial. A User who subsequently links to a study becomes a **Participant**.

Rationale

The dual-mode design serves two distinct populations from a single codebase: individuals tracking nosebleeds for personal health reasons (no clinical-trial context, no sponsor, no account) and clinical-trial participants whose data feeds the sponsor's regulatory submission. Personal mode is account-less by design — it removes onboarding friction for the first population, who would not benefit from an account they cannot use against any back-end, and it keeps the device the single point of control for their data. Linked mode adds the **Sponsor Portal** synchronization path; the explicit-consent gate on first sync ensures the **User** transitioning to **Participant** affirmatively chooses to share their previously-private local entries with the sponsor, rather than that data being silently uploaded on link. iOS and Android coverage is required because participants in any plausible clinical-trial population will hold devices on both platforms; restricting to one would exclude participants without a fallback.

Assertions

A. The System SHALL provide a mobile application for iOS platforms, available via the iOS app store.

B. The System SHALL provide a mobile application for Android platforms, available via the Android app store.

C. The mobile application SHALL support full offline operation for core diary functions in both personal use mode and linked use mode.

D. The mobile application SHALL NOT require account creation or login for personal use mode.

E. The **User** SHALL retain control over locally-entered data, including the right to delete it from the device while in personal use mode.

F. The mobile application SHALL obtain explicit **User** consent before synchronizing pre-existing local data to the **Sponsor Portal** upon linking.

6.1.2 Offline-First Data Entry

REQ ID: `DIARY-PRD-mobile-offline-first`

Overview

Offline-first architecture ensures reliable data collection regardless of network conditions. All **Users** benefit from local storage for immediate, reliable data entry. For **Participants** linked to clinical trials, automatic synchronization provides cloud backup and enables remote monitoring while maintaining the offline-first experience.

Rationale

Diary entries are time-sensitive: a nosebleed event must be captured as it occurs or shortly afterward to preserve recall accuracy and satisfy the ALCOA+ Contemporaneous principle. A network-required entry model would push participants toward either delayed entry (when connectivity is restored, by which time recall has degraded) or non-entry (when connectivity never recovers in the relevant window). Offline-first inverts that: every entry succeeds locally regardless of network state, and synchronization to the **Sponsor Portal** is a background concern handled by the platform whenever connectivity is available. The unsynchronized indicator preserves transparency for the participant — they can see whether their data has reached the sponsor — without making sync a gate on entry.

Assertions

A. The System SHALL allow **Users** to create diary entries without requiring internet connectivity.

B. The System SHALL allow **Users** to edit diary entries without requiring internet connectivity.

C. The System SHALL allow **Users** to view their complete entry history without requiring internet connectivity.

D. For linked **Participants**, the System SHALL indicate which entries have not yet synchronized to the **Sponsor Portal**.

E. For linked **Participants**, the System SHALL automatically synchronize unsynchronized entries when network connectivity becomes available.

6.1.3 Diary Start Day Definition

REQ ID: `DIARY-PRD-diary-start-day`

Overview

Clinical trial **Participants** may need to record nosebleeds that occurred before their first Mobile Application usage. The **Diary Start Day** establishes the earliest date for which diary entries are valid, balancing the need for historical data capture with the requirement to maintain reliable data quality. The **Diary Start Day** is set automatically based on the **Participant**'s actual entries, when a **Participant** records data for a date earlier than any prior entry, the **Diary Start Day** moves backward to that date. The **Diary Start Day** never moves forward, even if the earliest entry is subsequently deleted, to preserve the historical scope of the diary and ensure that documented gaps remain visible.

Definitions

Diary Start Day: The earliest date for which diary entries are valid for a given **User**.

Rationale

The **Diary Start Day** is the single boundary the **Mobile Application** uses to distinguish "no entry expected" from "missing entry" — every day from the **Diary Start Day** to today is part of the diary period and therefore a candidate for the missing-day display in the calendar. Automatic backward expansion (set to the earliest entry, can move backward but never forward) reflects two operational realities: participants discover the need to backfill historical events incrementally rather than declaring a start date up front, and once a date is recognized as part of the diary period, the visible gaps on dates within it are evidence the audit and sponsor reporting rely on — collapsing those gaps by moving the start day forward when the earliest entry is deleted would silently erase that evidence. The 365-day floor below installation date caps the historical window at a clinically meaningful range while leaving room for retrospective capture early in a trial. Future-date rejection is a contemporaneous-data integrity floor. The non-sync rule for pre-trial dates keeps the **Sponsor Portal** dataset bounded to the trial period regardless of how far back the personal diary extends.

Assertions

A. The System SHALL establish and maintain a **Diary Start Day** for each **User**.

B. The System SHALL set the **Diary Start Day** to the date of the earliest entry the **Participant** has ever recorded.

C. When a **Participant** records an entry for a date earlier than the current **Diary Start Day**, the System SHALL move the **Diary Start Day** backward to that date.

D. The System SHALL NOT allow the **Diary Start Day** to be set earlier than 365 days before mobile application installation on the current device.

E. The **Diary Start Day** SHALL only move backward, never forward, once set.

F. The System SHALL NOT prompt **Users** to set the **Diary Start Day** during onboarding.

G. Data entered for dates before the trial start date SHALL NOT synchronize to the **Sponsor Portal** or Rave EDC.

H. The **System** SHALL NOT allow the **Participant** to record an entry for a date in the future.

I. The **System** SHALL display each calendar day between the **Diary Start Day** and the current day, exclusive of any day for which a **Daily Status** has been recorded, as a missing day in the Calendar.

6.1.4 Clinical Trial Privacy Policy

REQ ID: `DIARY-PRD-privacy-policy`

Refines: `CAL-PRD-privacy-policy-configuration`

Overview

The Mobile Application SHALL provide access to a **Clinical Trial Privacy Policy** governing sponsor-side data handling for a linked study, distinct from the **Application Privacy Policy**.

Definitions

Application Privacy Policy: The privacy policy governing the Mobile Application itself, covering platform-level data handling.

Clinical Trial Privacy Policy: The privacy policy governing a Participant's participation in a specific clinical trial, covering sponsor-side data handling. Sponsor-configurable per deployment.

Rationale

Two privacy policies coexist in the **Mobile Application** because two distinct controllers handle distinct data flows: the **Application Privacy Policy** covers platform-level data handling (the **Mobile Application** itself, the platform vendor's data-processor role), while the **Clinical Trial Privacy Policy** covers the sponsor's clinical-trial data handling for participants in a specific study. Surfacing the trial-specific policy only when the participant is linked or actively linking prevents the application from showing sponsor-specific terms to **Users** who are using the app in personal mode and have no relationship with the sponsor. Retaining the version in effect at time of consent against the participant record satisfies the regulatory expectation that the document the participant actually consented to (which may change over the life of the trial) is preserved as evidence of informed consent.

Assertions

A. The System SHALL support a sponsor-configurable **Clinical Trial Privacy Policy** per deployment.

B. The System SHALL make the **Clinical Trial Privacy Policy** accessible to the Participant only when the Participant is linked to a study or is in the process of linking to a study.

C. The System SHALL retain the **Clinical Trial Privacy Policy** version that was in effect at the time of consent against the Participant record.

6.1.5 Clinical Trial Privacy Policy Configuration

REQ ID: `CAL-PRD-privacy-policy-configuration`

Rationale

The Callisto deployment binds the platform-configurable **Clinical Trial Privacy Policy** to a single hosted URL. The URL placeholder will be replaced with the production policy location before deployment goes live; the obligation is that the policy is hosted at a stable, retrievable URL the **Mobile Application** can link to from the **Linking Consent** text on the **Join the Study** screen and from the **Participation Status Badge**. A platform-configurable URL (rather than embedded content) lets the legal text be updated independently of mobile-app release cycles, while the version-retention rule in the parent REQ ensures every participant's consent is anchored to the specific version they read at the time of consent.

> **Follow-up — configurability**: This requirement currently encodes the only option implemented in code. Future sponsors may require different rules; introduce a configurable seam (e.g. a parameter on the CAL-PRD-* parent, or a new platform-side template the CAL- REQ Satisfies) when the need arises. Until that seam exists, this REQ is normative for the Callisto deployment.

Assertions

A. The Clinical Trial Privacy Policy SHALL be made available at [URL placeholder].

6.1.6 Application Lock

REQ ID: `DIARY-PRD-application-lock`

Refines: `CAL-PRD-application-lock-configuration`

Overview

The Mobile Application contains Participant-reported clinical data and identifying information that must be protected from unauthorized access on the Participant's device. An Application Lock requires the Participant to authenticate before each use, ensuring that someone who picks up an unattended device cannot view or modify diary entries. The lock is configurable so each sponsor deployment can select the authentication method appropriate to its risk profile and Participant population. When the configured method is unavailable on a Participant's device, a fallback to Device Authentication ensures continuity of access without compromising the protective intent.

Definitions

Application Lock: The state in which the **Mobile Application** requires the **Participant** to authenticate before any **Participant**-facing screen, action, or data is accessible.

PIN: A numeric secret of configurable length set by the **Participant** and used to release the **Application Lock**.

Device Authentication: Authentication performed by the device operating system using the credential the **Participant** has configured at the device level, including but not limited to device passcode, fingerprint, or face recognition.

Authentication Method: The mechanism required to release the **Application Lock**. The configured **Authentication Method** is one of: **PIN**, **Device Authentication**, or none.

Idle Timeout: The configurable elapsed time the **Mobile Application** may remain in the background before the **System** re-applies the **Application Lock** on return to the foreground.

Lock Application

Authentication Method

PIN Setup and Reset

Failed Attempt Handling

Configuration

Rationale

The **Application Lock** is the per-device gate that protects participant clinical data when a device is left unattended, lost, or stolen. The lock applies at two boundaries — fresh launch and resume-from-background-after-idle — because both are realistic re-entry points after an interval in which the device could have changed hands. The “no data, no action other than authentication” rule while locked closes the leakage channels that would otherwise exist (e.g., notification preview, deep-link routing into a partial workflow). The sponsor-configurable **Authentication Method** lets each deployment choose between three security-posture options: **PIN** (cheapest, works on every device, controlled by the platform), **Device Authentication** (delegates to the OS-managed credential, which may include biometrics), or none (when the deployment determines a separate lock is not required). The failed-attempt fallback to **Device Authentication** rather than account lockout preserves participant access — a participant who has forgotten their **PIN** can still recover by authenticating against the OS credential and resetting — while still raising the bar against a casual unauthorized attempt. Failed-attempt counter reset on success prevents accumulated rare typos from compounding into a fallback over normal use.

> OPEN: Should the **Application Lock** be bypassed for **Push Notification** taps that lead to time-sensitive flows (e.g., Ongoing Epistaxis Event Reminder), or always enforced?

Assertions

A. The **System** SHALL apply the **Application Lock** when the **Mobile Application** is launched from a fully closed state.

B. The **System** SHALL apply the **Application Lock** when the **Mobile Application** returns to the foreground after spending more than the **Idle Timeout** in the background.

C. While the **Application Lock** is applied, the **System** SHALL NOT display **Participant** data and SHALL NOT permit any **Participant** action other than authentication.

D. The **System** SHALL release the **Application Lock** only upon successful authentication using the configured **Authentication Method**.

E. When the configured **Authentication Method** is **PIN**, the **System** SHALL require the **Participant** to enter their **PIN** to release the **Application Lock**.

F. When the configured **Authentication Method** is **Device Authentication**, the **System** SHALL invoke the device operating system’s authentication prompt to release the **Application Lock**.

G. When the configured **Authentication Method** is none, the **System** SHALL NOT apply the **Application Lock**.

H. When the configured **Authentication Method** is **PIN** and the **Participant** has not yet set a **PIN**, the **System** SHALL require the **Participant** to set a **PIN** before any **Participant** data is accessible.

I. The **System** SHALL require the **Participant** to enter the **PIN** twice during setup and

SHALL reject the setup when the two entries do not match.

J. The **System** SHALL allow the **Participant** to change their **PIN** from the **Mobile Application** settings, after first authenticating with their existing **PIN** or **Device Authentication**.

K. The **System** SHALL track consecutive failed **PIN** entry attempts.

L. When the number of consecutive failed **PIN** entry attempts reaches the configured **Failed Attempt Threshold**, the **System** SHALL fall back to **Device Authentication**.

M. Upon successful **Device Authentication** following a fallback, the **System** SHALL require the **Participant** to reset their **PIN** before any **Participant** data is accessible.

N. The **System** SHALL reset the failed attempt counter upon successful authentication.

O. The **System** SHALL support sponsor-configurable selection of the **Authentication Method** per deployment.

P. The **System** SHALL support sponsor-configurable **PIN** length per deployment.

Q. The **System** SHALL support sponsor-configurable **Idle Timeout** per deployment.

R. The **System** SHALL support sponsor-configurable **Failed Attempt Threshold** per deployment.

6.1.7 Application Lock Configuration

REQ ID: `CAL-PRD-application-lock-configuration`

Overview

These values configure the platform mechanism defined in DIARY-PRD-application-lock for the Callisto deployment.

Rationale

The Callisto deployment selects **PIN** as the **Authentication Method** because it works uniformly across every iOS and Android device the participant population may carry, requires no device-level biometric or passcode configuration the participant might not have set up, and keeps the unlock UX consistent regardless of device capabilities. A 4-digit **PIN** is the everyday-credential length participants encounter elsewhere (bank cards, transit, app-level locks) and balances memorability against the brute-force surface, which is further bounded by the **Failed Attempt Threshold** fallback to **Device Authentication** after 5 attempts. The 15-minute **Idle Timeout** is long enough to cover routine interruptions (a participant putting the device down to answer a door, take a call, attend to a child) without forcing repeated re-authentication during an active diary session; it is short enough that an unattended device left in a public space cannot be picked up much later and still access participant data without authentication. The 5-attempt **Failed Attempt Threshold** matches the **Login Rate Limit Threshold** used on the Sponsor Portal, keeping the failed-attempt boundary consistent across the two authentication surfaces.

Assertions

A. The **Authentication Method** SHALL be configured to **PIN**.

B. The **PIN** length SHALL be configured to 4 digits.

C. The **Idle Timeout** SHALL be configured to 15 minutes.

D. The **Failed Attempt Threshold** SHALL be configured to 5 consecutive failed attempts.

6.2 Questionnaires Overview

The platform implements three clinical questionnaires: the **Daily Epistaxis Record** (the daily eDiary instrument), the **NOSE HHT** survey, and the **HHT-QoL** survey. Each requirement binds the instrument to its validated source document and to the computer-readable data file that is the single implementation reference.

6.2.1 Daily Epistaxis Record Questionnaire

REQ ID: `DIARY-PRD-questionnaire-daily-epistaxis`

Overview

Daily epistaxis tracking is the core data collection activity in HHT clinical trials because nosebleed frequency, duration, and severity are the primary outcome measures for evaluating treatment efficacy. participant self-reporting captures events as they occur rather than relying on retrospective recall during clinical visits, supporting the Contemporaneous principle of ALCOA+. “No nosebleeds” and “Don’t remember” entries are valid daily summaries that distinguish between confirmed absence of events and missing data, which is critical for data integrity in regulatory submissions.

Definitions

Daily Epistaxis Record: The primary nosebleed tracking instrument used in the HHT Clinical Diary, derived from Clark et al. (2018) with modifications.

Epistaxis Event: A single nosebleed occurrence recorded by the participant with timing, duration, and intensity data.

Rationale

The **Daily Epistaxis Record** is the primary instrument the diary platform exists to capture; every other surface (calendar, day view, recording flow, notifications, score calculations) is in service of getting reliable instances of this record into the clinical dataset. Binding the requirement to the Clark et al. (2018) source instrument anchors the diary content to a validated reference so the platform's implementation can be audited against an external definition rather than re-derived locally. Any modification to the source is required to carry its own documented justification because clinical instruments derive their interpretive validity from the published version; undocumented deviations would break the chain back to the validated source and undermine the dataset's regulatory standing. The single computer-readable data file is the implementation reference — the same artifact the application loads and the specification references — which prevents drift between what the spec says and what the application runs.

Assertions

A. The system SHALL implement the **Daily Epistaxis Record** questionnaire for capturing participant-reported nosebleed events.

B. The source instrument SHALL be documented with citation: Clark et al. "Nosebleeds in hereditary hemorrhagic telangiectasia: Development of a participant-completed daily eDiary." Laryngoscope Investig Otolaryngol. 2018. doi:10.1002/lio2.211.

C. All modifications to the source instrument SHALL be documented and traceable to their justification.

D. The instrument content, field definitions, and validation rules SHALL be implemented in a computer-readable data file that has been manually checked for accuracy.

6.2.2 NOSE HHT Questionnaire

REQ ID: `DIARY-PRD-questionnaire-nose-hht`

Overview

The NOSE HHT (Nasal Outcome Score for Epistaxis in Hereditary Hemorrhagic Telangiectasia) is a validated 29-question participant-reported outcome measure assessing the physical, functional, and emotional impact of nosebleeds on HHT participants. The platform implements this instrument faithfully from the validated source. Questionnaire content is defined in the source document and transcribed once into a computer-readable data file — the data file is the single implementation reference, eliminating duplication and ensuring the application and the specification never diverge.

Definitions

NOSE HHT: The Nasal Outcome Score for Epistaxis in Hereditary Hemorrhagic Telangiectasia, a validated clinical instrument published in JAMA Otolaryngology — Head and Neck Surgery, 2020.

True Copy: A complete, unaltered reproduction of the source document obtained directly from the original publisher URL.

Rationale

The **NOSE HHT** is a validated clinical instrument; its interpretive validity depends on faithful reproduction of the published questions, answer choices, and scoring rules. The True Copy obligation anchors the platform's implementation to the publisher's authoritative version — the same document any auditor can independently retrieve from the cited URL — and removes any ambiguity about which variant of the instrument the platform uses. Transcription into a computer-readable data file is the bridge between the human-readable source and the running application: the application loads the data file directly, so the spec's reference to the data file and the application's runtime behavior are guaranteed to match. The three-category presentation (Physical, Functional, Emotional) matches the source instrument's structure and is required for participant comprehension and downstream score interpretation.

Assertions

A. The system SHALL implement the **NOSE HHT** questionnaire as defined in the source document referenced in assertion B.

B. A **True Copy** of the source document SHALL have been obtained from: https://jamanetwork.com/journ

C. The **True Copy** SHALL be referenced by the computer-readable data file that implements this questionnaire.

D. The source document SHALL be transcribed into a computer-readable data file that has been manually checked for accuracy. This transcribed version will serve as the official reference for all questions, answer choices, and scoring rules used in the system.

E. The System SHALL present the 29 **NOSE HHT** questions organized into the three categories defined in the source instrument: Physical, Functional, and Emotional.

F. The Questionnaire Display Name for the NOSE HHT Questionnaire SHALL be “NOSE HHT Survey”.

6.2.3 HHT-QoL Questionnaire

REQ ID: `DIARY-PRD-questionnaire-hht-qol`

Overview

The HHT-QoL questionnaire is a validated 4-question participant-reported outcome measure assessing how nosebleeds and HHT-related problems affect participants’ daily activities and social engagement. The platform implements this instrument faithfully from the validated source. Questionnaire content is defined in the source document and transcribed once into a computer-readable data file — the data file is the single implementation reference, eliminating duplication and ensuring the application and the specification never diverge.

Definitions

HHT-QoL: A validated clinical instrument published in Blood Advances, 2022, measuring the impact of HHT symptoms on participants’ daily and social activities.

Rationale

The **HHT-QoL** is a short (4-question) validated quality-of-life instrument complementing the NOSE HHT’s symptom focus with an activities-and-engagement perspective. The True Copy and transcription requirements track the same faithfulness obligation that applies to the NOSE HHT (assertion-for-assertion): the published source defines the instrument, and the platform’s data file is the single reference the application consumes. The key-phrase emphasis (italics on “interrupted”, “avoided”, “had to miss”, “other than nosebleeds”) is a substantive part of the published instrument’s wording — these emphases are how the original instrument signals which words the participant should be asked to weigh most carefully — and dropping them would silently re-engineer the question stems away from the validated form. Preserving them keeps the platform’s rendering aligned with the source.

Assertions

A. The system SHALL implement the **HHT-QoL** questionnaire as defined in the source document referenced in assertion B.

B. A **True Copy** of the source document SHALL have been obtained from: https://ashpublications.org/blc

C. The **True Copy** SHALL be referenced by the computer-readable data file that implements this questionnaire.

D. The source document SHALL be transcribed into a computer-readable data file that has been manually checked for accuracy. This transcribed version will serve as the official reference for all questions, answer choices, and scoring rules used in the system.

E. The system SHALL emphasize the key phrases — interrupted, avoided, had to miss, and other than nosebleeds — in the **HHT-QoL** questions, matching the formatting of the original validated instrument.

F. The Questionnaire Display Name for the HHT-QoL Questionnaire SHALL be “HHT Quality of Life Survey”.

6.3 Participant Questionnaire Workflow

The participant-facing workflow over **Portal-Sent Questionnaires** comprises the rules that govern how a participant moves through Preamble, questions, Review, and Submission, the corresponding interface behavior, and the Session Timeout / Session Expiry mechanism that bounds how long an in-progress questionnaire can be left idle.

6.3.1 Portal-Sent Questionnaire Rules

REQ ID: `DIARY-PRD-questionnaire-portal-sent-rules`

Refines: `DIARY-GUI-questionnaire-portal-sent-workflow`, `DIARY-PRD-questionnaire-ses`

Overview

Portal-Sent Questionnaires are initiated by a **Study Coordinator** from the **Sponsor Portal** and delivered to the participant via push notification. The rules below define what the participant sees (Preamble), how they progress through the questionnaire, when their answers become a Submission, and the editing window between Submission and Finalization.

Definitions

Portal-Sent Questionnaire: A Questionnaire initiated by a Study Coordinator from the Sponsor Portal and delivered to the participant via push notification.

Preamble: The introductory content block presented to the participant before questionnaire questions begin informing the participant of the estimated completion time and that answers will not be saved until Submission.

Submission: The participant action that signals all questions are answered and the Questionnaire is ready for Study Coordinator review.

Finalization: The Study Coordinator action that permanently locks participant answers, triggers score calculation and pushes the data to Rave EDC.

Preamble

Completion Rules

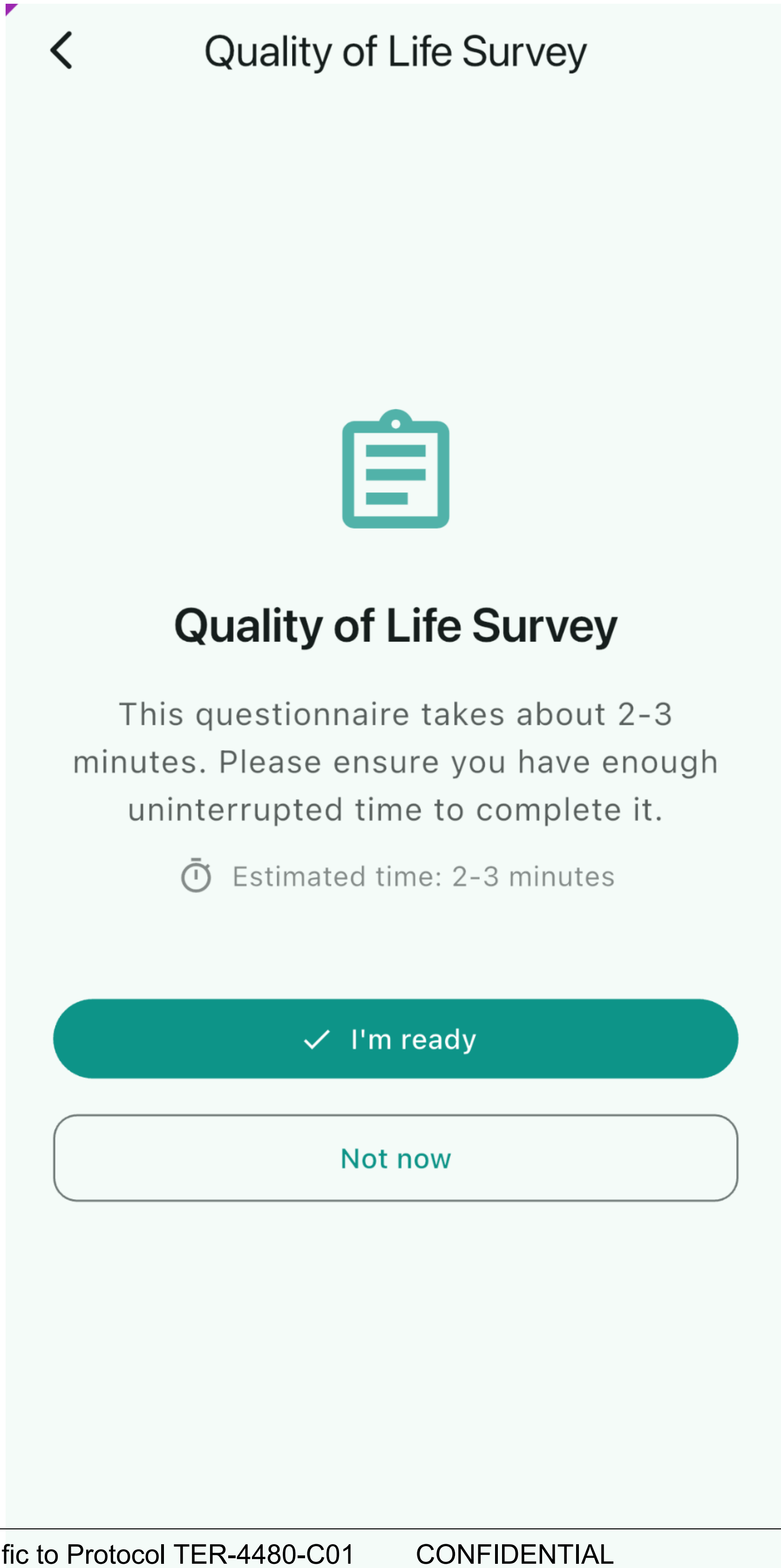
Submission

Edit Rules

Rationale

The Preamble exists because a **Portal-Sent Questionnaire** is a non-trivial commitment of participant time (the NOSE HHT has 29 questions; an ad-hoc questionnaire can be longer); telling the participant up front how long it will take and that progress is preserved between sessions reduces the rate at which participants start and abandon mid-flow. One-question-at-a-time presentation matches the validated-instrument format (the source documents present questions one at a time on paper) and prevents the participant from scanning ahead, which could bias later answers. Skipping is prohibited because the validated scoring algorithms require complete responses; partial questionnaires are not interpretable. The in-progress-preservation rule is the participant-side guarantee that “Exit” is safe — combined with the **Session Timeout** override (which can discard in-progress answers if the participant has been idle too long), it gives the participant flexible but bounded continuation. Editing is open between Submission and Finalization because Submission signals “participant is done”, but the **Study Coordinator** review may surface answer issues the participant should be able to correct without resubmitting from scratch; Finalization is the irreversible boundary because that is when the score is computed and the data ships to **Rave EDC**.

Screen reference



See:

Assertions

A. The System SHALL present the **Preamble** to the participant each time the participant opens a **Portal-Sent Questionnaire**.

B. The **Preamble** SHALL inform the participant of the estimated time required to complete the questionnaire.

C. The **Preamble** SHALL inform the participant that their progress will be retained while completing the questionnaire and that answers are submitted only when they complete **Submission**.

D. The System SHALL require the participant to confirm readiness before proceeding from the **Preamble** to the questions.

E. When the participant indicates they are not ready, the System SHALL return the participant to the Main home screen.

F. The System SHALL present one question at a time during **Portal-Sent Questionnaire** completion.

G. The System SHALL NOT permit the participant to skip any question in a **Portal-Sent Questionnaire**.

H. The System SHALL preserve in-progress answers locally while the participant is completing the questionnaire and SHALL NOT commit answers as a Submission until the participant completes **Submission**.

I. The System SHALL make completion progress available to the participant throughout the questionnaire.

J. The System SHALL allow the participant to navigate back and forth between the questions before **Submission**.

K. The System SHALL allow the participant to exit the questionnaire at any point before **Submission** without losing in-progress answers, subject to the **Session Timeout** defined in DIARY-PRD-questionnaire-session-timeout.

L. The System SHALL present a review of all answers to the participant before **Submission**, allowing the participant to modify any answer before proceeding.

M. The System SHALL require an explicit participant action - clicking the Submit button - to complete **Submission**.

N. The System SHALL allow the participant to edit their answers with no time limit after **Submission** until **Finalization**.

O. The System SHALL NOT permit the participant to edit their answers after **Finalization**.

6.3.2 Portal-Sent Questionnaire Workflow

REQ ID: `DIARY-GUI-questionnaire-portal-sent-workflow`

Refines: `CAL-GUI-questionnaire-call-back-participant`

Overview

The interface for a **Portal-Sent Questionnaire** spans four screens: the **Preamble**, the per-question screen, the **Review Screen** (after the final question), and the post-Submission Acknowledgement Dialog. The behavior on each screen tracks the PRD-level rules above and adds the visual affordances (progress indicator, navigation controls, action labels) participants use to move through the flow.

Definitions

Review Screen: The screen presented to the participant after answering the final question, displaying all questions and selected answers before Submission.

Preamble

Question Screen

Review Screen

Editing an Answer

Submission

Post-Submission Editing

Finalization

Rationale

The four-screen structure (Preamble, Question, Review, Acknowledgement) makes the questionnaire's logical structure visible to the participant: a clearly demarcated start (Preamble with I'm Ready / Not Now), per-question progress visibility (Question #X out of Y title), a final aggregated review before commit (Review Screen with per-question edit affordance), and an explicit confirmation that the submission landed (post-Submission Acknowledgement Dialog). The Next-disabled-until-answered rule combined with the prohibition on skipping is how the platform enforces the no-skipped-questions PRD assertion at the GUI layer. Returning to the **Review Screen** on re-open between Submission and Finalization keeps the participant's mental model aligned with the PRD-level edit window — they always re-enter through the aggregated review, not into a partial-flow ambiguity, so the next action is either "Submit again" or "I'm finished, close this". The read-only state after Finalization is the GUI-side counterpart of the PRD-level "no edit after Finalization" rule.

> **Follow-up — configurability**: This requirement currently encodes the only option implemented in code. Future sponsors may require different rules; introduce a configurable seam (e.g. a parameter on the CAL-PRD-* parent, or a new platform-side template the CAL- REQ Satisfies) when the need arises. Until that seam exists, this REQ is normative for the Callisto deployment.

Screen reference

See:

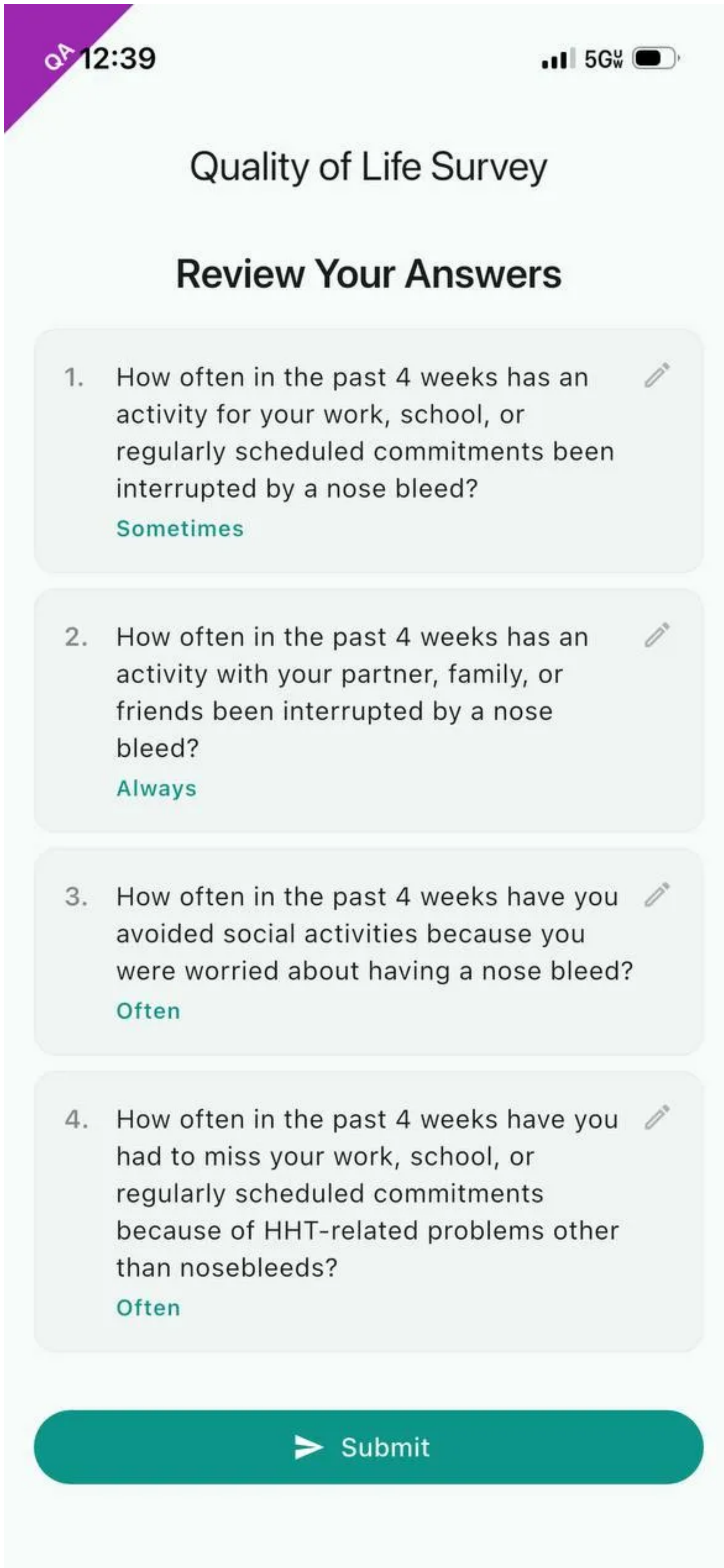


Figure 6.1: Questionnaire Review Screen

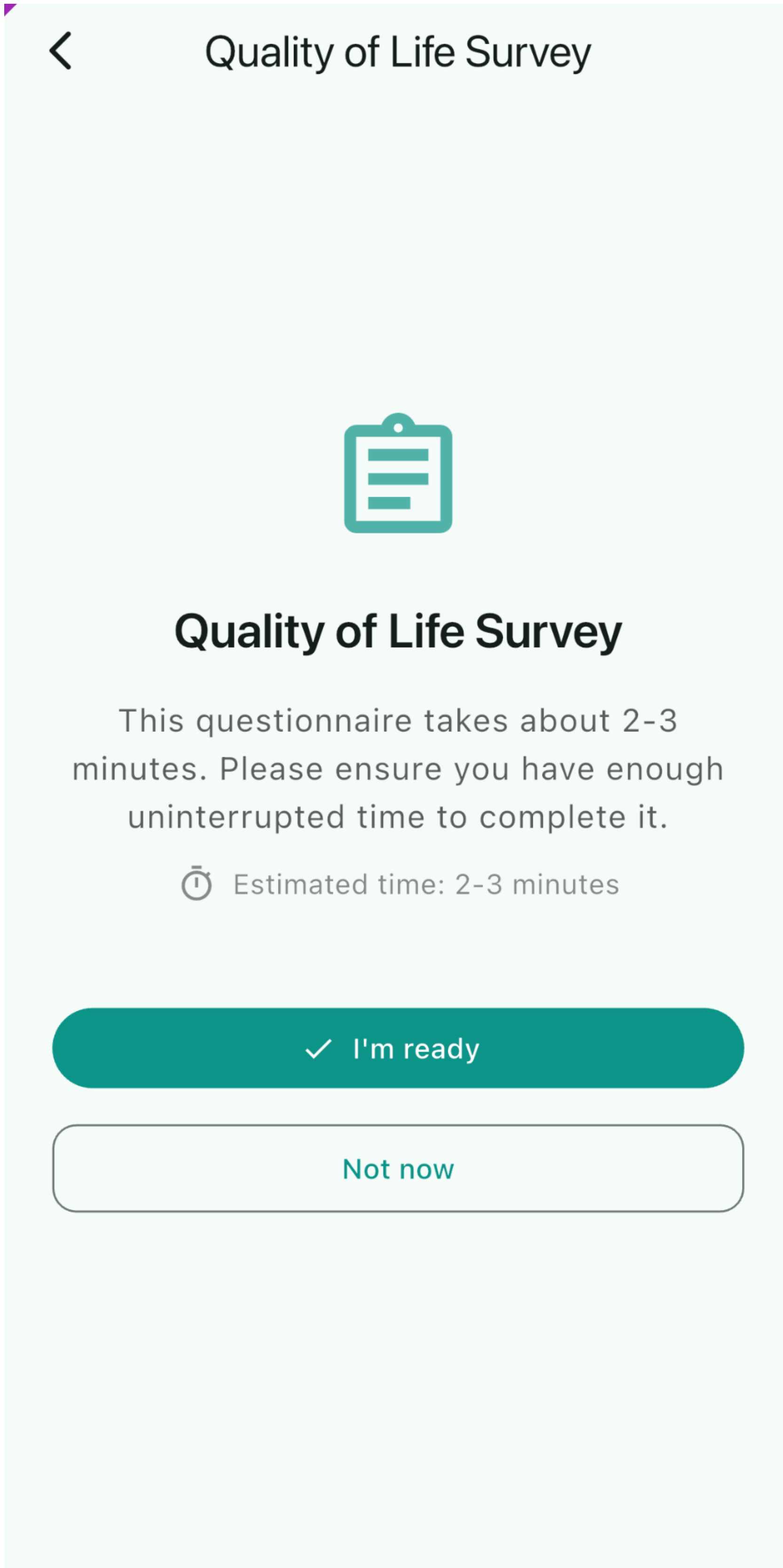


Figure 6.2: Questionnaire Preamble Screen

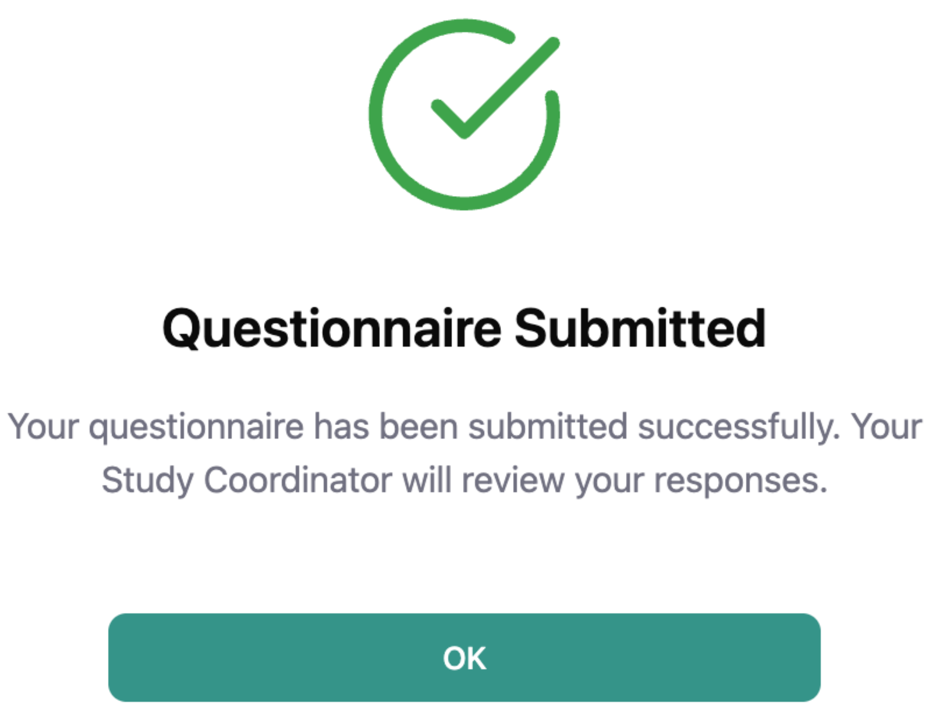


Figure 6.3: Post-Submission Acknowledgement Dialog

Assertions

A. The interface SHALL present the **Preamble** with two actions: an I'm Ready button and a Not Now button.

B. When the participant selects Not Now, the interface SHALL navigate the participant to the home screen.

C. When the participant selects I'm Ready, the interface SHALL navigate the participant to the first question.

D. The interface SHALL present a progress indicator showing the participant's current position within the questionnaire (Question #X out of Y).

E. The interface SHALL display the Questionnaire Display Name as the screen title on every question screen.

F. The interface SHALL present a back navigation control on every question screen.

G. The interface SHALL present a Next action on every question screen.

H. The Next action SHALL be disabled until the participant selects an answer.

I. When the participant selects an answer and selects Next, the interface SHALL advance to the next question.

J. When the participant navigates back to a previously answered question, the interface SHALL display the previously selected answer.

K. After the participant answers the final question, the interface SHALL present the **Review Screen**.

L. The **Review Screen** SHALL present all questions and the participant's selected answers.

M. The **Review Screen** SHALL allow the participant to navigate to any individual question to change their answer.

N. The **Review Screen** SHALL present a submit action.

O. When the participant navigates to an individual question from the **Review Screen** and changes an answer, the interface SHALL present a Save button.

P. When the participant selects Save, the interface SHALL return the participant to the **Review Screen**.

Q. When the participant confirms **Submission**, the interface SHALL display an Acknowledgement Dialog confirming the questionnaire has been submitted successfully.

R. When the participant re-opens a submitted questionnaire before **Finalization**, the interface SHALL present the **Review Screen**.

S. When the participant opens a finalized questionnaire, the interface SHALL present the questionnaire in a read-only state with no edit or submit actions available.

6.3.3 Portal-Sent Questionnaire Call-Back — Participant Experience

REQ ID: `CAL-GUI-questionnaire-call-back-participant`

Overview

Participants need clear, immediate feedback when a Portal-Sent Questionnaire is called back so they do not waste effort on an inactive questionnaire or become confused about its disappearance.

Definitions

Call-Back Notice: An Acknowledgement Dialog informing the **Participant** that a **Portal-Sent Questionnaire** assigned to them has been called back by the study team.

Open Questionnaire Interruption

Task List

Next Session Notification

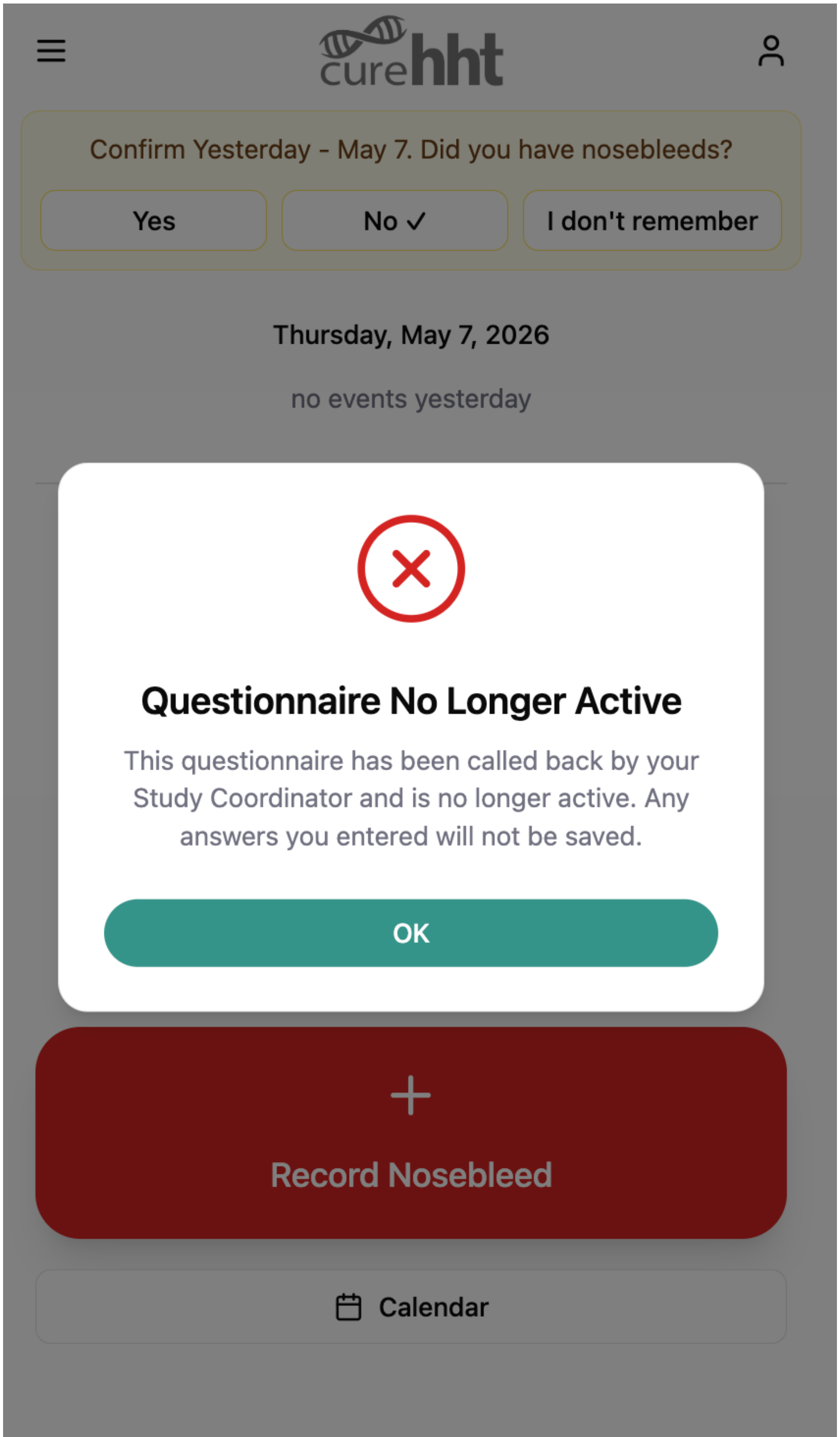
Offline Submission Attempt

Rationale

A call-back removes the questionnaire from the participant's active task set, but if the participant has the questionnaire open at the moment of call-back (or attempts to submit one whose call-back happened while they were offline), they need explicit feedback that the questionnaire is no longer assigned to them. The Call-Back Notice provides that feedback in three contexts: live interruption (open now, called back now), deferred notification (called back while not open, surface on next session), and offline submission attempt (called back while offline, surface when submission tries to land). The non-dismissible-by-outside-tap rule (assertion B) prevents the participant from accidentally clearing the notice without reading it. Removing the Questionnaire Task from the Task List immediately on call-back closes the entry point on the **Main Screen** so the participant does not navigate into the called-back questionnaire from there. Routing acknowledgement to the **Main Screen** rather than back into the now-defunct questionnaire flow is the only sensible destination — the questionnaire is no longer interactable, so returning the participant to it would be a dead end.

Follow-up — configurability: This requirement currently encodes the only option implemented in code. Future sponsors may require different rules; introduce a configurable seam (e.g. a parameter on the CAL-PRD-* parent, or a new platform-side template the CAL- REQ Satisfies) when the need arises. Until that seam exists, this REQ is normative for the Callisto deployment.

Screen reference



See:

Assertions

A. When a **Portal-Sent Questionnaire** is called back while the **Participant** has it open, the interface SHALL display the Call-Back Notice.

B. The Call-Back Notice SHALL NOT be dismissible by tapping outside the dialog.

C. When the **Participant** acknowledges the Call-Back Notice, the interface SHALL navigate the **Participant** to the **Main Screen**.

D. When a **Portal-Sent Questionnaire** has been called back, the interface SHALL remove the corresponding Questionnaire Task from the Task List.

E. When a **Portal-Sent Questionnaire** has been called back and the **Participant** did not have it open at the time of call-back, the interface SHALL display the Call-Back Notice on the **Participant's** next Application session.

F. When the **Participant** attempts to submit a **Portal-Sent Questionnaire** that has been called back, the interface SHALL display the Call-Back Notice.

6.3.4 Questionnaire Session Timeout

REQ ID: `DIARY-PRD-questionnaire-session-timeout`

Refines: `DIARY-GUI-questionnaire-session-expiry`, `CAL-PRD-questionnaire-session-res`

Overview

A configurable session timeout allows each questionnaire to enforce appropriate completion constraints based on its length and clinical requirements.

Definitions

Session Timeout: The configurable maximum duration of inactivity allowed during questionnaire completion. When exceeded, the session expires and the answers selected so far are not retained.

Session Expiry: The state reached when a participant has not completed a questionnaire within the configured Session Timeout duration.

Timeout Warning Notification: Push notification delivered to the participant when the Session Timeout is approaching.

Session Expiry Notification: Push notification delivered to the participant when Session Expiry has occurred.

Timeout Tracking

Expiry Behavior

Notifications

State Preservation

Configuration

Rationale

Some clinical questionnaires (e.g. the NOSE HHT, the HHT-QoL) require contemporaneous answering — a participant's frame of reference shifts measurably if they pause for hours between questions, and the resulting answers no longer reflect a single moment of self-report. A configurable **Session Timeout** lets each questionnaire enforce a "complete this in one sitting" constraint without baking a single duration into the platform. Discarding in-progress answers on expiry rather than retaining them is what gives the timeout its clinical meaning: the next attempt starts fresh from the **Preamble**, with a participant frame of reference that the sponsor can interpret. The opt-out (no **Session Timeout** configured) covers the case where in-progress preservation is fine (e.g. an ad-hoc questionnaire with no contemporaneous requirement). The two notifications — Warning before expiry, Expiry on the event — give the participant a chance to return before answers are discarded (Warning) and clear feedback when answers have already been discarded (Expiry).

Screen reference

See:

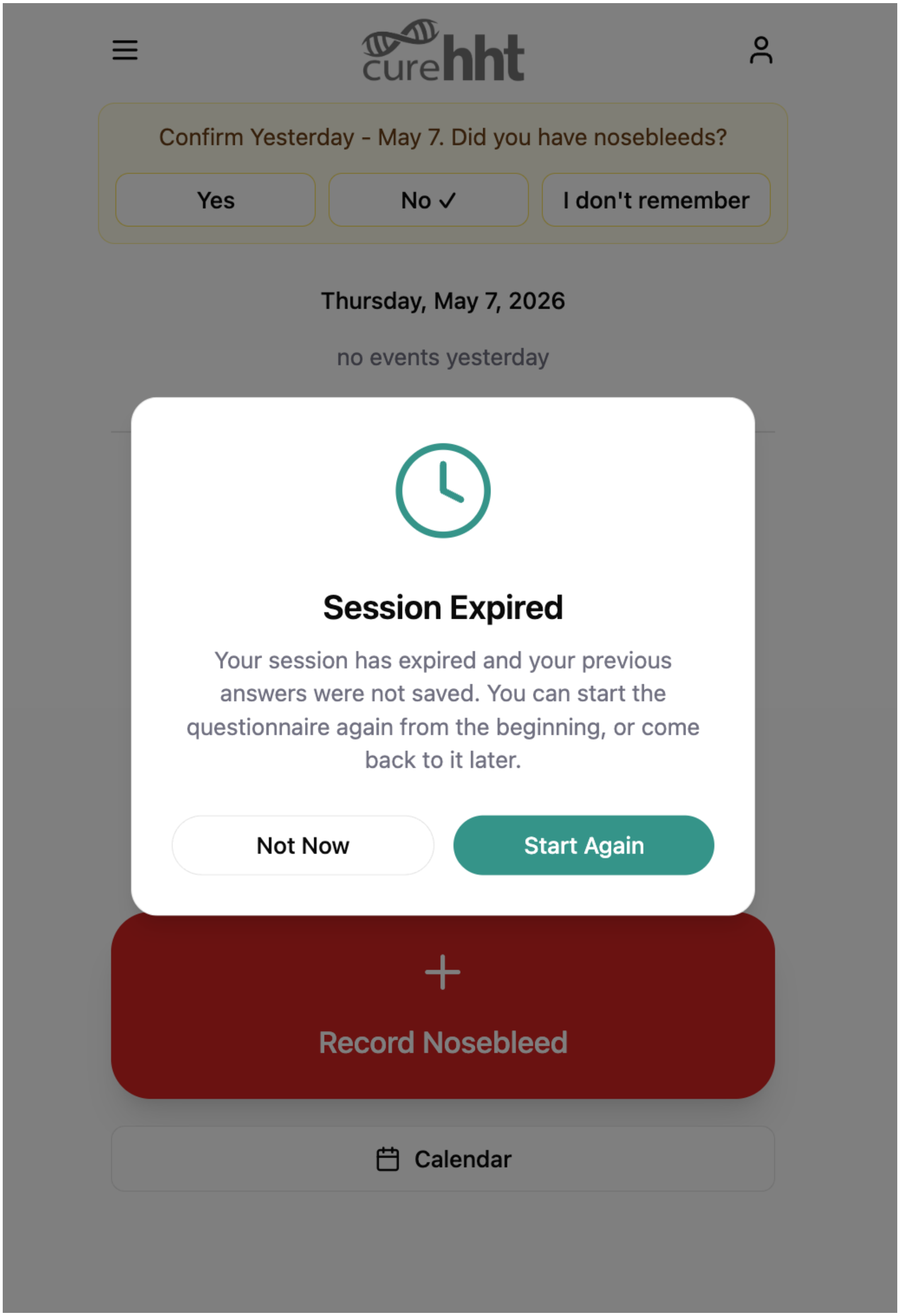


Figure 6.4: Session Expiry Dialog

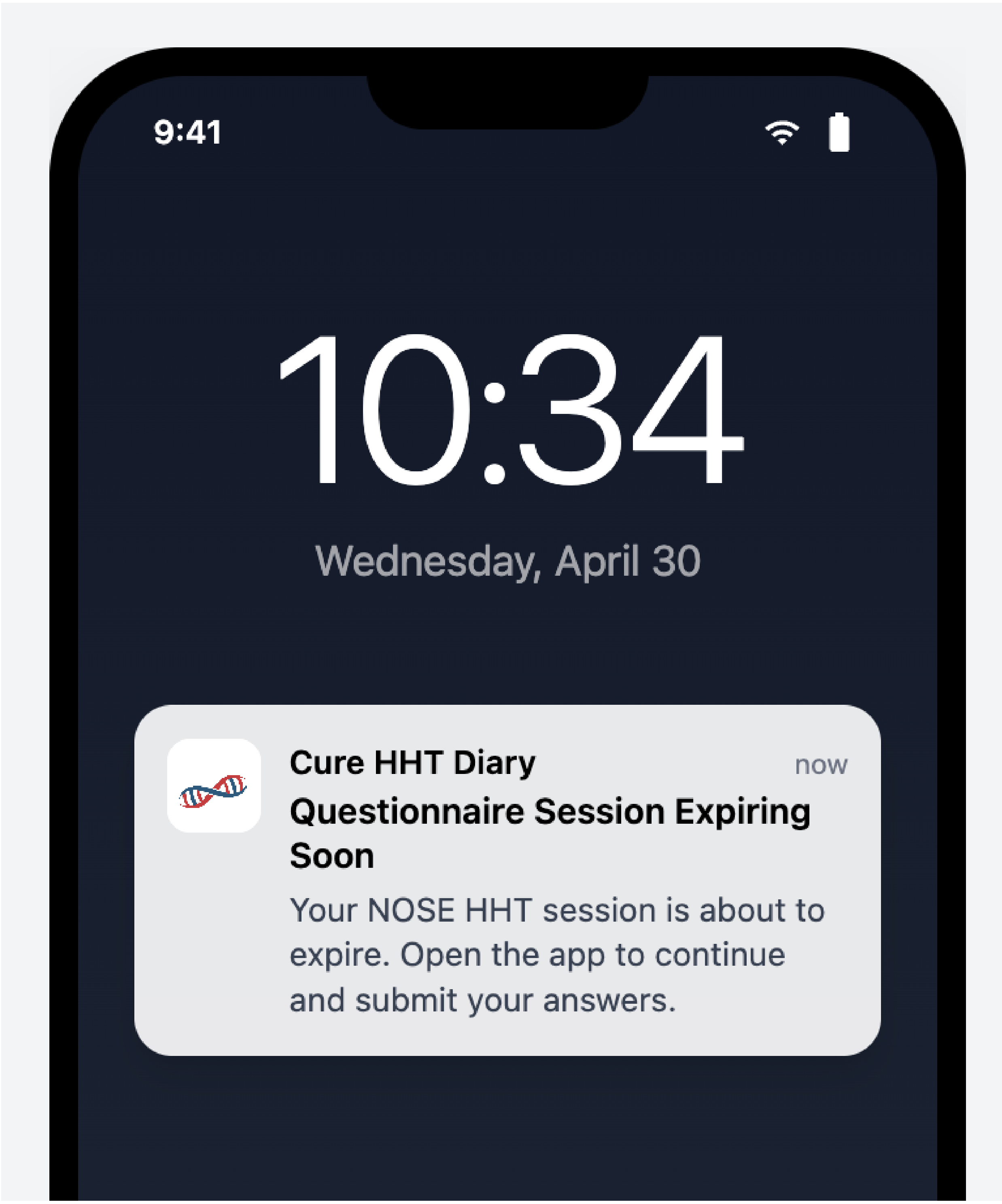


Figure 6.5: Timeout Warning Notification

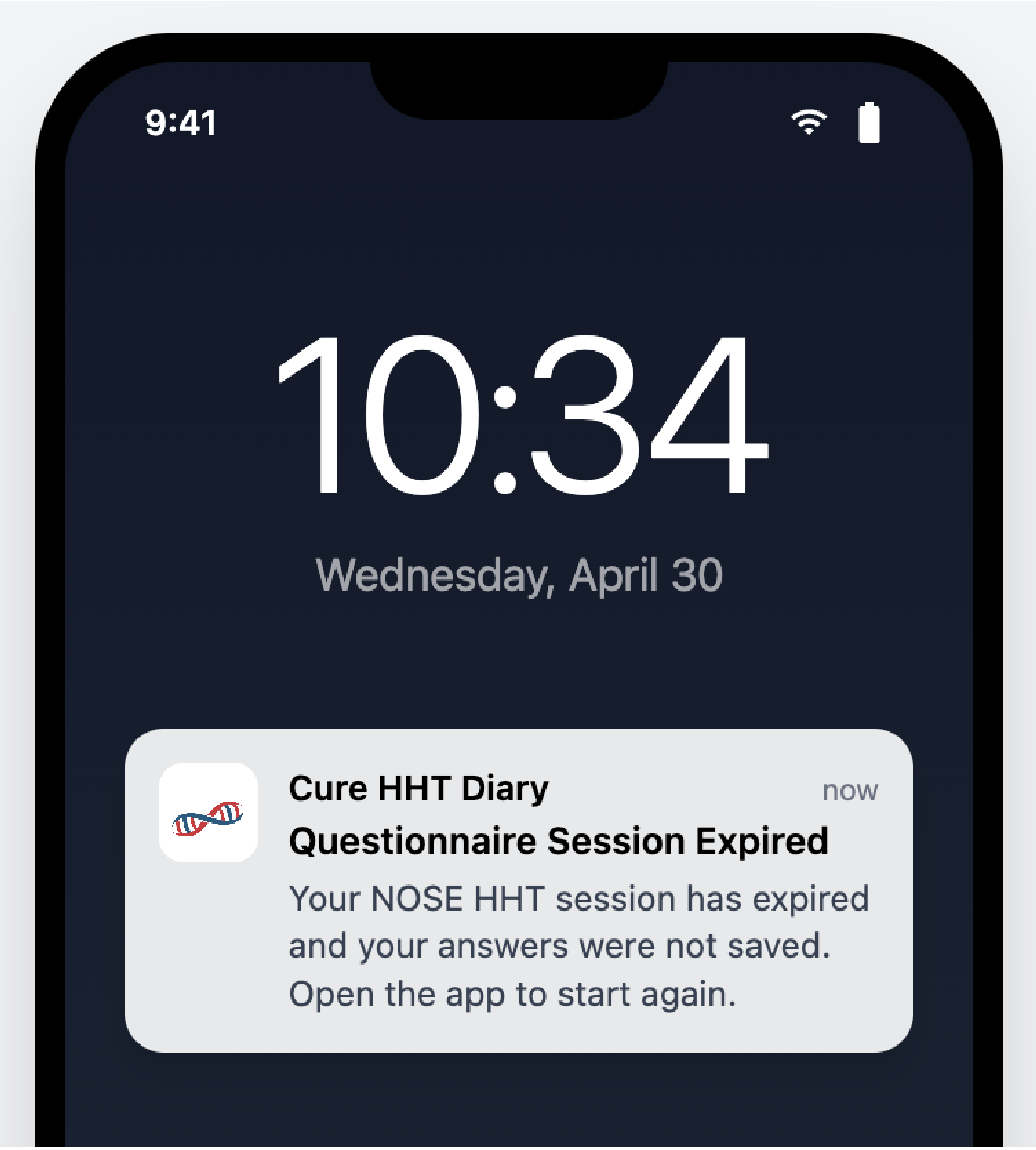


Figure 6.6: Session Expiry Notification

Assertions

A. When a questionnaire is configured with a **Session Timeout**, the System SHALL track elapsed inactivity from the participant's most recent interaction with the questionnaire.

B. The System SHALL only advance the inactivity timer while the participant is not actively interacting with the questionnaire.

C. When the **Session Timeout** is exceeded before the participant submits the questionnaire, the System SHALL discard all answers selected so far.

D. When a participant returns to a questionnaire in a state of **Session Expiry**, the System SHALL present the questionnaire from the beginning including the **Preamble**.

E. When a questionnaire is configured with a **Session Timeout**, the System SHALL deliver a **Timeout Warning Notification** to the participant when the **Session Timeout** is approaching.

F. When a questionnaire is configured with a **Session Timeout**, the System SHALL deliver a **Session Expiry Notification** to the participant when **Session Expiry** has occurred.

G. When a questionnaire is not configured with a **Session Timeout**, the System SHALL preserve and restore the participant's in-progress answers on return with no timeout constraint.

H. When a questionnaire with a **Session Timeout** has not yet expired, the System SHALL return the participant to their in-progress questionnaire on return.

I. Each questionnaire definition SHALL support optional configuration of **Session Timeout** duration.

J. Each questionnaire definition SHALL support configuration of the threshold before expiry at which the **Timeout Warning Notification** is delivered.

6.3.5 Questionnaire Session Expiry

REQ ID: `DIARY-GUI-questionnaire-session-expiry`

Overview

The interface for **Session Expiry** consists of three surfaces: the **Timeout Warning Notification** (push notification before expiry), the **Session Expiry Dialog** (in-app dialog on return after expiry), and the **Session Expiry Notification** (push notification when expiry occurs). On return to an active (not-yet-expired) session, the interface restores the participant to where they left off.

Definitions

Session Expiry Dialog: The Acknowledgement Dialog displayed to the participant when they open a questionnaire that has reached Session Expiry.

Timeout Warning

Expired Session on Return

Session Expiry Notification

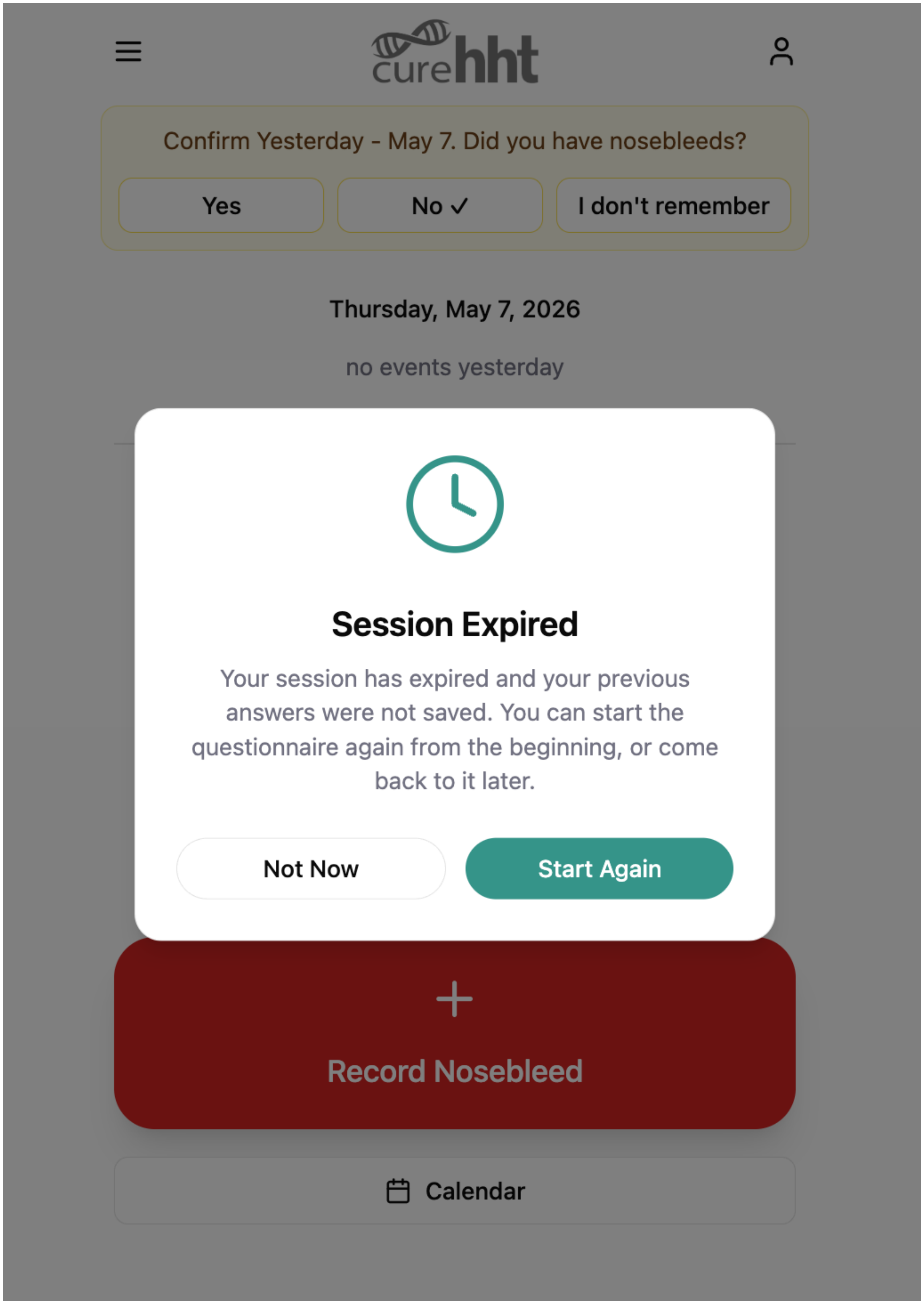
Active Session on Return

Rationale

The **Session Expiry Dialog** is the in-app companion to the **Session Expiry Notification**: the notification tells the participant the session expired while they were away from the app; the dialog confirms it when they return and tells them their prior answers are gone. Start Again / Not Now are the two realistic next-actions — restart now from the Preamble, or defer until later — and both are made explicit so the participant does not face a silent reset on next open. Restoring the active-session participant to their last question (with in-progress answers intact) is the GUI-side guarantee that the PRD's in-progress-preservation rule actually surfaces to the participant; if the interface always re-entered through the Preamble, the preservation would be invisible and the participant would lose confidence that pausing is safe.

> **Follow-up — configurability**: This requirement currently encodes the only option implemented in code. Future sponsors may require different rules; introduce a configurable seam (e.g. a parameter on the CAL-PRD-* parent, or a new platform-side template the CAL- REQ Satisfies) when the need arises. Until that seam exists, this REQ is normative for the Callisto deployment.

Screen reference



Assertions

A. When a **Timeout Warning Notification** is delivered, the interface SHALL present it as a push notification indicating that the questionnaire session is approaching expiry.

B. When the participant opens a questionnaire that has reached **Session Expiry**, the interface SHALL display a **Session Expiry Dialog** informing the participant that their session has expired and their previous answers were not saved.

C. The **Session Expiry Dialog** SHALL present a Start Again button and a Not Now button.

D. When the participant selects Start Again, the interface SHALL dismiss the **Session Expiry Dialog** and present the **Preamble**.

E. When the participant selects Not Now, the interface SHALL navigate the participant to the home screen.

F. When a **Session Expiry Notification** is delivered, the interface SHALL present it as a push notification indicating that the questionnaire session has expired.

G. When the participant returns to a questionnaire with a session that has not yet reached **Session Expiry**, the interface SHALL restore the participant to the question they were on when they left, with their in-progress answers intact.

6.4 Daily eDiary: HHT Epistaxis Data Capture

The daily eDiary data-capture flow for **Epistaxis Events** comprises the platform-level data capture standard, the participant-facing recording flow (Start, Max Intensity, End), and the deletion flow (in-flight during recording and post-save from event history).

6.4.1 HHT Epistaxis Data Capture Standard

REQ ID: `DIARY-PRD-epistaxis-capture-standard`

Refines: `DIARY-GUI-epistaxis-record`, `DIARY-GUI-epistaxis-delete`

Overview

The HHT Epistaxis Data Capture Standard defines the requirements for participant-reported nosebleed event collection in HHT clinical trials. Epistaxis frequency, duration, and severity are primary outcome measures for evaluating treatment efficacy. Timezone-aware timestamps ensure precise duration calculation regardless of participant location.

Definitions

Daily Status: The participant's self-reported summary of nosebleed activity for a given day. Valid values are Had Nosebleed, No Nosebleed, and Don't Remember.

Intensity: A participant-reported measure of nosebleed severity captured on a defined six-level scale ordered from least to most severe.

Incomplete Record: An **Epistaxis Event** that has been saved but is missing one or more required fields: start time, **Intensity**, or end time.

Rationale

The Daily Status / Epistaxis Event two-level structure exists because the clinical question is two-leveled: "Did anything happen today?" (a daily-grain answer that can be Had Nosebleed, No Nosebleed, or Don't Remember) and, when something did happen, "What were the events?" (per-event start, end, intensity). Mutually exclusive **Daily Status** values prevent contradictory records (a participant cannot simultaneously have had a nosebleed and not had one). Required start, end, and **Intensity** on each **Epistaxis Event** capture the three primary outcome measures (frequency by count, duration by interval, severity by intensity); allowing any of these to be missing without an explicit incomplete-record state would silently downgrade dataset quality. Timezone-aware capture is essential because participants travel and durations computed from naive timestamps across a DST boundary or international move can be off by an hour or more. The **Incomplete Record** classification lets the platform distinguish "in progress, missing data" from "fully captured" so the participant can be prompted to finish and the dataset can flag what is not yet complete.

Assertions

A. The System SHALL require a participant to record a **Daily Status** for each calendar day within the diary period.

B. The System SHALL enforce mutual exclusivity of the three **Daily Status** values — a participant SHALL NOT record more than one **Daily Status** per diary entry.

C. When a participant records a **Daily Status** of Had Nosebleed, the System SHALL require the participant to provide start time and end time for the **Epistaxis Event** with timezone information.

D. The System SHALL calculate the duration of an **Epistaxis Event** from the recorded start time and end time, accounting for timezone differences.

E. The System SHALL require the participant to record an **Intensity** value for each **Epistaxis Event**.

F. The System SHALL present notes options from a predefined list (defined in CAL-PRD-entry-time-restrictions-configuration) for each **Epistaxis Event**; the available options SHALL be sponsor-configurable per study protocol.

G. The System SHALL classify an **Epistaxis Event** as an **Incomplete Record** when it has been saved with one or more required fields not recorded.

6.4.2 Record Nosebleed Event

REQ ID: `DIARY-GUI-epistaxis-record`

Overview

The nosebleed recording flow walks the participant through three steps — **Start** time, **Max Intensity**, **End** time — visible at all times in a persistent **Progress Indicator** at the top of every screen. The participant can revisit any completed step. Validation at the End Time step prevents impossible records (end before start, end after current real time) and prompts for confirmation on entries that look like they may have been recorded too quickly after the event.

Definitions

Progress Indicator: The persistent bar displayed at the top of every screen in the recording flow showing the three steps — Start, Max Intensity, and End — and their completion state.

Time Picker: The time selection component presenting the current time as default, a date selector, a timezone selector, and minute adjustment controls of -15, -5, -1, +1, +5, and +15 minutes.

Flow Structure

Nosebleed Start Time

Max Intensity

Nosebleed End Time

Rationale

The three-step structure (Start, Max Intensity, End) matches the clinical model of an event (when it began, how bad it got, when it ended); making the **Progress Indicator** persistent across every screen of the flow keeps the participant oriented and allows direct navigation to any completed step without back-tracking. The **Time Picker**'s minute-adjustment chips (-15/-5/-1/+1/+5/+15) target the realistic adjustment granularity for recording a recently-completed event and avoid the precision-mismatch of a free-text time entry. The auto-advance on intensity selection (vs. requiring a separate Next tap) shaves friction from the most common path (set start, pick intensity, set end) while still letting the participant revisit via the **Progress Indicator** if they want to change their intensity choice. End-time validation (no earlier than start, no later than now) catches the two impossible-time inputs at the GUI layer rather than relying on later data validation. The two-minute confirmation prompt is a soft-stop against premature finalization — a participant who is still actively having a nosebleed when they set the end time is reminded to check before committing.

> **Follow-up — configurability**: This requirement currently encodes the only option implemented in code. Future sponsors may require different rules; introduce a configurable seam (e.g. a parameter on the CAL-PRD-* parent, or a new platform-side template the CAL- REQ Satisfies) when the need arises. Until that seam exists, this REQ is normative for the Callisto deployment.

Assertions

A. The interface SHALL display the **Progress Indicator** on every screen throughout the recording flow showing the Start, Max Intensity, and End steps.

B. The interface SHALL highlight the active step in the **Progress Indicator** and show the recorded value for each completed step.

C. The interface SHALL allow the participant to navigate to any previously completed step by tapping it in the **Progress Indicator**.

D. The interface SHALL present a Back navigation action on every screen throughout the recording flow.

E. The interface SHALL present a **Time Picker** for the participant to set the nosebleed start time.

F. The interface SHALL present a Set Start Time button that confirms the start time and advances to the Max Intensity screen.

G. The interface SHALL present the six intensity options as a selectable grid: Spotting, Dripping, Dripping quickly, Steady stream, Pouring, and Gushing.

H. The interface SHALL NOT advance to the End Time screen until the participant has selected one of the intensity options.

I. When the participant selects an intensity option, the interface SHALL highlight the selected option and automatically advance to the End Time screen.

J. The interface SHALL present a **Time Picker** for the participant to set the nosebleed end time.

K. When the participant selects Set End Time, the interface SHALL save the record and return the participant to the home screen.

L. The System SHALL validate that end time is greater than or equal to start time.

M. The System SHALL restrict end time selection to times not exceeding the current real time.

N. The System SHALL require confirmation when creating entries less than 2 minutes old.

6.4.3 Nosebleed Event Delete

REQ ID: `DIARY-GUI-epistaxis-delete`

Overview

Deletion of a nosebleed event can happen in two contexts: in-flight during the recording flow (before the record has been saved) and post-save from event history. Both paths require a **Reason Dialog — Predefined List** before the deletion is applied, and the participant may cancel at any point.

Delete During Recording Flow

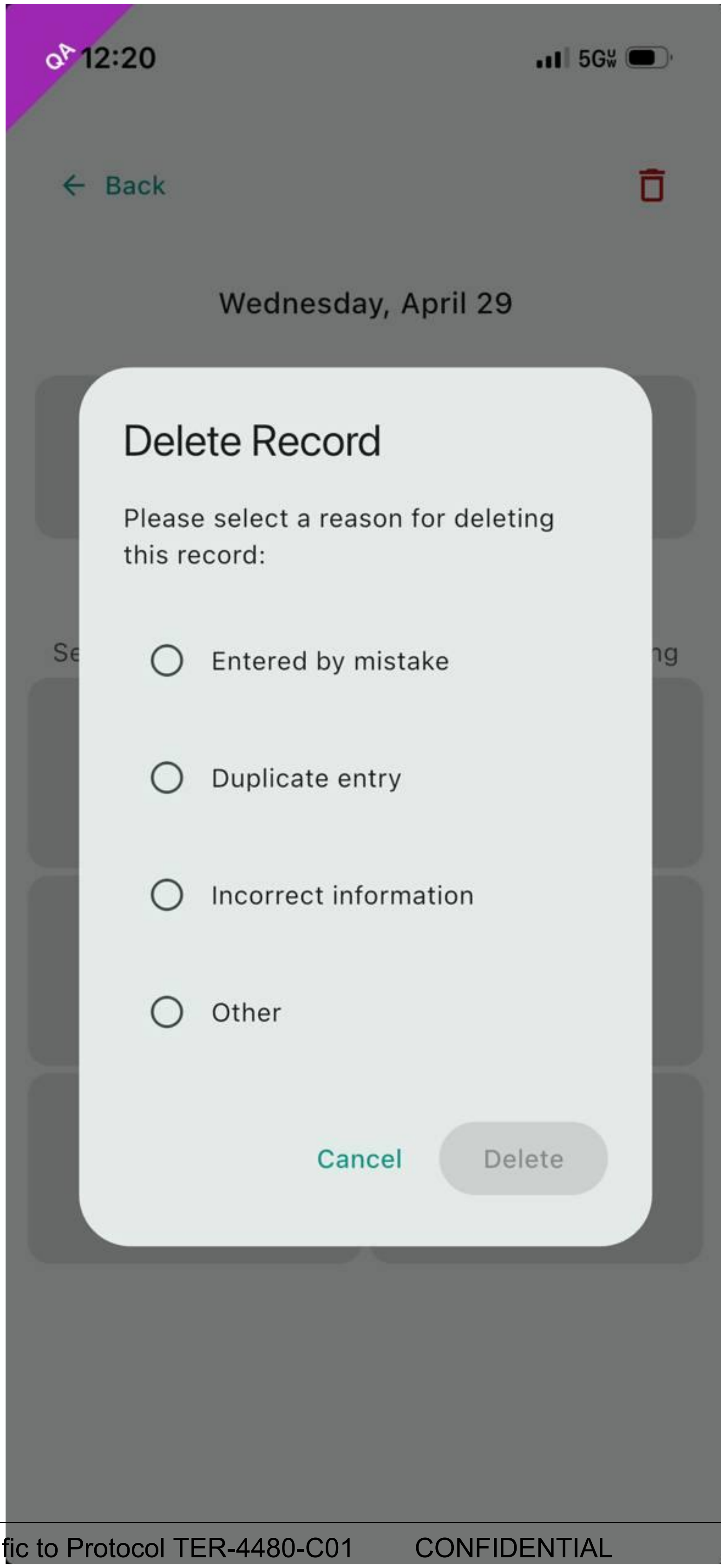
Delete From Event History

Rationale

The delete action is exposed on every screen of the recording flow because the participant may realize partway through that they started the record by mistake (wrong date, wrong event, accidental tap) and there is no benefit to forcing them to complete the flow before deleting. Routing both in-flight delete and post-save delete through a **Reason Dialog — Predefined List** keeps the audit trail's deletion-reason capture consistent across both contexts — a deletion at any point produces a categorized reason record. Predefined-list (rather than free-text) reasons match the dataset analyzability goal: deletion reasons are the kind of metadata downstream analysis aggregates over, and a controlled vocabulary makes that analysis tractable. The cancel path on either delete returns the participant to where they were (in-flight: the same screen; post-save: the previous screen, with the record unchanged) so an accidental delete-tap is fully recoverable.

> **Follow-up — configurability**: This requirement currently encodes the only option implemented in code. Future sponsors may require different rules; introduce a configurable seam (e.g. a parameter on the CAL-PRD-* parent, or a new platform-side template the CAL- REQ Satisfies) when the need arises. Until that seam exists, this REQ is normative for the Callisto deployment.

Screen reference



Assertions

A. The interface SHALL present a delete action on every screen throughout the recording flow.

B. When the participant selects the delete action during the recording flow, the interface SHALL display a **Reason Dialog — Predefined List** before proceeding.

C. When the participant confirms the delete reason, the interface SHALL discard the in-progress record and return the participant to the home screen.

D. When the participant cancels the Reason Dialog, the interface SHALL return the participant to the screen they were on.

E. When the participant selects the delete action on a saved **Epistaxis Event**, the interface SHALL display a **Reason Dialog — Predefined List** before proceeding.

F. When the participant confirms the delete reason, the interface SHALL remove the event from the participant's diary and return the participant to the previous screen.

G. When the participant cancels the Reason Dialog, the **Epistaxis Event** SHALL remain unchanged.

6.5 Score Calculation

The platform's score-calculation obligations comprise the general rule that scores are computed per a validated algorithm and stored with the questionnaire record, plus the specific algorithm bindings for the **HHT-QoL** and **NOSE HHT** instruments.

6.5.1 Questionnaire Score Calculation

REQ ID: `DIARY-PRD-questionnaire-score-calculation`

Refines: `DIARY-PRD-score-hht-qol`, `DIARY-PRD-score-nose-hht`

Overview

Validated clinical questionnaires require scoring according to published, protocol-defined algorithms to ensure data integrity and comparability across participants and studies. Storing scores with survey records ensures traceability and supports downstream Rave EDC synchronization and clinical analysis.

Rationale

Scores are the analytical artifact downstream clinical research derives outcomes from — frequency, severity, quality-of-life impact. Computing scores per the validated published algorithm rather than per a platform-local re-derivation is what gives the resulting numbers their interpretive validity: a NOSE HHT score from this platform is comparable to a NOSE HHT score from any other study only because both implement the published JAMA Otolaryngology algorithm faithfully. Storing the score with the questionnaire record (rather than recomputing on demand) preserves the value the participant's answers actually produced at the time of submission — even if the algorithm reference is later updated, the historical record reflects what was computed and shipped to **Rave EDC** for that participant. Traceability to source for each **Questionnaire Type** is the audit-trail counterpart of the True Copy obligation on the questionnaire definitions: any auditor must be able to follow the chain from a stored score back to the published algorithm that produced it.

Assertions

A. The System SHALL calculate **Questionnaire** scores according to the validated algorithm defined for each **Questionnaire**.

B. The System SHALL store the calculated score with the associated questionnaire record.

C. The scoring algorithm SHALL be traceable to its source definition for each **Questionnaire Type**.

6.5.2 HHT-QoL Score Calculation

REQ ID: `DIARY-PRD-score-hht-qol`

Rationale

The **HHT-QoL** scoring algorithm is fully specified by the cited Blood Advances 2022 publication; the platform's role is to implement that algorithm faithfully against the questionnaire's four answer fields. Anchoring the requirement to the DOI rather than restating the algorithm here preserves the property that the published reference is the single source of truth — any platform-side restatement would create a second authority that could drift from the original. The DOI is stable across editorial revisions, so an auditor can retrieve exactly the version the platform's implementation was checked against.

Assertions

A. The System SHALL calculate the **HHT-QoL** questionnaire score according to the algorithm defined in: Development and performance of a hereditary hemorrhagic telangiectasia-specific quality-of-life instrument, Blood Advances, 26 July 2022, Volume 6, Number 14, DOI 10.1182/bloodadvances.2022007748.

6.5.3 NOSE HHT Score Calculation

REQ ID: `DIARY-PRD-score-nose-hht`

Rationale

The **NOSE HHT** scoring algorithm is fully specified by the cited JAMA Otolaryngology 2020 publication, which is the same publication the questionnaire definition itself cites for the question and answer choices. Anchoring the score requirement to the same DOI keeps the questionnaire and its score on a single source-of-truth chain: the published instrument defines what to ask and how to score the answers, and the platform's two requirements (one for the questionnaire content, one for the score) both reference it. As with **HHT-QoL**, the DOI is the stable retrieval handle an auditor uses to confirm the platform's implementation matches the validated reference.

Assertions

A. The System SHALL calculate the **NOSE HHT** questionnaire score according to the algorithm defined in: Development and Validation of the Nasal Outcome Score for Epistaxis in Hereditary Hemorrhagic Telangiectasia (NOSE HHT), JAMA Otolaryngology – Head & Neck Surgery, November 1, 2020, Volume 146, Number 11, DOI 10.1001/jamaoto.2020.3040.

6.6 Diary Entry Rules

The platform-level validation rules that apply to every diary entry comprise the two-tier (justification, lock) time-based restrictions, the duration reasonableness check, and the overlapping-event detection-and-resolution flow.

6.6.1 Time-Based Entry Restrictions

REQ ID: `DIARY-PRD-entry-time-restrictions`

Refines: `CAL-PRD-entry-time-restrictions-configuration`

Overview

Clinical trial data integrity requires that participant-reported data be contemporaneous. As time passes after an event, recall accuracy degrades and the risk of unreliable data increases. A two-tier restriction model balances the need for late data capture (with documented justification) against the need to prevent entries that are too old to be clinically reliable. The justification tier preserves ALCOA+ compliance by requiring documented reasons for non-contemporaneous entries. The lock tier enforces a hard boundary beyond which no data modification is permitted.

Definitions

Justification Threshold: The configurable elapsed time from midnight (00:00) of the event date in the **Participant's** local timezone after which the **Participant** must provide a reason before creating or editing an entry for that date.

Lock Threshold: The configurable elapsed time from midnight (00:00) of the event date in the **Participant's** local timezone after which the date is fully locked and no creation, editing, or deletion of entries is permitted.

Entry Justification: A reason selected from a sponsor-defined predefined list that the **Participant** must provide when creating or editing an entry that has exceeded the **Justification Threshold**.

Justification Tier

Lock Tier

Configuration

Rationale

The two-tier model exists because two distinct integrity concerns argue for two distinct boundaries. Recall accuracy degrades gradually after an event, so for some window after the event date the data is still useful but the participant should explicitly acknowledge they are entering it late (this is the **Justification Threshold** — a soft boundary that captures a categorical reason on the record). Beyond a longer window, the data is no longer clinically reliable regardless of justification, so the platform makes it impossible to modify (the **Lock Threshold** — a hard boundary that disables create/edit/delete). Predefined-list justifications (rather than free-text) match the analyzability goal: the audit trail aggregates over a controlled vocabulary of reasons rather than over unbounded prose. Deletion within the justification window does not require justification because the participant is removing data they reported, not modifying it; the audit trail still captures the deletion event and its own reason via the standard delete flow. The `Lock >= Justification` invariant ensures the two tiers compose correctly (a Justification-required state always precedes a Lock state for the same date). The "only after trial start date" qualifier on the **Lock Threshold** means pre-trial historical entries (covered by the **Diary Start Day**) are not subject to the lock, since those entries are explicitly retrospective and the lock semantics don't apply.

Assertions

A. When the elapsed time for an event date exceeds the **Justification Threshold** but has not exceeded the **Lock Threshold**, the System SHALL require the **Participant** to select an **Entry Justification** before saving a new entry or an edit to an existing entry for that date.

B. The System SHALL NOT require an **Entry Justification** when the **Participant** deletes an entry within the justification window.

C. The System SHALL NOT allow free-text entry for **Entry Justification**.

D. The System SHALL store the selected **Entry Justification** with the entry record.

E. When the elapsed time for an event date exceeds the **Lock Threshold**, the System SHALL NOT permit creation of new entries for that date.

F. When the elapsed time for an event date exceeds the **Lock Threshold**, the System SHALL NOT permit editing of existing entries on that date.

G. When the elapsed time for an event date exceeds the **Lock Threshold**, the System SHALL NOT permit deletion of existing entries on that date.

H. The System SHALL support sponsor-configurable **Justification Threshold** per deployment.

I. The System SHALL support sponsor-configurable **Lock Threshold** per deployment.

J. The **Lock Threshold** SHALL be greater than or equal to the **Justification Threshold**.

K. The System SHALL support sponsor-configurable definition of the **Entry Justification** predefined list per deployment.

L. When neither threshold is configured for a deployment, the System SHALL NOT apply time-based entry restrictions.

M. The **Lock Threshold** SHALL apply only to event dates on or after the trial start date.

6.6.2 Time-Based Entry Restrictions Configuration

REQ ID: `CAL-PRD-entry-time-restrictions-configuration`

Overview

The Callisto clinical protocol requires that nosebleed event entries older than one calendar day include documented justification for the timing discrepancy, and that entries older than 48 hours are locked entirely to maintain data reliability. These values configure the platform mechanism defined in DIARY-PRD-entry-time-restrictions.

Rationale

The 24-hour **Justification Threshold** binds the soft "explain why this is late" boundary to one calendar day, matching the Yesterday Entry Reminder's daily-cadence prompt — a participant who has not entered yesterday's data within 24 hours has crossed into territory where the protocol requires a documented reason for the delay. The 48-hour **Lock Threshold** doubles the justification window, recognizing that some justifiable-late entries take an additional day to surface (a participant who was traveling or unwell at the 24-hour mark gets a further day to backfill with justification) while still capping the editable window at two days for reliability. The four-option predefined list covers the realistic operational distribution of late-entry causes: transferring from paper records during onboarding, retrospective recall of a specific event, retrospective estimation when specifics are not recalled, and the residual "Other" for cases the three categories do not cover. The prompt text frames the obligation in plain language so the participant understands what they are being asked to explain.

Assertions

A. The **Justification Threshold** SHALL be configured to 24 hours.

B. The **Lock Threshold** SHALL be configured to 48 hours.

C. The **Entry Justification** predefined list SHALL contain exactly four options: "Entered from paper records," "Remembered specific event," "Estimated event," and "Other."

D. The **Entry Justification** prompt SHALL display the text: "This is an event more than one day old. Please explain why you are adding/changing it now."

6.6.3 Duration Reasonableness Check

REQ ID: `DIARY-PRD-entry-duration-check`

Refines: `CAL-PRD-entry-duration-check-configuration`

Overview

Unusually short or long event durations may indicate data entry errors. Confirmation prompts at configurable thresholds help ensure data accuracy while allowing legitimate outlier events to be recorded.

Definitions

Duration Reasonableness Check: A configurable validation rule that prompts the **Participant** to confirm the recorded duration of an event when it falls outside expected boundaries.

Short Duration Threshold: The configurable minimum duration below which a confirmation prompt is triggered.

Long Duration Threshold: The configurable maximum duration above which a confirmation prompt is triggered.

Short Duration Check

Long Duration Check

Prompt Timing

Configuration

Rationale

Duration outliers are a common data-entry-error signal: a typo or wrong-timezone selection can produce a one-second or twelve-hour nosebleed that the participant did not actually have. A soft confirmation prompt at configured boundaries catches these without rejecting legitimately-outlying events (a multi-hour nosebleed can be real for certain HHT participants). The two checks are independently enableable because a deployment may want only one direction (e.g. only the short-duration check, if long durations are clinically plausible in the protocol). The pre-save timing rule (assertion E) ensures the participant sees the prompt at the moment of decision rather than discovering after the fact that their entry was flagged. Reject-and-edit is the inverse of confirm-and-save and gives the participant a direct path to correct a typo without re-entering the event from scratch.

Assertions

A. When the duration of an event is less than the **Short Duration Threshold**, the System SHALL prompt the **Participant** to confirm the duration before saving.

B. The confirmation prompt SHALL provide a confirm option that saves the entry as entered and a reject option that returns the **Participant** to edit the duration.

C. When the duration of an event is greater than the **Long Duration Threshold**, the System SHALL prompt the **Participant** to confirm the duration before saving.

D. The confirmation prompt SHALL provide a confirm option that saves the entry as entered and a reject option that returns the **Participant** to edit the duration.

E. The System SHALL display the confirmation prompt before the save operation completes.

F. The System SHALL support sponsor-configurable **Short Duration Threshold** per deployment.

G. The System SHALL support sponsor-configurable **Long Duration Threshold** per deployment.

H. The System SHALL support sponsor-configurable enablement of each check independently per deployment.

I. When a check is not enabled for a deployment, the System SHALL NOT display the corresponding confirmation prompt.

6.6.4 Duration Reasonableness Check Configuration

REQ ID: `CAL-PRD-entry-duration-check-configuration`

Overview

The Callisto clinical protocol defines specific duration boundaries for nosebleed events to flag potential data entry errors. These values configure the platform mechanism defined in DIARY-PRD-entry-duration-check.

Rationale

The 1-minute **Short Duration Threshold** catches the most likely short-side data-entry error — a participant who tapped Set End Time immediately after Set Start Time, producing a near-zero duration that almost certainly reflects a misclick rather than a real event. The 60-minute **Long Duration Threshold** flags the long-side counterpart — a duration over an hour is clinically possible for some HHT participants but unusual enough that a confirmation prompt is worthwhile against the much more common case of a wrong end-date or wrong-timezone entry producing a multi-hour or multi-day duration. Both checks are enabled for the Callisto deployment because the cost of an undetected outlier (a corrupted primary outcome measure) substantially exceeds the friction cost of a one-tap confirmation when a real outlier event occurs. The prompt text is phrased as a question ("is that correct?") rather than a warning, matching the soft-confirmation intent — the participant is invited to confirm or correct, not blocked.

Assertions

A. The **Short Duration Threshold** SHALL be configured to 1 minute.

B. The **Long Duration Threshold** SHALL be configured to 60 minutes.

C. Both the short duration check and the long duration check SHALL be enabled.

D. The short duration confirmation prompt SHALL display the text: "Duration is under 1 minute, is that correct?"

E. The long duration confirmation prompt SHALL display the text: "Duration is over 1 hour, is that correct?"

6.6.5 Overlapping Event Detection and Resolution

REQ ID: `DIARY-PRD-entry-overlap-resolution`

Refines: `DIARY-GUI-entry-overlap-resolution`

Overview

Nosebleed events cannot physically overlap — a participant cannot have two simultaneous, independent nosebleeds. When a new record's time range intersects with an existing record, the system must detect the conflict and guide the participant through resolution before the record is committed. This ensures data integrity and prevents logically impossible entries from reaching the clinical dataset.

Definitions

Overlap: A state in which the time range of a new or edited entry intersects with the time range of one or more existing entries for the same **Participant**.

Conflicting Record: An existing entry whose time range intersects with the entry being created or edited.

Resolution: The **Participant's** choice of how to handle an **Overlap** between two entries.

Detection

Pre-Resolution State

Resolution Sequencing

Resolution Options

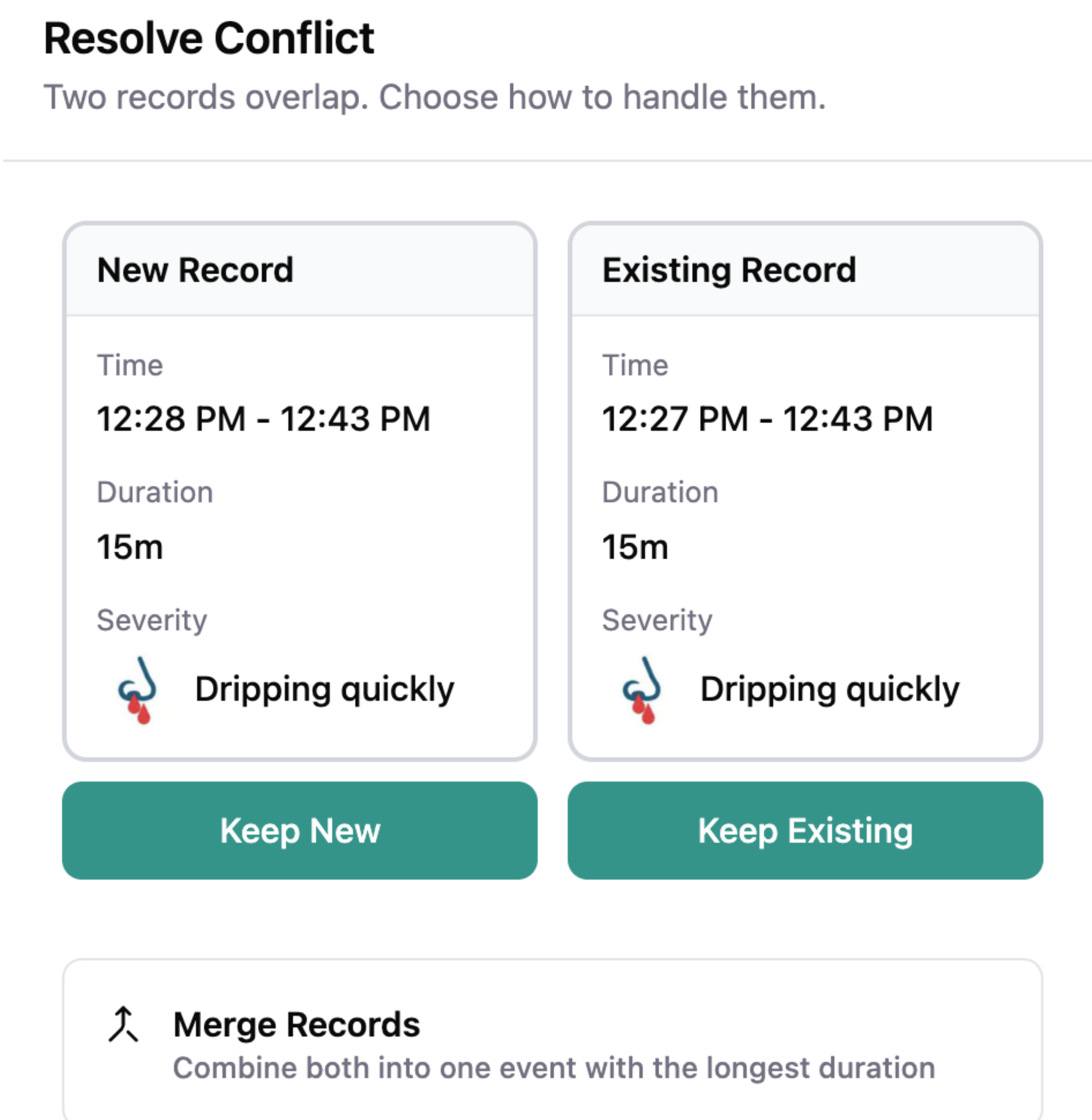
Merge Behavior

Post-Resolution Confirmation

Rationale

Two nosebleeds cannot physically overlap; allowing overlapping records into the dataset would be silent data corruption. The two-stage detection (predictive on start-time, confirmatory on end-time) gives the participant early notice that an overlap is forming without interrupting them mid-record, then escalates to mandatory resolution at the moment the conflict becomes a hard reality. Saving the new entry to persistent storage immediately on creation (assertion D) is the durability guarantee that the participant's work is not lost if the application crashes mid-resolution; hiding it from the diary history until resolution completes (assertion E) keeps the participant's view consistent with what has actually been resolved. One-conflict-at-a-time resolution starting with the oldest, repeated until no overlaps remain, is the only sequencing that handles the cascade case (a new record overlaps two existing records that themselves overlap each other once merged). Three resolution options cover the realistic choices: the new record was right and the old one was wrong (keep-new), the old record was right and the new one is a duplicate (keep-existing), or both were partial captures of one continuous event (merge). Merge using earliest-start / latest-end with the higher severity captures the union-by-time and max-by-severity that best represents one continuous event reconstructed from two partial captures. Pre-commit review is the final integrity gate — the participant sees the resulting entry as the platform will save it and can edit any field before confirming.

Screen reference



See:

Assertions

A. The System SHALL evaluate for potential **Overlap** when the **Participant** sets the start time of a new or edited entry, and SHALL confirm the **Overlap** when the **Participant** sets the end time.

B. The System SHALL evaluate **Overlap** by comparing the actual start and end timestamps of the new or edited entry against all existing entries for the same **Participant**.

C. The System SHALL evaluate **Overlap** against both completed entries and ongoing entries.

D. When an **Overlap** is detected, the System SHALL save the new entry to persistent storage immediately upon creation.

E. When an **Overlap** is detected, the System SHALL NOT display the new entry in the **Participant's** diary history until **Resolution** is complete.

F. The System SHALL present the **Participant** with one **Conflicting Record** at a time, starting with the oldest.

G. After each **Resolution** step, the System SHALL evaluate the resulting entry for further **Overlaps**.

H. The System SHALL repeat the **Resolution** flow until no **Overlaps** remain.

I. When all **Overlaps** are resolved, the System SHALL commit all resulting changes as a single transaction.

J. For each **Overlap**, the System SHALL offer three resolution options: keep the new entry and discard the **Conflicting Record**, keep the **Conflicting Record** and discard the new entry, or merge both entries into one.

K. When the **Participant** selects merge, the System SHALL construct the merged entry using the earliest start time and the latest end time across both entries.

L. When either entry in a merge is ongoing, the merged entry SHALL be ongoing.

M. When the **Participant** selects merge, the System SHALL assign the higher severity value from the two original entries to the merged entry.

N. After any resolution choice, the System SHALL present the resulting entry to the **Participant** for review and confirmation before committing.

O. The **Participant** SHALL be able to edit the resulting entry before confirming.

6.6.6 Overlapping Event Resolution Flow

REQ ID: `DIARY-GUI-entry-overlap-resolution`

Overview

When an overlap is detected, the **Participant** needs early warning and a clear resolution path. Displaying a warning as soon as the start time indicates a potential conflict gives the **Participant** awareness without interrupting their recording flow. Once the full time range is established, the resolution screen provides a side-by-side comparison and straightforward actions to resolve the conflict.

Definitions

Resolution Screen: The screen presented to the **Participant** when an **Overlap** is confirmed, displaying the new entry and the **Conflicting Record** side by side with available resolution actions.

Early Warning

Resolution Trigger

Resolution Screen

Post-Resolution

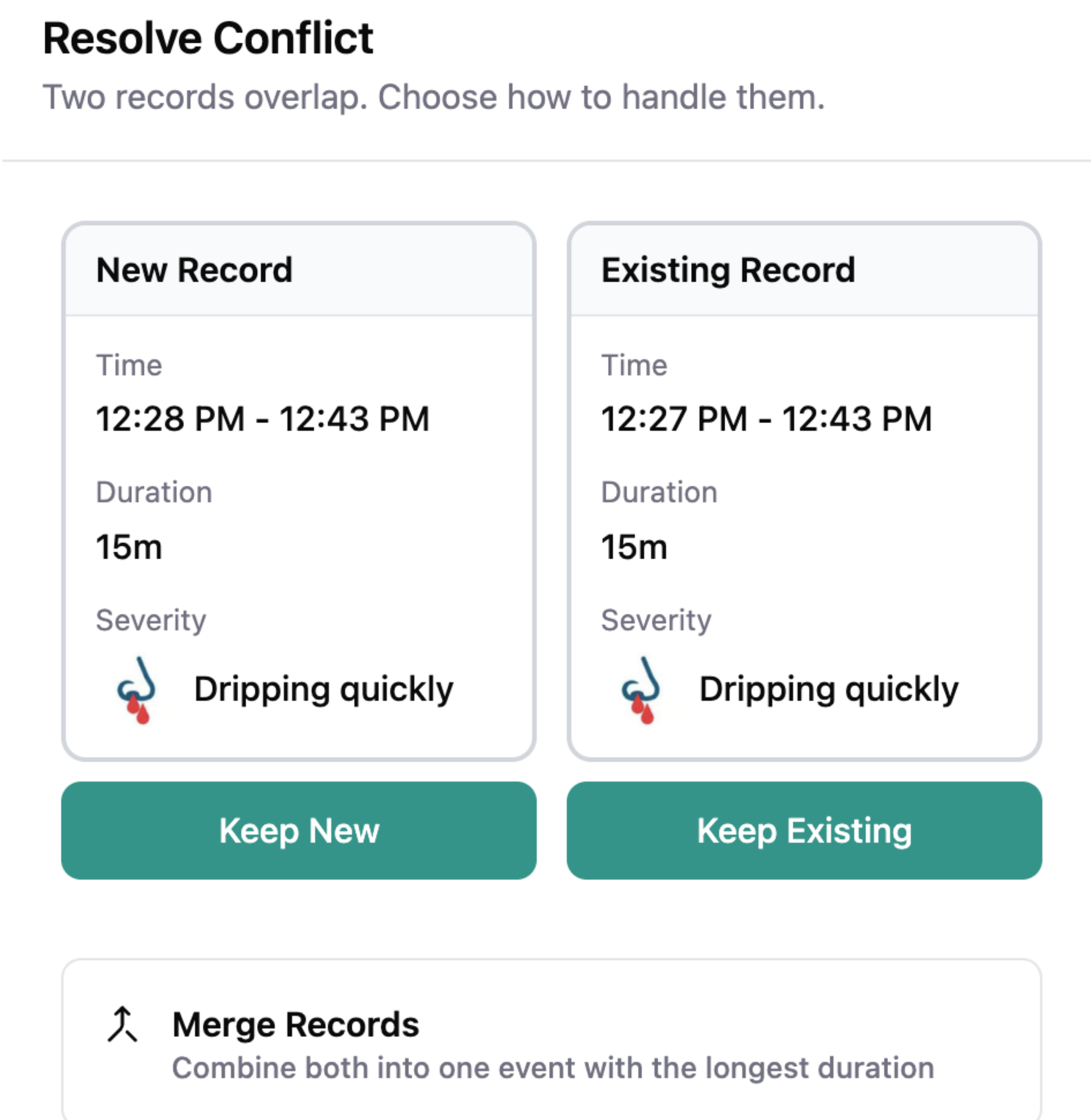
Multiple Conflicts

Rationale

The early warning on start-time is a non-blocking signal: the participant sees that something is going to need resolution but is not pushed into the resolution flow yet, because the end-time may still make the overlap go away (e.g. the participant moves the end time earlier than the conflicting record's start). Only when the end-time confirms the conflict does the GUI commit the participant to the **Resolution Screen**. The side-by-side display makes the comparison legible — same time-range / duration / severity rows for both entries — so the participant can make an informed choice between Keep New, Keep Existing, and Merge. The Merge-availability check against the **Lock Threshold** is required because merging would produce edits to the Conflicting Record's date, which the lock prohibits; surfacing the unavailability with an explanatory message (rather than silently disabling) keeps the participant oriented. Routing the post-resolution result through the standard **Record Nosebleed** screen (pre-filled) gives the participant a final edit-and-confirm opportunity on the same surface they used to enter the original record — a single interaction model end-to-end rather than a resolution-specific screen the participant has to learn separately. One-at-a-time presentation for multiple overlaps matches the PRD-level resolution sequencing rule, and the return-to-Main-Screen on completion closes the recording journey at its natural endpoint.

> **Follow-up — configurability**: This requirement currently encodes the only option implemented in code. Future sponsors may require different rules; introduce a configurable seam (e.g. a parameter on the CAL-PRD-* parent, or a new platform-side template the CAL- REQ Satisfies) when the need arises. Until that seam exists, this REQ is normative for the Callisto deployment.

Screen reference



See:

Assertions

A. When a potential **Overlap** is detected after the **Participant** sets the start time, the interface SHALL display a warning indicating that overlapping events have been detected and the number of conflicting records.

B. The warning SHALL NOT prevent the **Participant** from continuing the recording flow.

C. When the **Participant** sets the end time and the **Overlap** is confirmed, the interface SHALL navigate the **Participant** to the **Resolution Screen**.

D. The interface SHALL present the **Resolution Screen** displaying the new entry and the **Conflicting Record** side by side.

E. Each entry on the **Resolution Screen** SHALL display the time range, duration, and severity.

F. When the **Conflicting Record** is older than the **Justification Threshold**, the System SHALL require an **Entry Justification** before saving the resolution.

G. The **Resolution Screen** SHALL present a Keep New action and a Keep Existing action. The Resolution Screen SHALL present a Merge Records action only when the Conflicting Record's event date has not exceeded the **Lock Threshold**.

H. The Merge Records action SHALL display a description indicating that both records will be combined into one event.

I. When the Merge Records action is unavailable due to the **Lock Threshold**, the Resolution Screen SHALL display a message indicating that the conflicting record is locked and cannot be merged.

J. After the **Participant** selects any resolution option (Keep New, Keep Existing, or Merge Records), the interface SHALL navigate to the standard **Record Nosebleed** screen pre-filled with the resulting entry data.

K. The **Record Nosebleed** screen SHALL allow the **Participant** to review and edit any field of the resulting entry before confirming.

L. When multiple **Overlaps** exist, the interface SHALL present one **Resolution Screen** at a time.

M. When all **Overlaps** are resolved and the **Participant** confirms the final entry, the interface SHALL return the **Participant** to the Main Screen.

6.7 Mobile Application Navigation and Screens

The foundational navigation surfaces of the **Mobile Application** comprise the **Main Screen** layout (zones, task area, content area, fixed bottom actions), the two top-navigation-bar menus (User Menu, Application Menu) and their dependent screens, and the Calendar / Day View pair the participant uses to navigate to any past date.

6.7.1 Main Screen Layout

REQ ID: `DIARY-GUI-main-screen-layout`

Overview

The **Main Screen** is the **Participant's** primary interface for daily diary use. Organizing the screen into distinct zones ensures that urgent items (disconnection alerts and tasks) are always visible, diary content is scrollable, and the primary recording action is always accessible. When no tasks or alerts are active, the content area expands to use the full available space.

Definitions

Main Screen: The default screen displayed to the **Participant** upon opening the mobile application.

System Notice Area: A dedicated zone at the top of the Main Screen reserved for persistent, non-dismissable system notices that require the Participant's attention.

Screen Zones

Task List Area

Content Area

Fixed Bottom Actions

Rationale

The five-zone structure (System Notice Area, top navigation bar, Task List, content area, fixed bottom actions) reflects an attention-priority ordering: persistent system notices appear above everything because they typically encode a state the participant must acknowledge (e.g. **Disconnection Notification**), and the fixed bottom actions appear at the most-reachable thumb position because they are the two highest-frequency participant actions (Record Nosebleed, open Calendar). The Task List zone collapsing when empty (with the content area expanding to fill the space) keeps the screen from carrying empty chrome for participants in a quiet stretch of the trial — most days, when there are no incomplete records or pending portal-sent questionnaires, the content area uses the full middle of the screen. Yesterday / Today grouping in the content area matches the everyday read pattern: the most recent days are what the participant actually wants to see; the calendar is the path to anything older.

> **Follow-up — configurability**: This requirement currently encodes the only option implemented in code. Future sponsors may require different rules; introduce a configurable seam (e.g. a parameter on the CAL-PRD-* parent, or a new platform-side template the CAL- REQ Satisfies) when the need arises. Until that seam exists, this REQ is normative for the Callisto deployment.

Assertions

A. The Main Screen SHALL display the following zones in order from top to bottom: System Notice Area (when applicable), top navigation bar, Task List (when tasks are active), content area, and fixed bottom actions.

B. The System Notice Area SHALL appear above the top navigation bar.

C. When the System Notice Area is displayed, it SHALL NOT reduce the space available for any other Main Screen zone.

D. When the **Task List** contains more tasks than the available space allows, the **Task List** SHALL scroll within its allocated area.

E. When no tasks are active and no **Disconnection Notification** is displayed, the content area SHALL expand to fill the available space.

F. The content area SHALL display diary entries grouped by date, showing Yesterday and Today sections.

G. The content area SHALL be scrollable between the task area and the fixed bottom actions.

H. When a date section contains no entries, the content area SHALL display a message indicating no events for that date.

I. The **Main Screen** SHALL display a Record Nosebleed button fixed at the bottom, regardless of scroll position.

J. The **Main Screen** SHALL display a Calendar button fixed at the bottom below the Record Nosebleed button, regardless of scroll position.

6.7.2 Mobile Application Navigation and Screens

REQ ID: `DIARY-GUI-mobile-navigation`

Overview

The mobile application provides two menus accessible from the top navigation bar: a **User Menu** for account and study-related actions, and an **Application Menu** for application-level functions. Each menu leads to dedicated screens. The top navigation bar is visible on the **Main Screen** at all times.

Definitions

User Menu: The menu accessed from the right side of the top navigation bar, grouping account and study-related actions.

Application Menu: The menu accessed from the left side of the top navigation bar, grouping application-level functions.

Top Navigation Bar

User Menu

User Profile Screen

Application Menu

Rationale

The two-menu split (User Menu on the right, Application Menu on the left) groups actions by the question the participant is asking: "something about my study" (User Menu: Join the Study, User Profile, Help Center) vs. "something about the app itself" (Application Menu: Accessibility, Application Privacy Policy, Licenses). Single-menu-open-at-a-time is a standard mobile-UX safety (opening one menu closes the other) so the participant never sees two competing surfaces overlap. Hiding Join the Study when already linked (assertion G) removes the no-op action from the menu — once the participant is a **Participant**, the action that converted them from **User** to **Participant** is no longer meaningful. The User Profile's Clinical Trial section is the participant's anchor for their study status: empty + guidance when unlinked, **Participation Status Badge** when linked. The Application Menu's three items are the standard app-level metadata participants may need to reach but rarely do (privacy, accessibility, licenses).

> **Follow-up — configurability**: This requirement currently encodes the only option implemented in code. Future sponsors may require different rules; introduce a configurable seam (e.g. a parameter on the CAL-PRD-* parent, or

a new platform-side template the CAL- REQ Satisfies) when the need arises. Until that seam exists, this REQ is normative for the Callisto deployment.

Assertions

A. The **Main Screen** SHALL display a top navigation bar containing the **Application Menu** access on the left, the sponsor or Application logo in the center, and the **User Menu** access on the right.

B. The top navigation bar SHALL remain visible on the **Main Screen** at all times.

C. Only one menu SHALL be open at a time. Opening one menu SHALL close the other if it is open.

D. Tapping anywhere outside an open menu SHALL dismiss it.

E. The **User Menu** SHALL contain the following items: Join the Study, User Profile, and Help Center.

F. When the **User** clicks the "Join the Study" button SHALL navigate to the linking code entry screen.

G. When the **Participant** is linked to a study, "Join the Study" button SHALL not be visible on the menu.

H. The Help Center screen SHALL display contact information for support.

I. When the **User** is not linked to any study, the Clinical Trial section SHALL display a message indicating no active study link and guidance on how to join.

J. When the **Participant** is linked to a study, the Clinical Trial section SHALL display the **Participation Status Badge**.

K. The **Application Menu** SHALL contain the following items: Accessibility and Preferences, **Application Privacy Policy**, and Licenses.

L. The **Application Privacy Policy** screen SHALL display or link to the **Application Privacy Policy**.

M. The Licenses screen SHALL display open-source and third-party license information.

6.7.3 Calendar and Day View

REQ ID: `DIARY-GUI-calendar-day-view`

Overview

The **Calendar** provides **Participants** with a monthly overview of their diary activity and a way to navigate to any past date to view, add, or edit entries. The **Day View** is the screen displayed when a **Participant** selects a date from the **Calendar**.

Definitions

Calendar: The monthly view modal displaying date states and allowing the **Participant** to navigate to a specific date.

Day View: The screen displayed when the **Participant** selects a date from the **Calendar**, showing the entries for that date or prompting the **Participant** to record a **Daily Status**.

Calendar Modal

Day View — No Daily Status Recorded

Day View — Daily Status Recorded

Entry Selection

Day View Navigation

Locked Dates

Rationale

The Calendar is the participant's at-a-glance survey of their diary period: every date's visual indicator answers "have I recorded for this day, and if so, what?" without forcing the participant to drill into each date. The seven-state legend (recorded events, confirmed no-events, don't-remember, incomplete-or-missing, not-recorded, locked, today) covers every diary-state distinction the platform tracks; collapsing two states into one indicator would hide either incomplete-record warnings or locked-date evidence that the participant needs to see. The Day View bifurcation (no-status-recorded prompt vs. status-recorded list) reflects two distinct participant journeys — first-time entry for a date (Add / No / Don't recall three-action prompt) vs. revisiting an already-recorded date (entry list with Add new event affordance). Lock-state handling on a locked date suppresses every action that would attempt to modify the date, surfacing the lock explicitly rather than letting the participant tap an action and discover the rejection — the explanatory message ("can no longer be edited") is what makes the lock comprehensible. Submitted **Portal-Sent Questionnaires** appearing in the date's entry list grounds the questionnaire as a part of the diary record on the date it landed, rather than as an out-of-band artifact the participant has to navigate separately to find.

> **Follow-up — configurability**: This requirement currently encodes the only option implemented in code. Future sponsors may require different rules; introduce a configurable seam (e.g. a parameter on the CAL-PRD-* parent, or a new platform-side template the CAL- REQ Satisfies) when the need arises. Until that seam exists, this REQ is normative for the Callisto deployment.

Assertions

A. When the **Participant** selects the Calendar button, the interface SHALL display the **Calendar** as a modal overlay.

B. The **Calendar** SHALL display a monthly view with navigation arrows to move between months.

C. Each date in the **Calendar** SHALL display a visual indicator representing its diary state.

D. The **Calendar** SHALL display a legend identifying each visual indicator state.

E. The **Calendar** SHALL display the following date states: nosebleed events recorded, no nosebleeds confirmed, unknown / don't remember, incomplete or missing data, not recorded, **locked, and today**.

F. A date SHALL display the incomplete state when the **Participant** has one or more **Incomplete Records** for that date.

G. A date SHALL display the missing state when the date is within the diary period (from Diary Start Day to yesterday) and the Participant has not recorded any Daily Status for that date.

H. The **Calendar** SHALL present a close action that dismisses the modal and returns the **Participant** to the **Main Screen**.

I. When the selected date has no **Daily Status** recorded, the **Day View** SHALL display the date and the prompt "What happened on this day?" with three actions: Add nosebleed event, No nosebleed events, and I don't recall / unknown.

J. When the **Participant** selects No nosebleed events, the interface SHALL record a **Daily Status** of No Nosebleed for that date.

K. When the **Participant** selects I don't recall / unknown, the interface SHALL record a **Daily Status** of Don't Remember for that date.

L. When the **Participant** selects Add nosebleed event, the interface SHALL navigate the **Participant** to the nosebleed recording flow with the date set to the selected date.

M. When the selected date has a **Daily Status** recorded, the **Day View** SHALL display the date, the total event count, an Add new event action, and a list of all entries for that date.

N. Each nosebleed entry in the list SHALL display the start time, timezone, severity, and duration.

O. Each entry in the list SHALL be selectable.

P. Submitted **Portal-Sent Questionnaires** SHALL appear in the entries list on the date they were submitted.

Q. When the **Participant** selects an entry, the interface SHALL navigate to the nosebleed recording flow for editing if the entry has not exceeded the **Lock Threshold**, or display the entry details in a read-only state if it has.

R. The **Day View** SHALL present a back action that returns the **Participant** to the **Calendar**.

S. When a date has exceeded the **Lock Threshold**, the **Calendar** SHALL display that date in a greyed-out visual style distinct from other date states.

T. When the **Participant** selects a locked date with no **Daily Status** recorded, the **Day View** SHALL display a message indicating no record exists for that date and that the date can no longer be edited. The three action buttons (Add nosebleed event, No nosebleed events, I don't recall / unknown) SHALL NOT be displayed.

U. When the **Participant** selects a locked date with a **Daily Status** recorded, the **Day View** SHALL display the existing entries in a read-only state and SHALL display a message indicating the date can no longer be edited. The Add new event action SHALL NOT be displayed.

6.8 Participant Tasks and Notifications

The **Mobile Application** delivers a set of push notifications and reminders to the **Participant**, together with the on-screen **Task List** and **Participation Status Badge** that surface task-state information in the app itself. The push notifications defined here all conform to the cross-cutting `DIARY-PRD-notification-behavior` template (tap behavior, no in-notification actions, offline-deferred delivery).

6.8.1 Participant Task List

REQ ID: `DIARY-GUI-participant-task-list`

Overview

The task list provides **Participants** with a prioritized set of actions requiring their attention. Displaying tasks prominently on the **Main Screen** and linking directly to the relevant flows reduces friction and encourages timely data entry. Tasks are automatically resolved as the **Participant** addresses them, keeping the list current.

Definitions

Task List: The prioritized set of actionable items displayed on the **Participant's Main Screen** requiring **Participant** attention.

Incomplete Records Task: A task indicating the **Participant** has one or more saved entries with missing required data.

Questionnaire Task: A task indicating a **Portal-Sent Questionnaire** has been sent to the **Participant** and has not yet been submitted.

Yesterday Reminder Task: A task prompting the **Participant** to record a **Daily Status** for the previous day.

Task Types and Priority

Incomplete Records Task

Questionnaire Task

Yesterday Reminder Task

General Behavior

Rationale

The three task types correspond to the three categories of follow-up the platform needs to surface to the participant: data the participant already entered that is missing required fields (**Incomplete Records Task**), questionnaires the **Study Coordinator** has assigned that have not been submitted (**Questionnaire Task**), and the daily-status prompt for yesterday (**Yesterday Reminder Task**). The priority ordering reflects intervention urgency: incomplete records are at risk of becoming permanently locked (highest priority), questionnaires are due but not yet on a hard deadline (middle), and yesterday's status is a soft prompt that auto-resolves over time (lowest). The **Yesterday Reminder Task**'s three-action surface (Yes / No / Don't Remember) lets the participant resolve the task without leaving the Main Screen for the common case (Yes navigates to recording; No and Don't

Remember commit the **Daily Status** in place). Removing tasks the moment their removal condition is met (and not requiring a refresh) keeps the list honest — a participant who completes their incomplete records sees the task disappear immediately, rather than being told to refresh or re-open the app. **Incomplete Records Task** persistence regardless of age (assertion E) is the operational counterpart of the **Lock Threshold**: even after the underlying record is locked and can no longer be completed, the task stays visible so the participant is aware of the now-permanently-incomplete state.

> **Follow-up — configurability**: This requirement currently encodes the only option implemented in code. Future sponsors may require different rules; introduce a configurable seam (e.g. a parameter on the CAL-PRD-* parent, or a new platform-side template the CAL- REQ Satisfies) when the need arises. Until that seam exists, this REQ is normative for the Callisto deployment.

Assertions

A. The **Task List** SHALL support the following task types, displayed in the priority order listed: | Priority | Task Type | Trigger | Removal | | :—- | :—- | :—- | :—- | | 1 | **Incomplete Records Task** | **Participant** has one or more saved entries with missing required data | All incomplete entries are completed or deleted | | 2 | **Questionnaire Task** | Study Coordinator sends a **Portal-Sent Questionnaire** | Sponsor finalizes the questionnaire | | 3 | **Yesterday Reminder Task** | New day begins and the **Participant** has not recorded a **Daily Status** for the previous day | **Participant** records a **Daily Status** for the previous day |

B. The **Incomplete Records Task** SHALL display the count of incomplete entries.

C. When the **Participant** selects the **Incomplete Records Task**, the interface SHALL navigate to a screen listing all incomplete entries.

D. From the incomplete entries screen, the **Participant** SHALL be able to complete or delete each entry.

E. The **Incomplete Records Task** SHALL persist regardless of the age of the incomplete entries.

F. The interface SHALL display one **Questionnaire Task** per **Questionnaire Type**.

G. When the **Participant** selects a **Questionnaire Task**, the interface SHALL navigate to the questionnaire flow for that **Portal-Sent Questionnaire**.

H. After the Participant submits a **Portal-Sent Questionnaire**, the questionnaire SHALL be accessible as a record on the day it was submitted via the **Calendar**.

I. After the sponsor finalizes the **Portal-Sent Questionnaire**, the **Questionnaire Task** SHALL be removed from the **Task List**.

J. After **Submission**, the **Questionnaire Task** SHALL display a completed visual state indicating the questionnaire has been submitted and is awaiting sponsor review.

K. While the **Questionnaire Task** is in a submitted state, the Participant SHALL be able to select it to review and edit their answers.

L. The **Yesterday Reminder Task** SHALL present three response actions: Yes, No, and Don't Remember.

M. When the **Participant** selects No, the interface SHALL record a **Daily Status** of No Nosebleed for the previous day and remove the task.

N. When the **Participant** selects Don't Remember, the interface SHALL record a **Daily Status** of Don't Remember for the previous day and remove the task.

O. When the **Participant** selects Yes, the interface SHALL navigate the **Participant** to the nosebleed recording flow with the date set to the previous day.

P. The **Yesterday Reminder Task** SHALL NOT appear if the **Participant** has already recorded a **Daily Status** for the previous day.

Q. Tasks SHALL be removed from the **Task List** when their removal condition is met.

R. The **Task List** SHALL update as conditions change without requiring the **Participant** to refresh.

6.8.2 Disconnection Notification

REQ ID: `DIARY-PRD-notification-disconnection`

Refines: `DIARY-GUI-participation-status-badge`

Overview

Participants need immediate, clear notification when their mobile Application is disconnected from the sponsor portal. A persistent, non-dismissible notification ensures the **Participant** is always aware of their disconnected state and cannot accidentally or intentionally hide it.

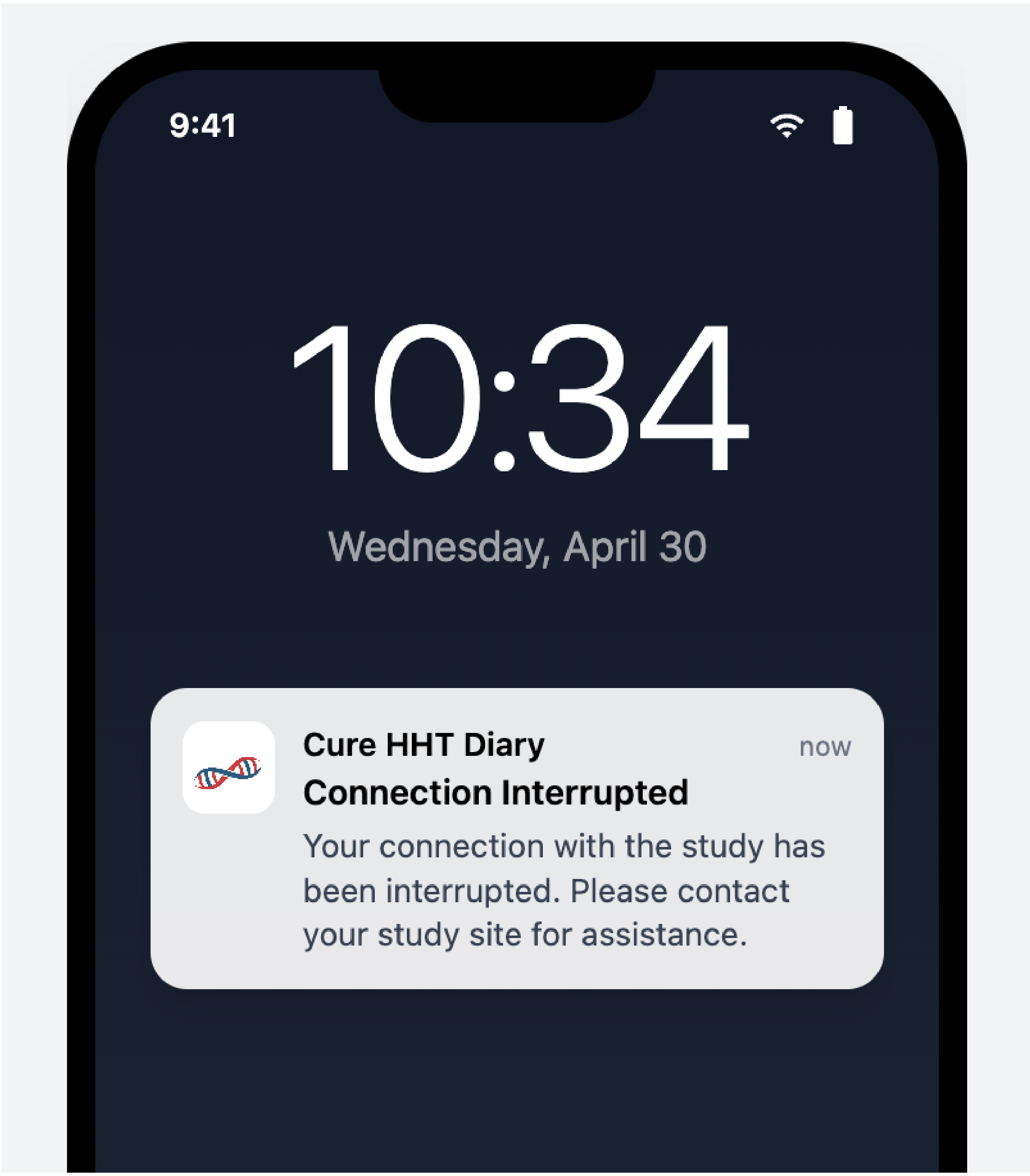
Definitions

Disconnection Notification: A persistent, non-dismissible notification displayed to the **Participant** when their mobile Application is disconnected from the **Sponsor Portal**.

Rationale

Disconnection is an operational state with clinical-data consequences: data the participant enters while disconnected does not reach the sponsor and may sit unsynced for an indeterminate period. The participant needs to know they are disconnected so they can either resolve it (contact their site for a new linking code) or at minimum understand why their data is not visible to the sponsor. Persistent + non-dismissible is the only display mode that survives the participant's natural impulse to clear unfamiliar notifications: a dismissible disconnection banner would be cleared within minutes by most participants, and the disconnection would then be invisible to them on every subsequent app open. The sponsor-configurable message + default text lets each deployment customize the language (e.g. site contact phone numbers, terminology consistent with the sponsor's other communications) while keeping a working default for deployments that don't configure it.

Screen reference

See: 

Assertions

A. When a **Participant's** status is **Disconnected**, the System SHALL display a **Disconnection Notification** on the **Main Screen**.

B. The **Disconnection Notification** SHALL persist until the **Participant** is reconnected to the **Sponsor Portal**.

C. The **Disconnection Notification** SHALL NOT be dismissible by the **Participant**.

D. The System SHALL support sponsor-configurable message content for the **Disconnection Notification**.

E. When no sponsor-specific message content is configured, the **Disconnection Notification** SHALL display the following default message: "Your connection with the study has been interrupted. Please contact your study site for assistance."

6.8.3 Participation Status Badge

REQ ID: `DIARY-GUI-participation-status-badge`

Overview

The **Participation Status Badge** gives **Participants** a persistent, at-a-glance view of their clinical trial involvement. The badge adapts its display based on the **Participant's** current status so they always know whether they are connected, disconnected, or no longer participating in a study.

Definitions

Participation Status Badge: The visual component displayed in the Clinical Trial section of the user profile screen showing the **Participant's** current study participation state and associated details.

Placement

Linked State

Disconnected State

Not Participating State

Automatic Update

Rationale

The badge consolidates “where do I stand with this study” into a single visual the participant always finds in the same place (Clinical Trial section of the User Profile). Three state variants cover the three operational situations: actively-linked (sponsor logo, linking code, join date — full participation context), disconnected (warning indicator, linking code, Enter New Linking Code action — the recovery path is reachable from the badge itself), and not-participating (inactive style with end date — the badge persists as historical evidence of past participation without competing for attention against current state). The Clinical Trial Privacy Policy link persists across status changes because the policy version the participant consented to is part of their permanent record, and they have a continuing right to retrieve it. Automatic update on status change is the standard freshness guarantee; sponsor-configurable logo display in the Not Participating state lets the deployment decide whether the badge in its end-of-participation state still carries sponsor branding (an end-of-trial communication preference).

> **Follow-up — configurability**: This requirement currently encodes the only option implemented in code. Future sponsors may require different rules; introduce a configurable seam (e.g. a parameter on the CAL-PRD-* parent, or a new platform-side template the CAL- REQ Satisfies) when the need arises. Until that seam exists, this REQ is normative for the Callisto deployment.

Assertions

A. The interface SHALL display the **Participation Status Badge** in the Clinical Trial section of the user profile screen.

B. When the **Participant’s** status is **Linked - Awaiting Start** or **Trial Active**, the **Participation Status Badge** SHALL display the sponsor logo, the **Participant’s** linking code, and the date and time the **Participant** joined.

C. The **Participation Status Badge** SHALL include a link to the **Clinical Trial Privacy Policy** from the moment the Participant links to the study, and the link SHALL remain available thereafter regardless of subsequent status changes.

D. When the **Participant's** status is **Disconnected**, the **Participation Status Badge** SHALL display a warning indicator, the **Participant's** current linking code, and a message that the connection has been interrupted.

E. When the **Participant's** status is **Disconnected**, the **Participation Status Badge** SHALL present an Enter New Linking Code action that navigates the **Participant** to the linking code entry screen.

F. When the **Participant's** status is **Not Participating**, the **Participation Status Badge** SHALL display in an inactive visual style with the end date of participation.

G. The **Participation Status Badge** SHALL update automatically when the **Participant's** status changes.

H. The System SHALL support sponsor-configurable display of the sponsor logo on the **Participation Status Badge** when the **Participant's** status is **Not Participating**.

6.8.4 Incomplete Record Lock Warning Notification

REQ ID: `DIARY-PRD-notification-incomplete-record-lock`

Refines: `CAL-PRD-notification-incomplete-record-lock-configuration`

Overview

Once an **Incomplete Record** reaches the Lock Threshold defined in DIARY-PRD-entry-time-restrictions, the Participant cannot complete, edit, or delete it — the record is permanently retained in its incomplete state. To give Participants a final opportunity to act before this irreversible lock, the **System** sends a push notification at a configurable time before the lock fires. This requirement defines the platform mechanism; the trigger offset and notification text are sponsor-configurable.

Definitions

Lock Warning Offset: The configurable elapsed time before the Lock Threshold at which the **System** sends a push notification to the Participant about an **Incomplete Record** that has not been completed or deleted.

Notification Behavior

Configuration

Rationale

The **Lock Threshold** produces an irreversible state — once exceeded, the **Incomplete Record** is permanently retained in its missing-field state and cannot be completed, edited, or deleted. The lock-warning notification exists because that irreversible outcome benefits from one final participant nudge before it lands; without the warning, a participant who simply forgot about an incomplete record discovers the permanent lock only on next app open, long after they could have done anything about it. Once-per-record (assertion B) prevents notification spam from a single long-lingering record; suppression on completion or deletion before the warning offset (assertion C) avoids notifying about records the participant has already resolved. The opt-out semantics (not configured = no notification) and the Lock-Warning-less-than-Lock-Threshold invariant compose with the parent **Entry Time Restrictions** REQ — if a deployment hasn't configured the parent thresholds, the warning has no boundary to anchor to and is therefore not sent.

Assertions

A. The **System** SHALL send a push notification to the Participant when an **Incomplete Record** reaches the Lock Warning Offset before the Lock Threshold defined in DIARY-PRD-entry-time-restrictions.

B. The **System** SHALL send the notification only once per **Incomplete Record**.

C. When a Participant completes or deletes an **Incomplete Record** before the Lock Warning Offset is reached, the **System** SHALL NOT send the notification for that record.

D. The **System** SHALL support a sponsor-configurable Lock Warning Offset per deployment.

E. The **System** SHALL support sponsor-configurable notification text per deployment.

F. When the Lock Warning Offset is not configured for a deployment, the **System** SHALL NOT send the notification.

G. The Lock Warning Offset SHALL be less than the Lock Threshold defined in DIARY-PRD-entry-time-restrictions.

6.8.5 Incomplete Record Lock Warning Notification Configuration

REQ ID: `CAL-PRD-notification-incomplete-record-lock-configuration`

Overview

The clinical protocol defines a 48-hour lock on diary entries (CAL-PRD-entry-time-restrictions-configuration). To give Participants a final opportunity to complete or delete an **Incomplete Record** before the lock fires, a warning notification is 24 hours prior. These values configure the platform mechanism defined in DIARY-PRD-notification-incomplete-record-lock.

Rationale

The 24-hour Lock Warning Offset places the warning notification at the midpoint of the 48-hour Lock Threshold window (CAL-PRD-entry-time-restrictions-configuration) — early enough that the participant has a full day to act, late enough that the record has been in incomplete state long enough for the warning to be useful rather than premature. The notification text names the dated record explicitly so the participant knows which entry needs action, states the 24-hour deadline so the participant understands the urgency, and spells out the consequence (permanent lock, no further changes) so the action's importance is clear without medicalized hedging.

Assertions

A. The Lock Warning Offset SHALL be configured to 24 hours.

B. The notification text SHALL be: "You have an incomplete nosebleed record from [date]. Complete or delete it within the next 24 hours, otherwise it will be permanently locked and you will not be able to change or remove it."

6.8.6 Portal-Sent Questionnaire Notification

REQ ID: `DIARY-PRD-notification-portal-sent-questionnaire`

Overview

When a **Study Coordinator** sends a **Portal-Sent Questionnaire** from the Sponsor Portal, the **Participant** must be made aware that a new questionnaire is available for completion. A push notification serves as the awareness mechanism. The Questionnaire Task on the **Main Screen** provides the **Participant**'s entry point into the questionnaire flow once the application is open.

Trigger

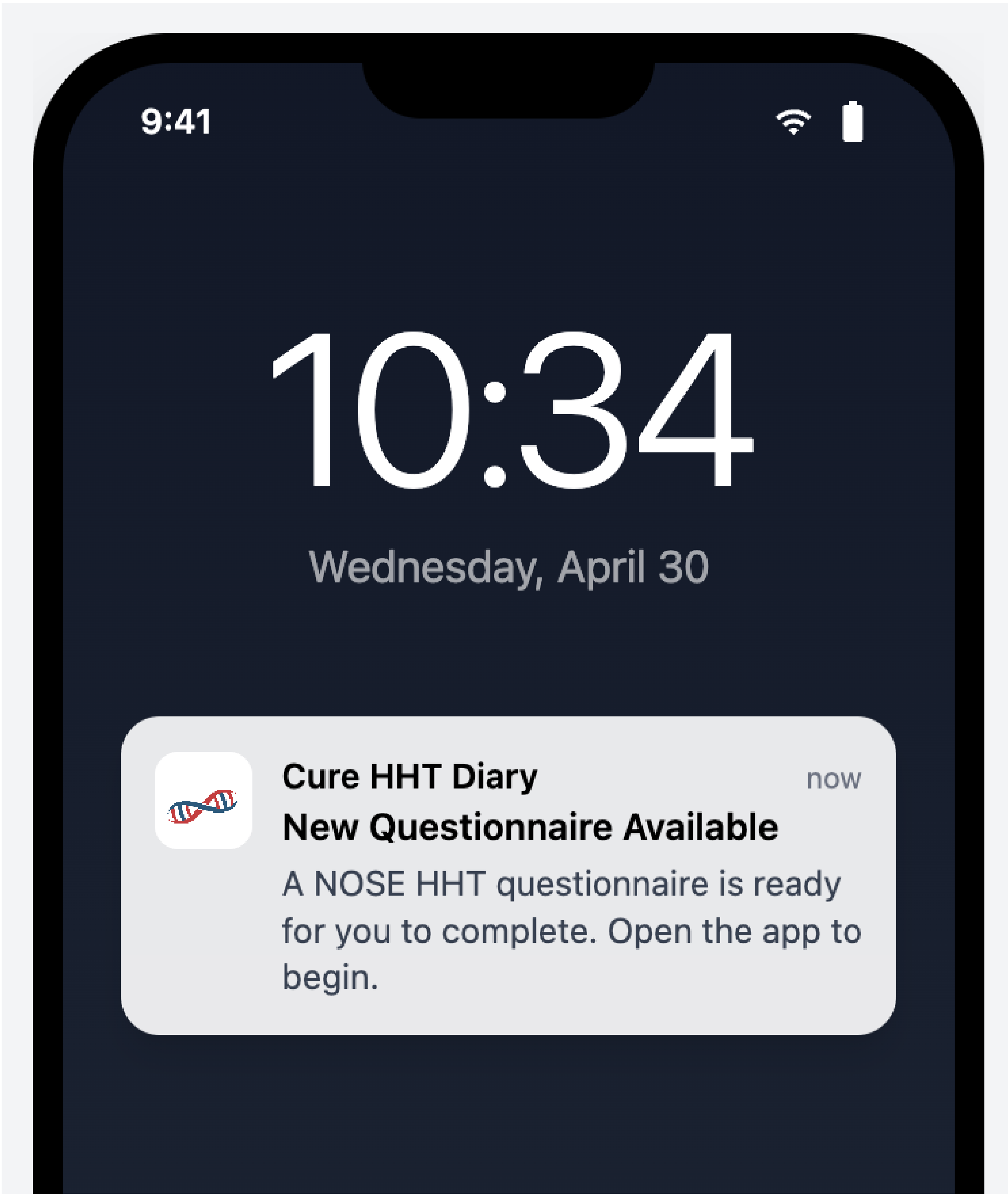
Offline Delivery

Suppression

Rationale

The push notification is the participant's awareness signal that a new **Portal-Sent Questionnaire** is waiting; without it, the participant would discover the questionnaire only when they happen to open the app, which can extend the response window beyond what the **Study Coordinator** intended. Trigger-on-delivery (rather than trigger-on-send-from-portal) is the correct boundary because delivery is what actually puts the questionnaire on the participant's device; sending from the portal to an offline device is a sponsor-side action that has no participant-side effect yet. Offline-deferred delivery follows the cross-cutting notification template — the OS deferred-delivery mechanism is sufficient for the nudge purpose. Submission and call-back suppression closes the two cases where the notification would be misleading: a questionnaire already submitted no longer needs the participant's attention, and a called-back questionnaire is no longer assigned to them and a notification would invite wasted effort on an inactive questionnaire.

Screen reference

See: 

Assertions

A. When a **Portal-Sent Questionnaire** is successfully delivered to the **Mobile Application**, the System SHALL deliver a **Push Notification** to the **Participant**.

B. When the **Mobile Application** is offline at the time a **Portal-Sent Questionnaire** is sent, the System SHALL deliver the **Push Notification** when the **Mobile Application** next establishes connectivity.

C. The System SHALL NOT deliver a **Push Notification** for a **Portal-Sent Questionnaire** that has already been submitted by the **Participant**.

D. The System SHALL NOT deliver a **Push Notification** for a **Portal-Sent Questionnaire** that has been called back by the **Study Coordinator**.

6.8.7 Yesterday Entry Reminder Notification

REQ ID: `DIARY-PRD-notification-yesterday-entry`

Refines: `CAL-PRD-notification-yesterday-entry-configuration`

Overview

Clinical trial data completeness under ALCOA+ principles requires a **Daily Status** for each calendar day in the diary period. A daily reminder fired at a configured time of day prompts the **Participant** to review the previous day and respond via the Yesterday Reminder Task on the **Main Screen**.

Definitions

Reminder Time: The configurable time of day, in the **Participant**'s local timezone, at which the Yesterday Entry Reminder Notification is delivered.

Trigger

Timezone

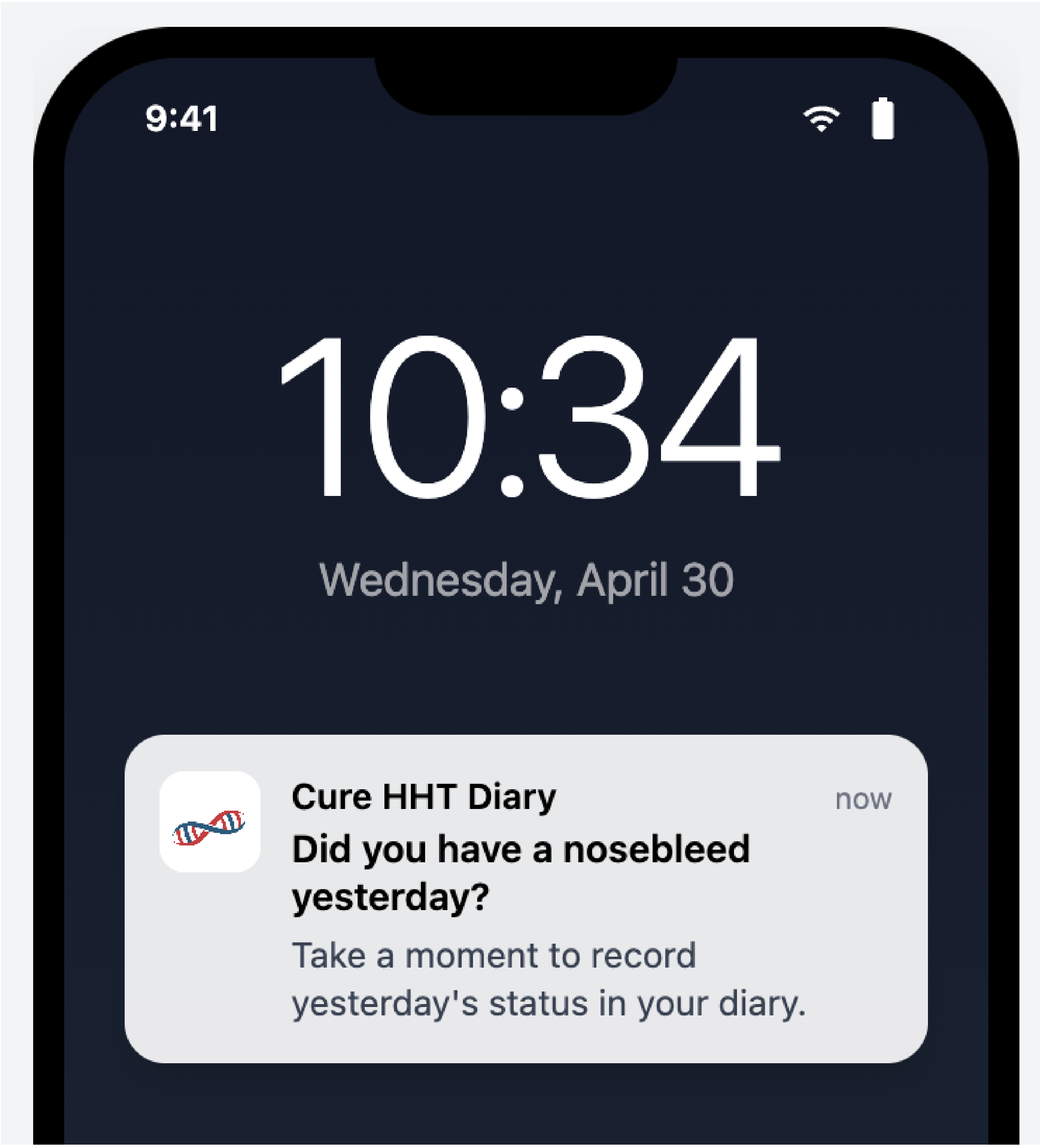
Suppression

Configuration

Rationale

ALCOA+ Complete principle requires a **Daily Status** for every day in the diary period; the Yesterday Entry Reminder is the platform's mechanism for prompting participants to maintain completeness day-by-day rather than discovering large gaps weeks later. Local-timezone evaluation (both for the **Reminder Time** and for "previous calendar day") is essential because participants travel — a UTC-anchored reminder would fire at unpredictable local times for a participant who has moved across timezones, undermining the time-of-day intent. Sponsor-configurable **Reminder Time** lets each deployment select a delivery time appropriate to its participant population (e.g. morning for a working-age population). Once-per-day cap (assertion E) prevents pathological repeat-fires; suppression-on-status-recorded prevents the notification from arriving after the participant has already addressed the yesterday gap via the Task List or Calendar.

Screen reference

See: 

Assertions

A. The System SHALL deliver a **Push Notification** at the **Reminder Time** when no **Daily Status** has been recorded for the previous calendar day.

B. The System SHALL evaluate the **Reminder Time** against the **Participant**'s device local timezone.

C. The System SHALL evaluate “the previous calendar day” against the **Participant**'s device local timezone.

D. The System SHALL NOT deliver the **Push Notification** when a **Daily Status** has been recorded for the previous calendar day.

E. The System SHALL deliver at most one Yesterday Entry Reminder Notification per calendar day.

F. The System SHALL support sponsor-configurable **Reminder Time** per deployment.

6.8.8 Yesterday Entry Reminder Configuration

REQ ID: `CAL-PRD-notification-yesterday-entry-configuration`

Rationale

The 09:00 **Reminder Time** for the Yesterday Entry Reminder targets a morning-routine delivery window: the participant is likely awake, has finished the previous day, and is at a natural reflect-and-record point before the day's activities begin. A morning fire is more reliable than evening for daily-status capture because the participant has full memory of yesterday and the day is not yet competing for their attention. Evaluating the time against the participant's device local timezone (per the parent REQ) keeps the 09:00 delivery consistent regardless of participant travel.

Assertions

A. The **Reminder Time** for the Yesterday Entry Reminder Notification SHALL be configured to 09:00 (24H format) in the **Participant**'s local timezone.

6.8.9 Ongoing Epistaxis Event Reminder

REQ ID: `DIARY-PRD-notification-ongoing-epistaxis`

Refines: `CAL-PRD-notification-ongoing-epistaxis-configuration`

Overview

A **Participant** who starts recording an **Epistaxis Event** but does not complete the record may forget to return and capture the end time, resulting in inaccurate duration data. A configured sequence of escalating reminders prompts the **Participant** to complete the record or confirm it is still ongoing. Once the sequence has fired, no further reminders are sent.

The **Participant** may configure their own reminder schedule for personal use. A sponsor-configured schedule overrides the personal setting for the duration of a clinical trial.

Definitions

Reminder Schedule: An ordered list of elapsed-time intervals defining the timing of successive reminder notifications for an Incomplete Record. Each interval is measured from the time of the previous notification, with the first interval measured from the Participant's most recent interaction with the record.

Trigger and Sequence

Reset on Interaction

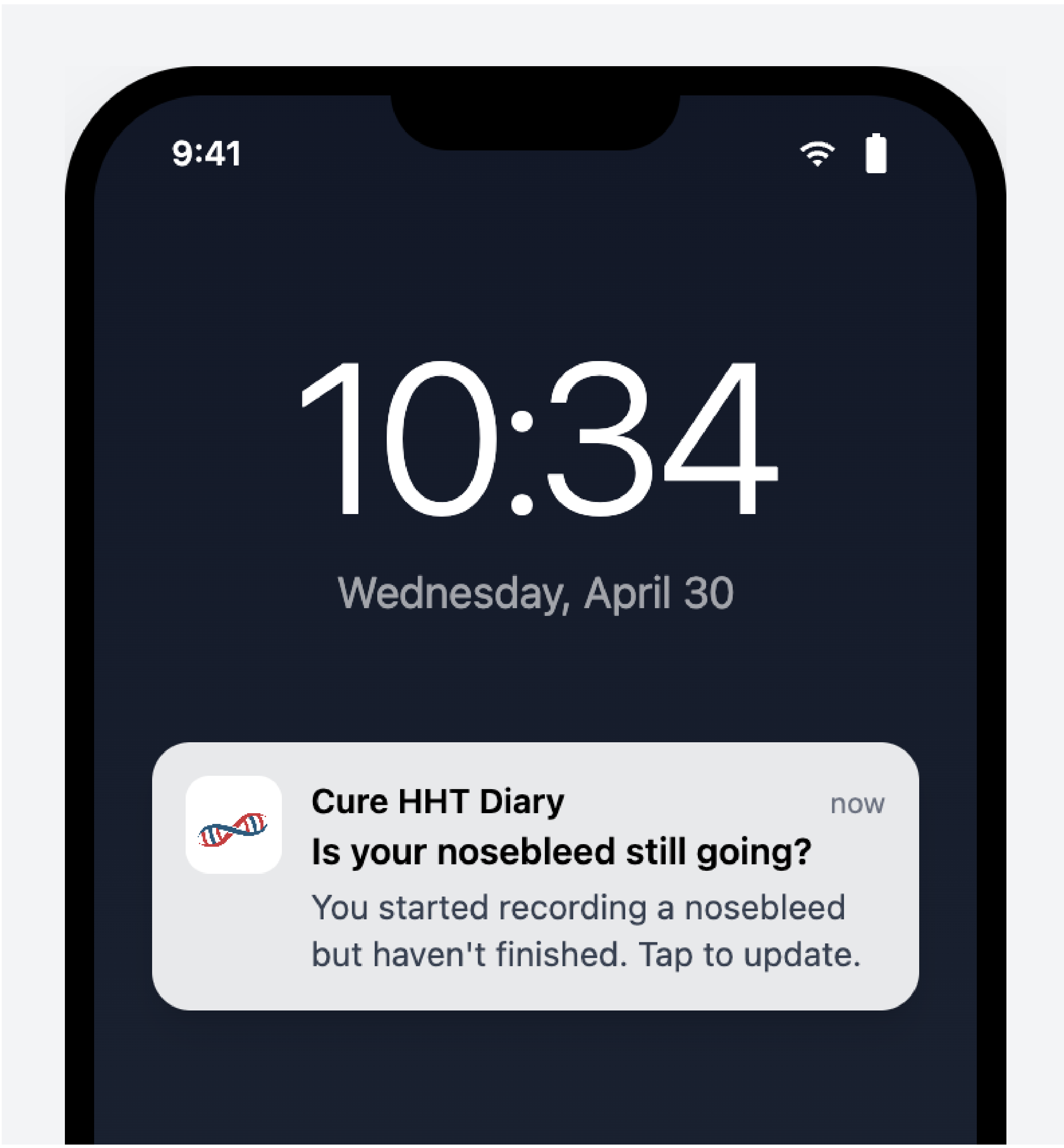
Termination

Configuration

Rationale

A nosebleed event captured with an accurate start time but a missing or wildly-late end time produces unreliable duration data — one of the three primary outcome measures. The Ongoing Epistaxis Event Reminder addresses the natural failure mode where the participant starts the record at the moment the nosebleed begins, then forgets to return when it stops because their attention has moved on. An escalating schedule (intervals measured from previous notification) gives the participant a reasonable chance to respond without saturating their notification surface — once the final interval has fired and no response landed, further nudging is unlikely to help and the data will be flagged via the **Incomplete Records Task** anyway. Reset-on-interaction (assertion D) restarts the schedule when the participant touches the record, on the inference that they are now engaged again and a fresh sequence is the right cadence. Sponsor override of the participant's personal schedule (assertion J) reflects the clinical-trial priority: while participating in a study, the sponsor's clinical protocol drives the reminder cadence; outside a trial, the participant controls their own experience. Empty default (assertion G) is the right “no policy yet” state — a deployment that has not configured a schedule produces no reminders rather than guessing at intervals.

Screen reference

See: 

Assertions

A. The **System** SHALL track the elapsed time since the **Participant**'s most recent interaction with each **Incomplete Record**.

B. The **System** SHALL deliver a **Push Notification** to the **Participant** at each interval in the **Reminder Schedule**.

C. After the final interval in the **Reminder Schedule** has elapsed, the **System** SHALL NOT deliver further reminders for that **Incomplete Record**.

D. When the **Participant** interacts with an **Incomplete Record**, the **System** SHALL restart the **Reminder Schedule** from the first interval.

E. When an **Incomplete Record** is completed, the **System** SHALL stop delivering reminders for that record.

F. When an **Incomplete Record** is deleted, the **System** SHALL stop delivering reminders for that record.

G. The default **Reminder Schedule** for the **Mobile Application** SHALL be empty, resulting in no reminders being delivered.

H. The **System** SHALL allow a **Participant** to configure their own **Reminder Schedule** in **Mobile Application** settings.

I. The **System** SHALL support sponsor-configurable **Reminder Schedule** per deployment.

J. When a sponsor-configured **Reminder Schedule** is in effect, the **System** SHALL apply the sponsor-configured schedule and SHALL NOT apply the **Participant**'s personally configured schedule.

6.8.10 Ongoing Epistaxis Event Reminder Configuration

REQ ID: `CAL-PRD-notification-ongoing-epistaxis-configuration`

Rationale

The Ongoing Epistaxis Event Reminder schedule (5/10/15/30 minutes) is a deliberate escalation: the first three reminders fire in the window where a typical nosebleed is still resolving (5-15 minutes) — most events end within this window, and the reminders help the participant remember to capture the end time as it happens. The 30-minute interval is the final attempt: if the participant has not responded by then, they are likely engaged with something else or the event has ended without active capture, and further reminders would saturate without informing. The four-interval cap respects the parent REQ's terminating-sequence rule.

Assertions

A. The **Reminder Schedule** for the Ongoing Epistaxis Event Reminder SHALL be configured to deliver notifications at the following intervals: 5 minutes, 10 minutes, 15 minutes, 30 minutes.

6.8.11 Historical Gap Reminder

REQ ID: `DIARY-PRD-notification-historical-gap`

Refines: `CAL-PRD-notification-historical-gap-configuration`

Overview

The Historical Gap Reminder addresses missing **Daily Status** entries on days older than yesterday but still within the editable window.

Definitions

Historical Gap: A calendar day between the **Diary Start Day** and the current day, exclusive of the current day and the previous day, for which the **Participant** has not recorded a **Daily Status**.

Historical Gap Reminder: A daily **Push Notification** delivered to the **Participant** prompting them to address one or more **Historical Gaps**.

Trigger

Editable Window

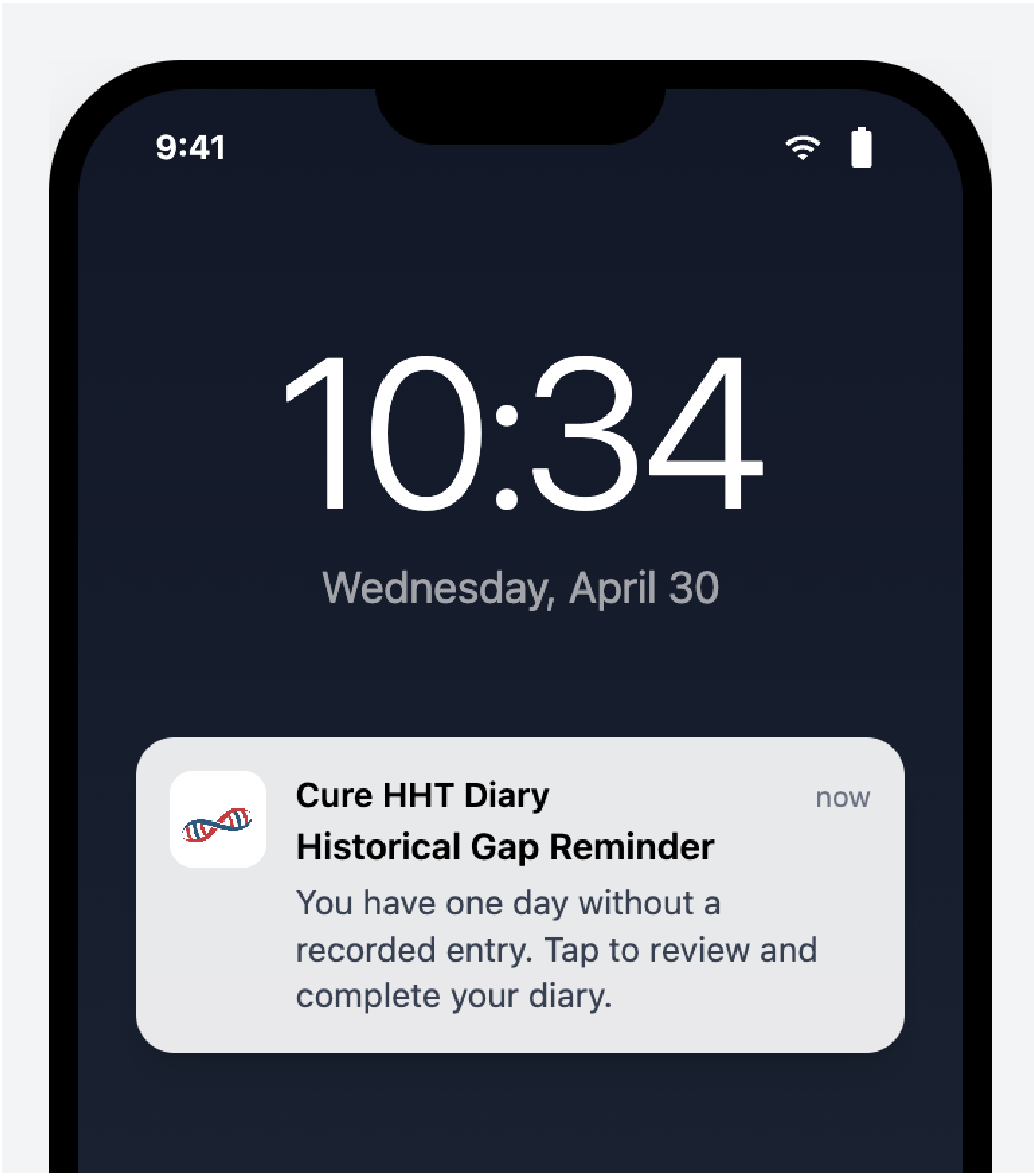
Mode-Dependent Default

Configuration

Rationale

Historical gaps (days within the diary period that have no **Daily Status**) accumulate when a participant misses the Yesterday Entry Reminder window across multiple days; without a separate prompt, those gaps stay invisible to the participant unless they happen to open the Calendar. The Historical Gap Reminder is the platform's once-daily nudge to address them, evaluated at the same configured **Reminder Time** as the Yesterday reminder so the participant gets a coherent daily reminder cadence. The editable-window exclusion (assertions D and E) prevents the reminder from prompting action on dates the participant can no longer modify — locked dates and pre-trial-start dates have no action available. The mode-dependent default (disabled in personal use, enabled in linked use) reflects the difference in motivation: a personal-use user has no external party expecting their data and may not want pressure to backfill; a linked **Participant** is held to a clinical-trial completeness standard. The participant retains the ability to override the default in personal mode (assertion G), while linked mode applies the sponsor-configured behavior.

Screen reference

See: 

Assertions

A. The **System** SHALL evaluate, once per calendar day at the configured **Reminder Time**, whether the **Participant** has any **Historical Gap** within the editable window.

B. When the **Participant** has at least one **Historical Gap** within the editable window, the **System** SHALL deliver a **Historical Gap Reminder** to the **Participant**.

C. The **System** SHALL deliver at most one **Historical Gap Reminder** per calendar day.

D. The **System** SHALL exclude from the **Historical Gap** evaluation any day for which the elapsed time has exceeded the **Lock Threshold**.

E. In linked use mode, the **System** SHALL exclude from the **Historical Gap** evaluation any day before the trial start date.

F. In personal use mode, the **Historical Gap Reminder** SHALL be disabled by default.

G. In personal use mode, the **System** SHALL allow the **User** to enable or disable the **Historical Gap Reminder** from the Mobile Application settings.

H. In linked use mode, the **Historical Gap Reminder** SHALL be enabled by default.

I. The **System** SHALL support sponsor-configurable **Reminder Time** per deployment.

J. The **System** SHALL support sponsor-configurable notification text per deployment.

6.8.12 Historical Gap Reminder Configuration

REQ ID: `CAL-PRD-notification-historical-gap-configuration`

Rationale

The 09:00 **Reminder Time** matches the Yesterday Entry Reminder time so the participant receives a single coherent daily-reminder cadence at the same hour rather than two notifications at unrelated times. The platform's once-per-day cap on each reminder type ensures the participant sees at most one Historical Gap Reminder per day even if multiple historical gaps exist; the notification text is therefore phrased to direct the participant to the diary (where they can address whichever gaps they choose) rather than naming any specific date. The "Tap to review and complete your diary" call-to-action sends the participant to the Calendar where they can see every gap visually and choose how to backfill.

Assertions

A. The **Reminder Time** for the Historical Gap Reminder SHALL be configured to 09:00 (24H format) in the **Participant**'s local timezone.

B. The **Historical Gap Reminder** notification text SHALL be: "You have one day without a recorded entry. Tap to review and complete your diary."

6.9 Device Linking

The participant-side linking flow comprises the platform-level rules for error handling and rate limiting on linking-code entry, the **Join the Study** screen the participant uses to enter a code and consent to the **Clinical Trial Privacy Policy**, and the **Successful Linking Confirmation** that completes the transition from **User** to **Participant**.

6.9.1 Linking Code Entry Error Handling

REQ ID: `DIARY-PRD-linking-code-entry-errors`

Refines: `DIARY-GUI-join-study-screen`, `DIARY-GUI-linking-confirmation`, `CAL-PRD-linki`

Overview

Security best practices require generic error messages that do not reveal whether a linking code exists, is expired, or has been used. This prevents malicious actors from enumerating valid codes or determining code status. Rate limiting protects against brute-force attacks while a configurable cooldown balances security with usability.

Definitions

Linking Code Validation Failure: Any unsuccessful attempt to enter a linking code, regardless of the specific reason for failure.

Rate Limit Threshold: The configurable maximum number of consecutive **Linking Code Validation Failures** permitted within a single rate limit window before further attempts are temporarily blocked.

Rate Limit Cooldown: The configurable duration after which the rate limit counter resets and the **User** may attempt linking again.

Error Messaging

Rate Limiting

Configuration

Rationale

Linking codes are short, human-typeable secrets that confer access to a clinical-trial participant identity; the platform's threat model has to assume that someone may attempt to guess or enumerate valid codes. Two mitigations compose: a generic error message that does not distinguish "code does not exist" from "code expired" from "code already used" (assertions A and B — distinguishing them would tell an attacker exactly which guesses are worth pursuing), and a rate limit that caps the number of attempts in a window. The site-contact guidance (assertion C) is the legitimate-user recovery path that survives the genericity rule — a real participant who has mistyped their code gets the same message as an enumerator, but they also get the actionable next step (contact the site). The cooldown anchored at the first failed attempt in the window (assertion E) prevents the trivial workaround of attempting just below the threshold, waiting one second, and continuing — the window is a fixed time interval from the first miss. Sponsor-configurable threshold, cooldown, and text let each deployment tune the security/usability trade-off and customize the message.

Assertions

A. When a **Linking Code Validation Failure** occurs, the System SHALL display a single generic error message regardless of the specific reason for failure.

B. The error message SHALL NOT reveal the specific reason for failure.

C. When a **Linking Code Validation Failure** occurs, the System SHALL provide guidance to contact the site for a new code.

D. When the number of consecutive **Linking Code Validation Failures** reaches the **Rate Limit Threshold**, the System SHALL block further linking attempts and display the **Rate Limit** error message until the **Rate Limit Cooldown** has elapsed.

E. The **Rate Limit Cooldown** SHALL be calculated from the first failed attempt in the current window.

F. The System SHALL support sponsor-configurable **Rate Limit Threshold** per deployment.

G. The System SHALL support sponsor-configurable **Rate Limit Cooldown** per deployment.

H. The System SHALL support sponsor-configurable error message text per deployment.

6.9.2 Linking Code Entry Error Handling Configuration

REQ ID: `CAL-PRD-linking-code-entry-errors-configuration`

Overview

These values configure the platform linking code error handling mechanism defined in DIARY-PRD-linking-code-entry-errors.

Rationale

The 10-attempt **Rate Limit Threshold** is calibrated for the linking-code surface — a longer typeable code than a password, where a participant making honest typos may go through several attempts on a single onboarding session before getting the code right. Setting the threshold below 10 (e.g. the Sponsor Portal's 5) would too often lock out legitimate participants on first-time linking; setting it much higher would give enumeration attempts useful headroom. The 1-hour **Rate Limit Cooldown** is long enough to make brute-force enumeration impractical at the platform's code-space (an attacker hitting the limit gets at most ~240 attempts/day against a single device, against a code space measured in millions) while keeping a realistic recovery window for a legitimate participant who has hit the limit through honest typos — they can return after lunch or after contacting their site for a fresh code. The generic error message "Invalid linking code." satisfies the parent REQ's no-information-leakage rule (the same text fires regardless of whether the code is unknown, expired, or used), and the Rate Limit message names the specific delay so the participant knows when they may try again rather than retrying indefinitely.

Assertions

A. The **Rate Limit Threshold** SHALL be configured to 10 attempts.

B. The **Rate Limit Cooldown** SHALL be configured to 1 hour.

C. The **Linking Code Validation Failure** error message SHALL display the text: "Invalid linking code."

D. The **Rate Limit** error message SHALL display the text: "Too many attempts. Please wait an hour."

6.9.3 Join the Study Screen

REQ ID: `DIARY-GUI-join-study-screen`

Overview

The **Join the Study** screen is presented to the User when initiating the linking workflow. It captures the **Mobile Linking Code** and the Participant's explicit consent to the **Clinical Trial Privacy Policy** before any link is established.

Definitions

Linking Consent: The explicit acknowledgement by the Participant that they have read and consent to the Clinical Trial Privacy Policy. This consent must be captured before the linking code submission is permitted.

Rationale

The **Join the Study** screen has to do two things at once: capture the linking code (the technical secret that binds device to participant) and capture **Linking Consent** (the regulatory acknowledgement that the participant has read the **Clinical Trial Privacy Policy**). Co-locating them on a single screen rather than splitting across two screens reflects that both are required before linking can proceed, and a two-step split would invite the participant to complete one without the other. The Submit-disabled-until-both-present rule is the GUI-side enforcement of the dual gate — there is no path forward without both. Retaining the consent record with the policy version against the participant record is the audit-trail requirement: the platform must be able to demonstrate which version of the policy each participant consented to, separately from the current version, for the lifetime of their participation. The configurable code format on the entry field is the platform-side accommodation for sponsor-specific code shapes (length, separator characters) without imposing a single format across deployments.

> **Follow-up — configurability**: This requirement currently encodes the only option implemented in code. Future sponsors may require different rules; introduce a configurable seam (e.g. a parameter on the CAL-PRD-* parent, or a new platform-side template the CAL- REQ Satisfies) when the need arises. Until that seam exists, this REQ is normative for the Callisto deployment.

Assertions

A. The **Join the Study** screen SHALL display a **Mobile Linking Code** entry field formatted according to the configured code format.

B. The screen SHALL display a **Linking Consent** checkbox with text confirming the Participant has read, understood, and consents to the **Clinical Trial Privacy Policy**.

C. The **Linking Consent** text SHALL include a link that opens the **Clinical Trial Privacy Policy**.

D. The **Submit** action SHALL be disabled until both a complete **Mobile Linking Code** is entered and the **Linking Consent** checkbox is checked.

E. The System SHALL retain the **Linking Consent** acknowledgement, including the **Clinical Trial Privacy Policy** version, against the **Participant** record upon successful link.

6.9.4 Successful Linking Confirmation

REQ ID: `DIARY-GUI-linking-confirmation`

Overview

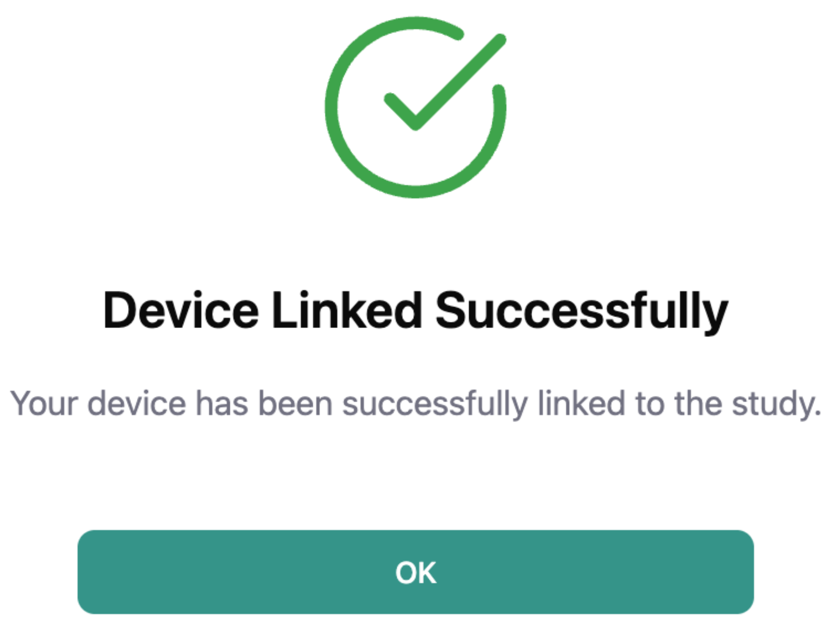
When a Participant successfully enters a valid Mobile Linking Code on the Join the Study screen, the interface needs to clearly confirm that the link was established and transition the Participant from personal use mode into linked use mode.

Rationale

A successful link transitions the participant from a personal-use **User** to a clinical-trial **Participant** — a state change that affects what data syncs, which notifications fire, and which screens display the sponsor logo. The Acknowledgement Dialog makes the transition explicit so the participant has a clear before/after rather than discovering the change passively. Routing to the **User Profile** screen after acknowledgement (rather than back to the **Main Screen**) lands the participant on the screen where the new linked state is most visible: the Clinical Trial section now displays the **Participation Status Badge** instead of the unlinked-state guidance, which serves as the visible confirmation of the new state.

> **Follow-up — configurability**: This requirement currently encodes the only option implemented in code. Future sponsors may require different rules; introduce a configurable seam (e.g. a parameter on the CAL-PRD-* parent, or a new platform-side template the CAL- REQ Satisfies) when the need arises. Until that seam exists, this REQ is normative for the Callisto deployment.

Screen reference



See:

Assertions

A. When the **Participant** successfully submits a valid **Mobile Linking Code**, the interface SHALL display an **Acknowledgement Dialog** confirming that the device has been linked to the study.

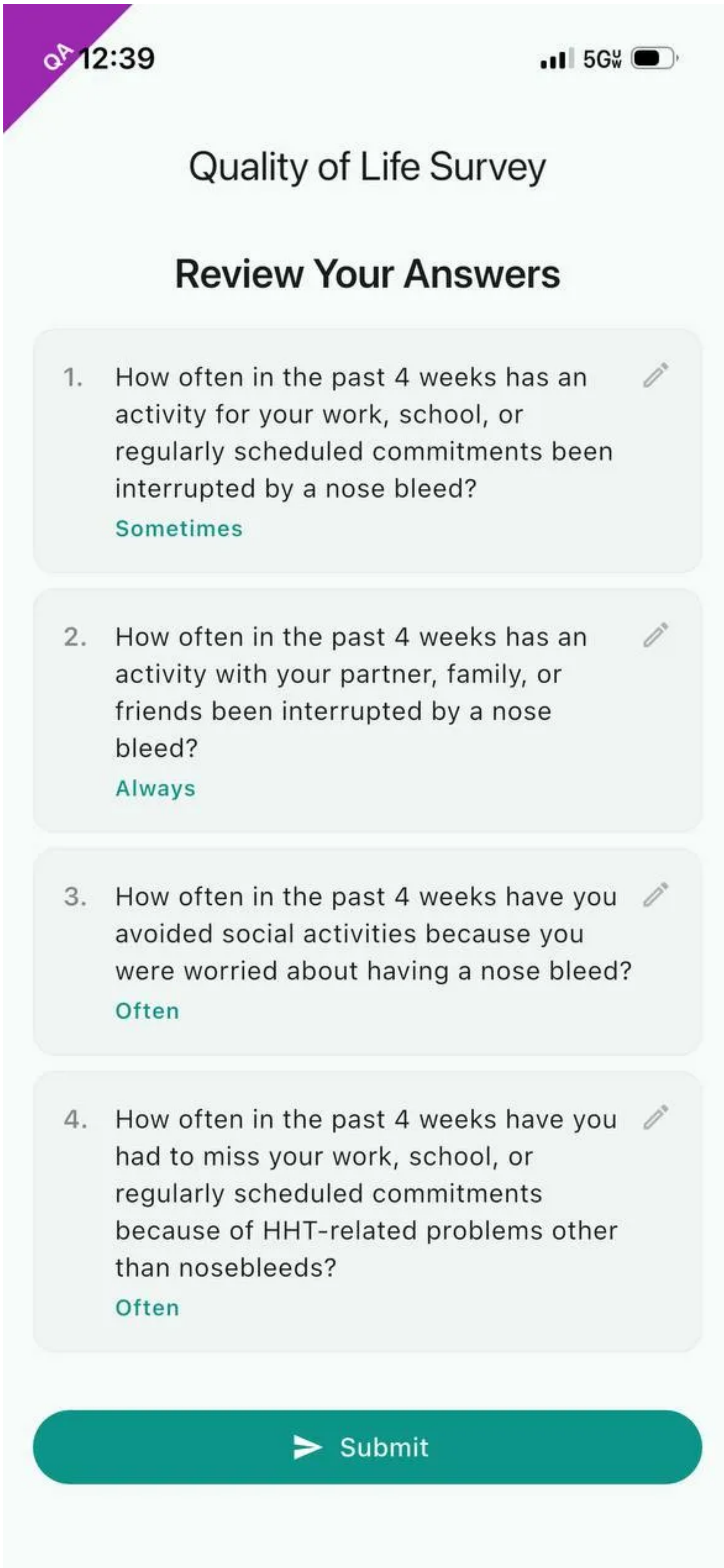
B. When the **Participant** acknowledges the dialog, the interface SHALL navigate the **Participant** to the **User Profile** screen.

Appendices

7.1 Screens

7.1.1 Questionnaire Review Screen

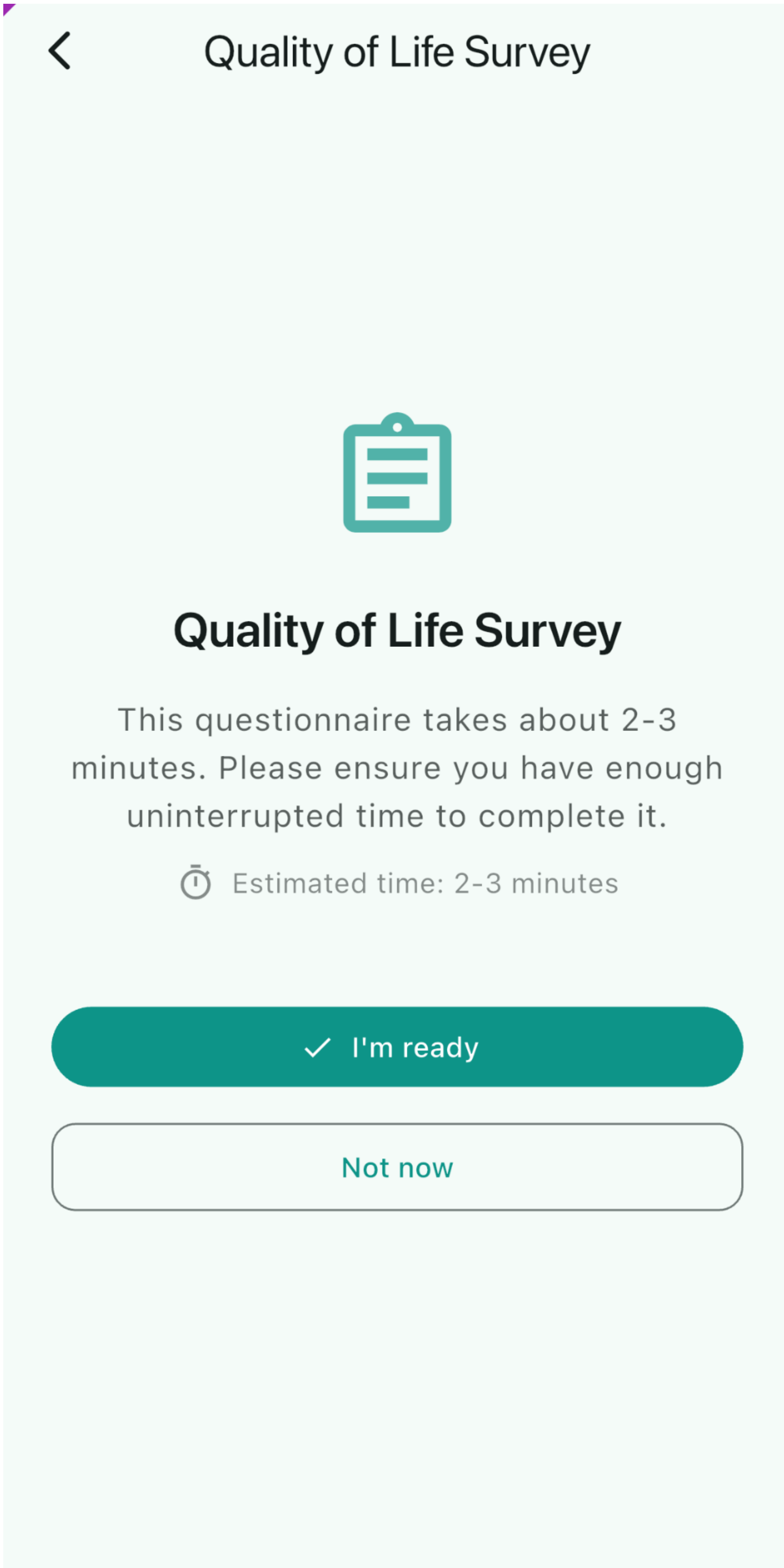
Trigger: **Participant** answers the final question of a **Portal-Sent Questionnaire**, before **Submission**.
Displays: **Questionnaire Display Name**, every question in original order, the **Participant**'s selected response beneath each question, an Edit affordance per question, and a **Submit** action.
Outcome: Selecting **Submit** completes **Submission**.
Reference: GUI-p00001.


QA
12:39
5G
Quality of Life Survey
Review Your Answers
1. How often in the past 4 weeks has an activity for your work, school, or regularly scheduled commitments been interrupted by a nose bleed?
Sometimes
2. How often in the past 4 weeks has an activity with your partner, family, or friends been interrupted by a nose bleed?
Always
3. How often in the past 4 weeks have you avoided social activities because you were worried about having a nose bleed?
Often
4. How often in the past 4 weeks have you had to miss your work, school, or regularly scheduled commitments because of HHT-related problems other than nosebleeds?
Often
Submit

7.1.2 Questionnaire Preamble Screen

Trigger: **Participant** opens a **Portal-Sent Questionnaire**.
Displays: **Questionnaire Display Name**, estimated time to complete.

Actions: **I'm Ready** (proceed to first question), **Not Now** (return to **Main Screen**). *Reference:* REQ-p02065, GUI-p00001.


Quality of Life Survey
Quality of Life Survey
This questionnaire takes about 2-3 minutes. Please ensure you have enough uninterrupted time to complete it.
Estimated time: 2-3 minutes
I'm ready
Not now

7.1.3 Resolve Conflict — Resolution Screen

Trigger: New or edited entry **Overlaps** with one or more **Conflicting Records**. **Displays:** New entry and **Conflicting Record** side by side, showing time range, duration, and severity for each.

Actions: **Keep New** (retain new entry, discard **Conflicting Record**), **Keep Existing** (retain **Conflicting Record**, discard new entry), **Merge Records** (combine both using earliest start time, latest end time, and higher severity). **Conditional behavior:**

- **Merge Records** is presented only when the **Conflicting Record**'s event date has not exceeded the **Lock Threshold**.
- When the **Conflicting Record** is older than the **Justification Threshold**, an **Entry Justification** is required before saving the resolution.

Outcome: **Participant** is navigated to the **Record Nosebleed** screen pre-filled with the resulting entry data for review and confirmation.

Reference: REQ-p05008, GUI-p05009.

Resolve Conflict

Two records overlap. Choose how to handle them.

New Record	Existing Record
Time	Time
12:28 PM - 12:43 PM	12:27 PM - 12:43 PM
Duration	Duration
15m	15m
Severity	Severity
Dripping quickly	Dripping quickly

Keep New

Keep Existing

Merge Records
Combine both into one event with the longest duration

7.1

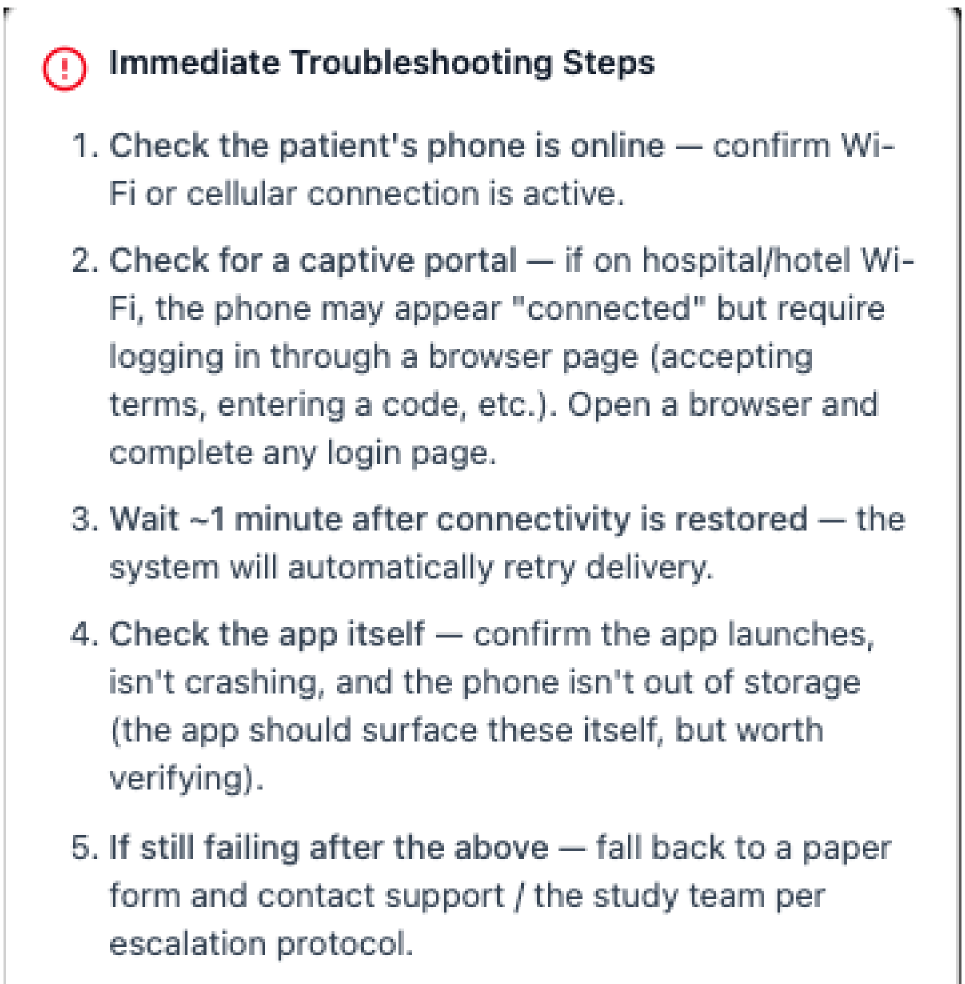
7.1.4 Troubleshooting Popover

Trigger: **Study Coordinator** selects the information icon adjacent to the **Status Badge** on a **Questionnaire Card** in **Delivery Failed** status.

Displays: Guidance for resolving delivery issues for the affected **Questionnaire**.

Dismissal: Click outside the popover.

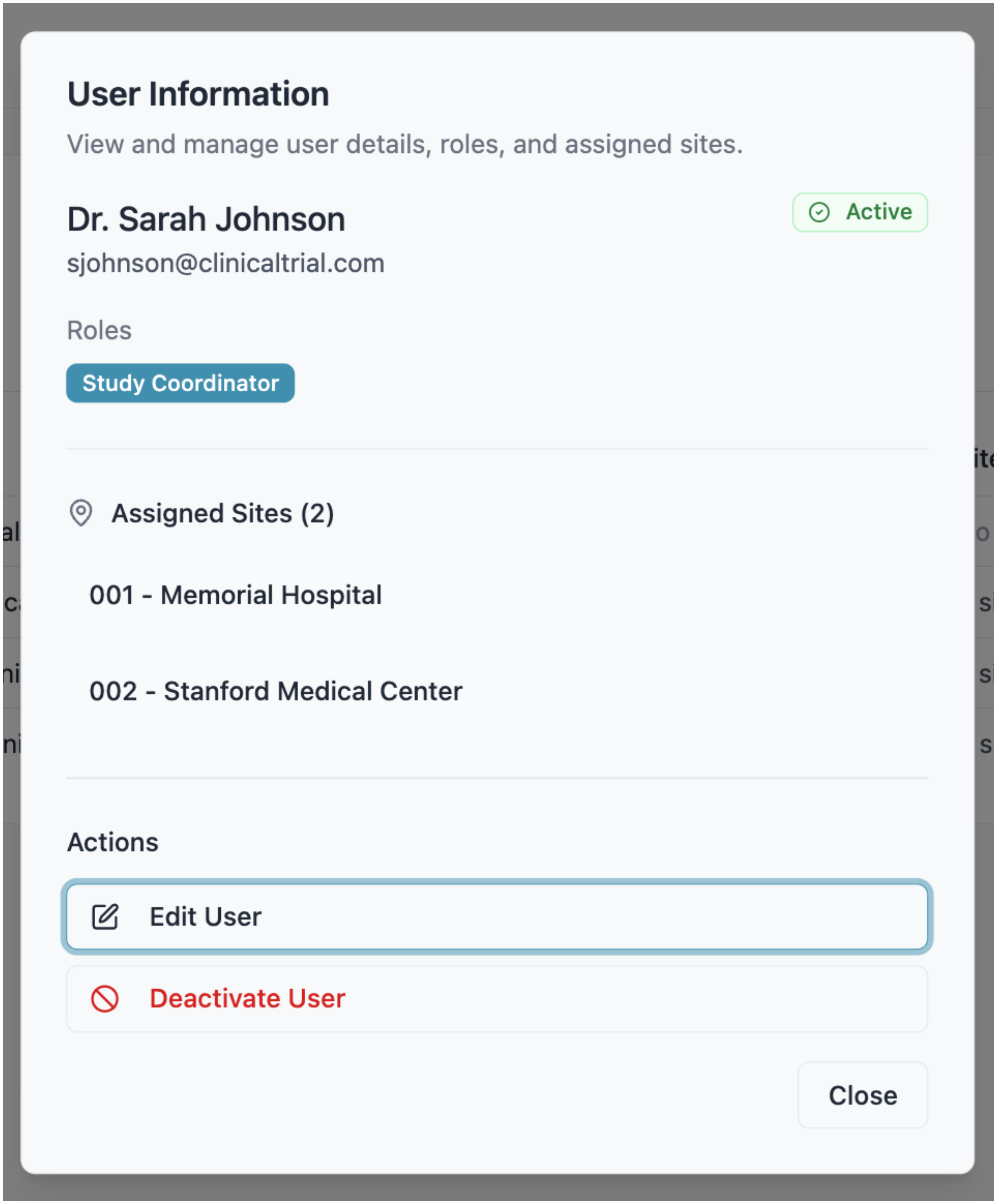
Reference: GUI-CAL-p00006.



7.2 Modals

7.2.1 User Information Modal

Trigger: **Administrator** selects a **User Account** row from the User Management interface.
Target identification: **Full Name**, **Email Address**, **Status**.
Displays: Assigned **Role**(s) and **Site** assignments. When the **User Account** holds one or more non-**Administrator Roles**, the **Administrator** may select a single **Role** to scope the displayed **Site** list. When the **User Account** is an **Administrator**, the modal indicates access to all **Sites**. *Actions (Active Users tab):* **Edit User**, **Deactivate User**, **Close**. The **Deactivate User** action is not presented for the currently authenticated user's own account.
Actions (Inactive Users tab): **Reactivate User**, **Close**.
Reference: GUI-p00033.


User Information
View and manage user details, roles, and assigned sites.
Dr. Sarah Johnson
Active
sjohnson@clinicaltrial.com
Roles
Study Coordinator
Assigned Sites (2)
001 - Memorial Hospital
002 - Stanford Medical Center
Actions
Edit User
Deactivate User
Close

7.1.1

7.2.2 Show Linking Code

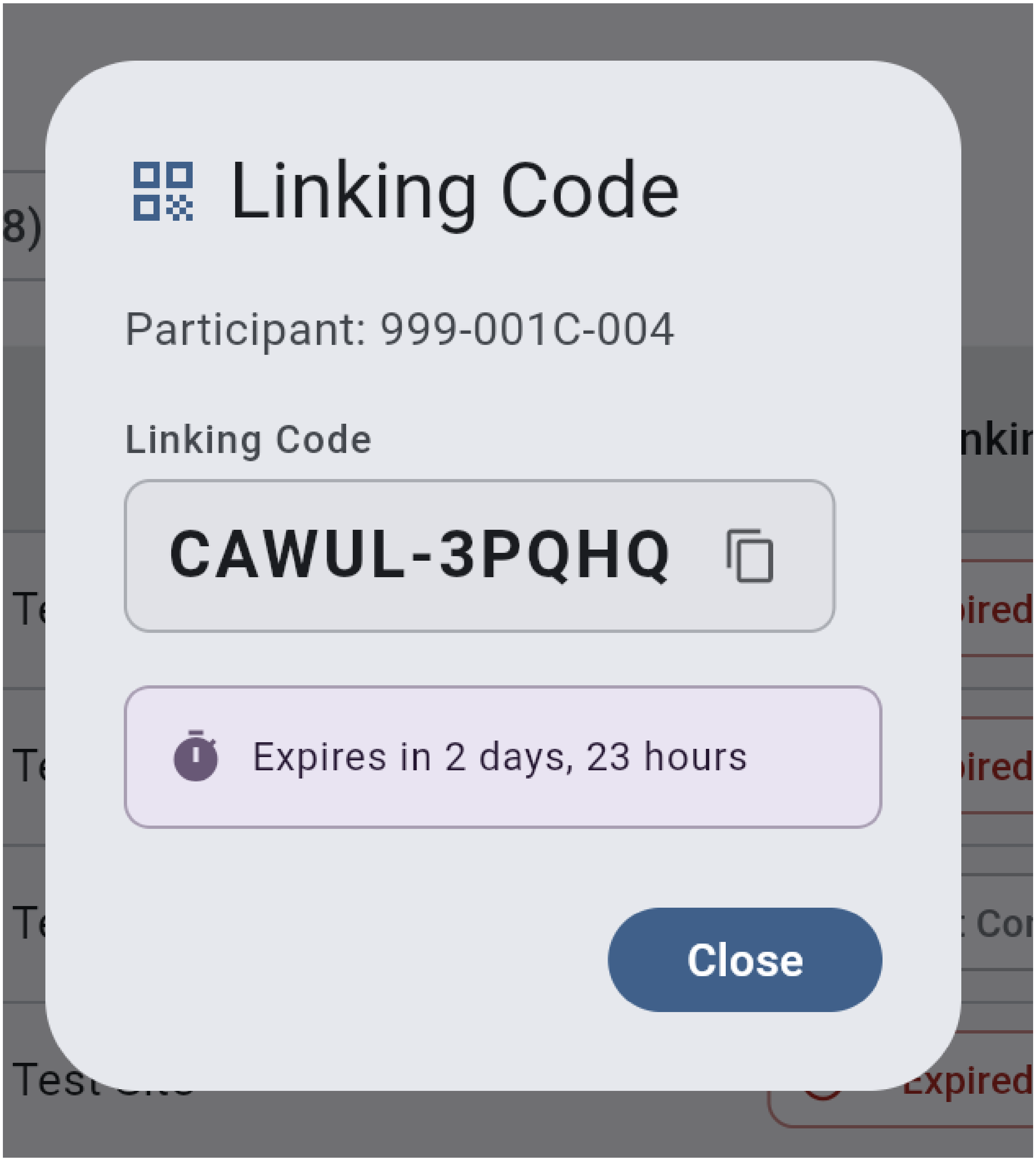
Trigger: **Study Coordinator** selects **Show Linking Code** for a **Participant**.
Target identification: **Participant** ID.

Variants:

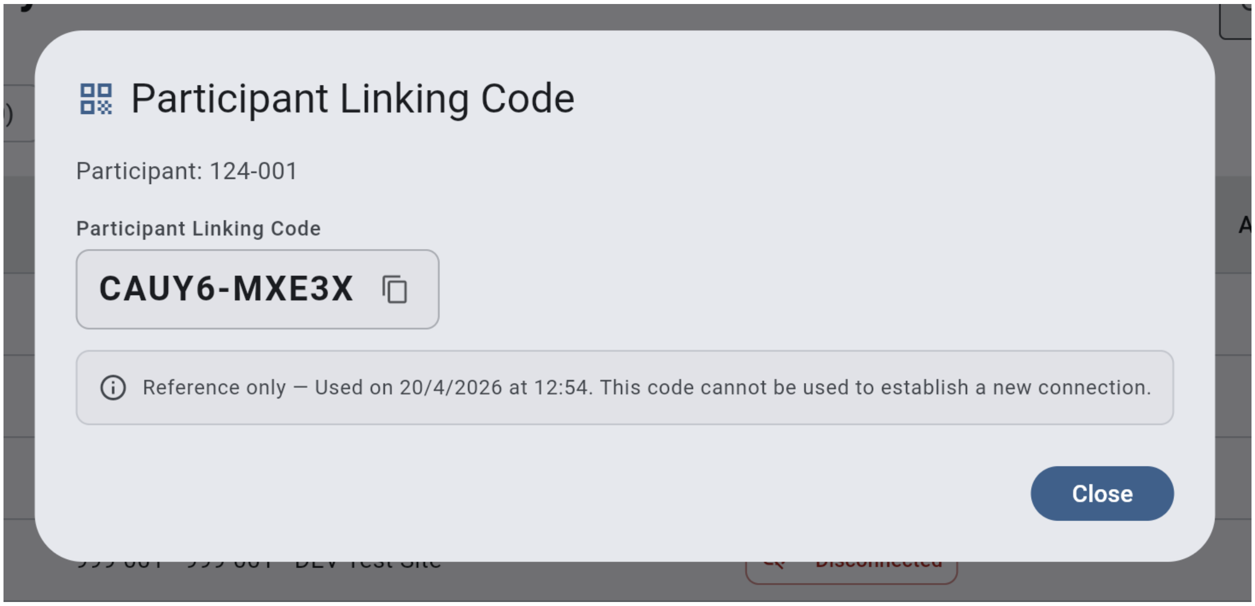
- **Pending** status: displays the active **Mobile Linking Code** with **Copy** and **Save as PDF** actions. **Save as PDF** generates a PDF containing the **Mobile Linking Code** and **Participant** instructions.
- Any other status: displays the **Participant Linking Code** (the code previously used to establish the connection) with a **Copy** action, and indicates the code is shown for reference only.

Dismissal: **Close**.
Reference: GUI-p03001.

Pending Status



After Participant was linked to the Mobile Application



7.2.3 Manage Questionnaires Modal

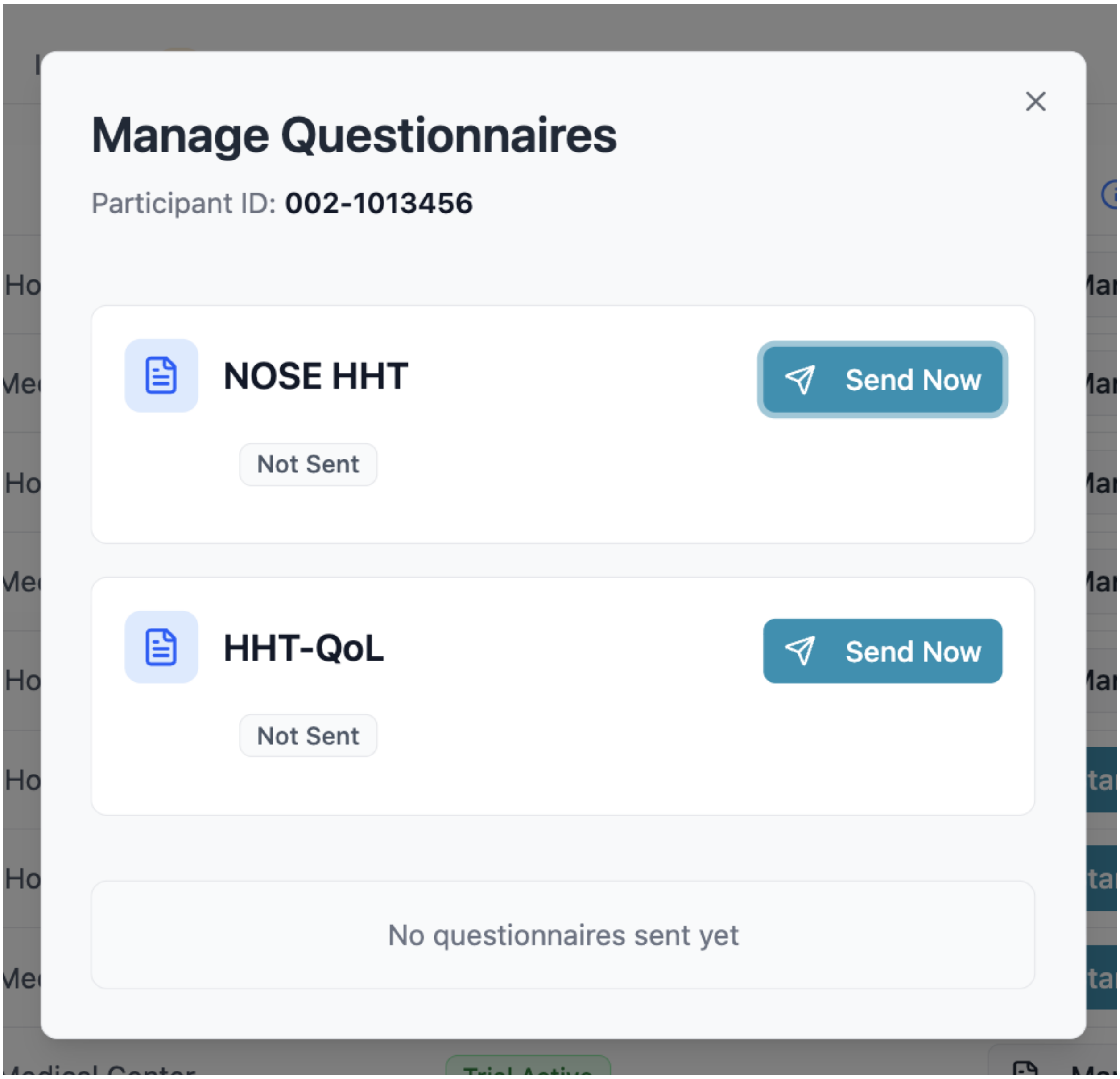
Trigger: **Study Coordinator** selects **Manage Questionnaires** for a **Participant** with **Trial Active** status.

Target identification: **Participant** ID in the modal header.

Displays: One **Questionnaire Card** per **Questionnaire Type** enabled for the deployment. Each card shows the **Questionnaire Type** name and current **Questionnaire Status**, with content and actions varying by status and **Cycle** state (e.g., **Send Now**, **Start Next Cycle**, **Finalize**, call back).

Dismissal: Close action.

Reference: GUI-CAL-p00006.



7.3 Dialogs

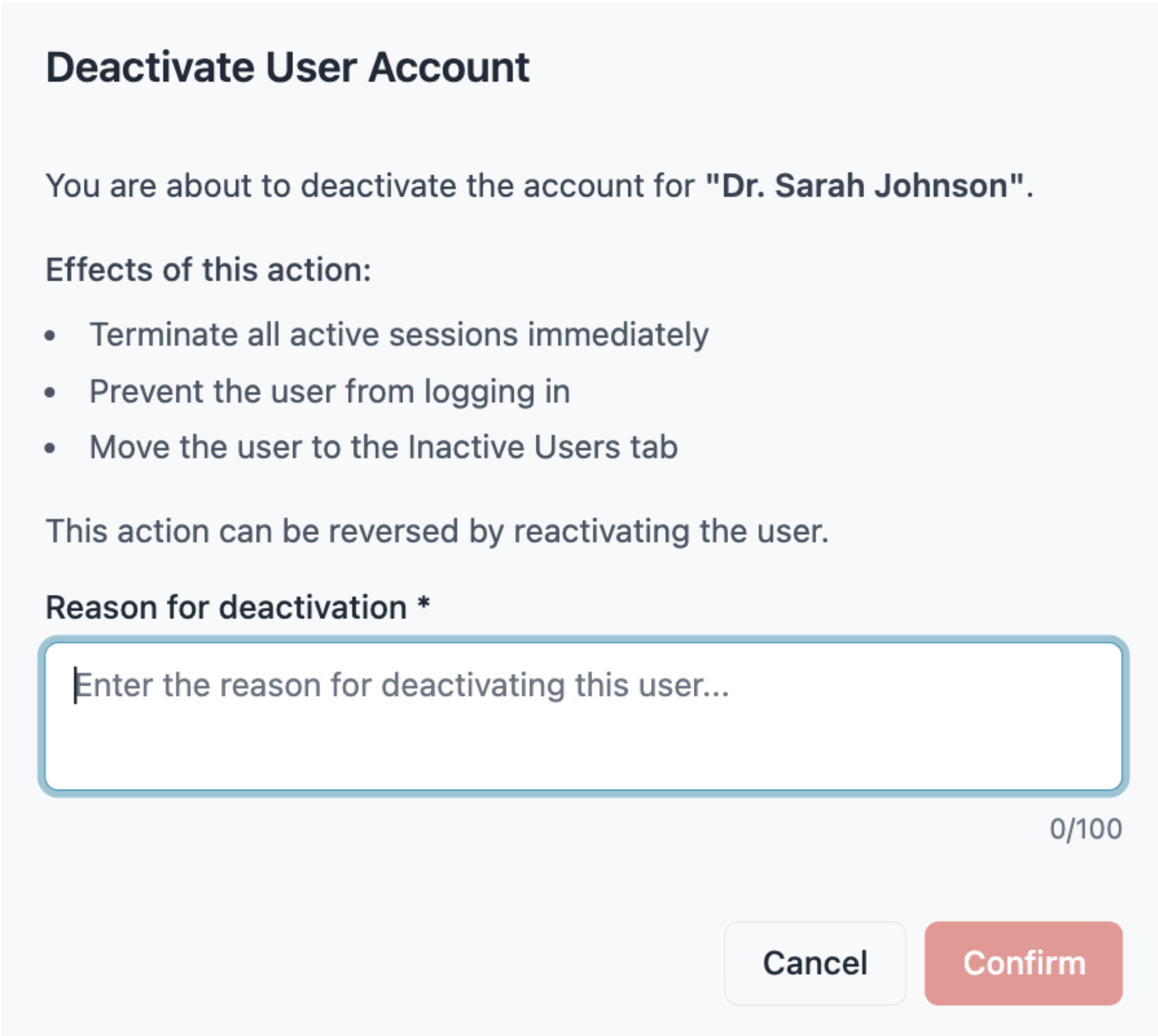
7.3.1 Deactivate User Account — Reason Dialog (Free Text)

Trigger: **Administrator** initiates deactivation of a **User Account** from the **Active Users** tab. *Target identification:* **Full Name**.

Consequence summary: The **User Account** will be deactivated and the user will no longer be able to access the **System**.

Pattern: Reason Dialog — Free Text (see §4.1).

Reference: REQ-p20031, GUI-p00031.



7.3.2 Reactivate User Account — Reason Dialog (Free Text)

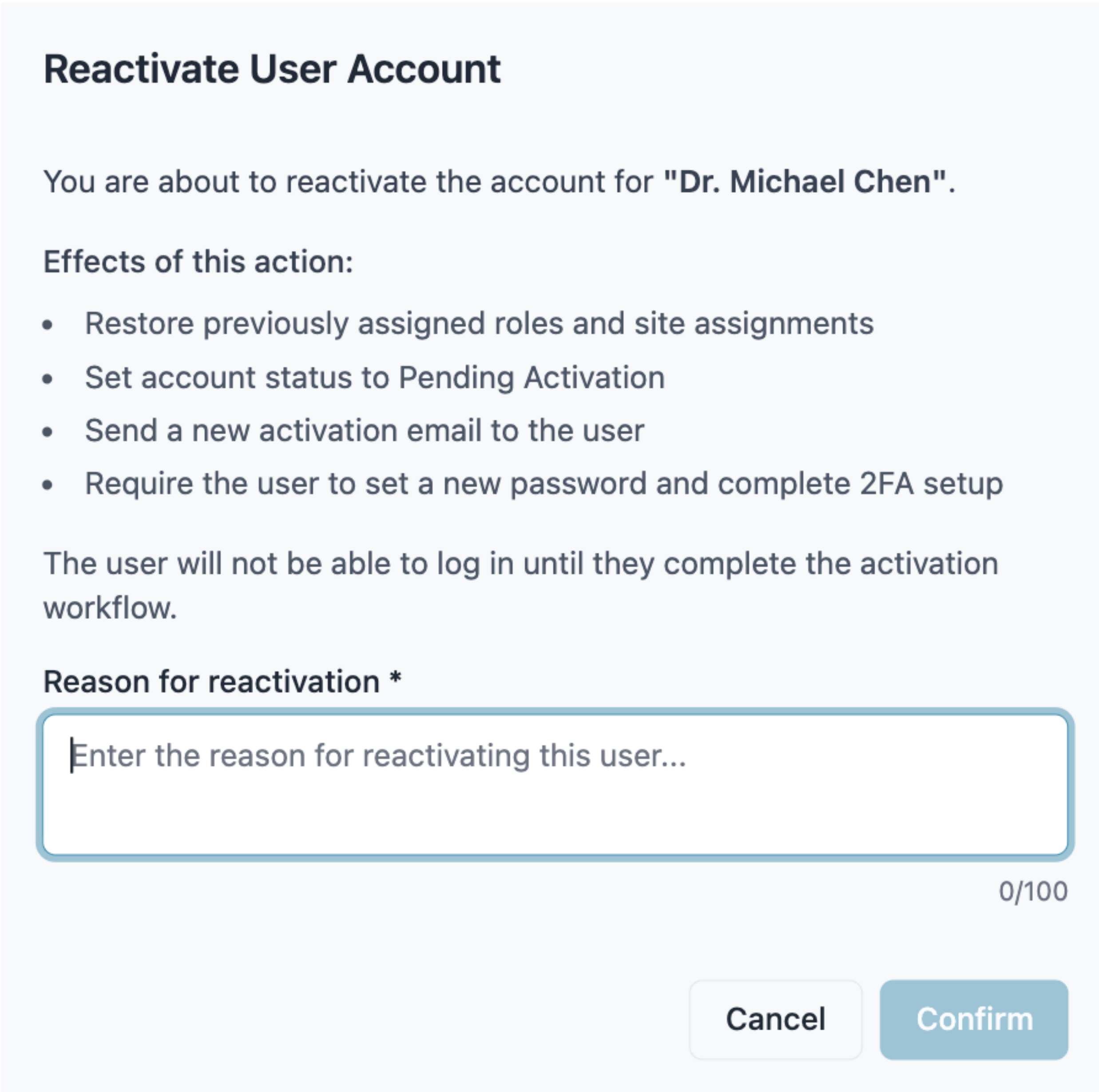
Trigger: **Administrator** initiates reactivation of a **User Account** from the **Inactive Users** tab.

Target identification: **Full Name**.

Consequence summary: The **User Account** will be set to **Pending Activation** and the user will be required to complete the activation workflow before regaining access.

Pattern: Reason Dialog — Free Text (see §4.1).

Reference: REQ-p20032, GUI-p00032.



7.3.3 Link Participant — Confirmation Dialog

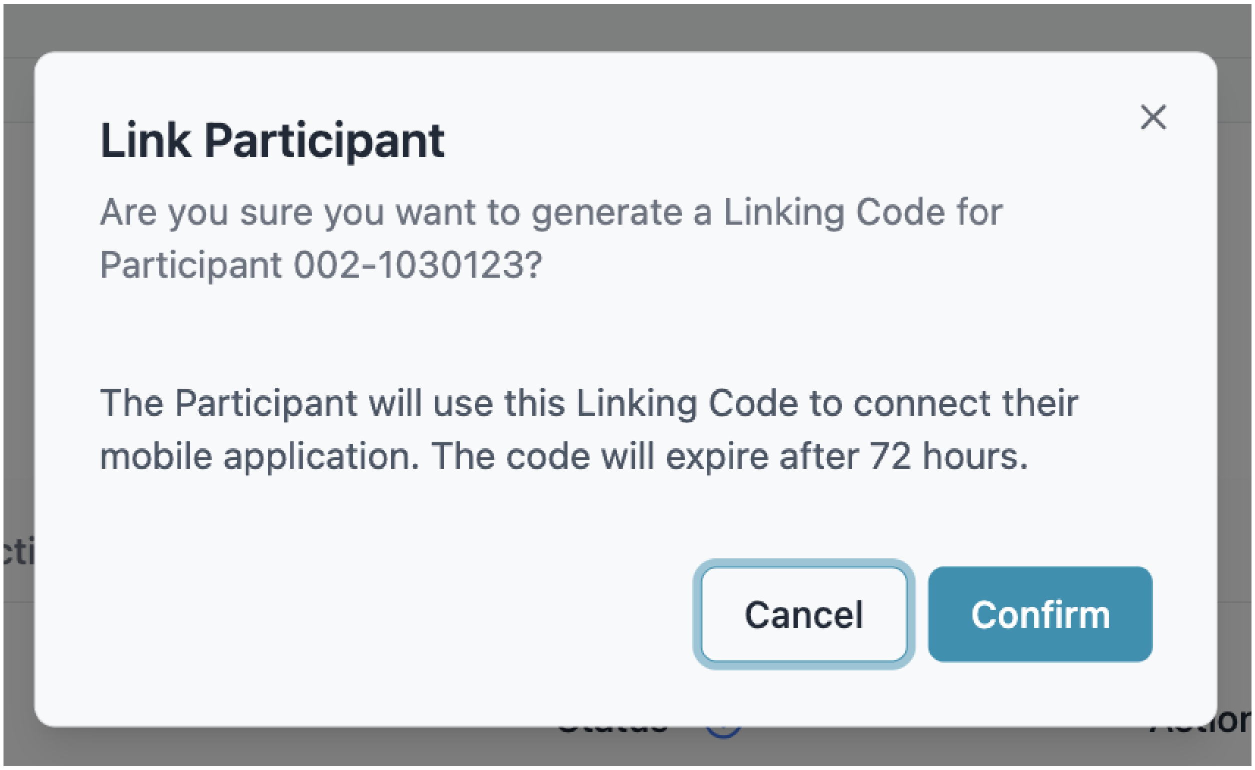
Trigger: **Study Coordinator** initiates **Link Participant** for a **Participant** with **Not Connected** status.

Target identification: **Participant** ID.

Consequence summary: A **Mobile Linking Code** will be generated for the **Participant** and will expire after the configured duration.

Pattern: Confirmation Dialog (see §4.1).

Reference: REQ-p70009, GUI-p03001.



7.1.2

7.3.4 Linking Code Generated — Acknowledgement Dialog

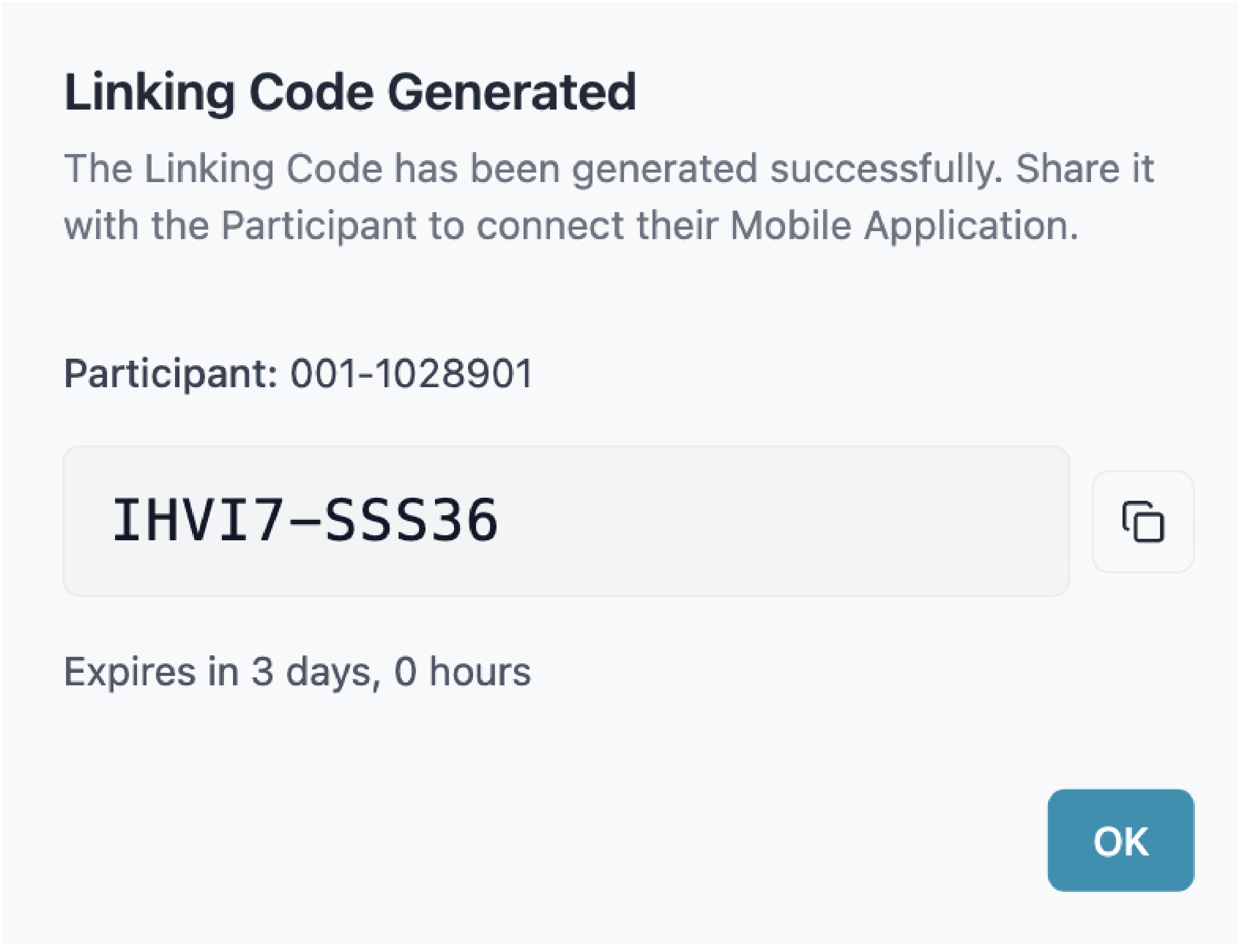
Trigger: **Study Coordinator** confirms generation of a **Mobile Linking Code** from the Link Participant Confirmation Dialog (§7.3.3).

Target identification: **Participant** ID.

Displays: Generated **Mobile Linking Code** with a **Copy** action, and remaining time until expiry. *Outcome:* On dismissal, the **Participant**'s **Status Badge** updates to **Pending**.

Pattern: Acknowledgement Dialog (see §4.1).

Reference: GUI-p03001.



7.3.5 Start Trial — Confirmation Dialog

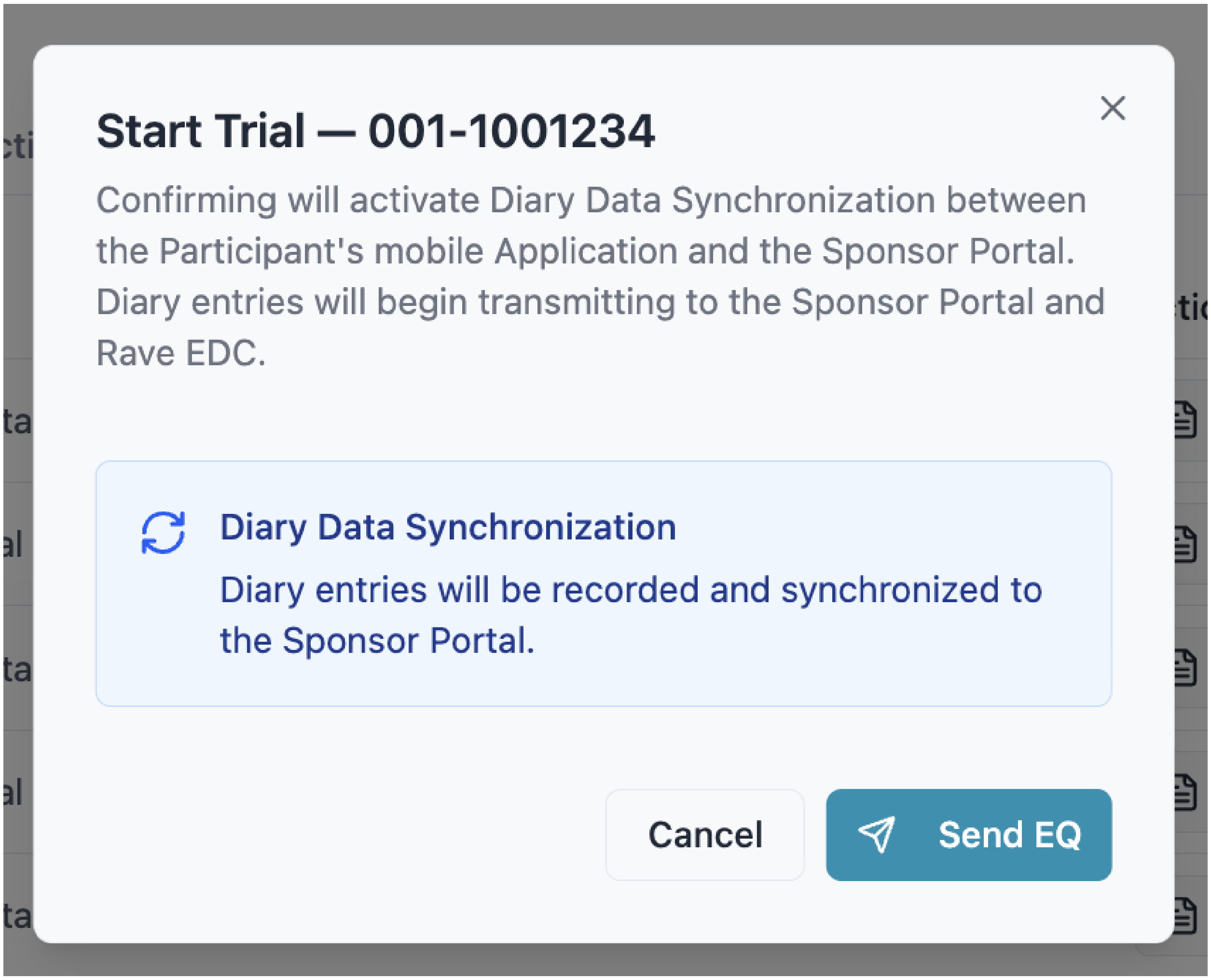
Trigger: **Study Coordinator** selects **Start Trial** for a **Participant** with **Linked - Awaiting Start** status.

Target identification: **Participant** ID.

Consequence summary: **Diary Data Synchronization** will be activated, transmitting diary entries to the **Sponsor Portal** and **Rave EDC**. The **Participant**'s status will update to **Trial Active**.

Pattern: Confirmation Dialog (see §4.1).

Reference: REQ-CAL-p00022, GUI-CAL-p00005.



7.1.3

7.3.6 Disconnect Participant — Reason Dialog

Trigger: **Study Coordinator** selects **Disconnect Participant** from the **Participant Actions Modal** for a **Participant** with **Linked - Awaiting Start** or **Trial Active** status.

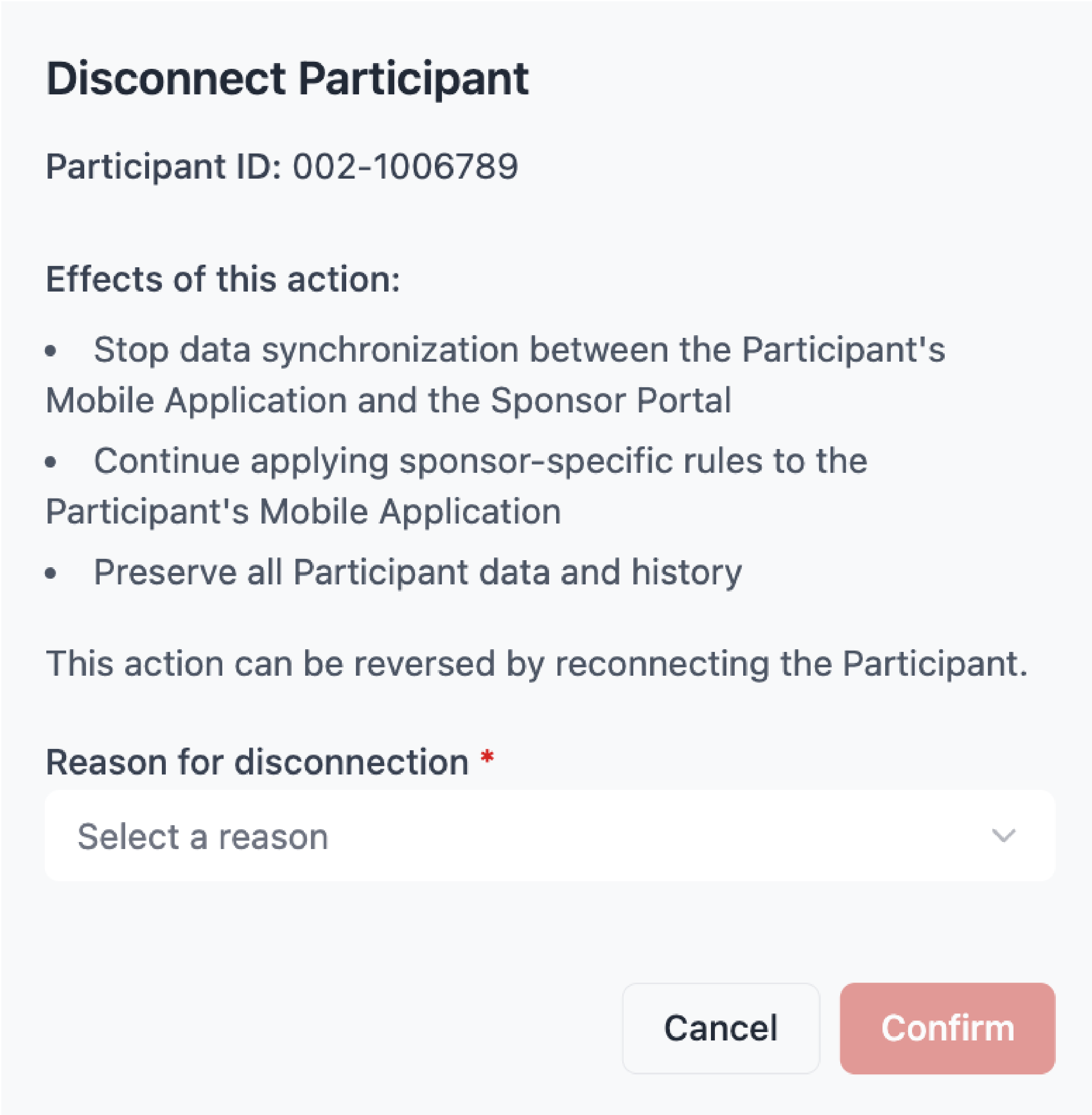
Target identification: **Participant** ID.

Consequence summary: Data synchronization between the **Mobile Application** and the **Sponsor Portal** will stop. All **Participant** data will be preserved.

Reason options: Device Issues, Technical Issues, Other.

Pattern: Reason Dialog — Predefined List (see §4.1).

Reference: REQ-p70010, REQ-CAL-p00020.



7.1.4

7.3.7 Reconnect Participant — Reason Dialog (Free Text)

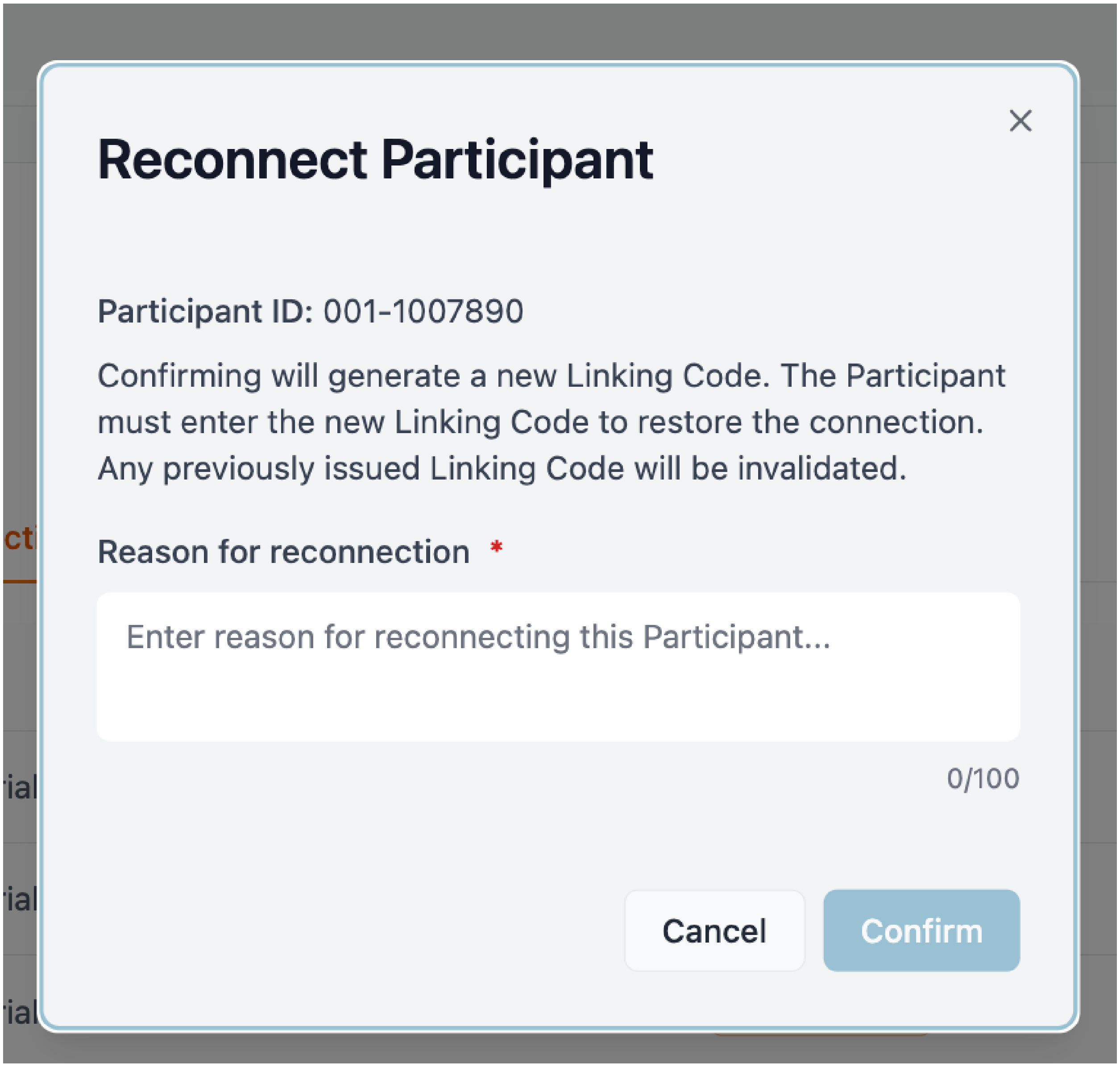
Trigger: **Study Coordinator** selects **Reconnect Participant** from the **Participant Actions Modal** for a **Participant** with **Disconnected** status.

Target identification: **Participant** ID.

Consequence summary: A new **Mobile Linking Code** will be generated. The **Participant** must enter the new code to restore the connection.

Pattern: Reason Dialog — Free Text (see §4.1).

Reference: REQ-p70011.



7.3.8 Mark as Not Participating — Reason Dialog (Predefined)

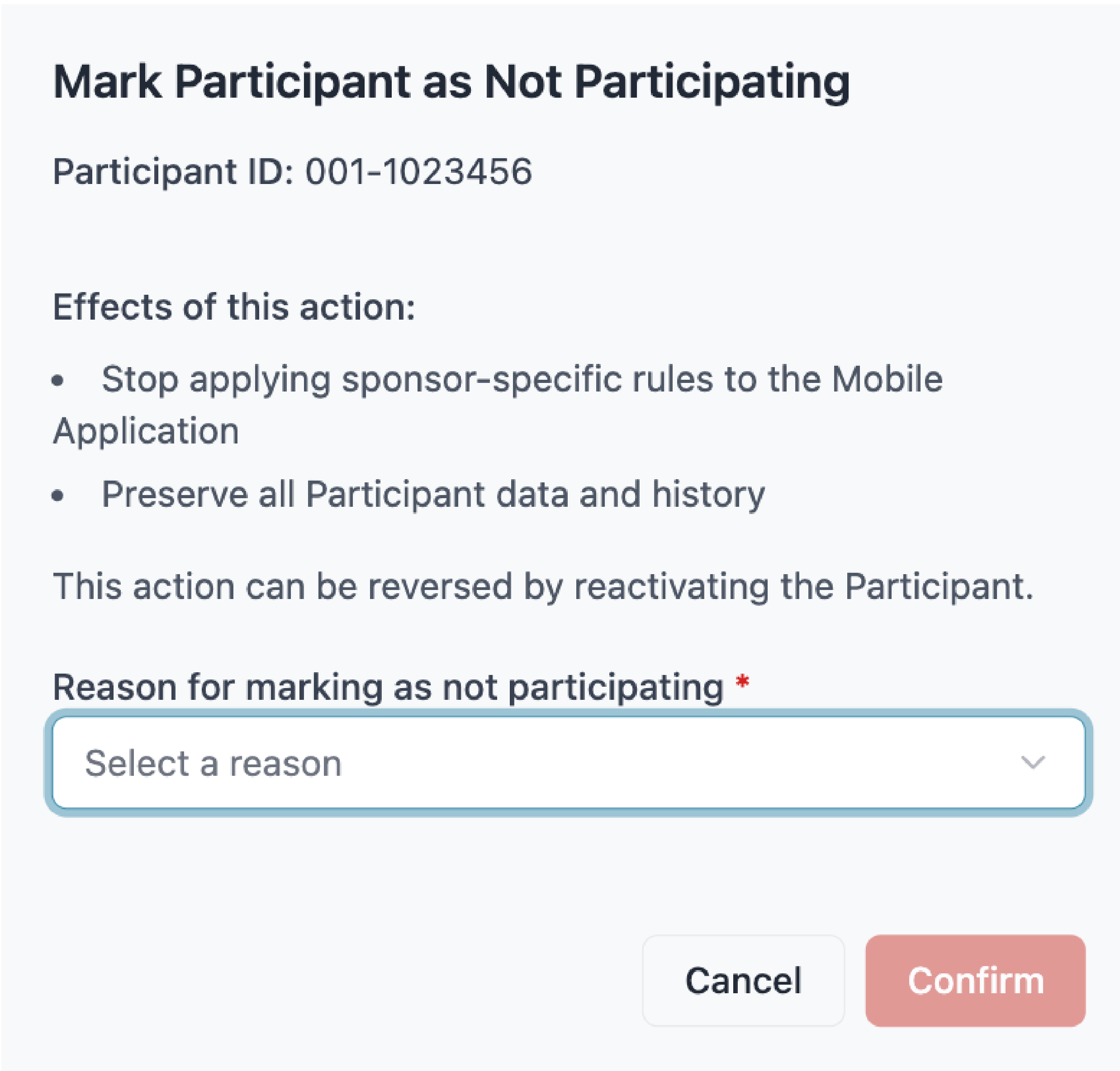
Trigger: **Study Coordinator** selects **Mark as Not Participating** for a **Participant** with **Disconnected** status.

Target identification: **Participant** ID.

Consequence summary: Sponsor-specific rules will no longer be applied to the **Participant**'s **Mobile Application**. *Reason options:* Subject Withdrawal, Death, Protocol Treatment/Study Complete, Other.

Pattern: Reason Dialog — Predefined List (see §4.1).

Reference: REQ-p70017, REQ-CAL-p00064.



7.1.5

7.3.9 Reactivate Participant — Reason Dialog (Free Text)

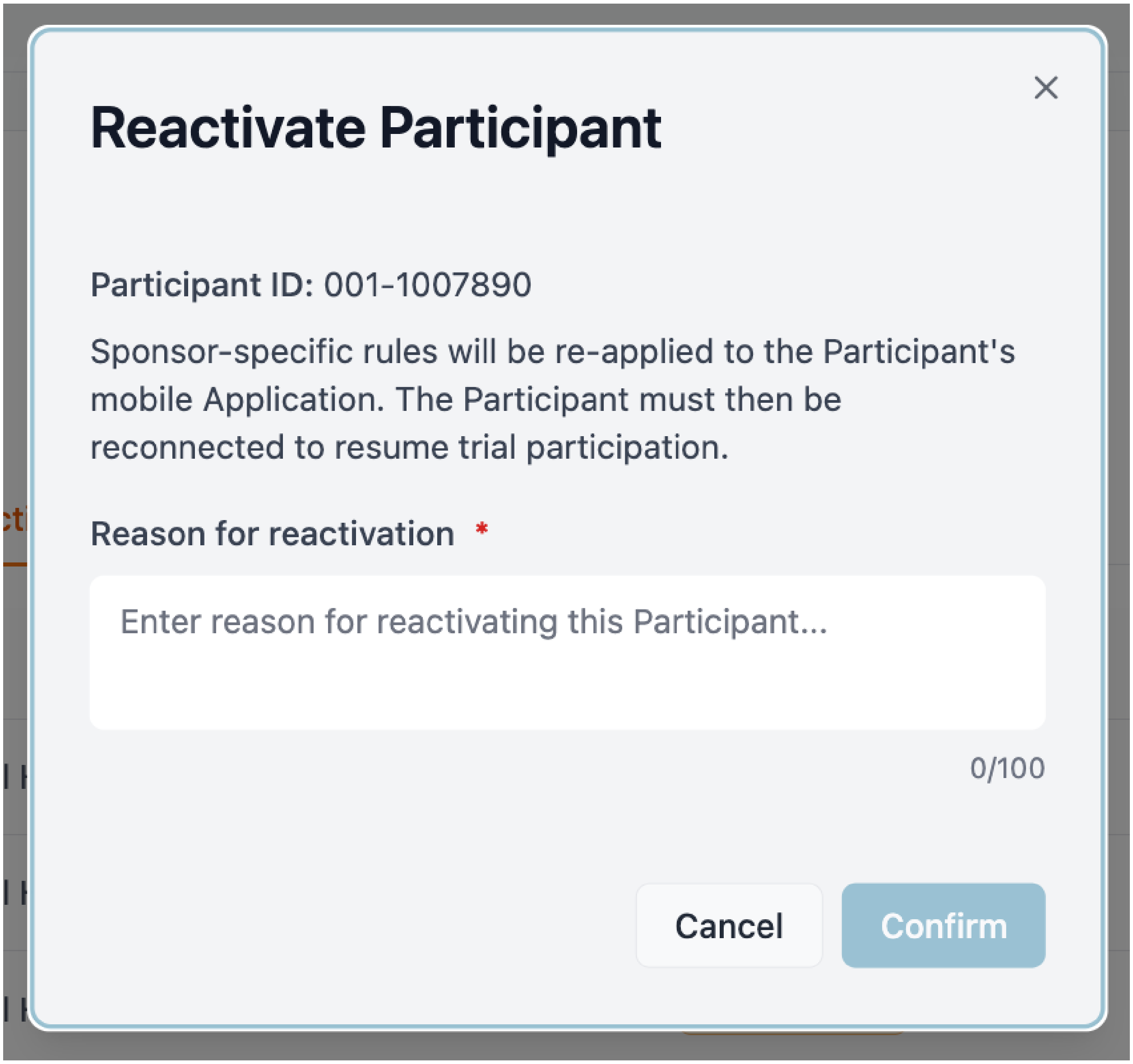
Trigger: **Study Coordinator** selects **Reactivate Participant** from the **Participant Actions Modal** for a **Participant** with **Not Participating** status.

Target identification: **Participant** ID.

Consequence summary: Sponsor-specific rules will be re-applied to the **Participant**'s **Mobile Application** and the standard reconnection workflow will become available.

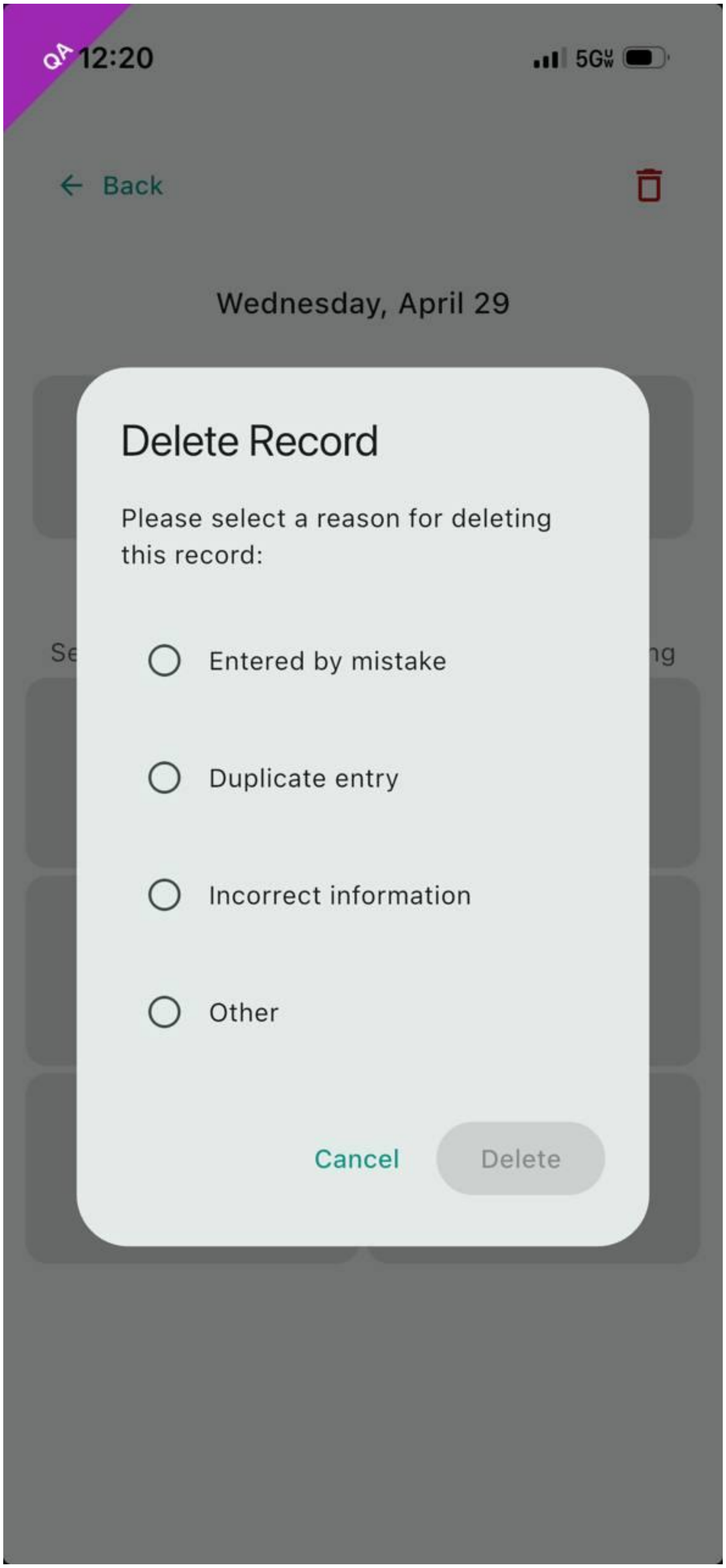
Pattern: Reason Dialog — Free Text (see §4.1).

Reference: REQ-p70016.



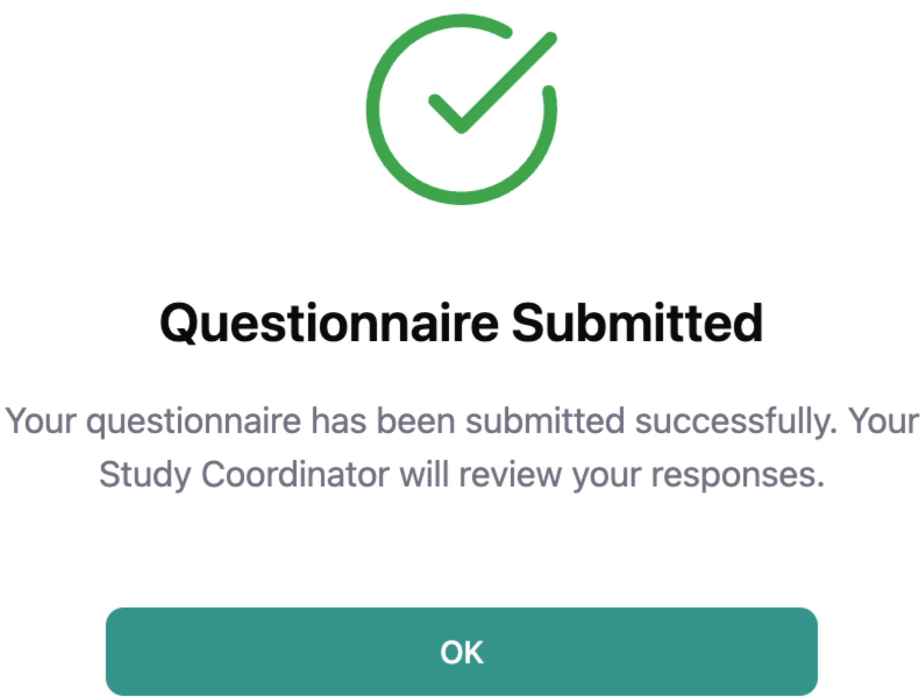
7.3.10 Delete Record — Reason Dialog — Predefined List

Trigger: **Participant** initiates the delete action on an **Epistaxis Event**, either during the recording flow or from event history.
Consequence summary: The **Epistaxis Event** will be removed from the **Participant**'s diary. *Reason options:* Entered by mistake, Duplicate entry, Incorrect information, Other.
Pattern: Reason Dialog — Predefined List (see §4.1).
Reference: GUI-p00003.


QA
12:20
5G
Back
Wednesday, April 29
Delete Record
Please select a reason for deleting this record:
Entered by mistake
Duplicate entry
Incorrect information
Other
Cancel
Delete

7.3.11 Post-Submission Acknowledgement Dialog

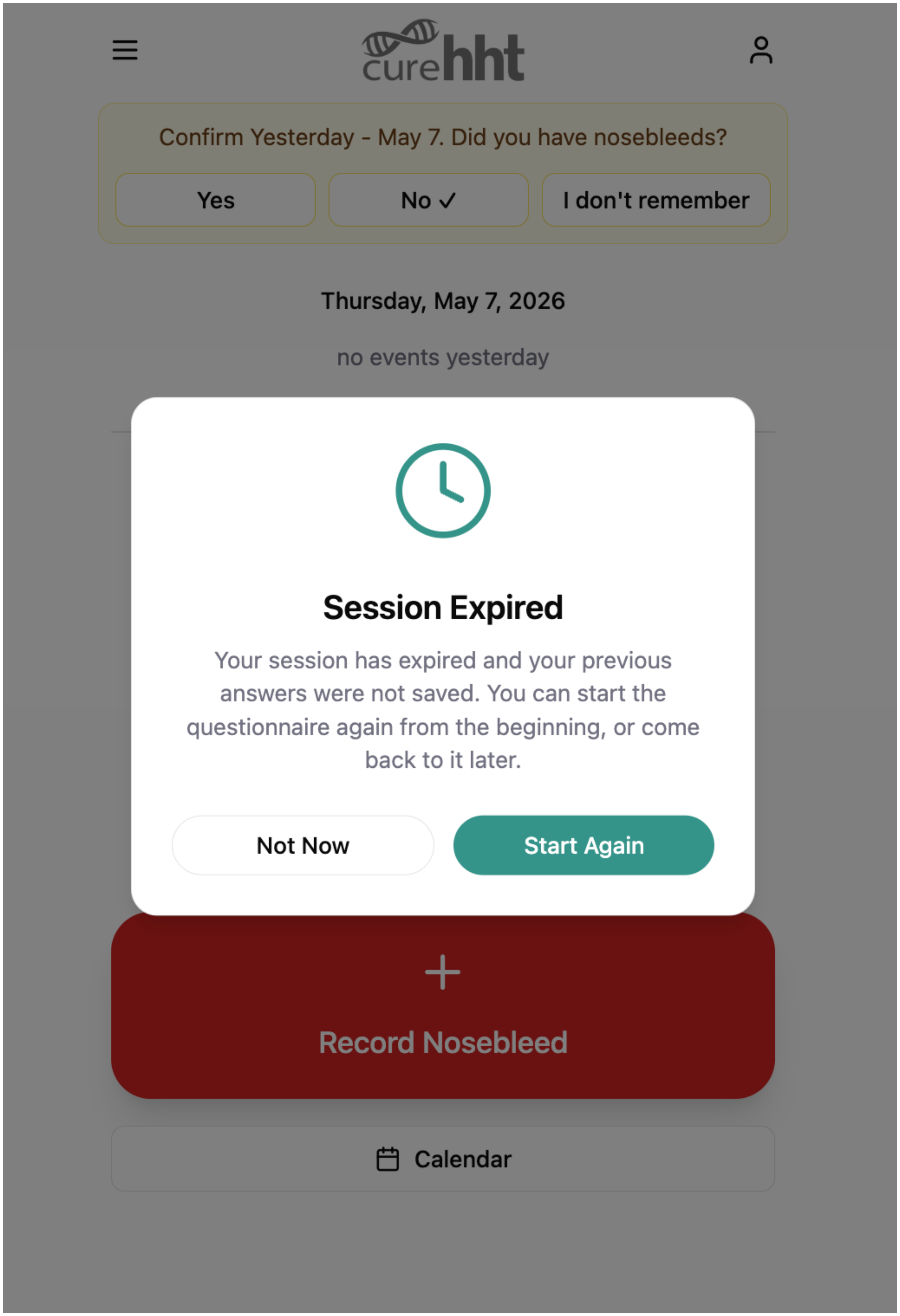
Trigger: **Participant** confirms **Submission** of a **Portal-Sent Questionnaire**.
Consequence summary: Confirms the **Questionnaire** has been submitted and informs the **Participant** that the **Study Coordinator** will review the responses.
Pattern: Acknowledgement Dialog (see §4.1).
Reference: GUI-p00001.


Questionnaire Submitted
Your questionnaire has been submitted successfully. Your Study Coordinator will review your responses.
OK

7.1.6

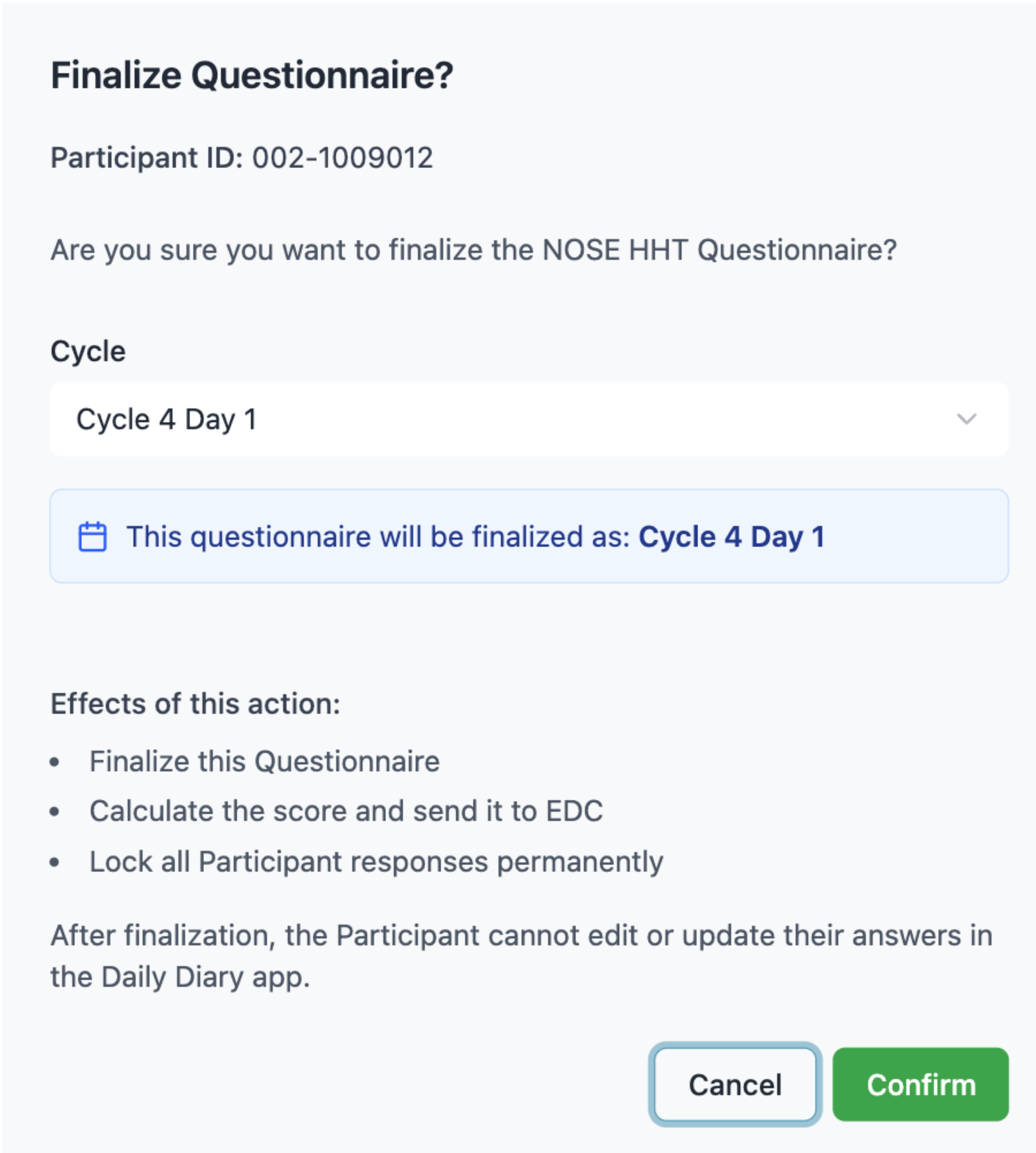
7.1.7 7.3.12 Session Expiry Dialog

Trigger: **Participant** opens a **Portal-Sent Questionnaire** that has reached **Session Expiry**.
Consequence summary: Informs the **Participant** that the session has expired and previous answers were not saved.
Actions: **Start Again** (begin the **Questionnaire** from the **Preamble**), **Not Now** (return to **Main Screen**).
Pattern: Acknowledgement Dialog (see §4.1).
Reference: REQ-p01073, GUI-p00004.


curehht
Confirm Yesterday - May 7. Did you have nosebleeds?
Yes
No ✓
I don't remember
Thursday, May 7, 2026
no events yesterday
Session Expired
Your session has expired and your previous answers were not saved. You can start the questionnaire again from the beginning, or come back to it later.
Not Now
Start Again
Record Nosebleed
Calendar

7.3.13 Questionnaire Finalization Dialog

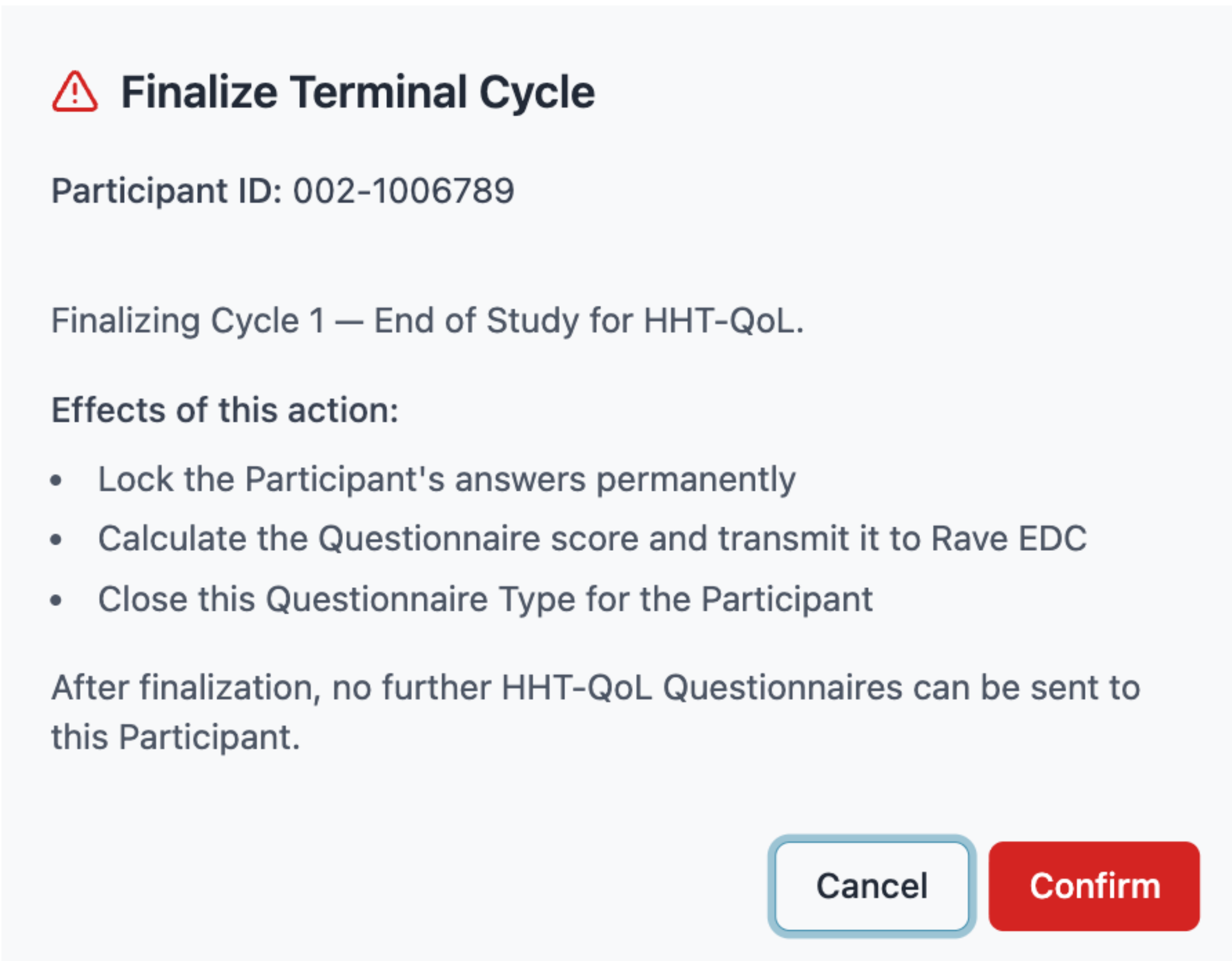
Trigger: Study Coordinator selects **Finalize** on a **Questionnaire Card** with **Delivery Failed** or **Ready to Review** status.
Consequence summary: Finalizes the Questionnaire — locks all Participant responses permanently, calculates the score, transmits it to Rave EDC, and prevents the Participant from editing their answers in the Mobile Application.
Actions: Finalize Questionnaire (apply finalization with the selected Cycle), Cancel (close the dialog with no changes).
Inputs: Cycle dropdown — selectable values are the Current Cycle N Day 1 value, End of Treatment, and End of Study.
Cross-flow: When a Terminal Cycle value (End of Treatment or End of Study) is selected, confirming opens the Terminal Cycle Warning Dialog before finalization is applied.
Pattern: Confirmation Dialog (see §4.1).
Reference: REQ-CAL-p00023 I–M, GUI-CAL-p00007 A–F.



7.3.14 Terminal Cycle Warning Dialog

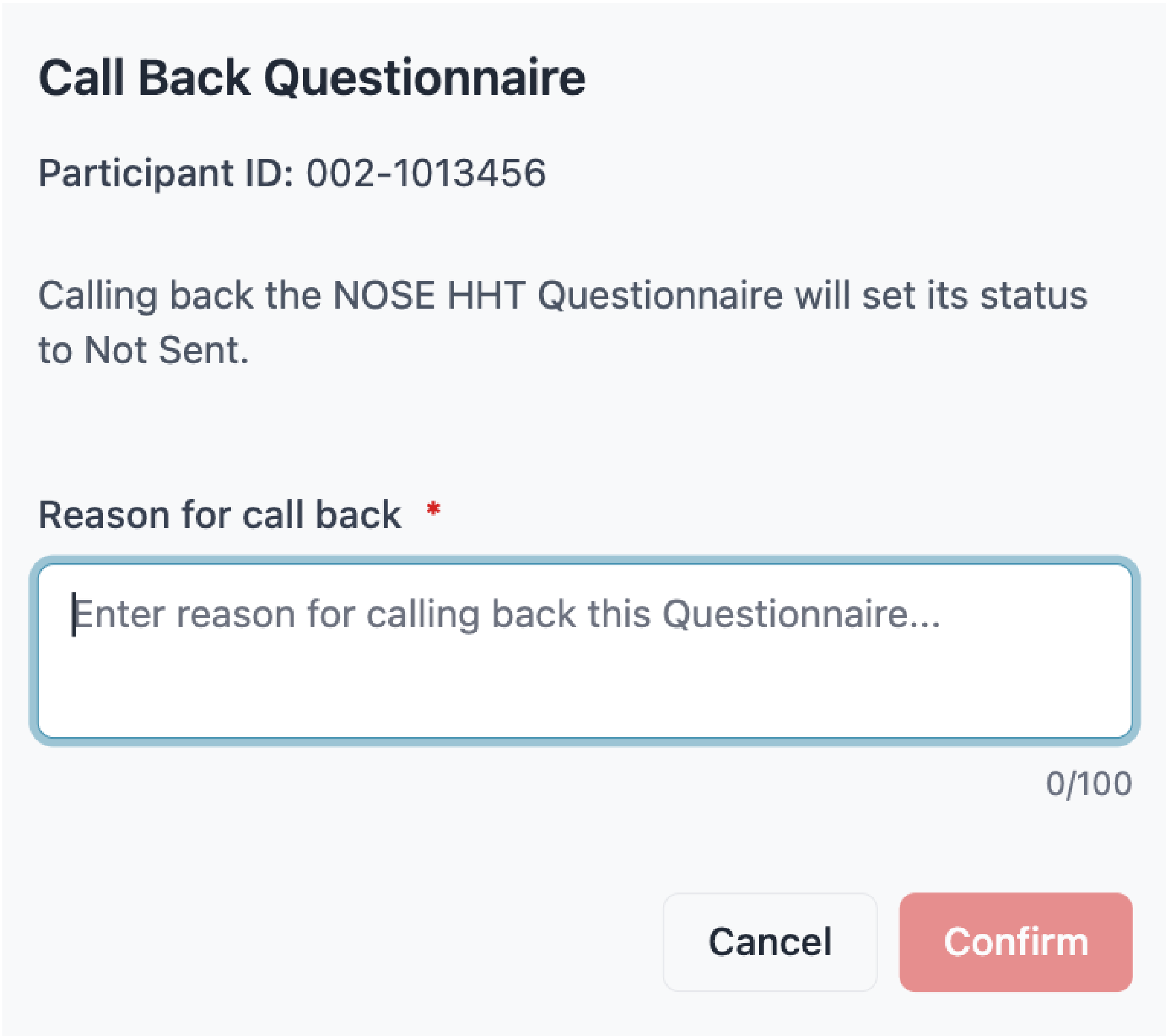
Trigger: **Study Coordinator** confirms finalization of a **Questionnaire** with a **Terminal Cycle** value (**End of Treatment** or **End of Study**) selected in the **Finalization Dialog**.
Target identification: **Participant** ID, **Questionnaire Type**, **Terminal Cycle** being assigned.

Consequence summary: The **Participant**'s answers will be locked, the score will be transmitted to **Rave EDC**, and the **Questionnaire Type** will be closed for the **Participant** so no further **Questionnaires** of that type may be sent.
Cancellation behavior: On cancel, the **Questionnaire** remains unchanged and the **Study Coordinator** is returned to the **Finalization Dialog**.
Pattern: Confirmation Dialog (see §4.1).
Reference: GUI-CAL-p00007.



7.3.15 Call Back Questionnaire — Reason Dialog (Free Text)

Trigger: **Study Coordinator** initiates the call back action on a **Questionnaire** with **Sent**, **Delivery Failed**, or **Ready to Review** status.
Target identification: **Participant** ID, **Questionnaire Type**.
Consequence summary: The **Questionnaire** status will be set to **Not Sent**.
Pattern: Reason Dialog — Free Text (see §4.1).
Reference: REQ-CAL-p00023, GUI-CAL-p00006.



7.2

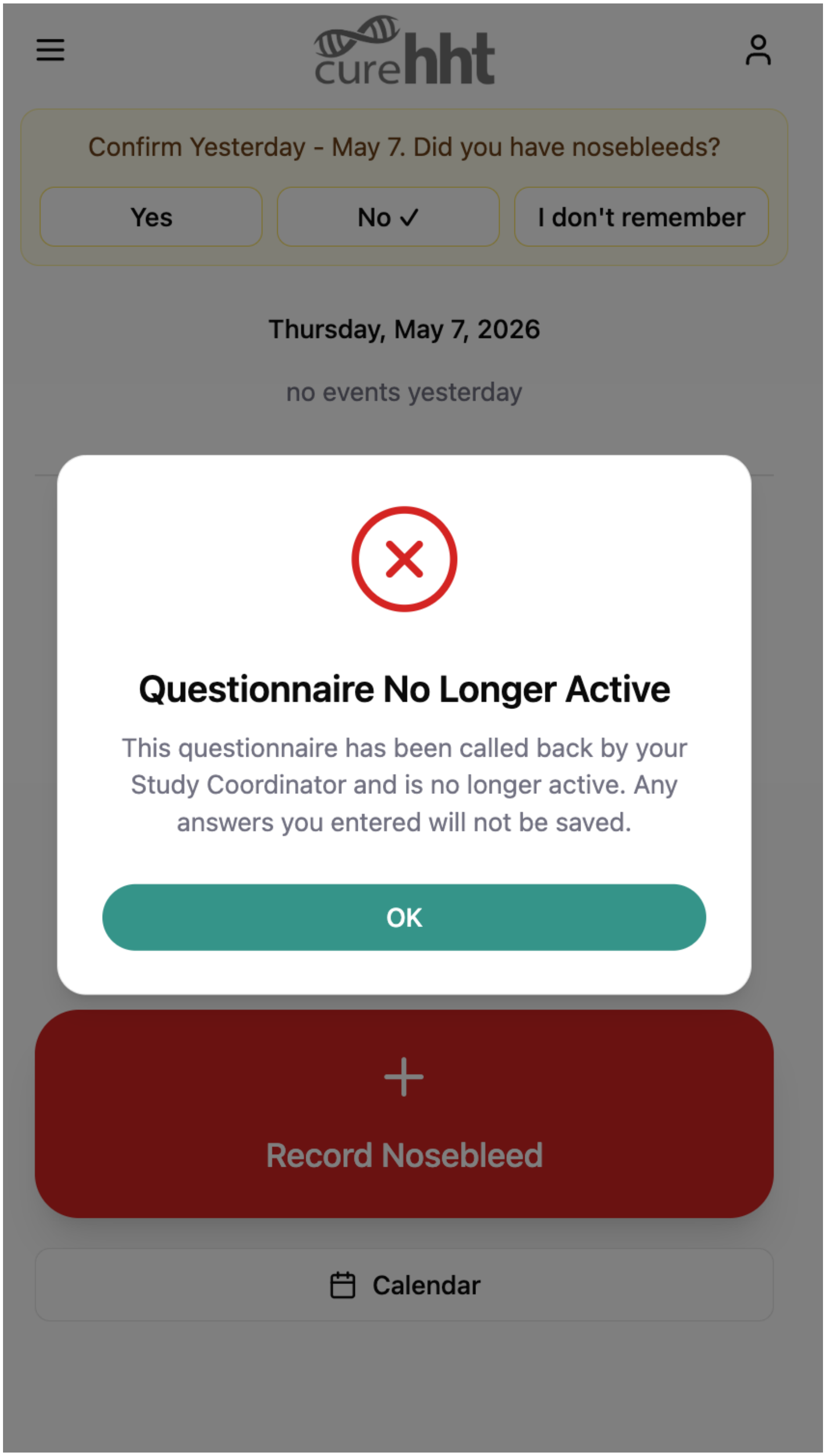
7.3.16 Call Back Notice

Trigger: A **Portal-Sent Questionnaire** assigned to the **Participant** has been called back by the **Study Coordinator**.
Consequence summary: Informs the **Participant** that the **Questionnaire** is no longer active and any answers entered will not be saved.
Presentation contexts:

- The **Portal-Sent Questionnaire** is called back while the **Participant** has it open.
- The **Participant** opens the **Mobile Application** after a call back has occurred.

- The **Participant** attempts to submit a **Portal-Sent Questionnaire** that has been called back while offline.

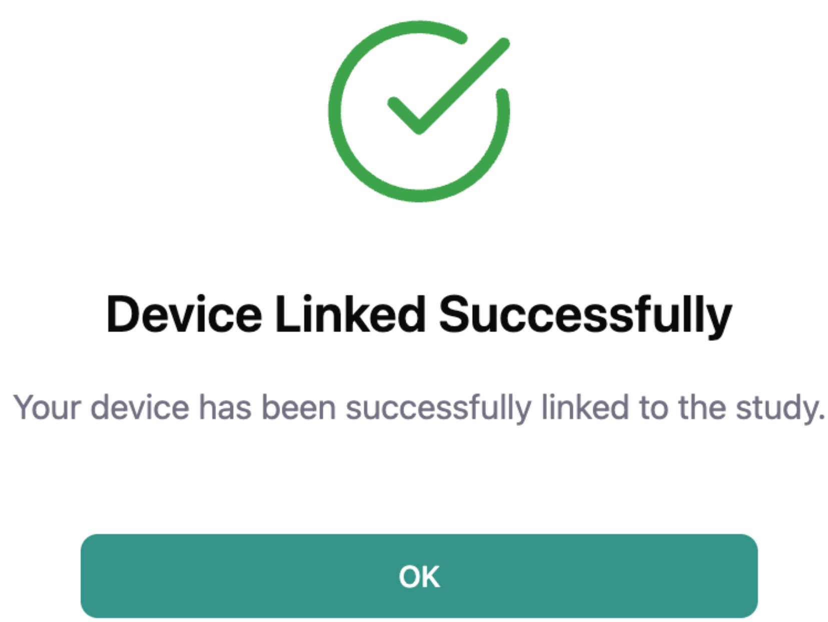
Behavior: Cannot be dismissed by tapping outside its bounds. On acknowledgement, the **Participant** is returned to the **Main Screen** and the corresponding **Questionnaire Task** is removed from the **Task List**.
Pattern: Acknowledgement Dialog (see §4.1).
Reference: GUI-CAL-p07002.


curehht
Confirm Yesterday - May 7. Did you have nosebleeds?
Yes
No ✓
I don't remember
Thursday, May 7, 2026
no events yesterday
Questionnaire No Longer Active
This questionnaire has been called back by your Study Coordinator and is no longer active. Any answers you entered will not be saved.
OK
Record Nosebleed
Calendar

7.3.17 Successful Linking Confirmation

Trigger: A **Participant** successfully submits a valid **Mobile Linking Code** on the **Join the Study** screen.
Consequence summary: Confirms to the **Participant** that the device has been linked to the study and transitions the **Participant** from personal use mode into linked use mode.

Behavior: On acknowledgement, the **Participant** is navigated to the **User Profile** screen.
Pattern: Acknowledgement Dialog (see §4.1).

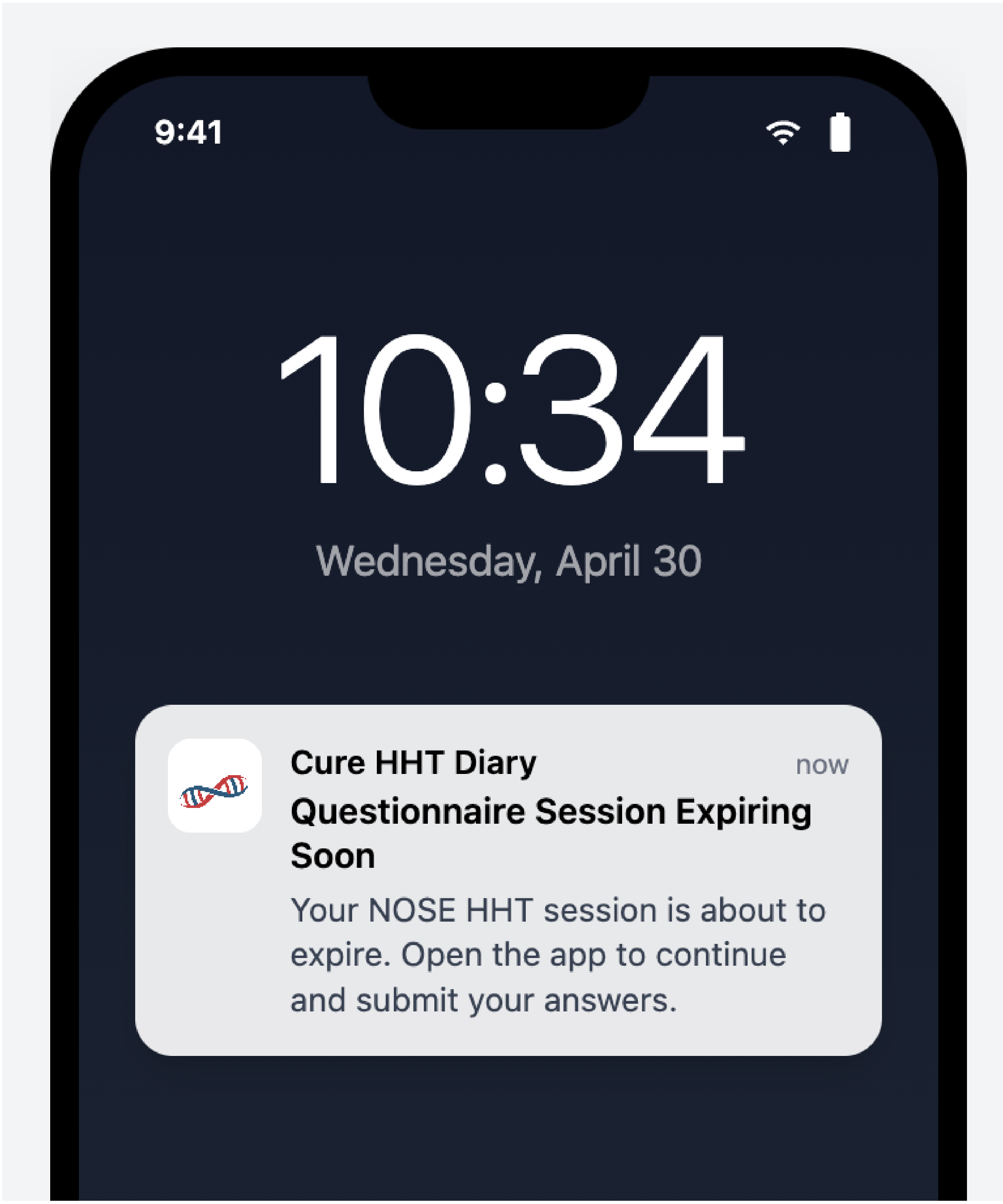


Reference: GUI-p05015.

7.4 Notifications

7.4.1 Timeout Warning Notification

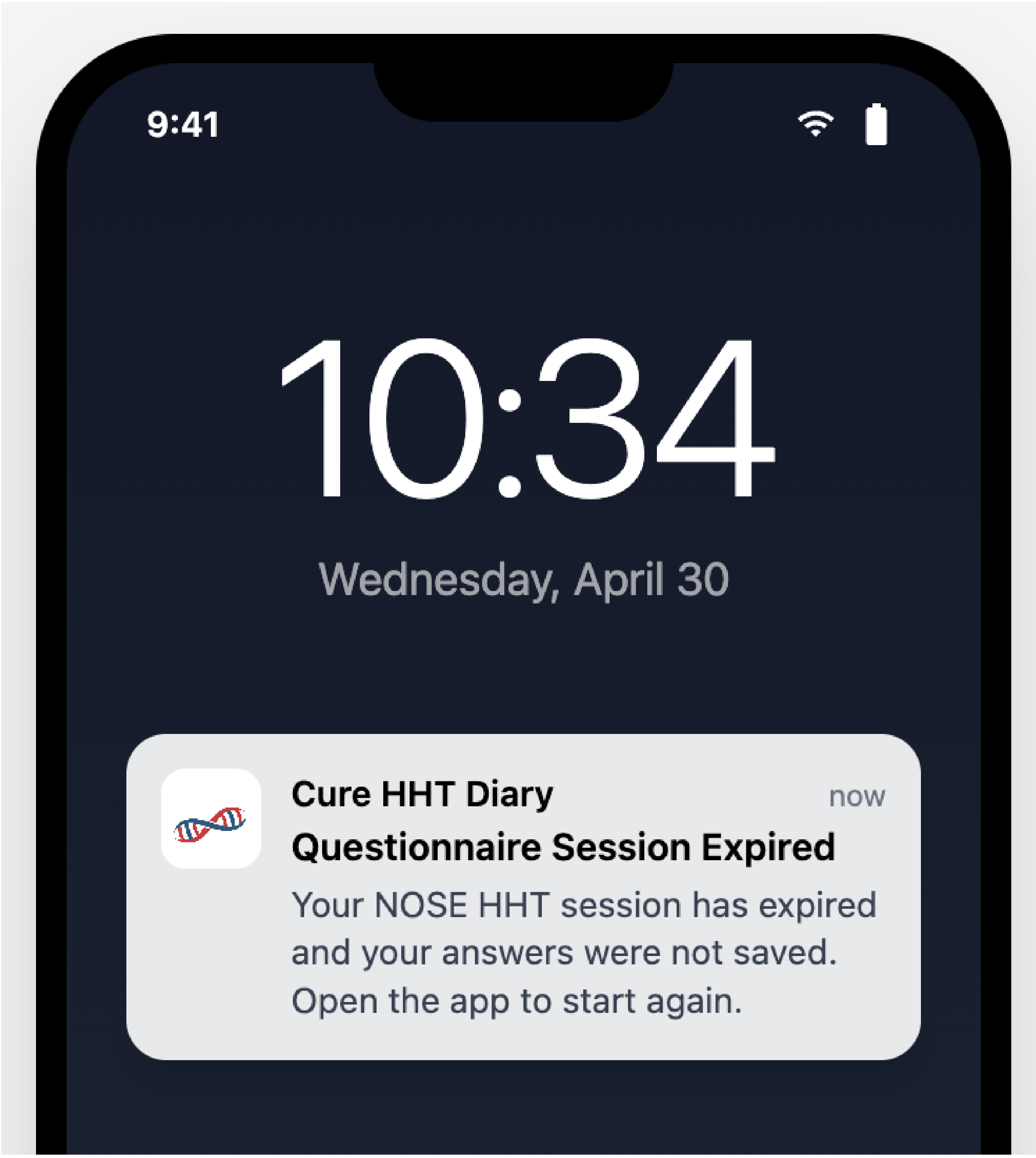
A push notification delivered to the Participant when a configured Questionnaire Session Timeout is approaching expiry. Tapping the notification opens the questionnaire in the Mobile Application. The wording is sponsor-configurable per Questionnaire definition; the default text is *[“Your [Questionnaire Display Name] session is about to expire. Open the app to continue and submit your answers.”]*


9:41
10:34
Wednesday, April 30
Cure HHT Diary
now
Questionnaire Session Expiring Soon
Your NOSE HHT session is about to expire. Open the app to continue and submit your answers.

7.2.1

7.4.2 Session Expiry Notification

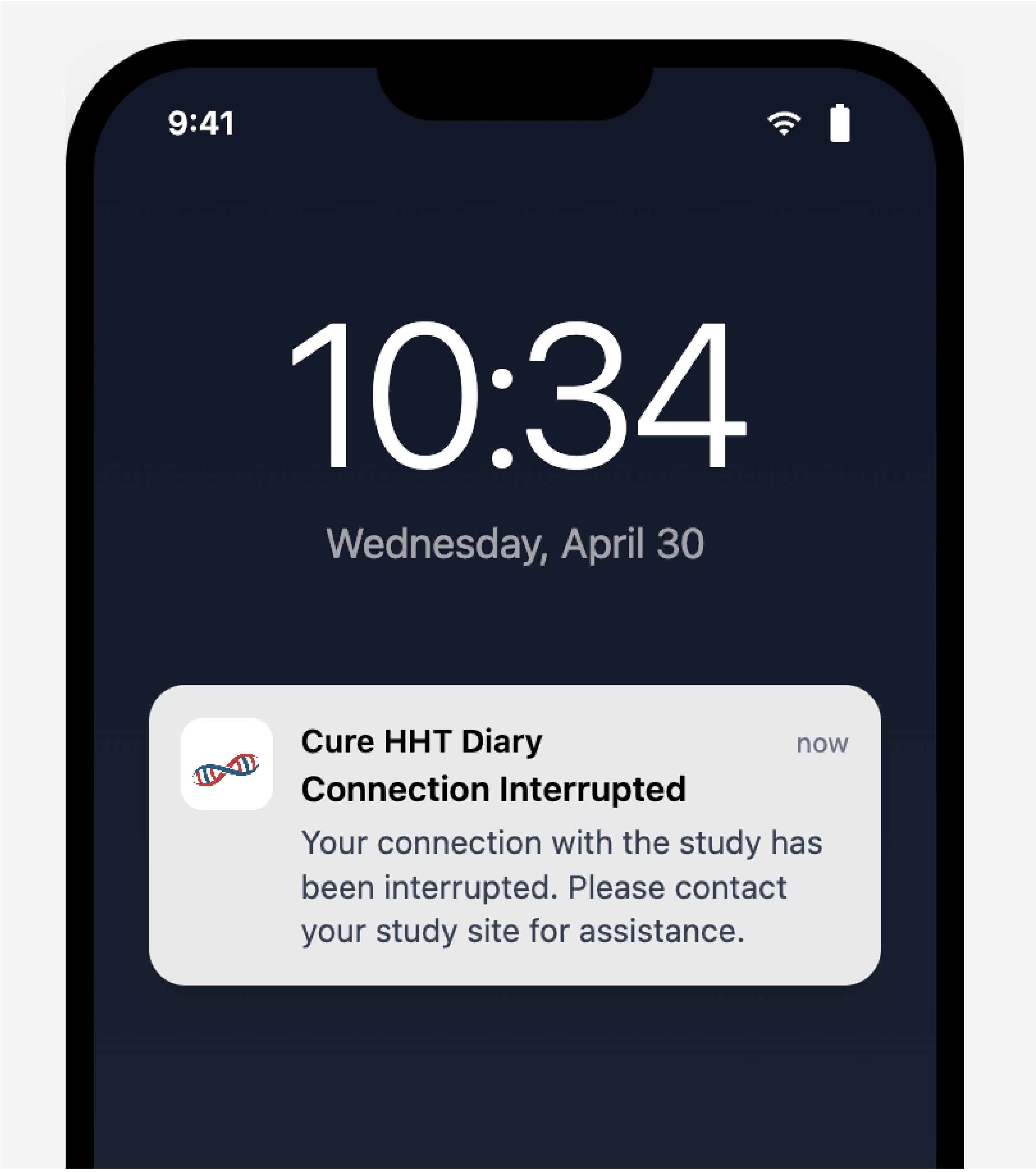
A push notification delivered to the Participant when the configured Session Timeout has been exceeded for an in-progress Questionnaire and the Participant's previously entered answers have been discarded. Tapping the notification opens the Mobile Application; if the Participant returns to the Questionnaire, the Session Expiry Dialog is displayed. The wording is sponsor-configurable per Questionnaire definition; the default text is *["Your [Questionnaire Display Name] session has expired and your answers were not saved. Open the app to start again."]*


9:41
10:34
Wednesday, April 30
Cure HHT Diary
now
Questionnaire Session Expired
Your NOSE HHT session has expired and your answers were not saved. Open the app to start again.

7.2.2

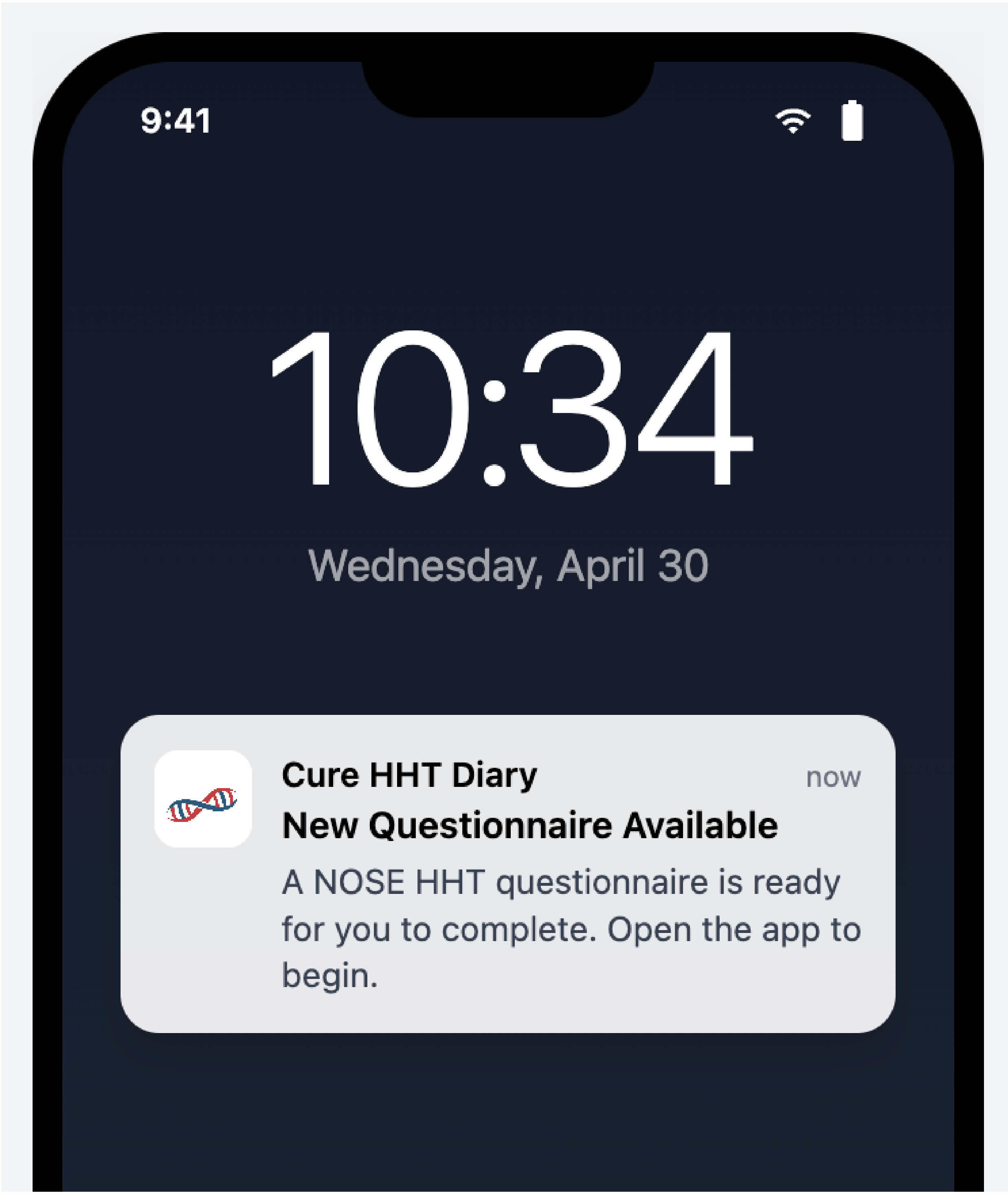
7.4.3 Disconnection Notification

A persistent, non-dismissible in-app notification displayed in the System Notice Area at the top of the Main Screen when the Participant's status is Disconnected. The notification persists until the Participant is reconnected to the Sponsor Portal. The wording is sponsor-configurable; the default text is *[“Your connection with the study has been interrupted. Please contact your study site for assistance.”]*



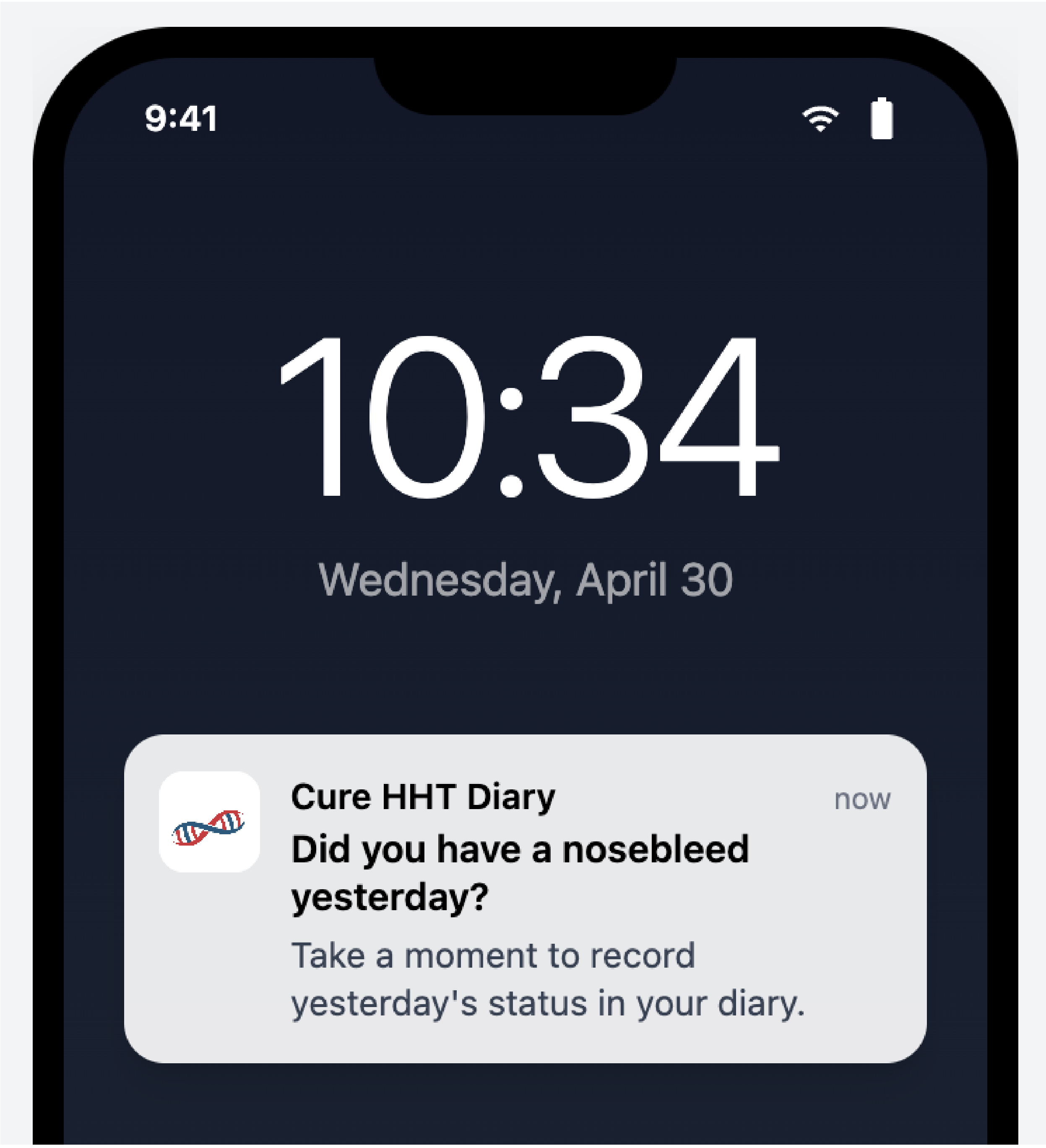
7.4.4 Portal-Sent Questionnaire Notification

A push notification delivered to the Participant when a Study Coordinator sends a Portal-Sent Questionnaire from the Sponsor Portal. Tapping the notification opens the Mobile Application; the Participant accesses the Questionnaire from the Questionnaire Task on the Main Screen. The wording is sponsor-configurable; the default text is *["A [Questionnaire Display Name] questionnaire is ready for you to complete. Open the app to begin."]*



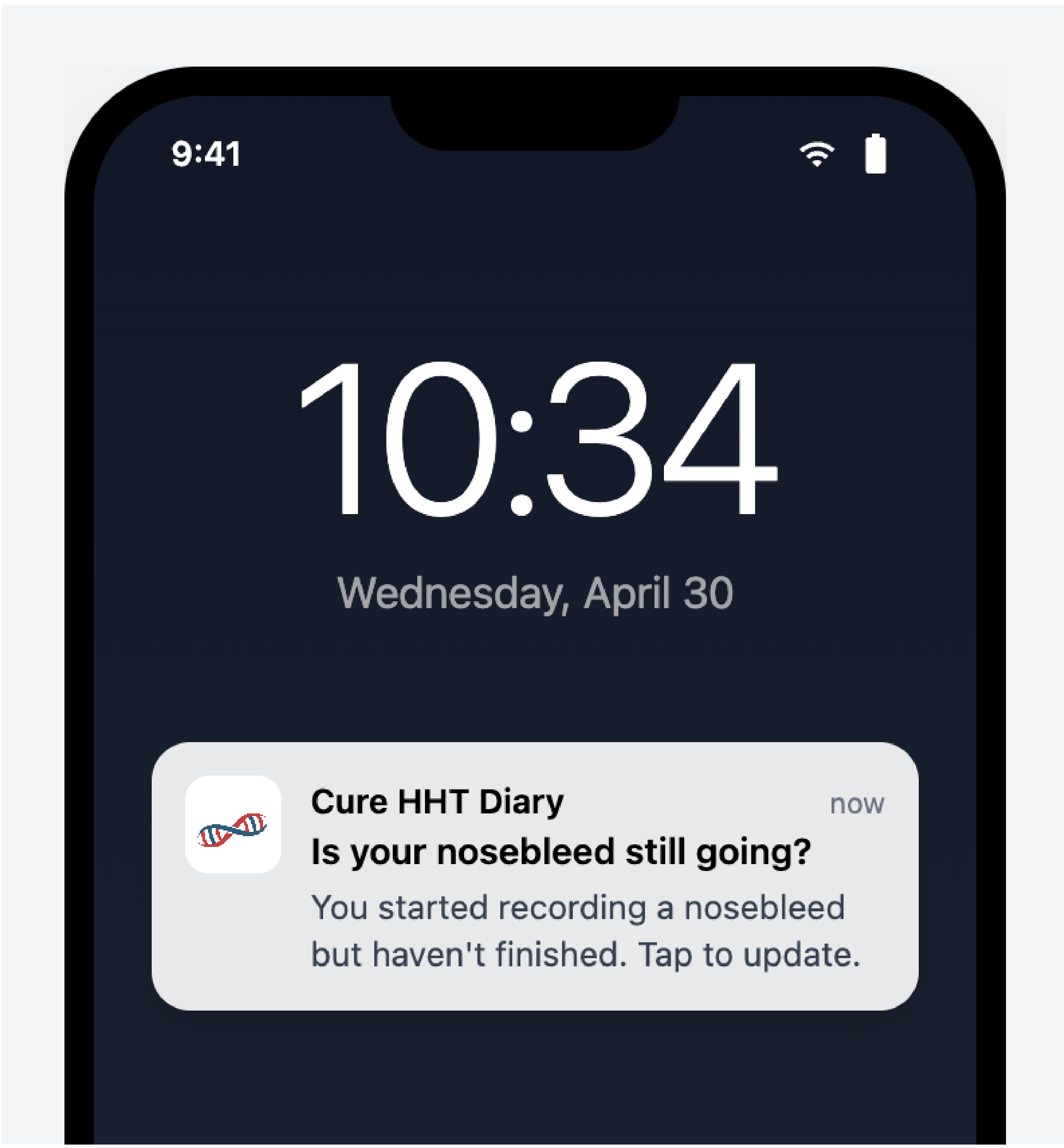
7.4.5 Yesterday Entry Reminder Notification

A push notification delivered to the Participant at the configured Reminder Time when no Daily Status has been recorded for the previous calendar day. Tapping the notification opens the Mobile Application; the Participant accesses the Yesterday Reminder Task on the Main Screen. The wording is sponsor-configurable; the default text is *["Did you have a nosebleed yesterday? Take a moment to record yesterday's status in your diary."]*



7.4.6 Ongoing Epistaxis Event Reminder

A push notification delivered to the Participant when an Incomplete Record has not been interacted with for the configured Reminder Interval. Tapping the notification opens the Mobile Application; the Participant returns to the Incomplete Record to either complete it or confirm the event is still ongoing. The wording is sponsor-configurable; the default text is *[“Is your nosebleed still going? You started recording a nosebleed but haven’t finished. Tap to update.”*



7.4.7 Historical Gap Reminder

A push notification delivered to the **Participant** at the configured Reminder Time when one or more Historical Gaps exist within the editable window. Tapping the notification opens the Mobile Application; the Participant accesses the missing days via the Calendar. The wording is sponsor-configurable; the default text is *[“You have one day without a recorded entry. Tap to review and complete your diary.”]*


9:41
10:34
Wednesday, April 30
Cure HHT Diary
now
Historical Gap Reminder
You have one day without a recorded entry. Tap to review and complete your diary.

Diary Platform - Glossary

Version: 2.0 **Audience**: Product Requirements **Last Updated**: 2026-02-09 **Status**: Draft

> **Purpose**: Establishes canonical terminology for the Diary Platform across all documentation, requirements, and communication.

8.1 Executive Summary

This glossary defines the standard terminology used throughout the Diary Platform project. Consistent terminology ensures clear communication among users, healthcare providers, clinical trial staff, developers, and regulatory auditors.

Key Principles: - Primary purpose: Personal health diary for tracking health observations - Secondary feature: Clinical trial support (extensive but not the core identity) - Use patient-friendly terms in user-facing contexts (avoid jargon like "ePRO", "eCRF") - Use regulatory-compliant terms in formal documentation and submissions - Use precise technical terms in development and architecture documentation - Maintain consistent terminology across all spec/ files and code

---

8.2 System Components

8.2.1 Diary Platform / Diary System

Definition: The complete system comprising all software, infrastructure, and data storage components that enable individuals to record and manage personal health observations, with optional clinical trial support features.

Components: - Diary (mobile application) - Sponsor Portal - Database - Supporting infrastructure and APIs

Primary Purpose: Personal health diary for tracking daily health observations

Secondary Features: Clinical trial support, data sharing with healthcare providers, research contribution

Usage Context: Use when referring to the entire system architecture or in formal documentation.

Avoid: "Clinical Diary Platform" (implies clinical trials are primary), "eSource system", "ePRO platform" (use only in sponsor-specific external documentation)

See: REQ-p00044 (Clinical Trial Compliant Diary Platform)

8.2.2 Diary / Mobile Diary

Definition: The iOS and Android smartphone application that individuals use to record daily health observations.

Preferred Terms: - **"Diary"** - Primary term, patient-friendly, simple and clear - **"Mobile Diary"** - When distinguishing from web portal - **"Diary app"** - Informal, acceptable when context is clear

Usage Context: - User-facing: "Diary" or "the app" - Formal documentation: "Diary mobile application" - Technical documentation: "Mobile Diary" or "Diary app"

Clinical Trial Context: May use "Clinical Diary" when emphasizing regulatory compliance

Avoid: - "ePRO application" (too technical, confusing to users) - "eCRF" (refers to different concept in clinical trials) - "eSource" (refers to the complete platform, not just the mobile app) - "Clinical Diary" as the primary name (clinical trials are a feature, not the identity)

Examples: - "Open the Diary and tap the + button to record a nosebleed" - "The Diary mobile application works offline" - "When used in clinical trials, the Diary serves as an eSource"

See: REQ-p00043 (Diary Mobile Application), prd-diary-app.md

8.2.3 Sponsor Portal / Portal

Definition: The web-based application used by healthcare providers, administrators, and (in clinical trial contexts) investigators, sponsors, auditors, and analysts to review user data, manage accounts, and administer features.

Preferred Terms: - **"Sponsor Portal"** - Primary term, used when distinguishing from mobile Diary - **"Portal"** - Short form when context is clear

Usage Context: Use to distinguish from the mobile Diary application.

Components: - User Data Review - Admin Dashboard - Healthcare Provider Dashboard - Investigator Dashboard (clinical trial feature) - Auditor Dashboard (clinical trial feature) - Analyst Dashboard - Sponsor Configuration (clinical trial feature)

Note: Portal serves both personal health tracking use cases (e.g., healthcare provider reviewing a user's diary) and clinical trial use cases (investigator monitoring trial participants).

Avoid: "Admin app", "web app" (too generic), "Clinical Trial Portal" as primary name

See: REQ-p70001 (Sponsor Portal Application), prd-portal.md

8.2.4 Database

Definition: The PostgreSQL database (hosted on Cloud SQL) that stores all diary entries, audit trails, user accounts, and configuration.

Architecture: - Event Store (immutable audit trail) - Record State (current values, derived from events) - Row-Level Security policies for access control

Usage Context: Use when discussing data storage, schema, or database architecture.

Clinical Trial Context: May be referred to as "Clinical Trial Database" when emphasizing regulatory compliance features.

See: REQ-p00046 (Clinical Data Storage System), REQ-p00003 (Separate Database Per Sponsor), prd-database.md

8.2.5 eSource

Definition: The Diary Platform as a whole, when used as the original electronic source of patient-reported data in a clinical trial (as opposed to transcribing from paper diaries).

Usage Context: Regulatory submissions, protocol documentation, when emphasizing the system is the primary data source in a clinical trial context.

Important: - "eSource" is a regulatory/clinical trial term, not the system's primary identity - Refers to the complete platform (Diary + Database + Portal), not just the mobile app - Use only when discussing clinical trial regulatory compliance

Regulatory Basis: FDA guidance on electronic source documentation

See: REQ-p00010 (FDA 21 CFR Part 11 Compliance)

8.3 Roles and Users

8.3.1 Personal Health Tracking Roles

User

Definition: Any individual who uses the Diary Platform. The default term for someone using the Diary for personal health tracking outside of a clinical trial.

Usage Context: - Personal health tracking: "User" is the standard term - Clinical trial contexts: Use specific role names instead (Study Participant, Investigator, etc.) since "User" is ambiguous when multiple roles exist - Technical documentation: Acceptable as a generic term when role is irrelevant

Note: "User" is acceptable in general documentation. In clinical trial contexts, always prefer specific role names to avoid ambiguity.

Healthcare Provider

Definition: A licensed healthcare professional (physician, nurse, specialist) who reviews a user's diary data to support clinical care. The healthcare provider typically accesses data through the Portal with the user's consent.

Usage Context: Personal health tracking context where the Diary data is shared with a care team.

Distinction from Investigator: A Healthcare Provider reviews diary data for clinical care purposes. An Investigator reviews diary data as part of a clinical trial protocol.

See: REQ-p70001 (Sponsor Portal Application)

Caregiver

Definition: A family member or trusted individual who has been granted delegated access to view or assist with a user's diary entries. Caregivers act on behalf of the user and are subject to the same privacy protections.

Usage Context: Personal health tracking context where the user shares diary access with someone who helps manage their health.

8.3.2 Clinical Trial Roles

Patient / Study Participant

Definition: An individual who uses the mobile Diary to record personal health observations.

Preferred Terms: - **"Patient"** - When emphasizing health tracking context (empathetic, widely understood) - **"Study Participant"** - When the individual is enrolled in a clinical trial (formal clinical trial context)

Usage Context: - Personal health tracking: "User" (see above) or "Patient" - Clinical trial context: "Study Participant" or "Patient" - Formal regulatory submissions: "Study Participant" (when required by convention)

Key Distinction: Not all Diary users are in clinical trials. Many use it purely for personal health tracking or to share with their healthcare providers.

Avoid: "Subject" (outdated, depersonalizing)

Investigator

Definition: A licensed clinical researcher responsible for enrolling patients, managing a clinical trial site, and ensuring protocol compliance. This is a clinical-trial-specific role.

Responsibilities: - Enroll patients using linking codes - Monitor patient engagement and data quality - Send questionnaires and reminders to patients - Review patient diary data for protocol compliance

Access: Site-specific data only (via Row-Level Security)

Avoid: "Site staff", "clinical user", "researcher" (be specific)

See: REQ-p00036 (Investigator Site-Scoped Access), REQ-p00037 (Investigator Annotation Restrictions)

Site

Definition: A physical location (hospital, clinic, research center) where clinical trial activities are conducted. Each site has one or more investigators. This is a clinical-trial-specific concept.

Usage Context: Sites are organizational units within a sponsor's clinical trial. Patients are enrolled at a specific site by an investigator at that site.

See: REQ-p00018 (Multi-Site Support Per Sponsor)

Sponsor

Definition: An organization (pharmaceutical company, foundation, research institution) that uses the Diary Platform to support a clinical trial. This is a clinical-trial-specific concept.

Multi-Sponsor Context: The Diary Platform supports multiple sponsors simultaneously, with complete data isolation between sponsors (separate databases, portals, and configurations).

Examples: Cure HHT Foundation (current), pharmaceutical companies developing HHT treatments

Key Distinction: Sponsors are organizations using the clinical trial features of the Diary Platform. Not all Diary Platform deployments involve sponsors—some may be purely personal health tracking.

See: REQ-p00001 (Complete Multi-Sponsor Data Separation), prd-architecture-multi-sponsor.md

Admin / Administrator

Definition: A staff member with privileges to create and manage user accounts, configure sites, and manage system settings. In clinical trial deployments, the Admin is typically a sponsor employee scoped to one sponsor's data.

Scope: Sponsor-wide access (all sites within one sponsor) in clinical trial context; organization-wide in non-CT deployments.

Avoid: "Sponsor admin" (redundant in CT context), "super admin" (no such role)

See: REQ-p00039 (Administrator Access with Audit Trail)

Auditor

Definition: An independent compliance reviewer with read-only access to all data and audit trails for regulatory compliance verification.

Access: Full read-only access across all sites within a sponsor

Key Distinction: Auditors can view but NEVER modify data.

See: REQ-p00038 (Auditor Compliance Access)

Analyst / Data Analyst

Definition: A researcher or data scientist with read-only access to data for analysis and reporting.

Access: Typically scoped to specific sites (may be sponsor-wide depending on assignment)

Key Distinction: Focused on data analysis, not compliance auditing.

See: REQ-p00022 (Analyst Read-Only Access)

Developer Admin

Definition: A system administrator with infrastructure-level access for deployment, monitoring, and maintenance.

Scope: Cross-sponsor system administration (database backups, infrastructure monitoring)

Data Access: Developer Admins have NO routine access to patient data. A documented break-glass procedure exists for emergency situations only, with full audit trail logging of all access.

Context: Internal operations only, not accessible to sponsors or clinical trial users.

See: ops-security.md for developer admin procedures

8.4 Data and Records

8.4.1 Diary Entry / Health Observation

Definition: A single user-reported record of a health event (e.g., nosebleed episode) including time, severity, duration, and other relevant details.

Preferred Terms: - **"Diary Entry"** - Standard term - **"Health Observation"** - More formal context - **"Entry"** - Short form when context is clear

Usage Context: Use "diary entry" in user-facing communication and general documentation.

Clinical Trial Context: When used in clinical trials, diary entries may capture protocol-specified data elements and serve as ePRO data.

Avoid: "Event" (too generic), "Record" (too technical), "Data point" (depersonalizing)

See: REQ-p00042 (HHT Epistaxis Data Capture Standard)

8.4.2 Event Store

Definition: The immutable log of all changes to diary entries and system data, implementing the Event Sourcing architectural pattern. Every create, update, or delete operation is stored as an event that is never modified or deleted (append-only). Current state is derived by replaying events, providing a complete change history.

Usage Context: Developer and architecture documentation.

Clinical Trial Context: The Event Store implements the Audit Trail required by FDA 21 CFR Part 11 and other regulations.

See: REQ-p00004 (Immutable Audit Trail via Event Sourcing), REQ-p01000 (Event Sourcing Client Interface), prd-event-sourcing-system.md

8.4.3 Record State / Current Values

Definition: The current values of diary entries and other data, derived from the Event Store by replaying all events. In Event Sourcing architecture, this is the "read model" (also known as a Materialized View in database terminology) optimized for queries.

Usage Context: Developer and technical architecture documentation.

Contrast: "Event Store" contains the history; "Record State" contains current values.

See: REQ-p01006 (Type-Safe Materialized View Queries), prd-event-sourcing-system.md

8.4.4 Audit Trail

Definition: The complete, tamper-evident record of all changes to health data, showing who made each change, when, why, and from what device.

Key Attributes: - Who: User identification - What: Change description - When: Timestamp (UTC) - Why: Reason for change (if applicable) - Where: Device/platform information

Usage Context: Compliance documentation, audit procedures, data integrity requirements.

Clinical Trial Context: Required by FDA 21 CFR Part 11, EU Annex 11, and ICH-GCP guidelines for regulatory compliance.

Implementation: Implemented using the Event Store architecture (technical term) to create an Audit Trail (regulatory term).

See: REQ-p00004 (Immutable Audit Trail via Event Sourcing), REQ-p70006 (Comprehensive Audit Trail), prd-clinical-trials.md

8.4.5 Linking Code

Definition: A unique, cryptographically random 10-character code used to securely connect a user's mobile Diary to a service. In clinical trial deployments, linking codes connect a user's device to their enrollment record at a trial site. In non-clinical-trial contexts, a similar mechanism could connect the Diary to a backup or data-sharing service.

Format: 10 characters, alphanumeric, case-insensitive

Purpose: Enables secure device connection without transmitting PII or creating accounts before the link is established.

Usage Context: Patient enrollment workflow (clinical trials), device provisioning.

See: REQ-p70007 (Linking Code Lifecycle Management), REQ-d00078 (Linking Code Validation)

8.4.6 De-identified Data

Definition: Health data with all personally identifiable information (name, email, date of birth, account identifiers) removed, leaving only a participant ID and health observations.

Purpose: Enables research analysis and data sharing while protecting user privacy.

Clinical Trial Context: De-identified data can be shared with research partners; identified data requires explicit consent. Subject to GDPR and HIPAA requirements.

See: REQ-p00016 (Separation of Identity and Clinical Data)

8.5 Architecture and Technical Terms

8.5.1 Multi-Sponsor Isolation

Definition: The architectural pattern ensuring complete data separation between different organizations (sponsors) using the Diary Platform for clinical trials.

Implementation: - Separate Cloud SQL database per sponsor - Separate GCP project per sponsor - Separate web portal per sponsor (sponsor-specific subdomain) - Shared mobile Diary app with automatic sponsor detection

Purpose: Enables a single codebase to serve multiple sponsors while maintaining regulatory compliance and data privacy.

Context: This is a clinical trial feature. Personal health tracking deployments may not use multi-sponsor architecture.

See: REQ-p00001 (Complete Multi-Sponsor Data Separation), REQ-p00008 (Single Mobile App for All Sponsors), prd-architecture-multi-sponsor.md

8.5.2 Offline-First

Definition: An architectural approach where the mobile Diary application functions fully without internet connection, storing data locally and synchronizing to the cloud when connectivity is available.

Benefits: - Users can make diary entries anywhere, anytime - No data loss if internet unavailable - Reduces dependency on network reliability - Improves user experience and app responsiveness

Technical Implementation: Local sembast NoSQL JSON database on device, background sync service.

See: REQ-p00006 (Offline-First Data Entry), REQ-p01001 (Offline Event Queue with Automatic Synchronization), dev-app.md

8.5.3 Event Sourcing

Definition: An architectural pattern where all changes to application state are stored as a sequence of immutable events, rather than overwriting data in place.

Benefits: - Complete history and timeline reconstruction - No data loss - Full audit trail for personal health records - Supports clinical trial regulatory compliance (FDA 21 CFR Part 11) when needed

Technical Details: Every change (create, update, delete) appends an event to the Event Store. Current state is derived by replaying events.

Usage Context: Technical architecture and developer documentation.

Note: Event Sourcing provides value for both personal health tracking (complete history) and clinical trials (regulatory compliance).

See: REQ-p00004 (Immutable Audit Trail via Event Sourcing), REQ-p01000 (Event Sourcing Client Interface), prd-event-sourcing-system.md

8.5.4 CQRS (Command Query Responsibility Segregation)

Definition: An architectural pattern separating write operations (commands that create events) from read operations (queries that read current state).

Implementation in Diary Platform: - **Write side**: Diary entries create events in Event Store - **Read side**: Portal queries read optimized Record State tables - Event Store and Record State are separate but synchronized

Benefits: Optimizes audit trail integrity (write side) and query performance (read side).

Usage Context: Technical architecture documentation.

See: dev-database.md

8.5.5 Row-Level Security (RLS)

Definition: Database-enforced access control that automatically filters data so users can only access records they are authorized to see, based on their role and sponsor/site assignment.

Purpose: Ensures data isolation and role-based access control at the database layer (defense-in-depth).

Example: An investigator at Site A cannot see diary entries from patients enrolled at Site B, even if both sites belong to the same sponsor — enforced by PostgreSQL RLS policies.

Relationship to RBAC: RBAC defines **what actions** users can perform; RLS defines **what data** users can access.

Usage Context: Security architecture, database design.

See: REQ-p00015 (Database-Level Access Enforcement), REQ-p00035 (Patient Data Isolation), prd-security-RLS.md

8.5.6 RBAC (Role-Based Access Control)

Definition: Access control paradigm where permissions are assigned to roles (User, Investigator, Admin, etc.) rather than individual users.

Roles: User, Healthcare Provider, Caregiver, Investigator, Admin, Auditor, Analyst, Sponsor, Developer Admin

Usage Context: Security architecture, access control documentation.

See: REQ-p00005 (Role-Based Access Control), prd-security-RBAC.md

8.6 Compliance and Regulatory Concepts

8.6.1 ALCOA+ Principles

Definition: Data integrity standards for electronic health records, widely adopted in both clinical and non-clinical contexts.

ALCOA+ Acronym: - **A**ttributable - Who created or modified the data - **L**egible - Readable and understandable - **C**ontemporaneous - Recorded at time of occurrence - **O**riginal - First recording, or certified copy - **A**ccurate - Correct and complete - **+C**omplete - All required data present - **+C**onsistent - Chronological order maintained - **+E**nduring - Preserved throughout retention period - **+A**vailable - Accessible for review and audit

Diary Platform Implementation: Event Sourcing architecture directly implements ALCOA+ principles: - **Attributable**: Every event includes user ID and timestamp - **Legible**: All data stored in readable formats - **Contemporaneous**: Events recorded immediately - **Original**: Events never modified (append-only) - **Accurate**: Validation enforced at entry - **Complete**: Full audit trail preserved - **Consistent**: Event Sourcing ensures consistency - **Enduring**: Retention policies enforced - **Available**: Query interfaces provide access

Clinical Trial Context: Required by FDA and EMA for clinical trial records.

Usage Context: Compliance documentation, audit procedures, data integrity requirements.

See: REQ-p00011 (ALCOA+ Data Integrity Principles), prd-clinical-trials.md

8.6.2 Electronic Signature

Definition: The automatic attribution of a data entry or change to a specific user account with timestamp and device information. In the Diary Platform, every diary entry and data modification is automatically signed with user account ID, timestamp (UTC), device information, and a cryptographic hash to prevent tampering.

Usage Context: All diary entries (personal health tracking and clinical trial) are automatically attributed to the user who created them.

Clinical Trial Context: Required by FDA 21 CFR Part 11. Electronic signatures are legally binding, equivalent to a handwritten signature. Users do not manually “sign” entries — the system automatically creates legally binding signatures.

See: REQ-p00010 (FDA 21 CFR Part 11 Compliance), prd-clinical-trials.md

8.6.3 FDA 21 CFR Part 11

Definition: U.S. Food and Drug Administration regulation (Title 21 Code of Federal Regulations Part 11) establishing requirements for electronic records and electronic signatures to be considered trustworthy, reliable, and equivalent to paper records.

Key Requirements: - Audit trails for all data changes - Electronic signatures with cryptographic integrity - Controlled system access - Data validation and integrity checks - Records retention for regulatory-mandated periods

Diary Platform Context: Core compliance framework for clinical trial features.

Usage Context: Regulatory submissions, compliance documentation.

See: REQ-p00010 (FDA 21 CFR Part 11 Compliance), prd-clinical-trials.md

---

8.7 Acronym Reference

Concise definitions of acronyms used across the project, sorted alphabetically. For detailed narrative definitions of ALCOA+, Electronic Signature, and FDA 21 CFR Part 11, see the Compliance and Regulatory Concepts section above.

8.7.1 AE / SAE (Adverse Event / Serious Adverse Event)

AE: Any undesirable medical event occurring in a patient during a clinical trial, whether or not related to the investigational treatment. **SAE**: A severe adverse event resulting in death, hospitalization, disability, or other serious outcomes. Formal AE/SAE reporting is handled through the sponsor's EDC system, not directly through the Diary Platform. SAEs require expedited reporting (typically within 24 hours).

8.7.2 CDISC (Clinical Data Interchange Standards Consortium)

Organization developing international data standards for clinical trial data exchange. Key standards: **CDASH** (data collection), **SDTM** (data submission format), **ADaM** (analysis datasets). Diary ePRO data can be exported in CDISC-compliant formats for regulatory submission.

8.7.3 CRA (Clinical Research Associate)

Staff employed by the sponsor or CRO who monitor trial sites to ensure compliance with GCP, the protocol, and regulatory requirements. CRAs perform source data verification (SDV). In the Portal, CRAs typically have **Auditor** role (read-only).

8.7.4 CRF / eCRF (Case Report Form / Electronic Case Report Form)

The primary tool used by investigators or site staff to record clinical trial data about a patient. **Key distinction**: CRF/eCRF data is entered by clinical trial staff, not by patients. The Diary Platform does not implement traditional eCRFs. Patient diary entries (ePRO data) may be exported to feed into a sponsor's EDC system. **Avoid**: Do not use "eCRF" to refer to patient diary entries.

8.7.5 CRO (Contract Research Organization)

Organization hired by a sponsor to manage aspects of a clinical trial. CRO staff typically have **Auditor** or **Analyst** roles in the Portal.

8.7.6 CTMS (Clinical Trial Management System)

Software for managing operational aspects of a clinical trial. The Diary Platform is not a CTMS — it is specifically a health diary / ePRO data collection system. Sponsors typically use a separate CTMS alongside the Diary Platform.

8.7.7 eCOA (Electronic Clinical Outcome Assessment)

Umbrella term for electronic capture of clinical outcomes: **ePRO** (patient reports), **ClinRO** (clinician assessment), **ObsRO** (observer/caregiver reports), **PerfO** (performance measurements). The Diary is specifically an ePRO tool.

8.7.8 EDC (Electronic Data Capture)

Software systems used to collect, manage, and store clinical trial data electronically, typically including eCRFs. The Diary Platform is not a full EDC system — it provides ePRO data that may be exported to a sponsor's EDC system (e.g., Medidata Rave, Oracle InForm).

8.7.9 EMA (European Medicines Agency)

European Union agency responsible for evaluating and supervising medicines. For EU clinical trials, the Diary Platform complies with EU Clinical Trial Regulation 536/2014 and EU GMP Annex 11.

8.7.10 ePRO (Electronic Patient-Reported Outcomes)

Electronic collection of health information reported directly by the patient, without interpretation by clinicians. The Diary functions as an ePRO tool when used in clinical trials. **Important**: While technically accurate, avoid "ePRO" in user-facing communication. Use "Diary" instead. **See**: REQ-p00043 (Diary Mobile Application)

8.7.11 ESS (Epistaxis Severity Score)

Standardized clinical assessment tool measuring the severity and impact of nosebleeds in HHT patients. Calculated from patient-reported diary data. Specific to HHT clinical trials.

8.7.12 FDA (U.S. Food and Drug Administration)

United States federal agency responsible for regulating pharmaceuticals, medical devices, and clinical trials. **See**: REQ-p00010 (FDA 21 CFR Part 11 Compliance)

8.7.13 GCP (Good Clinical Practice)

International ethical and scientific quality standards governing clinical trial conduct, including data collection, management, and participant protection. Primary guideline: ICH E6(R2). **Note**: "GCP" also stands for "Google Cloud Platform" — context determines meaning.

8.7.14 GDPR (General Data Protection Regulation)

European Union regulation governing data protection and privacy for EU residents. EU users have GDPR rights (access, rectification, erasure, portability). Clinical trial participation may limit some rights for data integrity reasons. Applies based on **user location**, not sponsor location. **See**: REQ-p01061 (GDPR Compliance), REQ-p01062 (GDPR Data Portability)

8.7.15 Google Cloud Platform (GCP)

Cloud infrastructure provider hosting the Diary Platform: Cloud SQL (PostgreSQL database), Identity Platform (authentication), Cloud Storage (backups), per-sponsor GCP projects for multi-sponsor isolation. **Note**: In clinical trial contexts, "GCP" typically refers to Good Clinical Practice.

8.7.16 HHT (Hereditary Hemorrhagic Telangiectasia)

Genetic disorder causing abnormal blood vessel formation, leading to frequent nosebleeds and other complications. The Diary Platform was initially developed for HHT patients but supports any health tracking use case. Current sponsor: Cure HHT Foundation.

8.7.17 HIPAA (Health Insurance Portability and Accountability Act)

United States federal law protecting the privacy and security of individuals' medical information. Key requirement: Business Associate Agreements (BAAs) with service providers. **See**: REQ-p01020 (Privacy Policy and Regulatory Compliance Documentation)

8.7.18 ICF (Informed Consent Form)

Signed document where a patient agrees to participate in a clinical trial. Must include how the Diary will be used, what data will be shared, privacy protections, and right to withdraw. The Diary Platform supports electronic ICF signatures compliant with 21 CFR Part 11.

8.7.19 ICH (International Council for Harmonisation)

Organization developing international standards for clinical trials. Key guidelines: **ICH E6(R2)** (Good Clinical Practice), **ICH E8** (General Considerations for Clinical Studies), plus Quality (Q), Safety (S), Efficacy (E), and Multidisciplinary (M) guidelines.

8.7.20 ICH-GCP / ICH E6(R2)

The foundational international ethical and scientific quality standard for clinical trials involving human subjects. The 2016 revision emphasizes risk-based monitoring and data integrity. The Diary Platform follows ICH-GCP guidelines for data integrity (ALCOA+), audit trails, source data verification, and participant protection. **See**: REQ-p00011 (ALCOA+ Data Integrity Principles)

8.7.21 IMP (Investigational Medicinal Product)

The drug, biological product, or device being tested in a clinical trial. Diary entries help assess the IMP's efficacy and potential side effects.

8.7.22 IND / NDA (Investigational New Drug / New Drug Application)

IND: Application to the FDA to test a new drug in humans. **NDA**: Application for FDA approval to market a drug. Sponsors submit Diary ePRO data as part of IND (during trials) and NDA (for approval) submissions.

8.7.23 IRB / IEC (Institutional Review Board / Independent Ethics Committee)

IRB (U.S.) / **IEC** (international): Independent ethics committee that reviews and approves clinical trial protocols, informed consent forms, and data privacy measures before a sponsor can use the Diary Platform at a trial site. Ongoing approval required for any changes to Diary data collection procedures.

8.7.24 NOSE-HHT (Nasal Outcome Score for Epistaxis in HHT)

Validated patient-reported outcome measure for assessing nosebleed impact in HHT patients. May be collected as a questionnaire within the Diary app during clinical trials. Specific to HHT clinical trials.

8.7.25 PHI (Protected Health Information)

Any individually identifiable health information protected under HIPAA (name, dates, contact information, medical record numbers, health data linked to an individual). PHI is encrypted at rest and in transit, subject to strict access controls via RBAC and RLS policies. **See**: REQ-p00016 (Separation of Identity and Clinical Data), prd-security-data-classification.md

8.7.26 PI / Sub-I (Principal Investigator / Sub-Investigator)

PI: Lead physician at a clinical trial site. **Sub-I**: Physicians or staff assisting the PI. Both have **Investigator** role access in the Portal. **See**: REQ-p00036 (Investigator Site-Scoped Access)

8.7.27 PII (Personally Identifiable Information)

Information that can identify, contact, or locate a specific individual, protected under GDPR and other privacy regulations. Overlap with PHI: all PHI is PII, but not all PII is PHI. **See**: prd-security-data-classification.md

8.7.28 PRO (Patient-Reported Outcome)

Any report of a patient's health condition that comes directly from the patient, without interpretation by a clinician. Diary entries are PROs. When collected electronically via the Diary app, they become ePROs.

8.7.29 SDV (Source Data Verification)

Process of verifying that data in the trial database matches original source documents. For the Diary Platform, diary entries **are** the source (eSource) — no paper transcription occurs. SDV focuses on verifying audit trails, timestamps, and data integrity.

8.7.30 UTC (Coordinated Universal Time)

Primary time standard used worldwide, not affected by time zones or daylight saving time. All database and audit trail timestamps are stored in UTC (with the patient's timezone offset for ePROs). Displayed in the user's local time zone in the Diary app and Portal.

8.8 Deprecated and Avoided Terms

8.8.1 Avoid: "ePRO" (Electronic Patient-Reported Outcomes)

Problem: Technical jargon, not meaningful to users, commonly confused with eCRF, implies clinical trials are the primary purpose.

Use Instead: "Diary" (user-facing and formal documentation)

Exception: May be used in sponsor-specific external documentation (privacy policies, marketing) when required by sponsor for clinical trial context.

8.8.2 Avoid: "eCRF" (Electronic Case Report Form)

Problem: In traditional clinical trials, eCRF refers to forms filled out by investigators, NOT patients. Using eCRF for patient diary entries creates confusion.

Correct Distinction: - **Diary entries**: User-reported health observations - **eCRF**: Investigator-completed case report forms (not implemented in current system)

Use Instead: "Diary entry", "Health observation"

8.8.3 Avoid: "Subject"

Problem: Outdated, depersonalizing terminology from earlier clinical research practices.

Use Instead: "User" (general), "Patient" (health context), or "Study Participant" (clinical trial context)

8.8.4 Avoid: "User" in clinical trial contexts (without qualifier)

Problem: In clinical trial contexts, "user" is ambiguous — could refer to patient, investigator, admin, auditor, or analyst.

Use Instead: Specific role name (Patient, Study Participant, Investigator, Admin, etc.)

Note: "User" is the standard term for individuals using the Diary for personal health tracking (non-clinical trial context).

8.8.5 Avoid: "App" (without qualifier in formal docs)

Problem: Ambiguous — could refer to mobile Diary or web Portal.

Use Instead: "Diary" for the mobile application, "Portal" for web application.

Exception: "App" is acceptable in user-facing communication when context is clear (e.g., "Open the app").

8.9 Sponsor-Specific Terminology

8.9.1 Cure HHT Foundation Context

Disease: Hereditary Hemorrhagic Telangiectasia (HHT)

Primary Health Observation: Epistaxis events (nosebleeds)

Sponsor-Specific Terms: - "Nosebleed Diary" - User-friendly name for the Diary - "Epistaxis events" - Clinical term for nosebleeds - "HHT-specific assessments" - Quality of life questionnaires, severity scores

Note: Terminology may vary for other sponsors. The core system uses generic terms (Diary, health observation) that adapt to sponsor-specific branding.

8.10 Usage Guidelines by Audience

8.10.1 User-Facing Communication

Use: - Diary, the app - Diary entry - Your health observations - Nosebleed (or other specific health event terms) - Your doctor, your care team

Avoid: - ePRO, eCRF, eSource - Event Store, Record State - Clinical trial jargon (unless user is in a trial) - Technical jargon

8.10.2 Healthcare Provider / Clinical Trial Staff

Use: - Diary - Portal - Patient (healthcare context), study participant (clinical trial context) - Investigator, site, sponsor (clinical trial context) - Linking code - Diary entry, health observation

Avoid: - Technical architecture terms (Event Store, CQRS) - Developer jargon

8.10.3 Regulatory and Compliance Documentation

Use: - Diary Platform, Diary System - eSource (when emphasizing clinical trial regulatory context) - Study participant (clinical trial context) - Audit trail, electronic signature - ALCOA+ principles - FDA 21 CFR Part 11 - Event Store (when discussing technical compliance implementation)

Avoid: - "Clinical Diary Platform" as primary name (clinical trials are a feature) - Informal terms without context

8.10.4 Developer and Technical Documentation

Use: - All technical terms (Event Store, CQRS, RLS, offline-first, etc.) - Mobile Diary, Portal, Database - Precise architectural terminology

Avoid: - Generic terms (app, user) without qualification - Regulatory jargon without explanation

---

8.11 Cross-References

Related Documentation: - **prd-architecture-multi-sponsor.md** - System architecture and multi-sponsor isolation (clinical trial feature) - **prd-diary-app.md** - Mobile Diary application features - **prd-portal.md** - Sponsor Portal features - **prd-database.md** - Database architecture, Event Sourcing, audit trails - **prd-privacy-policy.md** - Privacy policy and regulatory compliance documentation (REQ-p01020) - **prd-security.md** - Authentication, authorization, role definitions - **prd-security-RBAC.md** - Role-based access control specifications - **prd-security-RLS.md** - Row-level security policies - **prd-security-data-classification.md** - Data classification and PHI/PII handling - **prd-clinical-trials.md** - Regulatory compliance requirements (clinical trial feature) - **prd-event-sourcing-system.md** - Event Sourcing architecture

---

8.12 Document Control

Version History: - Version 2.0 (2026-02-09) - Consistency rewrite: reframed definitions to lead with personal health diary context, consolidated duplicates, merged regulatory sections, standardized format with REQ cross-references, resolved TODOs, added personal health tracking roles (User, Healthcare Provider, Caregiver), moved REQ-p01020 to prd-privacy-policy.md - Version 1.0 (2025-12-02) - Initial glossary creation

Review Schedule: Review quarterly and update when new terminology is introduced

Approval: This glossary establishes canonical terminology for all spec/ files and project documentation.